



**GREEN
CLIMATE
FUND**

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GCF/B.41/02/Add.03

27 January 2025

Consideration of funding proposals – Addendum III

Funding proposal package for FP255

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya
Country (ies):	Kenya
Accredited Entity:	FAO
Date of first submission:	[2023/07/26]
Date of current submission	[2024/10/31]
Version number	[V.006]



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NOTE TO ACCREDITED ENTITIES ON THE USE OF THE FUNDING PROPOSAL TEMPLATE

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Project	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	Not applicable		
A.4. Result area(s)		GCF contribution	Co-financers' contribution¹
	Mitigation total	Enter number %	Enter number %
	☐ Energy generation and access	Enter number %	Enter number %
	☐ Low-emission transport	Enter number %	Enter number %
	☐ Buildings, cities, industries and appliances	Enter number %	Enter number %
	☒ Forestry and land use	5 %	23 %
	Adaptation total	Enter number %	Enter number %
	☒ Most vulnerable people and communities	55 %	3 %
	☒ Health and well-being, and food and water security	35 %	50 %
	☐ Infrastructure and built environment	Enter number %	Enter number %
	☒ Ecosystems and ecosystem services	5 %	24 %
	A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	4,268,492 tCO ₂ eq over 20 years ²	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)
A.7. Total financing (GCF + co-finance³)	49,992,238 USD	A.9. Project size	Small (Upto USD 50 million)
A.8. Total GCF funding requested	29,215,309 USD		

1 Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

2 The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.

3 Refer to the [Policy of Co-financing](#) of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	<i>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</i>		
	☒ Grant <u>29,215,309</u> ☐ Loan <u>Enter number</u> ☐ Guarantee <u>Enter number</u>	☐ Equity <u>Enter number</u> ☐ Results-based payment <u>Enter number</u>	
A.11. Implementation period	6 years	A.12. Total lifespan	20 years
A.13. Expected date of AE internal approval	6/29/2023	A.14. ESS category	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g., GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	FAO and Agritererra will be Executing Entities for GCF proceeds and co-financing from FAO and the Government of Denmark. The host country, the Government of Kenya, acting through National Treasury, will execute sub-activities co-funded by the Government of Kenya. Please refer to Section 7 of the Feasibility Study Part C for further detail.		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
Kenya remains one of the most vulnerable countries to climate change and increased climate variability due to the continued importance of agriculture as an economic activity. The Lake Victoria Region, comprised of 14 counties, also referred to as Lake Region Economic Bloc (LREB), is one of the main agricultural producing regions of the country. It is facing impacts from the combination of increased temperatures, more frequent and intense droughts, and higher mean annual rainfall, leading to increased flooding and runoff. These are going to impact agriculture, which employs 72% of the working population in LREB. The local adaptive capacity is low to moderate at best, as most smallholder farmers are producing on small areas of land with limited means and without risk-reducing mechanisms. At the same time, low productivity creates pressure on land and natural resources, and cropland expansion is accelerating the degradation of ecosystems. As a result, emissions are also increasing, as marginal and fragile lands are being deforested. This project seeks to implement a transformation of agricultural value chains towards low-carbon, climate resilient pathways through locally-led approach. The project builds on the private sector, smallholder farming networks and the			

strength of the cooperative movement in Kenya and creates a mechanism for technology transfer and upscale, leveraging the knowledge and expertise that exists in national and international farming networks. Focusing on six priority value chains (dairy, poultry, coffee, tea, fruit trees and African leafy vegetables), the project intends to transfer and upscale both adaptation and mitigation technology at each value chain's production, harvesting, processing, and marketing stages.

The project is structured around 4 Outcomes. Outcome 1 builds on the devolution process to improve county-level government services that facilitate upscaling of climate resilient, low carbon, sustainable value chains with a particular focus on how governments support cooperatives and vulnerable farmers. The project also works with county administrations to invest in the community-based sustainable management and restoration of agricultural landscapes as a basis for climate resilience in the agriculture sector (Outcome 2). Activities under Outcome 3 transfer production, harvest, and processing technologies directly to farmers and their organizations in order to reduce emissions, strengthen resilience, and optimize natural resources management in response to climate variability and climate change impacts. Under Outcome 4, the project will improve the farmers' access to markets by working with value chain actors, and replication and scalability will be leveraged through private sector financing to support de-risking through Kenya Commercial Bank (KCB), a Direct Access Entity (DAE) in Kenya, Co-operative Bank of Kenya and Equity Bank.

Additional grant co-financing is mobilized through the Government of Denmark, FAO, and the Government of Kenya. The project will be delivered through FAO as an accredited entity, with FAO and Agriterro as co-Executing Entities, and the Government of Kenya will execute the activities they are co-financing.

The project targets 572,000 direct beneficiaries, who are mainly smallholder farmers (men and women). The financial internal rate of return of the project is 23% with a benefit-cost ratio of 1.11; and 25% and 1.13 respectively for RPC 8.5 and RPC 2.6 scenarios for a 20-years period at 18% discount rate. The project will be able to earn US \$ 10.7 million as a net present value under the same discount rate for 20 years under RPC 8.5 (US \$ 12.9 for RPC 2.6). The project can breakeven the project expenditure plus the beneficiaries' investments after 9 years (pay-back period) of project commencement for both climate change scenarios.

The project will directly contribute to the following targeted results in alignment with the GCF's Updated Strategic Plan (USP-2); T1 (by supporting the country's adaptation and mitigation goals), T4 Food (support 572,000 smallholder farmers adopting low-emission climate-resilient agricultural practices) and T5 Ecosystems (30,540 ha to be conserved, restored and sustainably managed) through T9 locally-led approach. The Project will also support dissemination of climate information to communities, directly contributing to T3 CIEWS. The Project also aligns with T10 (by linking supported cooperatives and SMEs with the partner private banks for loans).

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

B1.1 DEVELOPMENT PROBLEM OF KENYA: AGRICULTURE PRODUCTIVITY, VULNERABLE SMALLHOLDERS AND POVERTY.

1. Agriculture contributes to 26% of Kenya's GDP and 65% of Kenya's total exports⁴. Out of Kenya's 47.6 million people, 18.3 million (38%) are employed in the formal sector⁵ while the remaining 29.3 million people (62%) work in the informal sector, primarily in small scale agriculture and pastoralism⁶. More than two thirds of rural Kenyans who largely depend on agriculture and livestock sectors (67% or 18.7 million people) are experiencing multidimensional poverty. The Lake Region Economic Bloc (LREB), which consists of 14 counties located in the Lake Victoria Basin (LVB), is a vital agricultural region in Kenya and one of the most densely populated regions in Kenya where around 30% of the country's population lives in 5% of the country's territory⁷. In the LREB, 72% of the population relies on agricultural output for food security and livelihoods⁸.
2. Lake Victoria, the largest freshwater lake in Africa and the second largest in the world, is entirely fed by rainwater⁹ and provides the source of freshwater for Kenya. Most of the land surrounding the lake is used for cropland cultivation, in addition to some forested areas (**Error! Reference source not found.**¹⁰). While the LREB is historically known for its high agricultural potential, population pressure and reduced productivity in the agriculture sector have increased the vulnerability of smallholder farmers to climate change and reduce environmental resilience, affecting both the livelihoods of the population and GDP. Most crop farmers in LREB are smallholders, with chronically low productivity¹¹. Livestock farmers also have small herds with limited coping capacity and low productivity. Developed cash crop value chains, such as tea and coffee, are also produced by smallholders pooling their production through farmer organizations and cooperatives. For additional details on agriculture in LREB, please refer to Feasibility Study (FS), Part B, section 1.3.
3. The sector has witnessed reduction in productivity of the crop and livestock systems targeted by the project such as African leafy vegetables, tea, coffee, banana, as well as milk and egg, resulting in unstable incomes, and threatening food security. Overall, increasing rainfall variability has already led 10% of the population in the LVB to suffer chronic food insecurity, requiring short-term emergency food relief and long-term development programs¹². Key development challenges of the agricultural sector exacerbating vulnerability to climate change include poorly developed value chains, limited infrastructure, high rainfall dependence, and limited financial resources. For additional details on socio-economic development indicators, please refer to FS Part B section 1.2.

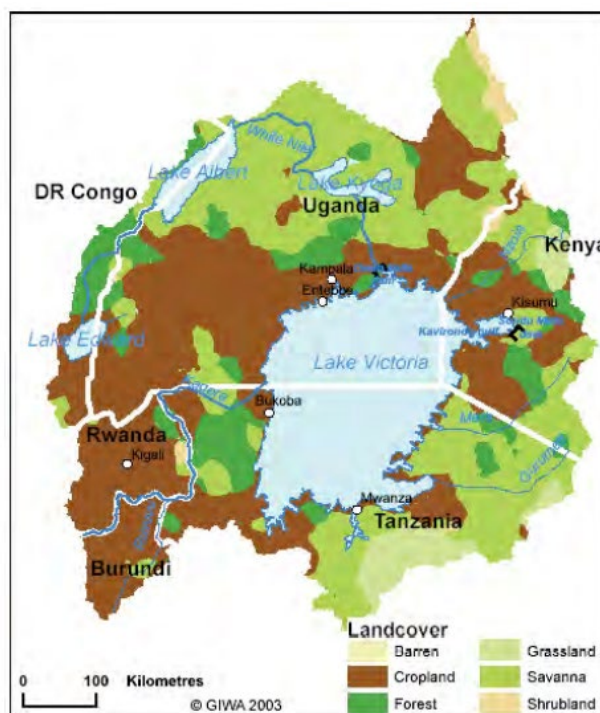


Figure 1: Lake Victoria basin

⁴ KIPPRA, 2013

⁵ Kenya National Bureau of Statistics. 2022. Economic Survey 2022. Retrieved from: <https://www.knbs.or.ke/wp-content/uploads/2022/05/2022-Economic-Survey1.pdf>

⁶ Kenya National Bureau of Statistics. 2022. Economic Survey 2022. Retrieved from: <https://www.knbs.or.ke/wp-content/uploads/2022/05/2022-Economic-Survey1.pdf>; page 54.

⁷ Kenya National Bureau of Statistics, 2019. The share of the population under 19 years is 44 percent with 45 percent male and 43 percent female; those aged between 15 and 64 constitute 49%, and those above 65 years make up 3 percent of the total population.

⁸ FAO, 2023

⁹ USAID. 2018. Lake Victoria Basin: Climate Change Adaptation Strategy and Action Plan. Retrieved from: <https://www.climatechange.org/resources/lake-victoria-basin-climate-change-adaptation-strategy-and-action-plan-2018-2023>

¹⁰ Eric O. Odada, et.al., 2004, Mitigation of Environmental Problems in Lake Victoria, East Africa: Causal Chain and Policy Options Analyses. The boundaries and names shown and the designations used on this map do not imply official endorsement or acceptance by the United Nations.

¹¹ Lake Region Economic Bloc, 2015

¹² USAID. 2018. Lake Victoria Basin: Climate Change Adaptation Strategy and Action Plan, 2018 - 2023. Retrieved from: https://www.climatechange.org/sites/default/files/asset/document/2018_USAID-PREPARED_Lake-Victoria-Basin-Adaptation-Strategy-Action-Plan.pdf

4. LREB aims to coordinate and facilitate service delivery; promote trade linkages, investment, and economic activity; enhance public investment and promote developmental research, and innovation in social planning and coordination of activities by creating communities of practice and centers of excellence. Importantly, the 14 counties of the LREB share a common vision of sustainable development and are mobilized to take joint action against environmental degradation, poverty, and climate change, recognizing potential large-scale impacts on people's livelihoods due to its high population density¹³. However, there has been less climate investment/support provided for this region compared to other regions such as arid and semi-arid (ASAL) regions.

B.1.2 CLIMATE-DEVELOPMENT NEXUS, CLIMATE DRIVERS AND CLIMATE CHANGE IMPACTS¹⁴

5. **The LREB is a vital agricultural region in Kenya with the largest source of freshwater and host almost 30% of the country's population. It plays a major role in ensuring food security and income for the entire country.** It is essential to establish sustainable and climate resilient pathways for agriculture in this region to benefit the national economy. However, there are increased observed and projected climate hazards in the region which negatively impact agricultural production and productivity. Smallholders and the value chain actors are vulnerable to the climate change impacts and have limited capacities in addressing these challenges. Please refer to Annex 2, FS, Section 6.1 for more on the geographical targeting of the project.

B.1.2.1 Climate Hazard

B.1.2.1a LREB Historical Climate Trends

6. **From 1981 to 2010, the annual average of both daily maximum and minimum surface temperatures increased by 0.8°C (0.03°C per year) in the eastern and southernmost areas, and by 1.1°C (0.04°C per year) in the western and northernmost areas**¹⁵ (Figure Error! Reference source not found.). The increases in daily minimums and maximums in the LREB are greater than in the central and easternmost areas of Kenya. These increasing trends from the W5E5 dataset were statistically significant.
7. **Northern Busia and Kakamega, Trans-Nzoia, Bungoma, and Nandi have experienced a total annual precipitation increase of 230 mm (15%) from 1981 – 2010. All other LREB counties have experienced a 60mm (-5%) decrease in precipitation.** While the northern regions have a net increase in precipitation, rainfall during the March-April-May (MAM) season has decreased by 14% and the October-November-December (OND) season has increased by 25% (Figure 9, Annex 2, Feasibility Study, Part A).
8. **During the MAM rain season (the long rain season), dry spells have increased**, with up to 50 dry days, with a maximum duration of 10 consecutive dry days (Figure 14, Annex 2, Feasibility Study, Part A). Southern LREB counties are experiencing more dry days during critical rainy seasons, despite receiving more total precipitation. In the northern LREB counties, the distribution of total precipitation is not equal. The number of days per year with precipitation exceeding 20mm has increased by 5 to 6 days from 1981 to 2010 (Figure 15, Annex 2, Feasibility Study, Part A). ENSO has intensified and increased the frequency of heavy rainfall events, leading to flooding in 2014, 2015, 2018, and 2019¹⁶.
9. **Flooding is historically the most common extreme event in the LREB. However, trends are evolving** (Figure 24, Annex 2, Feasibility Study, Part A). Northern LREB counties at higher latitudes are experiencing growing total precipitation, while central counties at lower altitudes and closer to Lake Victoria are seeing higher daily minimum and maximum temperatures increases. Southern counties are having more dry days. This increase in temperature is leading to 12-24 additional days per year with maximum surface temperatures above 35°C in Kisumu, Busia, and Siaya (Figure 12, Annex 2, Feasibility Study, Part A).

B.1.2.1b LREB Climate Projections - RCP 2.6 and 8.5

10. There is a model agreement (at least 60% of the models agree in the sign of the climate change signal) in daily maximum and minimum surface temperature change.
11. Table 1 and Table 2 below show the projected temperature increases under RCP 2.6 and 8.5 for the LREB counties (Figure 16, Annex 2, Feasibility Study, Part A) in the near term (2039) and long term (2100). All counties are expected to have increased annual average of daily maximum (
12. Table 1) and minimum surface temperatures (Table 2) from 2010 to 2100.

¹³ www.lreb.or.ke

¹⁴ A comprehensive analysis of climate trends, projected climate scenarios and risks and vulnerabilities was conducted by FAO, leveraging globally available data as well as locally sourced data. The full study, including value chain specific climate impact potential analysis and methodology, is found in Annex 2, Feasibility Study, Part A.

¹⁵ All data provided in this section is from an FAO-led climate change impact assessment conducted during 2022 for the project proposal.

¹⁶ MENR. 2016. Kenya National Adaptation Plan 2015-2030. <https://academia-ke.org/library/download/menr-kenya-national-adaptation-plan-2015-2030-2016/>

Table 1 Projected Climate Change Signal in Annual Average Daily Maximum Surface Temperatures in the Lake Region Economic Bloc (average of 6 CORDEX-CORE simulations) from the historical period (1976-2005).

Annual Average Daily Maximum Temperatures	2010 – 2039	2040 – 2069	2070 – 2100
RCP 2.6	+1.0°C	+1.5°C	+1°C
RCP 8.5	+1.0°C	+2.5°C	+4.0°C

Table 2 Projected Climate Change Signal in Annual Average Daily Minimum Surface Temperatures in the Lake Region Economic Bloc (average of 6 CORDEX-CORE simulations) from the historical period (1976-2005).

Annual Average Daily Minimum Temperatures	2010 – 2039	2040 – 2069	2070 – 2100
RCP 2.6	+1.0°C	+1.5°C	+1°C
RCP 8.5	+1.0°C	+2.5°C	+4.5°C

13. There is a high level of agreement among models regarding the climate change signal for total annual precipitation. It is projected to decrease by up to 50mm (-3%) in the mid-term (2040-2069); increase by 50mm (+3%) in the long-term (2070-2099) under RCP2.6; and increase by 100mm (+7%) in the long-term under RCP8.5, when compared to the baseline period of 1976-2005 (Figure 17, Annex 2, Feasibility Study, Part A). The changes in seasonal precipitation for the LREB counties vary (see Table 3, Figure 18, Annex 2, Feasibility Study, Part A). In both RCP 2.6 and 8.5 scenarios, the June-July-August-September (JJAS) rainy season decreases by 2070. However, the October-November-December (OND) rainy season, increases under both scenarios by 2070, with a much greater increase in the RCP 8.5 scenario. Despite the slight decreases in seasonal precipitation by 2040, the increase in total precipitation during the OND rainy season increase the likelihood of seasonal flooding.

Table 3 Projected Climate Change Signal in Total Seasonal Precipitation in the Lake Region Economic Bloc (average of 6 CORDEX-CORE simulations) from the historical period (1976-2005). Empty cells indicate no significant changes.

Total Seasonal Precipitation	2010-2039	2040 – 2069	2070 – 2100	Seasons
RCP 2.6		+7%	+7%	OND
		-5%		MAM
		-10%	-10%	JJAS
RCP 8.5	+7%	+7%	+15%	OND
	-5%	-5%		MAM
		-10%	-10%	JJAS

14. Extreme heat events, dry spells, and heavy rainfall events are projected to increase in the future in the LREB. Under RCP2.6, the annual number of extreme heat days ($T_{max} > 35^{\circ}C$) is expected to remain stable at 20 days as in the baseline period. Under RCP8.5, the number of extreme heat days will increase by 20 days by 2040 and 50 days by 2100 for all LREB counties except for Bomet, Nandi, Kericho, and Trans-Nzoia¹⁷ (Figure 20, Annex 2, Feasibility Study, Part A). Both RCP 2.6 and 8.5 show an increase in the annual number of dry days. Under RCP 2.6, the number of dry days will increase from 160 to 210 days by 2040 and to 220 days by 2100 (Figure 21, Annex 2, Feasibility Study, Part A). Seasonally, there will be 5 more dry days in the MAM season under both RCP2.6 and RCP8.5 by 2100. This will reach 18-20 days for Trans-Nzoia, Bungoma, Busia, Kakamega, and Nandi; and 12 to 16 days for Siaya, Vihiga, Kisumu, Kericho, Homa Bay, Migori, Kisii, Nyamira, and Bomet counties. Both RCP 2.6 and 8.5 show an increase in, the number of heavy rainfall events per year which will average around 10 days by 2040 and 20 days by 2100, compared to the current average of 5 days (Figure 23, Annex 2, Feasibility Study, Part A).

B.1.2.2 Climate exposure and vulnerability: observed and projected climate change impacts¹⁸

15. Smallholders are the beneficiaries of this project and are vulnerable to the current and future impacts of climate variability and climate change. The smallholders in the LREB are characterized as follows (please refer to Annex 2, FS, Part B, section 6.1.3 for details on selection of participants):

- Working small land parcels, typically less than 2 acres.
- Depending on a single or few commodities for livelihoods with limited diversification.
- With limited access to off farm employment.

¹⁷ All these counties are high-altitude, suggesting that elevation may mitigate the number of extreme heat days and by extension, annual surface temperature increases.

¹⁸ As per IPCC, vulnerability refers to "The propensity or predisposition to be adversely affected. Vulnerability encompasses a variety of concepts and elements including sensitivity or susceptibility to harm and lack of capacity to cope and adapt".

- Dependent on low input, rain-fed agriculture.
- Living in households of on average 5 persons.
- With low access to asset ownership.

16. **Smallholder farmers dominate Kenya's agricultural sector¹⁹ comprising 72% of the total working population in the LREB.** However, these farmers face chronic low productivity and limited alternative livelihoods, working on small land parcels with limited diversification, depending on low input, rain-fed production of a single or few commodities for livelihoods, with limited access to off farm employment and asset ownership²⁰. Even in developed cash crop value chains, such as tea and coffee, smallholders participate by pooling their production through farmer organizations and cooperatives. Further, the adoption of improved technologies and practices among smallholders, such as improved seed varieties, mechanization services, proper fertilizer use, input utilization, and access to extension services remains relatively low. Production systems are typically extensive and have high impact on natural resources per unit of output (please refer to Annex 2, FS, Part B, section 6.1.3 for details on selection of participants).
17. **Smallholders' livelihoods in the LREB are highly vulnerable to increasing temperatures to inter-annual rainfall anomalies. These anomalies result in dry spells, drought events, extreme rainfall and flooding, which cause compounded socio-economic impacts across the country. These impacts lead to losses of about 8% of the GDP every five years²¹.** For example, a 2007 study on the economic impacts of climate change on crops in Kenya found that decreased productivity from increasing temperatures could range between 32\$/ha and 178\$/ha²². The severe drought from 1998 to 2000 caused \$2.8 billion USD in losses through damages to crop yields, livestock production, and freshwater availability²³. Projected increasing maximum and minimum temperatures, along with changes in the length and intensity of the long rainy season (MAM), which is the main growing period responsible for 26% of the GDP, will have significant impacts on Kenya's rainfed agricultural and economic systems. These impacts particularly affect smallholder agri-businesses, leading to significant losses in revenue²⁴. Furthermore, riverine floods are responsible for GDP losses of up to 5.5% and 60% of weather-related casualties²⁵. The country lacks climate-proofed infrastructure, such as roads, bridges, dams, and water pipelines, which worsens the impacts of flooding.
18. In the 14 counties, **smallholders face high levels of exposure and sensitivity to climate variability and climate change. This is due to limited access to assets and services**, such as water (on average less than 30% have access to clean water and sanitation), food (on average, 16% of children suffer from malnutrition and 22% exhibit stunting), and energy (only 33% have access to electricity)²⁶. **Gender and cross-sectional inequalities persist**, with women, youth, and persons living with disabilities being more significantly affected by vulnerability. In general, women and youth earn less income from agriculture and often go unpaid for their labor. They are more likely to engage in informal economic activities, have less secure access to land, and have limited access to capital. Both in farmer organizations and in households, women and youth often have less influence on economically significant decision-making.
19. Crop and livestock production systems and related value chains in the LREB are vulnerable to climate change. Extreme weather impacts go beyond primary production and affect the entire value chain. Farmers, in particular, experience financial and income losses due to price disruptions and the loss of crops. Observed and projected climate hazards and their impacts are summarized in **Error! Reference source not found.a,b.** below and further detailed in the Feasibility study (Tables 8 to 15, Annex 2, Feasibility Study, Part A).

Table 4a. Summary of observed and projected climate hazards along the Lake Region Economic Bloc and their impacts on selected crop value chains.

¹⁹ <http://www.kenyagreece.com/sites/default/files/agricultural-sector-ds-2020.pdf>

²⁰ Lake Region Economic Bloc, 2015

²¹ World Bank Group. WBG. 2021. Climate Risk Profile: Kenya (2021): The World Bank Group.

https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15724-WB_Kenya%20Country%20Profile-WEB.pdf

²² Kabubo-Mariara and Karanja. 2007. The Economic Impact of Climate Change on Kenyan Crop Agriculture: A Ricardian Approach.

<https://openknowledge.worldbank.org/bitstream/handle/10986/7276/wps4334.pdf?sequence=1&isAllowed=y>

²³ World Bank Group. WBG. 2021. Climate Risk Profile: Kenya (2021): The World Bank Group.

https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15724-WB_Kenya%20Country%20Profile-WEB.pdf

²⁴ Ayugi, Brian Odhiambo, Wang Wen, and Daisy Chepkemai. "Analysis of Spatial and Temporal Patterns of Rainfall Variations over Kenya." *Journal of environment and earth science* 6 (2016): 69-83.

Kenya Meteorological Department, 2021. Extreme climate events in Kenya 2011 to 2020.

https://meteo.go.ke/sites/default/files/downloads/STATE%20OF%20THE%20%20CLIMATE%202020_14042021.pdf

²⁵ Kenya National Adaptation Plan: 2015-2030, Government of Kenya, July 2016.

https://www4.unfccc.int/sites/NAPC/Documents%20NAP/Kenya_NAP_Final.pdf

²⁶ Figures calculated from data contained in Table 1, Annex 2, FS, Part B Section 1.

Climate hazards (statistically significant changes)	Observed climate impacts	Future climate impacts
Temperature: Observed and projected increasing average maximum and minimum temperatures. Observed increase in extreme heat events in central counties. Projected increasing extreme heat events across the region.	Input supply, production, and harvest: High evapotranspiration risks. Loss of arable land. Observed coffee berry borer spread ^{27, 28} . Early flowering, flower abscission, decreased photosynthetic capacities in cowpeas ^{29, 30} , and spoilage, reducing the shelf-life of vegetables. Post-harvesting, handling, and trade: High temperatures and relative humidity: moderate to severe impacts on mycotoxin contamination during coffee storage ³¹ .	Input supply, production, and harvest: Future projections of FAO AEZ simulations: Decreased productivity of tea (10-30%), coffee (5-10%), cowpeas (5-10%), and banana (5-10%), under rainfed conditions, by mid-century under RCP2.6-RCP8.5, respectively, in the LREB. Decreased arabica coffee growing areas: crop unsuitability below 1700masl (coffee) and 2000masl (tea) ³² . Higher temperatures and dry spells will increase water demand, restricting banana cultivation to areas projected to receive more rainfall ³³ . Increase the likelihood of diseases like stem end rots, body rots (banana ³⁴), mold, mildew, and other pathogens.
Precipitation – Drought: Observed increasing dry days in southern counties and projected increasing dry days across the region. High interannual rainfall variability, negative rainfall anomalies influenced by ENSO and IOD phenomena during the OND season.	Input supply, production, and harvest: Drought and moisture stress: major impacts on tea ³⁵ and fruit trees ³⁶ . Reduced annual growth rate. Reduced soil moisture, soil compaction. Water scarcity and quality degradation through higher concentration of contaminants. Loss of arable land.	
Precipitation – Heavy rainfall and flooding: Observed and projected increasing heavy rainfall events. High interannual rainfall variability, positive rainfall anomalies influenced by ENSO and IOD phenomena during the OND season.	Input supply, production, and harvest: Heavy rainfall, hailstorms, and flooding: major impacts on tea, coffee, fruit trees, and African Leafy Vegetables. Water runoff, soil erosion and water logging, nutrient leaching, pests, and disease outbreaks. Hailstorms damage thousands of kilograms of tea production in one single day ³⁷ . Post-harvesting, handling, and trade: Food spoilage at storage, damaged transportation infrastructure, reduced food value at markets.	

Table 5b. Summary of observed and projected climate hazards along the Lake Region Economic Bloc and their impacts on selected livestock value chains.

Climate hazard (statistically significant changes)	Observed climate impacts	Future climate impacts
Temperature:	Input supply and production: Decreased productivity (meat, milk, eggs quantity and quality) ³⁸ .	Input supply and production:

²⁷ Pham et al., 2019. The impact of climate change and variability on coffee production: a systematic review. Climatic Change. <https://doi.org/10.1007/s10584-019-02538-y>

²⁸ Getachew WeldeMichael, Demelash Teferi, "The Impact of Climate Change on Coffee (Coffea Arabica L.) Production and Genetic Resources" International Journal of Research Studies in Agricultural Sciences (IJRSAS), 2019; 5(11), pp. 26-34, <http://dx.doi.org/10.20431/2454-6224.0511004>

²⁹ Mohammed et al., 2021. Effect of climate variation on the yield of cowpea (Vigna unguiculata). Vol. 17(3), pp. 456-462, March, 2021 DOI: 10.5897/AJAR2020.14960 Article Number: A77ACC766334 ISSN: 1991-637X

³⁰ Barros et al. 2021. Selection of cowpea cultivars for high temperature tolerance: physiological, biochemical and yield aspects. Physiol Mol Biol Plants (January 2021) 27(1):29–38 <https://doi.org/10.1007/s12298-020-00919-7>

³¹ FAO. 2006. Storage of Coffee: Good Hygiene Practices along the coffee chain. Rome. [www.ico.org/projects/ Good-Hygiene-Practices/cnt/cnt_sp/sec_3/docs_3.3/ Storage.pdf](http://www.ico.org/projects/Good-Hygiene-Practices/cnt/cnt_sp/sec_3/docs_3.3/Storage.pdf).

³² Adhikari, U. et al. 2015. Climate change and eastern Africa: a review of impact on major crops. Food and Energy Security. 4(2). <https://doi.org/10.1002/fes3.61>

³³ P.J.A. van Asten, A.M. Fermont, G. Taulya. 2011. Drought is a major yield loss factor for rainfed East African highland banana, Agricultural Water Management, Volume 98, Issue 4.

³⁴ Yahia, E. 2012. Avocado. In: Crop Post-Harvest: Science and Technology. <https://www.researchgate.net/publication/277697648>

³⁵ Cheserek, B. et al. 2015. Analysis of Links between Climate Variables and Tea Production in the Recent Past in Kenya. Donnish Journal of Research in Environmental Studies Vol 2(2) pp. 005-017 March, 2015. <http://www.donnishjournals.org/djres>

³⁶ Sabiiti et al., 2016. Empirical Relationships between Banana Yields and Climate Variability over Uganda. 7:03-13. Journal of Environmental and Agricultural Sciences (ISSN: 2313-8629)

³⁷ CGIAR-CIAT. Ministry of Agriculture, Livestock, Fisheries and Cooperatives. 2022. Kenya County Climate Risk Profiles. <https://ccafs.cgiar.org/resources/publications/kenya-county-climate-risk-profiles>

³⁸ Ngigi, M.W., Mueller, U. & Birner, R. Livestock Diversification for Improved Resilience and Welfare Outcomes Under Climate Risks in Kenya. Eur J Dev Res 33, 1625–1648 (2021). <https://doi.org/10.1057/s41287-020-00308-6>

Observed and projected increasing average maximum and minimum temperatures. Observed increase in extreme heat events in central counties. Projected increasing extreme heat events across the region.	Weather-related hazards: expansion and distribution of pathogens³⁹. Decreased feed intake leads to energy deficits, decreased fertility rates, animal fitness and longevity, and animal mortality⁴⁰. Post-harvesting, handling, and trade: Food spoilage at storage, damaged transportation infrastructure, reduced value at markets. Increased proliferation of spore-forming bacteria and milk evaporation⁴¹. High risk of egg spoilage⁴².	A projected increase in the number of consecutive extreme heat days in the LREB is expected to cause substantial heat stress on livestock, with declines in milk, poultry, and egg yields⁴³.
Precipitation – Drought: Observed increasing dry days in southern counties and projected increasing dry days across the region. High interannual rainfall variability, negative rainfall anomalies influenced by ENSO and IOD phenomena during the OND season.	Input supply and production: Drought stress on dairy and eggs production. Higher animal mortality. Indirect impacts on grassland, reduced quality and quantity of feed and water sources⁴⁴. Reduction of calving rates.	
Precipitation – Heavy rainfall and flooding: Observed and projected increasing heavy rainfall events. High interannual rainfall variability, positive rainfall anomalies influenced by ENSO and IOD phenomena during the OND season.	Input supply and production: Major impacts on poultry production. Weather-related hazards: water-borne diseases and increased prevalence of endo- and ecto-parasites. Indirect impacts: reduced quality and quantity of feed and water sources. Post-harvesting, handling, and trade: Food spoilage at storage, damaged transportation infrastructure, reduced value at markets. Increased proliferation of spore-forming bacteria⁴⁵. Spoilage, reduced sales and lower income for dairy farmers⁴⁶. High risk of egg spoilage⁴⁷.	

20. **Analysis was also conducted for the 6 selected value chains.** The 6 value chains are 3 commercial VC (coffee, tea, dairy) and three food security VC (poultry, african leafy vegetables and fruit trees). **The selection of the 6-value chain is through a stakeholder consultation involving all counties, where each county undertook its own prioritization exercise, based on best available knowledge⁴⁸.** The criteria used for prioritization included climate and environmental criteria, economic criteria, social criteria and additional criteria related to technology availability including the level and potential of participation in the value chains by economically and socially

³⁹ Kanyari, Kagira, and Mhoma, 2010. Prevalence of endoparasites in cattle within urban and peri-urban areas of Lake Victoria Basin, Kenya with special reference to zoonotic potential. *Sci Parasitol*, 11(4). <http://erepository.uonbi.ac.ke/handle/11295/33352>

⁴⁰ USAID. 2018. Climate Risk Profile. Kenya. Fact Sheet. https://www.climateinfo.org/sites/default/files/asset/document/2018_USAID-ATLAS-Project_Climate-Risk-Profile-Kenya.pdf

⁴¹ International Livestock Research Institute. 2020. Food safety landscape analysis: The dairy value chain in Kenya. <https://cgspage.cgiar.org/bitstream/handle/10568/108989/Food%20safety%20Kenya%20dairy%20value%20chain.pdf?sequence=1>

⁴² Mwongera, C., Nowak, A., Notenbaert, A.O.M., Grey, S., Osiemo, J., Kinyuwa, I., Lizarazo, M. & Girvetz, E. 2019. Climate-Smart Agricultural Value Chains: Risks and Perspectives. In: Rosenstock T., Nowak A., Girvetz E. (eds) *The Climate-Smart Agriculture Papers*. Springer, Cham. https://doi.org/10.1007/978-3-319-92798-5_20

⁴³ Rahimi, J., Mutua, J.Y., Notenbaert, A.M.O. et al. Heat stress will detrimentally impact future livestock production in East Africa. *Nat Food* 2, 88–96 (2021). <https://doi.org/10.1038/s43016-021-00226-8>

⁴⁴ Tadesse, G. and Dereje, M. 2018. Impact of Climate Change on Smallholder Dairy Production and Coping Mechanism in Sub-Saharan Africa-Review. *Advances in Life Science and Technology*. ISSN 2224-7181 (Paper) ISSN 2225-062X (Online) Vol.65, 2018.

⁴⁵ International Livestock Research Institute. 2020. Food safety landscape analysis: The dairy value chain in Kenya. <https://cgspage.cgiar.org/bitstream/handle/10568/108989/Food%20safety%20Kenya%20dairy%20value%20chain.pdf?sequence=1>

⁴⁶ Mwongera, C., Nowak, A., Notenbaert, A.O.M., Grey, S., Osiemo, J., Kinyuwa, I., Lizarazo, M. & Girvetz, E. 2019. Climate-Smart Agricultural Value Chains: Risks and Perspectives. In: Rosenstock T., Nowak A., Girvetz E. (eds) *The Climate-Smart Agriculture Papers*. Springer, Cham. https://doi.org/10.1007/978-3-319-92798-5_20

⁴⁷ Mwongera, C., Nowak, A., Notenbaert, A.O.M., Grey, S., Osiemo, J., Kinyuwa, I., Lizarazo, M. & Girvetz, E. 2019. Climate-Smart Agricultural Value Chains: Risks and Perspectives. In: Rosenstock T., Nowak A., Girvetz E. (eds) *The Climate-Smart Agriculture Papers*. Springer, Cham. https://doi.org/10.1007/978-3-319-92798-5_20

⁴⁸ Notes from the consultation are available in the Annex 7, Stakeholder Engagement Plan and further detail on selection process is included in the FS, Part B, section 6.1.2.

vulnerable groups such as women and youth. During the selection of the value chain, the project design team reviewed the prioritized value chain commodities in key national and regional documents including the County Integrated Development Plan (CIDP) of the 14 counties, the Bottom-Up Economic Transformation Agenda for Inclusive Growth (BETA) of the Kenya Government, Kenya County Climate Risk Profile (KCCRP). The purpose of the review is to establish the project's own criteria for the selection of the value chains and to ensure that the selected value chains fully align with the county's priorities in transferring the agriculture value chain to climate resilient and low carbon pathway. Please refer to section 6 of Annex 2, Feasibility Study for more detail. An overview of each Value Chain is provided in the FS, Part B, section 1.6. Detailed Value Chain Studies and Market Analysis are provided in Annex 23. **The selected value chains each exhibit vulnerability to climate change, but also have potential for resilience-building and/or emissions reductions through technology transfer.**

21. Farmers' socio-economic vulnerability exacerbated by climate impacts to each food value chain in the LREB are further summarized in Table 4 (Annex 2, Feasibility Study, Part A).

B1.3 CAPACITY TO ANTICIPATE AND RESPOND TO CLIMATE CHANGE

22. **Farmers' adaptive capacity to climate change is limited due to inconsistent access to water, limited income diversification, and lack of access to appropriate technologies and climate information.** Further, the majority of smallholders lack risk-sharing reducing mechanisms, such as social safety nets, financial resources, and alternative coping strategies, and to make more informed decisions about climate-responsive production choices.
23. **Despite previous interventions, the local adaptive capacity remains low, as most smallholder farmers continue to cultivate small plots of land with limited means, and without necessary the risk-reducing mechanisms to make climate-responsive production choices.** A survey conducted in Kakamega and Vihiga counties in the LREB, revealed that smallholder farmers heavily rely on rainfed, seasonal, and informal agricultural activities⁴⁹. Furthermore, the limited size of their landholdings also hampers their access to agricultural credit. Lessons learned from previous development programs implemented by various partners emphasize the importance of supplementing traditional extension-based interventions with risk-sharing mechanisms such as cooperatives. These cooperatives can help alleviate producer isolation, enhance adaptive capacity and improve access to adaptive assets (Refer to Annex 2, FS, Part B)⁵⁰.
24. **There is limited integration of climate and low carbon strategies into land use strategies and management plans.** Since the 1970s, land use change in the LREB has been driven by population pressure and low productivity, resulting in the expansion of agriculture into wetlands, grassland, and deforestation of hill slopes⁵¹. The risk of land degradation fueled by land use and land cover changes is considered high in many parts of LREB. These anthropogenic drivers lead to erosion, biodiversity losses and ecosystem diversification, land use changes, urbanization, and weak water management systems and pollution⁵². These changes are accelerating the degradation of ecosystems that are vital to supporting agricultural productivity. Deforestation hotspots have been identified in the southern areas of the LREB where tree cover has been reduced to make space for cropland (Figure 26, Annex 2, Feasibility Study, Part A).
25. **Impacts from interrelated climate hazards are exacerbated by anthropogenic pressure on key socio-economic activities in the region, such as the expansion of the production land causing deforestation.** Flooding impacts for example are exacerbated by land quarrying, sand harvesting, and wetland drainage to support agricultural development. These activities frequently increase soil erosion along riverbanks and farmlands⁵³.
26. **The destruction of catchment forests and wetlands is linked to the disruption of rainfall patterns, droughts, and floods.**⁵⁴ Prolonged droughts have negatively affected water availability and vegetation in the basin. Loss of

⁴⁹ Karugia et al. 2005. The Role of Rural Factor Markets in Reducing Poverty, Risks and Vulnerability in Rural Kenya: The Case of Kakamega and Vihiga Districts. Strategies and Analysis for Growth and Access.

⁵⁰ USAID. 2018. Lake Victoria Basin: Climate Change Adaptation Strategy and Action Plan, 2018 - 2023. Retrieved from: https://www.climateknowledgeportal.worldbank.org/sites/default/files/asset/document/2018_USAID-PREPARED_Lake-Victoria-Basin-Adaptation-Strategy-Action-Plan.pdf

⁵¹ Nyberg, Y., Wetterlind, J., Jonsson, M. and Öborn, I. (2021) Factors affecting smallholder adoption of adaptation and coping measures to deal with rainfall variability. International Journal of Agricultural Sustainability 19:2, pages 175-198.

⁵² World Bank Group. WBG. 2021. Climate Risk Profile: Kenya (2021): The World Bank Group. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15724-WB_Kenya%20Country%20Profile-WEB.pdf

⁵³ Agol D, Reid H, Crick F, Wendo H. 2021 Ecosystem-based adaptation in Lake Victoria Basin; synergies and trade-offs. R. Soc. Open Sci. 8: 201847. <https://doi.org/10.1098/rsos.201847>

⁵⁴ Odada, E. O., Olago, D. O., & Ochola, W. O. (2006). Environment for Development: An Ecosystems Assessment of Lake Victoria Basin Environmental and Socio-Economic Status, Trends and Human Vulnerabilities.

vegetation increases erosion⁵⁵, and some vegetation species take a long time to recover after extreme drought⁵⁶. Extreme flooding in parts of Busia, Homa Bay, and Kisumu counties submerges productive lands, destroying crops and properties, and causing erosion.

27. Climate change is partly responsible for crop failures, water shortages, reduced water quality, and decreased hydropower generation capacity in the basin due to low water levels⁵⁷. Scientific reports have confirmed that eutrophication and climate change are the leading causes of ecological degradation in LREB.
28. For more information on environmental conditions in the LREB, please refer to FS Part B, Section 1.4.
29. **The public services offered to farmers and cooperatives are insufficient to enable value chain transformation.** National and county-level climate policies, regulatory frameworks and strategies have been established in Kenya, reflecting a commitment to addressing climate challenges. However, the insufficient climate services provided to smallholders remain a concern. The technical capacity among county government staff is yet to be strengthened, resulting in a gap in translating these objectives into actionable plans that directly support farmers. Developing specific tools and methodologies tailored to the local context could be instrumental in bridging this gap.
30. **There is a lack of support from buyers and other actors in the value chain to help smallholders adopt climate actions and transition to low carbon pathways in their production. Additionally, there is a need to invest in sustainable supply chains and provide smallholders with access to markets and fair prices.** Market incentives may not currently align with climate action in agriculture. For example, there may be a lack of price premiums for sustainably produced agricultural products, making it challenging for farmers to adopt climate-smart practices. Additionally, the costs of environmental externalities associated with conventional agricultural practices, such as greenhouse gas emissions or soil degradation, are often not reflected in market prices. Limited access to credit, technical assistance, and market information can hinder efforts to adopt sustainable agricultural practices.

B1.4 GREENHOUSE GAS(GHG) EMISSION PROFILE

31. In 2015, the agriculture sector was the leading source of GHG emissions in Kenya. According to Kenya's Updated NDC (2020), GHG emissions are projected to reach 143 MtCO₂e by 2030, which is a 152% increase from 93.7 MtCO₂e in 2015⁵⁸. Kenya's Second National Communication⁵⁹ states that agriculture emissions, the largest source of GHG emissions in the country, are expected to increase from 30 MtCO₂e in 2010 to 39 MtCO₂e in 2030, primarily driven by livestock methane emissions (18.8 MtCO₂e in 2030) and emissions from agricultural soils (16.2 MtCO₂e by 2030).
32. Smallholder farming also utilizes woody biomass for charcoal production, resulting in deforestation, land degradation, biodiversity loss, and loss of soil organic matter. According to the findings from the cooperative census undertaken in 2022 for the development of this proposal, farmers appear to use a semi-diversified energy basket that varies according to costs, accessibility, and the needs of each value chain. Initial findings of the cooperative census indicate that only 41% of cooperatives exclusively use hydro-electricity, while the rest use a combination of hydroelectricity and other sources such as solar, firewood, LPG and kerosene. This is not a comprehensive census of energy uses, but it does provide a representative snapshot of the current situation in the LREB region. There is evidence in the area that hydro-electricity supply varies when rainfall is lower or there is a drought, highlighting the need to ensure adequate access to energy under projected climate conditions.

Finally, the region is also experiencing increasing emissions from deforestation, land degradation, and the expansion of agricultural land into forests. The annual rate of forest loss has been estimated at 6% since 2000⁶⁰. According to Global Forest Watch, "between 2001 and 2021, forests in Kenya emitted 8.56MtCO₂e/year and removed -13.7MtCO₂e/year. This results in a net carbon flux of -5.09MtCO₂e/year", with a significant proportion of forest emissions originations from the western region. While Kenya is not a net emitter, it is necessary to separate

⁵⁵ Mugo, R., Waswa, R., Nyaga, J. W., & Ndubi, A. (2020). Quantifying Land Use Land Cover Changes in the Lake Victoria Basin Using Quantifying Land Use Land Cover Changes in the Lake Victoria Basin Using Satellite Remote Sensing: The Trends and Drivers between 1985 and 2014. (September). <https://doi.org/10.3390/rs12172829>

⁵⁶ Awange JL, Aluoch J, Ogallo L, Omulo M, Omondi P (2007b) An assessment of frequency and severity of drought in the Lake Victoria region (Kenya) and its impact on food security. Climate Res 33:135–142

⁵⁷ Agency, C. (2013). October 2013 Japan international cooperation agency Nippon koei co., ltd. (October).

Environment, M. O. F. (2022). REPUBLIC OF KENYA MINISTRY OF ENVIRONMENT AND FORESTRY NATIONAL STRATEGY FOR ACHIEVING AND MAINTAINING OVER 10 % TREE COVER BY 2022.

The World Bank. (2013). Kenya Water Security and Climate Resilience Project. Retrieved from <https://projects.worldbank.org/en/projects-operations/project-detail/P117635?lang=en&tab=overview>.

⁵⁸ <https://unfccc.int/sites/default/files/NDC/2022-06/Kenya%27s%20First%20%20NDC%20%28updated%20version%29.pdf>

⁵⁹ <https://unfccc.int/sites/default/files/resource/Kennc2.pdf>

⁶⁰ <https://www.globalforestwatch.org>

agricultural growth from land expansion and deforestation. The Government of Kenya has recently implemented legal reforms this regard, including a ban on charcoal.

B1.5. Alignment with county priorities

33. The project aligns with relevant national laws, policies, and plans, and contributes to the development and climate change goals of the Government of Kenya. This includes the National Climate Change Act and Strategy, as well as laws and policies in the agriculture, land, water, and environment sectors, existing institutional frameworks for climate risk and disaster risk management as well as their equivalent at county level. The project also supports the implementation of Kenya's National Adaptation Plan (NAP) and Nationally Determined Contribution (NDC). Relevant policies, laws and institutional frameworks include:

- Kenya's Vision 2030 development plan
- The National Climate Change Act (2016) ⁶¹
- Kenya's Second National Communication⁶²
- Kenya's National Adaptation Plan 2015 – 2030⁶³
- National Climate Change Action Plan 2018 - 2022⁶⁴
- National Climate Change Response Strategy (NCCRS), 2010⁶⁵
- Nationally Determined Contribution (updated, 2020)
- The Cooperative Societies Act, Chapter 490, Revised 2005
- The Water Act, 2016
- The Energy Act, 2019⁶⁶
- The Kenya Climate Smart Agriculture Strategy (CSAS)⁶⁷, 2017-2026
- The Forest Conservation and Management Act (FCMA), 2016⁶⁸
- The Environmental management and Coordination Act (1999)
- The National Disaster Risk Management Policy (NDRMP), 2017
- The National Environment Policy (NEP)⁶⁹, 2013

34. The country has put in place several other key policies to support implementation of climate change adaptation and mitigation actions. Some of these policies include The National Policy on Climate Finance, which emphasizes how climate investment can support measures like reducing deforestation, diversifying livelihoods, and promoting research and innovation. Furthermore, the Climate Risk Management Framework aims to harmonize its climate change and disaster risk policies²⁴².

35. Regarding key agricultural policies, the National Livestock Policy of 2015 recognizes the increased frequency and severity of droughts as a result of climate change. This policy outlines measures to enhance the livestock subsector's contribution to food and nutritional security, provide raw materials for agro-based industries, and contribute to improved livelihoods. The Agricultural Sector Transformation and Growth Strategy (ASTGS) (2019-2029) and the Kenya Climate Smart Agriculture Strategy (KCSAS) (2017-2028) further prioritize addressing the impact of climate change on all agricultural sub-sectors (crops and livestock). The KCSAS highlights adaptation and resilience as one of four broad strategic areas²⁴⁴.

36. At the national level, coordination and implementation of climate change adaptation policy and action plans are overseen by the National Climate Change Council (NCCC), headed by the President, the Cabinet Secretary of Climate Affairs, and the Climate Change Directorate⁷⁰. County level coordination and implementation of climate change adaptation policy and action plans are overseen by the county governor and the climate change unit (CCU). Most county governments have enacted climate change laws, Climate Change policies, created climate change

⁶¹ Retrieved from: <https://faolex.fao.org/docs/pdf/ken160982.pdf> . The new National Climate Change Act is under developed, and this proposal will update the Act once it is launched.

⁶² And the developing Third National Communication

⁶³ https://www4.unfccc.int/sites/NAPC/Documents%20NAP/Kenya_NAP_Final.pdf

⁶⁴ <https://napglobalnetwork.org/wp-content/uploads/2022/01/napgn-en-2022-kenya-NCCAP-2018-2022-Implementation-Status-Report.pdf>

⁶⁵ https://cdkn.org/sites/default/files/files/National_Climate_Change_Response_Strategy_Executive_Brief.pdf

⁶⁶ <https://www.epra.go.ke/download/the-energy-act-2019/>

⁶⁷ <https://www.adaptation-undp.org/resources/plans-and-policies-relevance-naps-least-developed-countries-ldcs/kenya-climate-smart>

⁶⁸ <https://faolex.fao.org/docs/pdf/ken160882.pdf>

⁶⁹ <https://storage.g.oogoleapis.com/cclov-staging/aq199ci3atqwp92rm7vemvs3cw5k?GoogleAccessId=laws-and-pathways-staging%40soy-truth-247515.iam.gserviceaccount.com&Expires=1675118684&Signature=sYjEqcME%2Bv0LCAc6ldZe7UHDd%2Fk83LIZ4figZpW4gpSdp52TmTJ2A4Ngeus6xFMXcdZrtixkCizNYWWuvGAZxqXmEOVUHksWJXSpqxGWsejbDkRuDftbXtKdWeNnNWO6gEnx8v6Ygs89FFR2qrPhm>

[azmLsryUfuc48DfK%2F84IK7KIIOB%2Fcf3RRjOiSaBmBnVqr6Z8E9M0fNW2P4iROM3kxkQMESDgDnLBSfszVD7q3Nvd15yGFdBqq7QKPJpJJSjSCE0FIAIp3NDNXEuoGBU%2B6KZLraVGZLzRYHiYrZPidmiB3oCvtk5O1V9CyK44ftl3EzrvZFEbNnRklTBg%3D%3D&response-content-disposition=inline%3B+filename%3D%3B+filename%2A%3DUTF-8%27%27f&response-content-type=application%2Fpdf](https://storage.g.oogoleapis.com/cclov-staging/aq199ci3atqwp92rm7vemvs3cw5k?GoogleAccessId=laws-and-pathways-staging%40soy-truth-247515.iam.gserviceaccount.com&Expires=1675118684&Signature=sYjEqcME%2Bv0LCAc6ldZe7UHDd%2Fk83LIZ4figZpW4gpSdp52TmTJ2A4Ngeus6xFMXcdZrtixkCizNYWWuvGAZxqXmEOVUHksWJXSpqxGWsejbDkRuDftbXtKdWeNnNWO6gEnx8v6Ygs89FFR2qrPhm)

⁷⁰ Kenya National Adaptation Plan 2015.

funds, established Climate Change Units, and developed climate change adaptation action plans, as well as County Climate Information Services Plans (CISP), and 5-year County Integrated Development Plans (CIDP). However, capacity gaps remain, particularly in terms of supporting farmers, cooperatives, and value chains in their transition towards climate resilience and decarbonization.

37. For further detail on legal and policy coherence and an assessment of county level institutional capacity for climate change, please refer to Chapter 3 of the Feasibility Study Part B.

B1.6. Related projects/interventions

38. There are several ongoing initiatives in Kenya and in the LREB that support climate adaptation and mitigation in the agriculture sector. This project builds on these initiatives as well as on the devolution and decentralized climate governance systems set up by the Government of Kenya.
39. The list includes two GCF Readiness initiatives that ended in 2022. **The FAO-supported NAP initiative**⁷¹ delivered studies, information and institutional capacity that is being directly used as a contribution to this project. This NAP Readiness grant was implemented from 2018 to 2022 with FAO as a Delivery Partner. The project enhanced technical and institutional capacities for adaptation planning, built the NAP knowledge base, improved evidence-case for climate change adaptation, and promoted private sector investment in the adaptation process. The Readiness project conducted a ten-cohort training that reached 338 climate change units across all departments and ministries in the agricultural sector. The NAP readiness also delivered training to build the capacity of technical staff at the county level to integrate climate change risks and opportunities in planning and budgeting, and delivered data analysis and information management systems in six national institutions that were linked across all the 47 counties and with national planning processes. The activities under Outcome 1 are therefore designed to take these processes and capacities to the next stage, for example by broadening the trainings and technical assistance to the departments of cooperatives in each county, and refining the data collection processes, as well as collecting decentralized data, that will feed into the systems established under the NAP project.
40. The list below outlines several other initiatives:
- NDA Strengthening and Country Programming support for Kenya through the National Treasury (GCF Readiness)..
 - Agricultural Climate Resilience Enhancement Initiative (ACREI) (USD\$4,636,793, 2018-2024).
 - Integrated Landscape Management for conservation and restoration of the Mt. Elgon in Western Kenya (FSP) (USD\$ 5,354,587, 2022-2027).
 - Institutionalization and Scale-up of the Kenya Integrated Agriculture Management Information System (USD 4,932,408, 2023-2026,).
 - Government Financing of Locally Led Climate Action (G-FLLOCA)
 - The IFAD-supported Livestock Commercialization Project (KeLCoP)
 - The Rural Kenya Financial Inclusion Facility (RK-FINFA, IFAD)
 - Africa Rural Climate Adaptation Finance Mechanism (ARCAFIM)
 - IFAD is also developing a Global Net Zero Dairy project, which is at early stages of design and will be submitted to the GCF for financing.
41. The project is collaborating closely with the **Kenya Commercial Bank**⁷², a Direct Access Entity that is joining the project as a partner. The project will leverage loan financing from the KCB to support of smallholders and cooperatives. The KCB is currently preparing a project concept note for submission to the GCF, which will build on the capacity developed under this project.
42. For a full list and analysis of synergies with ongoing or pipelined programs and projects, please refer to Feasibility Study Part B section 2.3. Mechanisms for collaboration are explained in section 2.4 of the Feasibility Study, Part B. Further detail on value-chain specific programs and projects can also be found in Annex 23.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

43. The climate analysis shows that the anticipated changes in climate would make smallholder farmers more vulnerable by reducing production and productivity, and that these farmers are already experiencing vulnerability and climate risks from current variability. The climate analysis also shows that lack of adaptive capacity and sensitivity to climate variability and risks, are high among smallholders in LREB. LREB has one of the highest

⁷¹ Enhancing capacity for planning and effective implementation of climate change adaptation in Kenya through FAO.

⁷² Due to confidentiality of the project concept note, we are not able to share information on its content here.

population densities in Kenya. Together with the unsustainable natural resource management practices used by smallholders, climate change could have dramatic effects on the entire economy of the LREB. Furthermore, the drive to increase production in response to climate risks could, under the business-as-usual scenario, lead to increased emissions from extensification and land use change. These trends would culminate in less access to food and a visible impoverishment of local farmers.

44. There have been few climate finance initiatives in the LREB. Previous development programmes and investments also did not fully achieve maximum benefits for resilience and decarbonization of the agriculture sector because they missed a few crucial ingredients: to foster the emergence of locally-driven adaptation⁷³ solutions, building on the strengths and potential of local organizations such as cooperatives is key. Creating flexible demand driven technology transfer mechanisms that are based on peer support, and ensuring the financial viability – and even profitability – of any climate solution proposed, are key to ensuring continuous endogenous adaptation even after project execution.
45. The project proposes a set of adaptation practices designed to improve adaptive capacity, access to adaptive assets and social safety nets, prevent losses and sustainably increase productivity as a means of improving resilience to climate change. To ensure that any gains in resilience are not made at the cost of increasing emissions and the degradation of the ecosystems, the project proposes adaptation solutions focused on intensification and solutions focused on mitigation of emissions from the AFOLU sector.
46. To disseminate these adaptation and mitigation practices at scale and to promote rapid transformation, the project sets up a long-term, lasting mechanism to support technology transfer that leverages the strengths of the cooperative movement in Kenya as well as the devolution process. The project will increase the climate resilience of producers, facilitate both public and private finance investment in technology transfer and up-scale, and generate climate resilient, low-carbon business models.
47. Several mutually-reinforcing barriers have, however, prevented the deployment of adaptation solutions, which this project must now address in order to facilitate the paradigm shift.

Barrier 1: The public services offered to farmers and cooperatives are insufficient to enable value chain transformation.

48. At national level, there are adequate climate change and sector related policies and regulatory frameworks to guide climate action in all sectors. At county level, there is also notable progress in mainstreaming of climate change in relevant laws and regulations: over 33 counties out of 47 have enacted a climate change Act and a significant number of counties have developed their own county climate change funds. All counties have set up Climate Change Units. However, there is evidence that the County Integrated Development Plans (CIDPs,) which are the principal county policy and development plans, are only beginning to mainstream climate change issues⁷⁴.
49. Moreover, counties have yet to develop specific tools and methodologies that would enable operationalization of their CIDP's climate objectives into concrete action in support of farmers. A key example is the delivery of agro-meteorological advisory, climate information and extension, which still follows traditional, cost-heavy methods that have proven unsuccessful in reaching the full scope of last mile users. Given the dynamic nature of the agriculture sector in the LREB, there is scope for introducing innovation in the way that county governments support farmers as private sector agents.
50. The delivery of such improved, enabling services from government to private sector farmers would, however, require much improved information systems. To enable a local understanding of climate risks and vulnerabilities, facilitate rapid climate response and support the socio-economic performance of LREB farmers, the data and information basis at county level needs to be improved. This includes the nature of data (e.g., climate-related crop data, or accurate price data) and information services, the efficiency and effectiveness of collection and dissemination methods, and the way in which the information and data is used and leveraged for climate resilient, low-carbon growth.
51. Among county government staff, technical capacity to deal with climate variability and climate change is nascent but growing. Many training programs have taken place that provide a baseline of understanding, knowledge, and capacity. However, the departments of extension and cooperatives typically have not yet been fully included in capacity development for climate change. As a result, they are not yet fully able to deliver support programs, extension or to develop innovative guidelines that would facilitate adaptation and mitigation at the farm level.

⁷³ The 8 Principles of LLA are: 1.devolution of decision-making to the lowest appropriate level; 2. addressing structural inequalities; 3. providing predictable and accessible funding; 4. Investing in local institutional capacity; 5. building an understanding of climate risk; 6. flexible programming and learning; 7. transparency and accountability; 8. collaborative action and investment. Although the project makes an indirect linkage to LLA, the principles are upheld in all outputs and activities.

⁷⁴ FAO supported NAP Readiness, 2022, final report.

Barrier 2: Limited integration of climate resilient and low carbon strategies into land use planning and landscape management at county level.

52. Current environmental degradation trends show that cropland will continue to expand at the expense of the natural ecosystems unless critical changes occur to halt it. This situation is complex to address due to several interrelated factors:
53. While much is currently being done to strengthen county government capacity in all aspects of climate change, there remain key policy gaps in terms of managing land use, preventing further land use change, and conserving ecosystem services. There do not exist any integrated approaches tying together adaptation and mitigation priorities, and the territorial approach to managing natural resources is un-coordinated. As a result, natural parks and reserves remain disconnected from the broader landscape; and because enforcement is difficult, there are serious encroachment threats onto fragile areas, including wetlands. There is a need for a more integrated territorial approach that would bring together all sectors under common goals of agricultural productivity, environmental management, natural resources conservation, climate adaptation and emissions reductions.
54. There are, however, missing data and information at local level that would help inform the development of integrated landscape management strategies. While there have been major efforts at mapping key environmental resources (including for example the Kenya atlas of biodiversity, the Atlas of wetlands), these are not regularly updated. These reports do not always inform county policies on land use, and counties may not be entirely capacitated to integrate all the information into coherent, climate-informed, landscape management strategies and plans.

Barrier 3: Smallholders have limited capacity to access productive assets and face risks in adopting climate resilient, low-carbon technologies.

55. Farmers must first acquire the technical knowledge, know-how as well as the material assets required to be able to change their habitual patterns of production towards climate resilient, low-carbon methods. While many of the farmers, as evidenced in the Climate Change Survey and Cooperative Census⁷⁵, are aware of the impacts of climate variability and potential impacts of climate change on their livelihoods, few are adopting the full suite of available options for adaptation or mitigation. Without proper support from governments and without a mechanism to support the dissemination and broader adoption of climate resilient, low-carbon technologies, farmers will continuously be left to use maladapted extractive practices and think in terms of short-term results. Yet traditional climate extension methods have not proven entirely effective in getting the knowledge to farmers, and in accelerating adoption of best practices. A public-private alliance is needed that will foster farmer innovation and creativity, while supporting appropriate climate action through better rules, regulations, knowledge, and norms. Traditional “top-down” extension methods often perpetuate barriers to accessing technologies, because the technologies are not well suited to farmer priorities and needs. Peer- to peer networks have in the past, and elsewhere, proven effective means through which farmers can self-identify climate challenges and select solutions that make economic, social as well as climate sense.
56. However, the adoption of climate resilient, low-carbon technologies is also hindered by the sub-optimal organization of value chains and farmers in the LREB region, particularly for the non-traditional cash value chains. Too many smallholders work on very small, fragmented plots, in isolation from one another, or in informal groupings that bring little added value in terms of risk reduction. Existing cooperatives exhibit a higher level of organization that enables them to act as vehicles for risk sharing when adopting climate resilient, low-carbon technologies, save costs and reduce transaction costs for farmers (in the acquisition of knowledge and assets). Yet even formed cooperatives face organizational challenges that prevent them from fully realizing their potential for members, in terms of economic and climate gains: it is organizational gaps (such as lack of adequate climate-oriented business planning) that prevent cooperatives from leveraging finance that would enable scaling of climate resilient, low-carbon practices.
57. The low access to formal rural finance for adoption of climate resilient, low-carbon technologies prevents smallholders from accessing knowledge and assets they require. This low access is due not only to the organizational gaps listed above, but also the risk aversion of financial institutions who, themselves, lack the technical and operational capacity to deliver suitable products to their potential clients. In this regard, cooperatives also play a key role as intermediary, as a cooperative will always be in a better solvency position than individual farmers. In addition, access to financial services that would enable climate resilient, low-carbon practices is unequal among men, women, and vulnerable groups. As a result of these gaps, commodities produced using climate resilient, low-carbon technologies are not yet reaching markets to their full potential.

⁷⁵ The Climate Change Survey and Cooperative Census were conducted by FAO and Agriterria in 2022.

Barrier 4: There is a mismatch between supply and demand of finance to support smallholders and their organizations in the transition towards climate resilience/low carbon pathways.

58. There are clear information gaps that prevent climate resilient, low-carbon commodities from reaching markets and being profitable. In the more organized value chains such as coffee, tea and dairy, market exchanges and buyers are not yet aware of the potential value added of such products. Financial institutions are also not fully informed of the potential profitability of climate resilient, low-carbon value chains. There are no labelling schemes for climate resilient, low-carbon products, and as such, because they are not identified, they do not necessarily fetch a better price unless their comparatively better quality can be demonstrated. This then feeds into the risk aversion of farmers, who face costs for adoption of technologies, but no guarantee of increased income – and of financial institutions as well. The comparison has not yet been attempted, mainly because the awareness and knowledge among buyers is not yet sufficient – and in turn this lack of awareness is also partly driven by the policy gaps above (national and county level), that are not yet driving markets towards climate resilient, low-carbon products.
59. In the less organized value chains, because farmers operate in a less aggregated manner, links to markets are also not fully developed. Individual farmers or small groups must then contend with prices, supply, and quality on their own, and this creates risks and costs, which are compounded by the inefficiencies in public service delivery mentioned above. Cooperatives, big and small alike, must be empowered to reach buyers and customers and to accurately market their climate resilient, low-carbon products for the transition to be financially viable. Furthermore, financial institutions also face barriers in reaching potential smallholder clients: high transaction costs and the difficulties in identifying suitable business cases also compound accessibility barriers (such as the requirements for collateral and high interest rates). Financial institutions need to be strengthened so they can identify and cater to potentially successful business organizations and farmers and support them with accessible finance to satisfy market demand.
60. Taken together, these barriers and gaps require a coordinated approach where governments and private sector learn to work towards a shared purpose, each according to their comparative advantage, towards climate resilient, low-carbon value chains and ultimately, a transformation of the local economy.

Theory of Change

This project is premised on the following theory of change:

61. **Smallholder farmers in the LREB are currently and will be vulnerable to the impacts of climate variability and climate change**, regardless of the climate scenario that materializes. This vulnerability is due to the over-reliance on rain fed agriculture, the land and ecosystem degradation that undermines agricultural productivity, limited access to climate-appropriate production and processing technologies, practices, information and services, and underlying multidimensional poverty.
62. However, the same vulnerable farmers have at their disposal a set of **adaptive mechanisms that, if used properly and disseminated at scale, could deliver significant resilience-building benefits**. Among these adaptive mechanisms, cooperatives serve as risk sharing and risk reduction mechanism for members, and access to growing county government capacity in dealing with climate risks. The cooperative movement is quite vibrant in Kenya. Agricultural cooperatives play a significant role in Kenya's agricultural development, accounting for 46% of all registered cooperatives with approximately 7 million members.⁷⁶ Playing various roles, agricultural cooperatives help access inputs, improve members' position in the market chain, help farmers achieve economies of scale, increase production and productivity of members, provide extension support services, and market access for small-scale farmers, help households diversify livelihoods and crops, all of which contribute to boosting farmer resilience.⁷⁷ (please refer to FS, Part B, section 6.1.3 for more about cooperatives).
63. In addition to this, **agricultural development in the region runs the risk of becoming a significant source of emissions** given the current pressures towards the expansion of cropland at the expense of forests, the degradation of existing agricultural landscapes and marginal areas due to unsustainable land use practices, and inefficient practices in key emitting sectors such as livestock.
64. To adequately steer the agriculture sector and development pathway towards a climate resilient, low carbon trajectory, taking advantage of the existing strengths of the cooperative movement will be key – while filling the remaining capacity gaps and barriers, at individual, institutional and ecosystem levels.
65. Therefore, IF vulnerable smallholders and value chain actors have access to technologies, markets and finance to support production, processing and marketing of targeted VCs, and are supported by improved climate information, extension, and climate-resilient landscapes, THEN they will become more resilient to the impacts of climate change

⁷⁶ ASDS, 2010-2020, p.47

⁷⁷ JoAnn Jaffe and Terra Brockett, 2016

and emissions from the AFOLU sector will be reduced **BECAUSE** the priority value chains will be reoriented towards climate-resilient and low-carbon pathways, and farmer-led adaptation and mitigation actions will be supported by adequate gender-responsive public agro-climate services and public and private investment.

66. The **objective** of the project is to foster the emergence of climate-resilient, low-carbon, environmentally sustainable, and financially viable agriculture value chains by accelerating the transfer of technology, knowledge, assets, and services with a focus on agri-food cooperatives as key agents to leverage rural change.

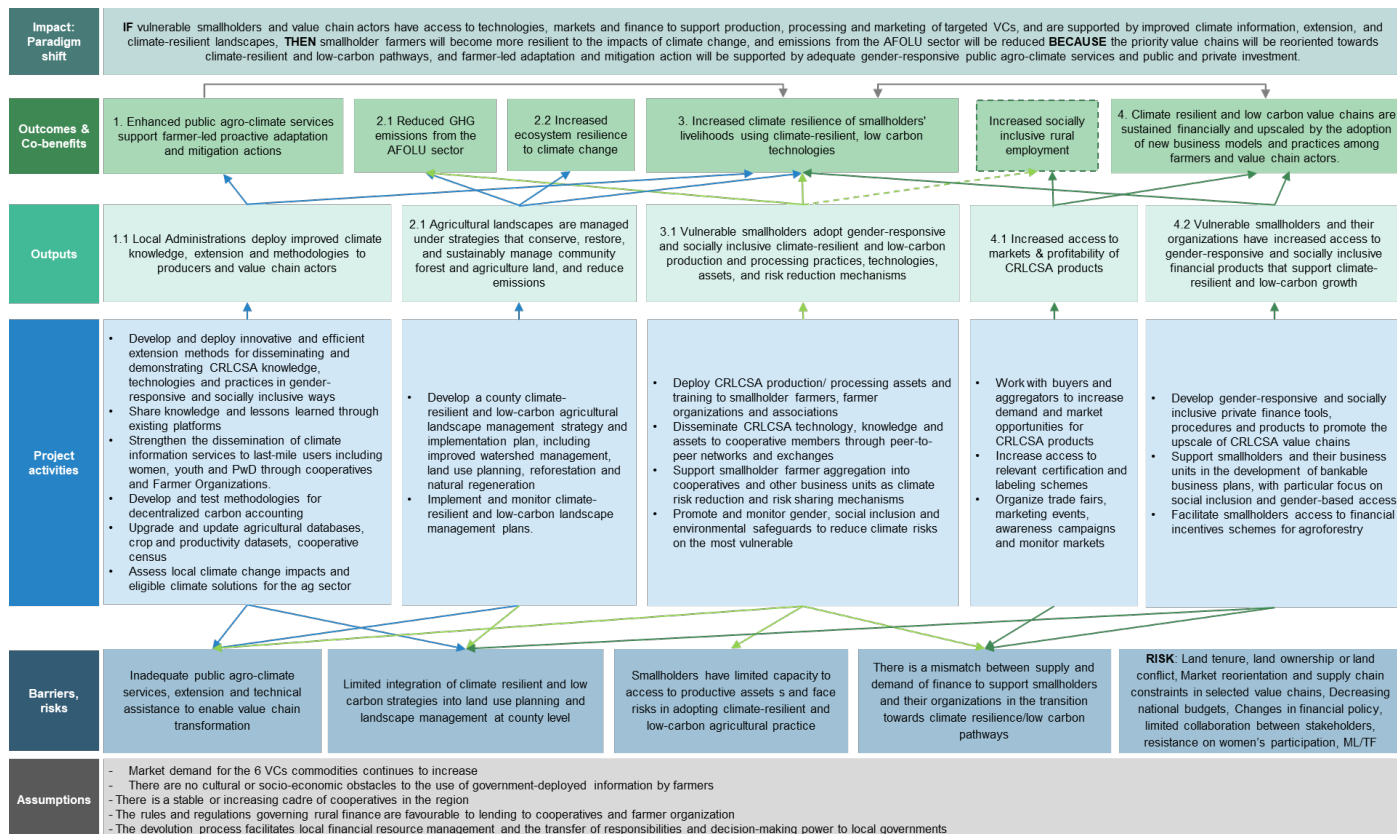


Figure 2: Theory of Change

PROPOSED CLIMATE SOLUTIONS:

67. The project will support **community-based adaptation** through **climate technology upscale and transfer** by empowering farmers' cooperatives and groups as the service centers. Further, the project will facilitate the integration and coordination of the value chain actors by providing **market access to the producers and financial incentives to the farmers and other value chain actors**. The project will ensure the **climate solutions promoted by the project are gender and social inclusion**. Table 5 shows the specific climate solutions, the climate impacts and vulnerabilities they address. Please refer to Annex2, FS Part A, Table 1 for a more detailed list of recommended interventions per value chain, and Annex 2, FS Part B, for a list of climate resilient technologies proposed that respond to the listed impacts and vulnerabilities.

Table 6 Climate change hazard/barriers; proposed interventions; and impacts/vulnerabilities addressed by the project

Climate Hazards/Barriers	Proposed Interventions	Impacts/vulnerabilities addressed
Increased temperature	- Promotion of climate-resilient technologies (Output 3.1): - Agroforestry, integrated production systems, - Climate resilient varieties/breeds - Climate-proofed storage facilities - Improved post-harvest/storage management	- Decreased growing area and arable land - Heat stress leading to decrease in productivity - Food spoilage during storage
Increased drought events & high interannual rainfall variability	- Sustainable landscape management (Output 2.1) - Promotion of climate-resilient technologies (Output 3.1): - Climate resilient varieties/breeds - Integrated water management;	- Decrease in productivity due to drought and moisture stress - Unstable supply of input (feed) - Water scarcity and degradation

	- Improved livestock management system (construction of feed storage facilities, feed diversification) - Integrated production system (agroecology) - Protective cultivation systems	
Increased heavy rainfall/ flooding events & high interannual rainfall variability	- Sustainable landscape management (Output 2.1) - Promotion of climate-resilient technologies (Output 3.1): - Integrated water management; - Integrated production system (improved drainage systems and anti-erosion systems) - Improved livestock management system (construction of feed storage facilities, animal disease control) - Enhance access to climate services in agriculture including rainfall forecasts, flooding advisory, weather-informed land, crop and livestock, and water management recommendations at production and post-harvest stages.	- Decrease in productivity and loss/damage due to floods - Water runoff, soil erosion - Pest/disease outbreaks - Damages to transportation infrastructure, reduced food value at markets
Unsustainable management & usage of wood and diesel as energy sources	- Sustainable landscape management (Output 2.1) - Promotion of low-emission technologies (Output 3.1): renewable energy technologies for production, harvesting, processing and storage, Waste to Energy loops (e.g. waste management, manure management)	Increase in GHG emissions from AFOLU sector
Limited public services to enable VC transformation	Improved database, extension systems and climate information services (Output 1.1)	Limited adoption of climate resilient and low carbon practices
Limited integration of climate resilient and low carbon strategies into landscape management	Development of county/regional climate resilient and low carbon agricultural landscape management strategies (Output 2.1)	Inconsistent approach for landscape management
Limited capacity to access finance, market and information	- Capacity building for farmers/cooperatives in business development (Output 3.1) - Technology & knowledge dissemination (Output 3.1) - Improved market access and financial incentives (tracability systems, certifications) (Output 4.1)	Limited capacity to adapt to the changing climate among farmers and extension workers
Information gaps between supply and demand	- Increase demand and market opportunities for climate resilient and low carbon commodities (Output 4.1) - Development of improved financial products (Output 4.2)	Less financial incentives for promotion of climate resilient and low carbon practices

68. This project applies the following key principles and concepts:

69. *Locally-driven approach*: This project will invest in vulnerable smallholder farmers through involving local communities in a participatory seasonal planning process and in identifying effective climate solutions and enhancing their access to technologies, market and finance. The project will enhance the local government's capacities to serve the targeted communities including in providing weather information and agro-climate services. The project aims at increasing adaptive capacity and strengthening public agro-climate services, which would support locally-led climate action.
70. *Financial viability for sustainability*: This project considers that farmers, be they smallholders, will seek visible benefit from their activities, be they food security or monetary. The aim is to ensure that smallholders are resilient, that agriculture does not generate emissions increases, and that farming becomes financially profitable. Financial viability and sustainability will be key to ensuring the continuation of locally-driven climate action, and to accelerating the pathway to broader adoption and upscaling.
71. *Technology transfer*: A key pillar of this project is the opportunity to accelerate the transition towards CRLCSA through technology transfer, leveraging the knowledge, know-how, tools, and expertise of farmers in Kenya and abroad and deploying farmer to farmer peer support systems. The project will include a mechanism through which farmers located in other parts of Kenya, Cooperative Partnership countries or elsewhere, will be able to transfer their knowledge and technology on adaptation and mitigation to local farmers in the targeted value chains.
72. *Smallholder aggregation*: Leveraging cooperatives, Farmer Organizations (FO), and their members, will help accelerate the diffusion of suitable practices and creates a strong resilience-building social network that will outlast the project. This can be accomplished by building on existing networks and using data-driven, evidence-based

tools and approaches. This project aims to create the conditions for complete transformation of key value chains. This includes working with value chain actors and stakeholders such as processors and buyers in addressing the barriers to adaptation and emissions reductions. The project will implement an inclusive and participatory approach to all activities.

73. *Sustainable Intensification*: At the heart of the climate problem in the LREB is the need to ensure a minimum level of ecological integrity and resilience, to allow for continued ecological service provision to the agriculture sector and beyond. This project will pay particular attention to trade-offs between the various dimensions of agriculture. The project will promote *sustainable intensification*, balancing the demands of a growing agri-food sector in terms of water, energy, and land, with the need to sustain ecological services for future generations throughout the broader landscape. This project will promote practices that increase production and productivity without further land expansion, while restoring or rehabilitating ecosystem services that support agricultural production.
74. *Gender and Social inclusion*: The project emphasizes empowering those who are most vulnerable in participating meaningfully in the transformation of agriculture, to ensure that they also derive resilience benefits. For women, youth, persons with disabilities, and social minorities, cooperatives offer inclusion that otherwise may not have been possible. The project will not only integrate Gender and Social inclusion perspectives in all its processes, activities and outputs (mainstreaming), it will also dedicate specific activities and budgets towards addressing gender-based or socially-based inequities, in the particular context of addressing climate risks.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1 Enhanced public agro-climate services support farmer-led proactive adaptation and mitigation actions.	☐	☐	☐	☒	☒	☒	☐	☐
Outcome 2.1 Reduced emissions from AFOLU sector	☐	☐	☐	☒	☐	☐	☐	☐
Outcome 2.2 Increased ecosystem resilience to Climate Change	☐	☐	☐	☒	☐	☐	☐	☒
Outcome 3 Increased climate resilience of smallholders' livelihoods using climate-resilient, low carbon technologies.	☐	☐	☐	☐	☒	☒	☐	☒
Outcome 4 Climate resilient, low carbon value chains are sustained financially and upscaled by the adoption of new business models and practices among farmers and value chain actors.	☐	☐	☐	☐	☒	☒	☐	☐

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Increased socially inclusive rural employment	☐	☒	☒	☐	☐	☐
	☐	☐	☐	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

75. The project is structured around four components and **5 connected outcomes**. Component 1 (Outcome 1) improves the institutional context that would support climate resilient, low carbon, and environmentally sustainable value chains. The purpose of this outcome is to support local stakeholders, including particularly county administrations, in becoming facilitators for the upscaling of CRLCSA value chains. This will require strengthening the capacity of county government stakeholders in the extension and cooperatives departments, adding to climate information flows and knowledge systems, and filling the remaining capacity gaps in technical capacity in line with devolution and the climate change priorities. Activities under Outcome 1 support the upscaling and broader adoption of activities under Outcome 2.1, 2.2 and Outcome 3 by ensuring that local communities, farmers, and other value chain actors have sufficient knowledge and adoption capacities climate resilient practices and awareness the needs for the climate change actions.
76. Component 2 (Outcomes 2.1 and 2.2) achieves direct emission reductions and support the conservation of ecosystem services in the face of multiple threats, including climate variability and climate change impacts, population pressure and agricultural land expansion. Under this Component, which is mainly implemented by the Government of Kenya through the county administrations themselves, a vision for resilient, low carbon agriculture landscapes will be formulated to inform future land use planning and land allocation based on climate change impact and risk assessments. This strategy will then be implemented and monitored in targeted areas.
77. Component 3 (Outcome 3) delivers adaptation and emissions benefits from the agri-food sector through the dissemination and upscale of technologies, assets, knowledge, practices, and services related to climate-resilient, low-carbon, environmentally sustainable agriculture practices in the 6 value chains. This is reaching smallholder farmers through community-based associations and organizations, and cooperatives.
78. Finally, Component 4 (Outcome 4) builds on the realization that financial viability of interventions is key to ensuring long-term adoption, scaling and replication, something that had been missing in previous interventions. Activities support the reorientation of business and market practices towards the adoption of climate resilient and low carbon models, and creates conditions for long-term sustainability of the project by supporting cooperatives and farmer organizations in becoming financially sound, autonomous, and profitable climate resilient and low carbon businesses.
79. Locally, although the project covers a large territory, project activities will be deployed in a coordinated manner. Vulnerability hotspots will be identified through Outcome 2, which will guide the prioritization of landscape management strategies and the selection of participating smallholders and farmer organizations. Smallholders will be reached through existing aggregations, such as informal groups, farmer field schools or cooperatives. Linkages between cooperatives working in complementary value chains will also be pursued. This targeting approach ensures linkages between farmers, cooperatives and the overall agricultural landscapes in which they are nested, for mutually reinforcing adaptation and mitigation benefits. Please refer to the FS, Part C, section 6 for more information on targeting.
80. Further detail on each activity and sub-activity, and a detailed description of the baseline for each output, is found in the Feasibility Study, Part C, Section 5.

Component 1 – Enabling local government support for adaptation and mitigation

Outcome 1 - Enhanced public agro-climate services support farmer-led proactive adaptation and mitigation actions.

Output 1.1 Local administrations deploy improved climate knowledge, extension, and methodologies to support producers and value chain actors.

81. The purpose of activities under this output is to help county governments serve farmers, farmer organizations and cooperatives in a manner that contributes to the transformation of the agriculture sector from current pathways to climate resilient, low-carbon trajectories.
82. All activities in this output build on a significant baseline of existing capacity, resources, information, and networks, and they are therefore carefully targeted to filling identified gaps. The six activities proposed here are based on existing county capacities, opportunities such as the availability of digital extension channels, and lessons learned from other projects with similar objectives (Refer to section 2.3 of the FS Part B). The expectation is that the approaches deployed here for the 6 targeted value chains will be scalable to all other value chains, regions, and sectors as well, transforming the way in which county governments interact with private sector to address climate change in Kenya.

Activity 1.1.1 Develop and deploy innovative and efficient extension methods for disseminating and demonstrating CRLCSA knowledge, technologies, and practices in gender-responsive and socially inclusive ways.

83. The first step (sub-activity 1.1.1.1) will be to examine and analyze the state of extension services in the agriculture sector and in the six value chains. This stocktaking exercise will analyze both public and private extension services. The purpose of this assessment is to determine gaps and challenges within and between counties and to develop more efficient and cost-effective solutions to respond to climate change to support transition towards climate resilient, low-carbon production and value chains. This assessment will be conducted in close collaboration with the agricultural extension departments in each county and in the national government, engaging input by farmers, FO, and other value chain actors. The assessment will also consider gender gaps in delivering extension services.
84. Based on this assessment, counties will come together to participate in the participatory design of new, innovative, and more efficient CRLCSA extension methods (sub-activity 1.1.1.2). Following the selection of methods or of improvements they wish to implement, each county will receive technical assistance for the development of operating procedures, manuals, and guidance (sub-activity 1.1.1.3). Each county will receive training and capacity building, including material support, for the delivery and effective deployment of extension methods for the dissemination of climate resilient, low carbon, sustainable agricultural knowledge, technologies, and practices (sub-activity 1.1.1.4).
85. The project will also ensure that gender and social inclusion issues are integrated within extension programs and within any support provided to cooperatives. This will ensure that existing inequalities are not exacerbated (including inequalities related to decision-making, labour burden, and access to agricultural information, finance, inputs), so that no one is left behind from efforts to address climate change. This will include designing and delivering specific modules within and in addition to agronomic training, which will include guidance on service provision to marginalized groups and the adaptation of services to persons living with disabilities.
86. The deployment of selected methods of extension will be made in parallel with, and complementary to, the dissemination of climate resilient, low-carbon technologies to farmers under Outputs 3.1. In the final year of the project, an assessment of effectiveness and efficiency will inform the modalities for continuation by the county government based on both independent evaluation and feedback mechanisms between recipients and providers (sub-activity 1.1.1.5).
87. Activity 1.1.1 will be executed by FAO. The project team will coordinate with county departments of agriculture, extension, climate change, environment, and cooperatives, with support from the project in the form of technical assistance and training.

Activity 1.1.2. Strengthen the dissemination of climate information services to last-mile users including women, youth and Persons living with disabilities (PLWD) through cooperatives and Farmer Organizations.

88. A similar process will be followed under activity 1.1.2, focused on the delivery of climate information services. The baseline assessment undertaken through the Cooperative Survey and Climate Change Survey in 2022 illustrated that men and women farmers are not yet receiving the full scope of necessary climate information services to enable adequate climate risk management and timely choices at both short (e.g., intra-seasonal) and long planning horizons (e.g., multi-year business planning).
89. Therefore, the project will, in line with improvements to extension services, seek to improve the quality, reach, usefulness and timeliness of climate information transmission to last mile-users. This will include building the capacity of devolved climate change units and decentralized meteorological offices to develop and disseminate locally tailored, user-friendly and value chain specific climate information services.
90. Based on a detailed stock-take and gap analysis in each county for the 6 value chains (sub-activity 1.1.2.1), the project will provide assistance towards the development or revision of existing climate information services (CIS)

including short- and long-term projections, seasonal forecast, agro-weather based forecast and nowcasting. and for improvements in the quality and availability of real-time weather and climate information, in cooperation with national and county meteorological departments (sub-activity 1.1.2.2). The stock-taking report will also identify the most effective and cost-efficient method for dissemination in the 6 value chains, and the project will (under Outcome 3) ensure that last-mile users are enabled to access the CIS and to understand the information. Training will be provided to agro-meteorological department staff on the development and delivery of climate information services to FOs and cooperatives using the extension methods developed under activity 1.1.1 (sub-activity 1.1.2.3).

91. To increase the use of the climate and weather information, the project will support seasonal planning at the ward level to develop locally relevant and tailored seasonal advisories in the targeted value chains and natural resources management (sub-activity 1.1.2.4). Climate information will be communicated through FOs and cooperatives, and by making use of digital agriculture solutions such as cell-phone based groups (e.g., WhatsApp distribution lists) or any extension services designed under activity 1.1.1 (sub-activity 1.1.2.5).
92. This dissemination will require the development of a list of intermediaries for transmission of climate information to groups including women, elderly and illiterate or low literate farmers (sub-activity 1.1.2.6), which will also be linked to the update of the agricultural databases under activity 1.1.4. This activity will include regular updates of farmers and farmer organizations distribution lists and maintenance of the cooperative census. Monitoring of the dissemination of climate information services will be conducted by county meteorological departments and will include surveillance of indicators such as type of product disseminated, method of dissemination, number of people reached with particular emphasis on traditionally marginalized groups (women, youth, PLWD and elderly), and frequency of use (sub-activity 1.1.2.7).
93. The gender equality and social inclusion objective for Activity 1.1.2 is to strengthen the dissemination of climate services and information to women, youth, and PLWD through cooperatives and farmer organizations. Detailed actions for achieving this objective are to liaise with county-level and cooperative gender specialists and leadership in selecting the climate information service dissemination strategies that are accessible to women, youth, and PLWD (i.e., radio, videos, public campaigns, in-person demonstrations). This process will be based on consultations with women, youth, and PLWD on the types of climate services and information needed during scoping. Key among this will be to consider the digital gender divide in implementation, and whether women (especially elder women and women living with disabilities) have additional barriers to CIS that require additional support - for example, direct access to finance. Consultations with cooperative members from marginalized groups will ensure their differentiated needs, priorities, and preferences are accounted for in strengthening the appropriate methods for dissemination.
94. Activity 1.1.2 will be executed by FAO.

Activity 1.1.3. Develop and test methodologies for decentralized carbon accounting.

95. In support of future upscaling and in direct relation to the implementation of the integrated landscape management strategies (Activities 2.1.1 and 2.2.2), the project will mobilize technical assistance to conduct training and demonstrations of carbon accounting methodologies at county level. It is expected that this activity will lead to the better integration of emissions reductions practices in future land use planning activities, green accounting, and may also support the creation of linkages between counties and carbon markets. As noted in the previous sections, locally specific data on emissions and emissions sources in the AFOLU sector are not readily available, and therefore cannot inform suitable mitigation strategies.
96. Among the methodologies being explored, the project will draw on IPCC guidance and any previous work undertaken in Kenya such as the System for Land based Emissions Estimation (SLEEK) platform under the National Communications processes, to downscale these at county level, with the supporting capacity building required. This will enable counties to undertake their own carbon balance accounting in line with the agricultural landscape management strategy.
97. A first step in this activity will be to conduct a participatory design workshop with the 14 counties (Meteorological, Environment, Agriculture, Forests, and Land departments and CCU) to select and develop a regionally specific and well adapted AFOLU carbon accounting methodology (sub-activity 1.1.3.1). A process will then be established for accessing and collecting local data, including remote sensing data, and computation of calculations (sub-activity 1.1.3.2). This will leverage existing capacity in the climate change units, including such that was provided by the Readiness Project and G-FLLOCA. Operating manuals and guidance documents will also be produced to support county administrations and to ensure harmonized comparable reports across the 14 counties (sub-activity 1.1.3.3). The project will then support the deployment of two decentralized carbon accounting exercises, using a learning-by-doing approach (sub-activity 1.1.3.4): at year 2, which will establish a baseline, and another at year 5, which

can begin to track changes that may have been induced by the project, including the changes resulting from the implementation of the landscape management strategies referred to below.

98. The gender equality and social inclusion objective for Activity 1.1.3 is to develop and test methodologies for decentralized carbon accounting in ways that are gender-responsive and socially inclusive. The participatory workshop with the 14 counties mentioned above will aim to have equal gender representation (50% women and 50% men) and youth representation and PLWD representation equivalent to their populations at national level (20% and 2%, respectively). In addition, the workshop will include a presentation and facilitated discussion session on the importance of integrating a gender equality and social inclusion lens to carbon accounting and project programming (e.g., enables inequalities in access to climate services, responses to climate change to be reduced).
99. Activity 1.1.3 will be executed by FAO.

Activity 1.1.4 Upgrade and update agricultural databases, crop and productivity datasets, cooperative census.

100. As noted during the development of this project, reliable data on production, prices, sales, cooperatives, and climate, are scattered across departments and vary among value chains and counties. To further enable county governments to better serve the transition to climate resilient, low-carbon pathways within the targeted value chains and beyond, data systems need to be improved and made more efficient. Complete and updated price and sales data informs development of climate resilient business plans for cooperatives and FOs, investment at both farmer and private sector levels and county plans in infrastructure and market development; crop and livestock production data that can be effectively related to climate data on a regular basis can also inform local climate change risk assessment and management in activity 1.1.5 and help steer farmers decisions away from maladapted practices⁷⁸. Finally, accurate and up to date data on farmers, land users, cooperatives and cooperative unions in the target value chains can help strengthen governance, government outreach to end users, and facilitate investment in the region.
101. Therefore, the project will first conduct a detailed data systems needs assessment (sub-activity 1.1.4.1). This will review the type of available data, channels for sharing the dataset, usage for decision making and barriers to access, and determine data gaps (coverage, accuracy, storage). The project will make recommendations to county governments on the improvement of these data systems in support of climate resilient, low-carbon agriculture. The next step will be to ensure that the database infrastructure is adequate to support its upgrading – this may require the purchase of additional servers or computers, as well as the development of technical guidelines and the deployment of training for relevant staff (sub-activity 1.1.4.2). The project will make sure that sufficient resources for the operation and maintenance of servers/platforms, as well as internet access, have been allocated.
102. Starting in the second year and up to year 5, the project will support counties in collecting, compiling, synthesizing, and analysing various datasets for the 6 value chains (sub-activity 1.1.4.3), such as crop production (losses and productivity) in relation to seasonal climate, land use per crop/livestock, processing and value addition data, and market data. This initial demonstration will serve as an illustration of how better data services can contribute to the reorientation of value chains. It can be extended to other value chains and sectors after the project is over. At the end of the 5th year, the counties will undertake an effectiveness review of these datasets through consultation with stakeholders and will continue supporting the priority data products using national budgets after the end of the project.
103. In addition to this, because the cooperative movement is crucial to this project and to the transformation of value chains, the project will support the county governments to update the cooperative census on a biennial basis by recruiting consultants to revise the interview protocols, sharing available data, and creating data infrastructure, including lists of contacts (sub-activity 1.1.4.4). The first time, the project will assist counties operationally to conduct the census to ensure accuracy of baseline and create linkages between counties and cooperatives; following this, the project will support the counties in developing an automated process for updating the census through an online surveying of cooperatives that is tied to the annual registration process. The cooperative census included a thorough list of indicators that covered membership and participation, social inclusion, sales and income, environmental and climate challenges and resilience, and access to assets and inputs (land, water, energy).
104. The gender equality and social inclusion objective for Activity 1.1.4 is to upgrade and update agricultural databases, crop and productivity datasets and cooperative census in ways that are gender-responsive and socially inclusive. The project will update databases with sex- and gender-disaggregated data and ensure that data on marginalized and invisible groups (i.e., women respondents living in male-headed households, widows) are included in updating datasets. Furthermore, women, youth, and PLWD will be trained on the technical aspects of

⁷⁸ For example, comparing the performance of a specific breed of cow, or variety of tree against another in the context of climate sensitivity would assist farmers in making more informed choices about breed and variety selection balancing long-term sustainability vs short-term gains.

data collection and management. Additional detailed actions for achieving this objective are listed in the Gender Action Plan.

105. This activity will be executed by FAO.

Activity 1.1.5. Assess eligible climate solutions for the agriculture sector in relation to climate impacts.

106. The proposed climate solutions included in this project are based on current knowledge of climate trends and projections, and on currently available technologies for the 6 targeted value chains. However, there is some uncertainty in the way climate impacts will materialize in the long term, and there also will be evolution in locally-specific vulnerability to the impacts of climate change, given the dynamic and diverse nature of the LREB economy. Technologies are also expected to evolve, as new approaches and techniques become available. The project creates, under Outcome 3, a continuous mechanism for technology transfer that facilitates adoption of adaptation and mitigation technologies by farmers and that will last beyond the project. However, this mechanism requires sound knowledge on climate impacts and suitable climate responses, else farmers may be led into maladaptive practices, or to make ill-informed short-term choices. Therefore, the purpose of this activity is to enable county governments to provide services related to the screening, assessment and participatory selection of suitable climate technologies that respond to the way in which climate impacts and risks are materializing locally.

107. As a first step, the project will build on the results of the GCF NAP Readiness project and the G-FLLOCA program⁷⁹ and deliver training for county climate change units, meteorological and agriculture departments on the selection, screening, and prioritization of climate solutions (sub-activity 1.1.5.1) considering available climate risk and vulnerability assessments (CRA), such as those conducted under G-FLLOCA, and the screenings that were conducted during this project's feasibility assessment. The project will support training in the identification of adaptation and mitigation climate solutions for the 6 value chains at county level, though it is expected that the process will be expanded to other value chains after the project.

108. Available Climate Risk Assessments (CRA), such as those conducted under G-FLLOCA, will then be used to inform a participatory screening of acceptable climate solutions for the targeted value chains, which will be conducted with the participation of the Departments of Agriculture and Extension, Kenya Agriculture, Livestock and Research Organization (KALRO), and other research partners (e.g., CGIAR), as well as farmer organizations and cooperatives. The result of this will be a government-validated list of climate technologies (Climate Technology Green List) that either promote resilience or help reduce emissions in the agriculture sector (sub-activity 1.1.5.2). The list will include technologies, approaches, practices, processes, and knowledge, with costs, documentation of benefits and links to potential providers. The list may be maintained and updated regularly; in the 5th year of the project, the list will be updated based on any new incoming climate impacts, scenarios, and projections for the region.

109. These "green lists" will be used under activities 3.1.1 and 3.1.2 by cooperatives and farmers to independently identify, and source climate solutions adapted to their own challenges, and (under activity 4.2) to guide decisions by financial institutions on lending towards climate resilient, low-carbon practices (reducing the risk of lending). Thus, the project helps build local capacity for autonomous adaptation, leveraging government technical capacity and private sector innovation and rapid pace of adaptation. These lists will also be shared across counties, and with the Council of Governors for broader uptake and replication (through activity 1.1.6)

110. The gender equality and social inclusion objective for Activity 1.1.5 is to assess eligible climate solutions for the agriculture sector in relation to climate impacts in ways that are gender-responsive and socially inclusive. The project will include women, youth, and PLWD in climate solution workshops for futures planning and mobilize women's and youth groups in generating climate solutions and county-level advocacy. Additional detailed actions for achieving this objective are listed in the Gender Action Plan.

111. This activity will be executed by FAO.

Activity 1.1.6. Share knowledge and lessons learned through existing platforms.

112. The purpose of this activity is to support the upscaling and broader adoption of project outputs and outcomes. In particular, the project will work with counties, cooperatives and smallholder farmers and other value chain actors to document and identify upscaling opportunities of lessons learned and factors of success that would enable the transfer of project approaches to other value chains, sectors, and regions.

113. To do this the project will leverage the already established Climate Change Multi-Sectoral Knowledge Platform, the Maarifa center and the Council of Governors, based on a Knowledge and Learning strategy that will be

⁷⁹ Under the FLLOCA program, counties will be supported in conducting county-wide risks and vulnerability assessments during this foreseen project.

developed in the first year of the project (sub-activity 1.1.6.1⁸⁰). As lessons and results from implementation are identified (starting year 3), the project will produce lessons learned documents, information products, knowledge and outreach material, and other such products to be discussed and disseminated across the various platforms. The lessons learned documents will include how the project created synergies with the National Adaptation Plan (NAP) and other national development plans, sectoral plans, and policies at both national and county level, where relevant. (sub-activity 1.1.6.2).

114. Knowledge sharing events will be organized bi-annually, bringing together project participants at county, regional and national levels as well as policy makers, experts in climate change, climate resilient agriculture and climate finance investors (sub-activity 1.1.6.3). Meetings will be held at county level during the first 5 years of the project, with regional and national level sharing events during the last two years to support upscaling. At the end of the project the activity will contribute synthesis reports and other documentation of lessons learned in support of a broader upscaling strategy. These knowledge sharing events will improve the regional coordination on responding to climate change impacts and management of agriculture landscape by facilitating dialogue and joint planning among the 14 county governments in the Lake Region Economic Bloc.
115. The gender equality and social inclusion objective for Activity 1.1.6 is to share knowledge and lessons learned through existing platforms in ways that are gender-responsive and socially inclusive. The project will ensure existing platforms are accessible to women, youth, and PLWD through consultations with local women's and youth groups and make the necessary modifications to existing platforms to increase access to marginalized and invisible groups in knowledge sharing events, in existing platforms, and in the development of strategies, sectoral plans and lessons learned documents (ensure representation of marginalized groups). Additional detailed actions for achieving this objective are listed in the Gender Action Plan.
116. This activity will be executed by FAO.

Component 2 – Sustainable Resilient Agricultural Landscapes

Outcome 2.1 - Reduced emissions from and increased adaptive capacity of the AFOLU sector

Outcome 2.2 – Increased ecosystem resilience to climate change

Output 2.1 Agricultural landscapes are managed under strategies that conserve, restore, and sustainably manage community forest and agriculture land, and reduce emissions.

Activity 2.1.1- Develop county and regional climate-resilient and low-carbon agricultural landscape management strategies, including improved watershed management, land use planning, reforestation, and natural regeneration.

117. Under this activity, FAO will provide technical assistance support to each county government towards the development of a climate resilient, low-carbon landscape management strategy and action plan, which will then be brought together in a LREB-wide integrated landscape management strategy.
118. The project will begin by conducting an assessment and mapping of agricultural landscape degradation in agricultural landscapes in each county, to obtain the latest portrait of the land in targeted areas and to determine possible vulnerability hotspots (sub-activity 2.1.1.1). Existing data will be correlated with physical observation and ground-truthing with communities will occur. The methods used for characterizing landscapes will be the Land Degradation Surveillance Framework (LDSF)⁸¹ method and/or the Restoration Opportunities Assessment Model (ROAM), which provide a biophysical baseline at landscape level and assesses processes that contribute to land degradation neutrality, including changes in land use and land cover, as well as help prioritize restoration opportunities and activities. This will be used to assist counties in prioritizing sites and types of interventions, to be implemented under activity 2.2.2.
119. In addition, the project will develop and deliver a stakeholder-tailored training on climate resilient, low-carbon landscape management to ensure harmonization of approaches (sub-activity 2.1.1.2). The training program will include aspects related to monitoring of landscape and ecosystem health, related to the carbon accounting methodologies developed under activity 1.1.3. This training will emphasize the benefits of integrated landscape management or integrated watershed management as both an ecosystem-based adaptation approach and as a method that helps reduce AFOLU emissions locally.
120. Based on the training and as a learning-by-doing undertaking, the project will support each county administration in the development of its own climate resilient, low-carbon landscape management strategy (LMS) (sub-activity 2.1.1.3).

⁸⁰ This sub-activity is financed by the budget under M&E

⁸¹ <https://samples.ccafs.cgiar.org/land-degradation-surveillance-framework-ldsf-field-guide/>

121. Participatory mechanisms will be built-in to the development and subsequent revisions of the LMS to make sure local populations, particularly smallholders who are currently using land for agriculture, have an opportunity to voice concerns. Attention will also be paid to the identification of tenure rights and holders, and any land conflicts that may hinder the achievement of project objectives. The LMS will include interventions on public or communal land (executed by county administrations) or interventions on private land (executed by communities with funding from counties and FAO) and will focus on areas where such interventions can facilitate or leverage improved productivity through ecosystem services. LMS will be submitted to County Governing Councils for adoption, which will create a further opportunity to raise awareness among county administrations and elected officials on the benefits of improved ecosystem management for economic outlook in each county, as well as integration into the CIDPs and climate change policies. The project will also support community awareness raising of the benefits of ensuring restoration, conservation, and sustainable management of their agriculture landscape (sub-activity 2.1.1.4), which will feed into the capacity of communities to formulate proposals for financing under activity 2.1.2.
122. Counties will then come together to develop a LREB-wide landscape management strategy (sub-activity 2.1.1.5). This strategy will give particular attention on upstream-downstream linkages, fragile ecosystems that straddle county borders, population movements and ecological trends nested within the broader Lake Victoria Basin. The purpose of this LREB strategy will be to harmonize action under a common objective and to pool resources for implementation later on. Counties will then be brought together at regular intervals for lessons sharing. The Council of Governors will participate in these events to make sure the benefits from integrated planning are integrated into the best practices for devolution. During the last year, the project will support an evaluation and lessons learned initiative that will inform the development of the next iteration of the LMS.
123. The project will ensure women, youth, and PLWD are consulted in the development of the landscape management strategy and implementation plan. The project will also train women, youth, and PLWD to implement and monitor landscape management plans. Additional detailed actions for achieving this objective are listed in the Gender Action Plan.

124. This activity will be executed by FAO in consultation with the Government of Kenya, including the National Treasury and the agriculture, environment, and forestry ministries at national and county levels.

Activity 2.1.2. Implement and monitor climate-resilient and low-carbon landscape management plans.

125. This activity is financed through co-financing from the Government of Kenya (through National Treasury governments, leveraging resources from the G-FLLOCA program) and FAO (through FAO-GEF7 Mt. Elgon Restoration Project). The identification of sites in which to implement priority activities will be undertaken by counties based on the results of the assessments of landscape degradation undertaken in activity 2.1.1. Methods used for these assessments include the Restoration opportunities Assessment Methodology (ROAM) which is participatory.
126. Selection and prioritization of the sub-projects is conducted using pre-established inclusion and exclusion criteria as detailed in the G-FLLOCA operations manuals, and requests for finance are submitted to the National Treasury G-FLLOCA unit for approval. Reporting on implementation also occurs through the National Treasury. The interventions that address climate change challenges and are eligible for inclusion in a LMS (GCF) or a Climate Change Action Plan (G-FLLOCA) include:
- Reforestation at spring heads or in degraded sensitive areas including gazetted forests.
 - Reforestation and creation of woodlots
 - Restoration of water catchments and watershed management
 - Community-Based Sustainable Land Management
 - Forest Landscape Restoration
 - Community-Based sustainable forestry
 - Conservation and assisted natural regeneration.
 - Conversion to agro-forestry or permaculture
 - Construction or upgrade of anti-erosive structures
 - Afforestation, erosion control, gully healing in extremely degraded areas
 - Construction or upgrade of water conservation, irrigation efficiency infrastructure
 - Wetland conservation, reconstruction, or rehabilitation
 - Restoration of riparian areas under cultivation (private land)
 - Construction/repair of community water retention ponds, weirs, dikes
 - Setting aside land (public community) as micro-reserves e.g., smaller papyrus wetlands around LVB and other water bodies
 - Removal of wastes, reduction of effluents and pollution

- Run-off management
- Land zoning
- Live-fencing, contouring, anti-erosive measures

127. The project will support the development of implementation plans and monitoring and evaluation plans for each landscape management strategy (sub-activity 2.1.2.1) while ensuring local community buy-in and participation through consultations (sub-activity 2.1.2.2).
128. Using GCF funding, each county will then implement selected reforestation, conservation, restoration, afforestation, watershed management, forest landscape restoration and/or improved land management actions in degraded hotspots of 200 ha per county around the agricultural landscapes targeted by this project (sub-activity 2.1.2.3), and in gazetted forests and reserves (10,000 ha) (sub-activity 2.1.2.4) over the 6 years of project implementation for a total of 12,800 ha⁸². Environmental and Social Safeguards and risk analyses will be conducted before implementation and environmental management plans will be developed as needed. Project proceeds will only be used to support the implementation of activities that fall within the Category B of risk assessment.
129. Modalities for implementation will include community-based delivery and direct county-government execution. Counties will also report annually on the implementation of their LMS (sub-activity 2.1.2.5) to the National Treasury, including on the number and type of projects under implementation and funds disbursed, social and environmental indicators such as number of participating women and youth, area set aside for conservation, area under restoration, type of ecological system under improved management (e.g., wetland, forest). After 6 years, counties will undertake a new survey and assessment of the landscape to fully capture the results and inform the next iteration of the strategy.
130. The Government of Kenya acting through National Treasury will execute the sub-activities co-financed by the Government of Kenya given its mandate to mobilize resources to counties, determine procedures and requirements for accessing resources, eligibility and disbursement, and ensure fiduciary standards. The counties will develop proposals to apply for the finance through the National Treasury. The National Treasury will review the proposal and decide if the requested finance will be disbursed to the counties to support activities in the selected sites. To access G- FLLoCA finance from the National Treasury, County Governments need to meet several criteria⁸³ and demonstrate Minimum Performance Conditions (MPC).
131. FAO will execute the FAO co-financed sub-activity 2.1.2.4 while the rest of sub-activities under Activity 2.1.2 will be founded and executed by the Government of Kenya.

Component 3 – Resilient livelihoods

Outcome 3 - Increased climate resilience of smallholders' livelihoods using climate-resilient, low carbon technologies.

Output 3.1 Vulnerable smallholders adopt gender-responsive and socially-inclusive climate resilient and low carbon production and processing practices, technologies, assets, and risk reduction mechanisms.

132. The project has pre-identified a suite of adaptation and mitigation technologies that will be promoted for uptake by smallholder farmers in the 6 value chains (Refer to FS section 5.3). These technologies will be included in the curriculum of farmer field schools under activity 3.1.1 and transferred through cooperatives under activity 3.1.2. This initial “Green-List” of technologies, which is based on best available knowledge, consultations, and research, is to be updated by counties regularly throughout the project, also leveraging consultations with smallholder farmers to ensure buy-in and increase adoption rates.

Activity 3.1.1. Deploy CRLCSA production/ processing assets and training to smallholder farmers, farmer organizations and associations.

133. Through the FFS approach, the project will allow farmers to experiment with and uptake climate-resilient practices, technologies, and farming systems. Farmers will receive training through FFS for 2 seasons (2 years) followed by monthly visits from the FFS facilitators throughout the project to ensure that knowledge the farmers

⁸² Sub-activity 2.1.2.2 is entirely financed by the county governments, whereas sub-activity 2.1.2.3 is financed by FAO through the GEF-7 Mount Elgon project.

⁸³ These are the selection criteria: (i) shall have put in place a CCCU that is adequately staffed; (ii) carried out a participatory climate change risk assessment (exposure to hazards and social vulnerability); (iii) prepared a CCAP; and (iv) allocated, on top of the CCRI, some 1.5 percent of its development budget to the CCCF. For the second and third grant cycles, counties must meet several additional MPCs such as (v) timely financial and implementation reporting, showing (vi) absorption capacity; (vii) use of the funds for eligible investments; (viii) compliance with social and environmental rules and regulations; and (ix) dissemination of information on the investments to communities.

learned during the FFS is reinforced, and that practices and technologies the farmers adopt are suitable to address climate variability and climate risks as well as the other community specific conditions, such as availability of inputs. All of the aspects of the FFS will be climate oriented, meaning that the farmers will be called upon to make use of climate information services to select climate resilient and low carbon interventions, and to evaluate their productivity results against assumptions related to climate conditions, as well as against productivity and income results.

134. A key aspect of Farmer Field Schools is to ensure the sustainability of the approaches promoted in current as well as future conditions. The FFS interventions are anticipated to enhance farmers' adaptive capacity and contribute towards the development of more resilient farming systems. In this regard, the services provided by the government through Outcome 1 will be crucial in helping define what technologies each FFS group will adopt at a given time.
135. In the first year, FAO will undertake, along with activity 1.1.5 and the Resilience Survey (see FS Section 6.3), a survey of participating smallholder farmers to provide a detailed understanding of climate resilient, low-carbon practices currently being implemented, assess technology and capacity needs (sub-activity 3.1.1.1). At the same time, a targeting and group formation exercise will be initiated, along with awareness raising and Free, Prior Informed Consent processes to ensure participating farmers do so willingly and in full understanding of their rights and responsibilities under the project (sub-activity 3.1.1.2).
136. Participating farmers will be selected from the areas in each county that are considered as "vulnerability hotspots," following activity 1.1.1 and the rapid CRA, which will be completed with the collaboration of counties. For criteria and methods for selecting participating farmers, please refer to FS Part C Section 5.3.
137. During Year 1, FAO will complete the development of the FFS curriculum and programs, manuals, and procedures of the FFS for the selected value chains (sub-activity 3.1.1.3), in close collaboration with the county extension department. This will include detailed information and guidance on climate risks and climate change scenarios, information on the recommended technologies (linked to activities 1.1.1 to 1.1.5) and how to implement them in the field, guidance on how to train smallholders, particularly those more vulnerable to exclusion such as women, PLWD and people with low levels of literacy. Documentary support, visual illustrations, posters, video clips and other teaching aides will also be produced in local language to support adoption and dissemination.
138. FAO will train 3 Master Trainers (sub-activity 3.1.1.4) on the climate resilient, low-carbon sustainable agricultural practices promoted in this project; local climate change risks, impacts, and vulnerability; as well as Farm Business School Modules including leadership skills and entrepreneurship such as book-keeping or contract farming. Gender modules aiming at achieving gender-transformative outcomes will also be included, to ensure that all trainings build the skills and improve the position of women smallholders (and other marginalized groups) within the value chain (not only at production stage). FAO's subject matter experts will then deliver a training of FFS facilitators in each county together with Master Trainers. FFS deployment will be organized with local non-governmental organizations (NGO) and community-based organizations (CBO), whose tasks will be to schedule FFS training sessions and demonstration activities, managing the FFS facilitators, and reporting on a daily basis in the FFS sites. For roles and responsibilities of and selection criteria of the Master Trainers and Facilitators, please refer to section 5.3 of the FS.
139. FFS roll-out will begin at year 2 (sub-activity 3.1.1.5). In the first year of the FFS training, farmers participate in field observation, analysis of the climate impacts in the communities, and identify solutions through learning and discussions led by the facilitators. In the second year, the facilitators conduct supervisory checks and visits, and organizes field days and farm visits among members to accelerate learning and dissemination. The FFS will use a participatory action-research approach for the adoption of climate resilient and low-carbon practices. The FFS participants will also have opportunities to join exchange and learning visits (at least twice for each FFS during the implementation period). The project will support FFS for the value chains selected by each county over 5 years. Each FFS will comprise of 25 men and women, though in some VC it may be possible to find groups made up of only men or only women. FFS will be sequenced in three cohorts (2520 FFS). Each FFS group (25 people) will receive around 35 trainings a year for two years. FFS groups will also receive targeted material support (sub-activity 3.1.1.6) such as inputs, tools and materials for applying the various climate resilient, low-carbon technologies. For sequencing and roll out of the FFS, please refer to section 5.3 of the FS. At the end of a cohort, farmers who are in a position to do so (e.g those who can pay coop membership dues) will then be supported to become members of existing cooperatives (see activity 3.1.3).
140. Activity 3.1.1 will be executed by FAO using GCF funding.

Activity 3.1.2. Disseminate CRLCSA technology, knowledge, and assets to cooperative members through peer-to-peer networks and exchanges.

141. Activity 3.1.2 is designed to transfer technology to farmers through cooperatives and will be executed by Agriterra using GCF and Danish MFA funding (sub-activities 3.1.2.1 and 3.1.2.3). The process envisaged for this activity follows a tried and tested process of technology transfer already piloted in Kenya by Agriterra, in which cooperatives receive multi-year support according to their needs and business plans, through a milestone-based action plan.
142. The project will begin by identifying and on-boarding participating cooperatives. For selection criteria please refer to section 5.3 of the FS. Following the initial screening and selection, each cooperative will undergo a comprehensive Cooperative Assessment and Climate Clever Check (CA-CCC) which includes the following elements (sub-activity 3.1.2.1):
143. Cooperative Assessment (CA): The assessment includes an analysis of financial data, production, sales, and employment data, governance and transparency, benefit sharing, and will also include an analysis of processes used to transfer knowledge, services, and technology to individual members, with a focus on vulnerable groups. The assessment will also consider practices and barriers to social inclusion and gender sensitivity.
144. Climate Clever Check (CCC): The CCC provides an assessment of climate risks on production and processing as experienced by each cooperative (current and future) and helps evaluate the soundness of any climate response plans. It includes an analysis of the climate change challenges from both adaptation and mitigation perspectives along the entire value chain, from input supply to consumption. Please refer to section 5.3 of the FS for additional detail on screening procedures.
145. Agriterra will benefit from the climate risk assessments conducted by counties under Activity 1.1.5. Challenges and the solutions applied by the cooperative are ranked and compared according to a point system. The impact of challenges and solutions on the business case of the cooperative is also reviewed (intensity of challenge and appropriateness of response). Each cooperative receives a “mark” or scoring against the different criteria at the end of the initial assessment; the assessment is then repeated once the cooperative is exited from project support to measure progress objectively and in a participatory manner.
146. The CA/CCC serves as a basis for recommending solutions to the members across the full spectrum of cooperative activities, focused primarily on climate interventions, extension services, governance, management, and finance interventions if they help to sustainably implement the climate interventions. The solutions will be taken from the list of adaptation and climate solutions proposed in each county under Activity 1.1.5.
147. Support received by cooperatives takes many forms:
 - Support in the identification and implementation of selected climate resilient, low-carbon technology, including: hiring and training cooperative extension officers, consultants and agri-poolers, training, and technical backstopping (sub-activity 3.1.2.2)
 - Mobilization of inputs, materials, equipment, supplies (sub-activity 3.1.2.3). For example, the project may assist a cooperative in acquiring renewable energy technology or processing or packaging material, provided it will benefit the entire membership of the cooperative. The project may also support the full cost of the rehabilitation or upgrade of cooperative infrastructure such as water reservoirs, storage facilities, shelters, barns, and hatcheries, particularly when those are assessed as maladapted to the impacts of climate variability and climate change.
 - Support for the development of climate resilient, low-carbon business plans and business cases towards and mobilization of capital and finance (under output 4.1)
 - Support for the improvement of governance and management particularly for meeting climate-related requirements, onboarding new members, disseminating information and benefit sharing, social and gender inclusion. (Activity 3.1.3)
148. The targeted number of cooperatives is 130, onboarded in the first 4 years of the project. To support the implementation of multi-year support workplans under sub-activity 3.1.2.2, the project will deploy service providers or holders of technologies, knowledge, processes and goods or services that respond to specific needs identified. These partners will be identified using the Agri-Pool roster of experts or through open procurement. Screening criteria for inclusion in the Agri-Pool are listed in section 5.3 of the FS.
149. Because the demand for support is likely to exceed the available resources of the project, and to ensure equitable access to project support among beneficiaries, the support workplans will include a measure of prioritization. In this regard, a key part of the project's upscaling and exit strategy will also rely on the provision of loans and finance services by participating financial institutions (as leveraged financing) to the more mature cooperatives under output 4.2.

150. In addition to leveraging expertise, goods and services, the project will ensure that the technologies are suitably disseminated to the entire cooperative membership (sub-activity 3.1.2.4). When training is delivered by Agripoolers or service providers to sub-groups of cooperative members, or to the executive or lead farmers, requirements will be in place to ensure that these then transfer all knowledge, processes, information, and capacity to other cooperative members. These include specific terms of reference for lead farmers and the maintenance of registries.

151. In order to track adaptation benefits from output 3.1, smallholder participants will also participate in the Resilience Survey (Refer to Monitoring and Evaluation Plan in Annex 11, and section 6 of the FS).

Activity 3.1.3. Support smallholder farmer aggregation into cooperatives and other business units as climate risk reduction and risk sharing mechanisms.

152. Activity 3.1.3 will be cofinanced by the Danish Ministry of Foreign Affairs. The purpose of this activity is to nourish the cooperative movement as a key mechanism to share and reduce risks among farmers, particularly climate risks and the financial risks farmers shoulder when adopting climate resilient, low-carbon technologies.

153. Training and capacity development will be provided to farmer organizations with a focus on the management and governance of FO and cooperatives (sub-activity 3.1.3.1). This will help increase smallholders' business capacity, a key requirement to ensure market access regardless of the form the FO takes.

154. The project will encourage farmers to become members of existing cooperatives or to create new cooperatives (sub-activity 3.1.3.2). At the end of an FFS cohort training, smallholders who are able to meet the minimum requirements for cooperative membership (e.g. payment of dues, or coop-specific requirements) will be invited to join. The project aims to support 6,300 farmers to join existing cooperatives, and the project will also provide technical assistance and material support to cooperatives to facilitate onboarding of new members (sub-activity 3.1.3.3).

155. Field visits and awareness raising events will be organized to highlight the benefits of cooperative membership, by documenting specific advantages participating farmers have gained. It is expected that the better performance of production and increased income seen among members of cooperatives will serve as a powerful incentive for other individual farmers who may be working in isolation. Knowledge sharing mechanisms such as those leveraged under Output 1.1 (Maarifa Centers, multistakeholder platforms, Lake Region Economic Bloc, Lake Victoria Basin Authorities) and other awareness raising mechanisms will be used to raise awareness of documented best practices within and outside cooperatives.

156. To support integration of new members and the efficient management of cooperatives (sub-activity 3.1.3.3), Agriterro will deploy the My.Coop⁸⁴ tool among beneficiary cooperatives. It was developed using International Labour Organisation's (ILO) Materials and Techniques for Cooperative Management (MATCOM) Program.

157. This activity will be executed by Agriterro.

Activity 3.1.4. Support improvements in social inclusion and women's meaningful participation in climate resilient, low-carbon value chains

158. This activity is designed to ensure that project activities, technology transfer and support to cooperatives also contribute to the achievement of improvements in social inclusion and in the participation of women, youth and PLWD in the 6 value chains. As noted in the Gender Assessment and Gender action plan, ensuring equitable participation and inclusion requires a more subtle approach that goes beyond fixing "quotas" and considers barriers to participation at various stages. The activity therefore aims to influence change in the way in which women, youth and PLWD participate in agriculture (beyond the primary production stage), benefit from their work (ensuring they receive appropriate remuneration and socio-economic benefits), and influence decision-making in households, farmer organizations, cooperatives and community or county policies, where gender relations and women's positions are improved, and women's entrepreneurship is boosted, contributing to SDG 5.1.

159. Gender focal points will be identified in counties, and in each EE, as well as within farmer organizations. At the start of the project, staff in the executing entities, project coordination unit, and counties will receive mandatory training in the prevention and management of Sexual Exploitation, Abuse, and harassment (SEAH), Gender-Based Violence (GBV) and in the application of the FAO Grievance and Redress Mechanism (GRM) to handle such incidents and ensure safe working conditions for women, PLWD, and vulnerable groups (sub-activity 3.1.4.1). Trainings will be delivered to men and women on gender and social inclusion-related barriers and on ways to identify and address them. The sub-activity also includes delivering awareness raising and training for women, PLWD and other vulnerable groups on their rights in the context of the project.

⁸⁴ https://www.ilo.org/global/topics/cooperatives/publications/WCMS_644824/lang-en/index.htm

160. The project will also deliver targeted trainings through FFS and cooperatives in the context of the dissemination of agricultural technologies, following the tested Agriterra and FAO approaches to develop leadership, participation and meaningful inclusion of women, youth and persons living with disabilities (sub activity 3.1.4.2). The approaches are the Women's Leadership-Youth leadership trainings developed by Agriterra as well as the Gender Action Learning System (GALS). These trainings will focus on accelerating the inclusion of vulnerable groups in production, processing and value addition, and trade. For example, during Women's leadership workshops (which are also attended by men), the role of women as cooperative members, entrepreneurs and leaders are discussed and action plans are developed to improve women's meaningful participation in the cooperative's activities. Solutions proposed in such action plans could include, among others: changing bylaws in cooperatives to facilitate women's accession to leadership roles, offering solutions to issues related to transferring ownership of trees and land to women, equipping women leaders with more knowledge, resources and support, ensuring recruitment of women extension workers and in non-traditional positions.

161. At the end of the project, the expected result of these efforts is that more women, youth and PLWD can more easily accede to roles of meaningful participation in the targeted value chains. The project also intends to monitor gender objectives through all other activities, including through a gender impact assessment at end of project and the collection of gender-and social-inclusion related data throughout implementation (sub-activity 3.1.4.3). Please refer to the Gender Action Plan (Annex 8) for detail.

162. This activity will be delivered by FAO using GCF funds, in coordination with Agriterra and the Government of Kenya at national and county levels, and in particular the beneficiary cooperatives and farmer organizations.

Component 4 – Scaling through CRLCSA market and finance

Outcome 4 - Climate resilient, low carbon value chains are sustained financially and upscaled by the adoption of new business models and practices among farmers and value chain actors.

Output 4.1 Increased access to markets and profitability of climate smart, low carbon sustainable agricultural products

163. Many smallholder farmers are producing using practices that are not well adapted to climate change, or that generate undue emissions, in particular in cases where agricultural land encroaches onto forests and marginal lands. Under Outcome 3, the project supports smallholders in producing more (and better) without resorting to land expansion, and to assist them in developing coping and risk reduction mechanisms in response to climate variability, climate change and climate shocks. However, at this stage the market incentives to sustain climate resilient, low-carbon production are not in place – that is, farmers are not aware they can obtain price premiums for climate resilient, low-carbon commodities.

Activity 4.1.1 Work with buyers and aggregators to increase demand and market opportunities for CRLCSA commodities.

164. The project will work on the demand side of the value chain to increase uptake of CRLCSA commodities produced by project beneficiaries. Profitability of adopting climate resilient, low-carbon technologies will be key to encouraging continued sustainability, replication, and upscaling. Therefore, the project will work with county governments and cooperatives, as well as other actors along the value chain, to develop market opportunities for climate resilient, low-carbon products.

165. Demand for the products included in the 6 value chains has long been established, including through the targeting exercise undertaken at the start of project design (demand was a criteria for selection) and through the market study and value chain analyses (refer to Annexes 23a-f). Initial data shows that fair trade, organic coffee and tea, and vegetables fetch a higher price on both local and international markets. In all other value chains, prices are currently set based on product quality and grading (e.g. avocado size, nutritional content). However, more information is needed to establish rational grading and price structures for producers to negotiate fair prices for their climate resilient, low carbon products. Therefore, the project will begin by carefully analyzing the demand, quality standards, grading norms and prices for climate resilient, low-carbon products, to inform quality-based equitable price scales and premiums for products produced with climate resilient, low-carbon technologies (with or without certification or labeling) (sub-activity 4.1.1.1). This will feed into a set of recommendations in each value chain on how to increase the market share of climate resilient, low-carbon products. The analysis will also help identify buyers of products supported by the project during project execution (examples for each value chain are listed in the Market Study and Value chain analyses under Annex 23a-f). It is expected that at least 2 buyers will be identified for each value chain.

166. This process will also be supported by the development of information products, knowledge events, and awareness campaigns to increase aggregators,' market intermediaries,' and buyers' awareness of the existence of climate resilient, low-carbon and sustainable commodities. Research and comparative analysis of quality, grading

and pricing of products supported under the project will be conducted, to benchmark climate resilient, low-carbon produce against existing quality standards (e.g., the coffee grading system) (sub-activity 4.1.1.2). This analysis will feed into the development of information products and into the development of the marketing campaigns for each value chain.

167. To create further market opportunities for targeted commodities, starting year 3, the project will team up with relevant value chain actors (aggregators, intermediaries, and farmer organizations) as well as county governments, to develop and deliver a targeted marketing campaign for each of the 6 value chains, highlighting the benefits and attractiveness of the products under climate resilient, low-carbon technologies (sub-activity 4.1.1.3). This will include developing technical briefs for corporate buyers and market exchange (e.g., The Coffee Exchange), but also consumer-oriented marketing products, video and print profiles of producers and cooperatives. The project will also work with county governments to develop their own ongoing marketing services in support of climate resilient, low-carbon value chains. Participation in trade fairs, trade missions, and funding to support buyer's visits to beneficiary farms and facilities will also be supported.
168. Finally, throughout its duration, the project will provide technical assistance and business mentoring, to the most mature cooperatives to negotiate sales agreements and contracts with identified buyers (sub-activity 4.1.1.4). This will include negotiation training, advice on marketing selected products, legal agreement review, price setting advice and branding. This service will be on-demand only once the cooperative has sufficiently demonstrated that it has adopted the climate resilient, low-carbon practices promoted in the project, and just prior to being exited from Agriterro support. The development and incubation of small agribusinesses among less mature farmer organizations (sub-activity 4.1.1.5) will also be supported through co-financing from FAO by leveraging the GEF-7 Project, which deploys innovative business hubs to promote market access and services, including business incubators for smallholder farmers pursuing nature-based livelihoods.
169. The Counties will then use the updated agricultural databases (activity 1.1.5) to identify and invite participants to marketing events, which will include market fairs, awareness campaigns, special trade shows, and showcases (sub-activity 4.1.1.6). Using the data collected in the agricultural databases, the counties will develop seasonal market insights that will include sales trends, production trends (including qualitative and quantitative trends), and price evolution for each of the 6 value chains (sub-activity 4.1.1.7). After the end of the project, this service may be extended to other value chains. This sub-activity creates a bridge between the county government and Kenyan governmental institutions and the work of the project, ensuring that county services to cooperatives also encompass market-oriented activities.
170. This activity will be delivered by FAO using GCF funds.

Activity 4.1.2. Increase access to various third-party certification and labeling schemes.

171. The purpose of this activity is to remove market access barriers and create a distinct market pathway for commodities produced or processed under climate resilient, low-carbon processes. Certification creates added value and ensures better prices are fetched for goods produced at a higher cost for the farmers and processors. This activity, which will be delivered jointly with the above, will provide technical assistance to farmer organizations and cooperatives in accessing and complying with relevant certification schemes on a voluntary basis. Third party certification schemes that are being considered include Encert Organic, Kenya Organic Agriculture Network or IFOAM, EcoCert, Global GAP, Rainforest Alliance, Fair Trade, and others, depending on the commodity and its market. First, the farmer organizations will be identified based on current market access, maturity, inclusiveness, and performance across the CA and CCC assessments, and of course the rate of adoption of CRLCSA practices among members. A selection of 30 cooperatives and FO is expected during the project, starting at year 3.
172. These organizations will be supported in identifying the most suitable certification scheme – i.e., the one that would bring the most added value both economically and from a climate resilience or decarbonization point of view. It is understood that some such certification schemes may take years to achieve, therefore only the cases where certification is realistically achievable within 3 years of project implementation will be taken on board. (sub-activity 4.1.2.1).
173. Each organization will then receive targeted technical assistance, training, and expert advisory services (sub-activity 4.1.2.2) to understand and apply the requirements, to strengthen reporting and transparency mechanisms, and to meet and document any legal or institutional requirements. The annual costs of maintaining certification will be borne by the farmer organization or cooperative. As part of the support provided under activity 4.2, the project will assist in ensuring that these costs are also included in the price-setting and business planning processes of each cooperative to ensure sustainability and continued uptake.
174. This activity will be delivered by Agriterro using GCF funds.

Output 4.2 Vulnerable smallholders and their organizations have increased access to gender-responsive and socially inclusive financial products that support climate resilient, low carbon growth.

Activity 4.2.1. Develop gender-responsive and socially inclusive private finance tools, procedures, and products to promote the upscale of CRLCSA value chains.

175. Under this activity, which will be implemented by FAO in cooperation with Kenya Commercial Bank (KCB), Cooperative Bank and Equity Bank, the project will support the development of capacity within financial institutions to support climate resilient, low-carbon value chains. This activity will help strengthen and increase the supply and accessibility of financial products available to support climate resilient, low-carbon agriculture in the region.
176. Working with the three FI, the project will support financial institutions, through consultancies and expert advice, in the development and roll out of new financial products or in the revision of existing products and services (sub-activity 4.2.1.1) that are more tailored to climate resilient, low-carbon practices and local stakeholders across the 6 value chains.
177. Each FI will benefit from support for the development of procedures, guidance material and other documentation, as well as for the development of data collection processes to enable disaggregated reporting and tracking of benefits and beneficiaries. Training will be developed for FI at headquarter as well as in decentralized branches in the LREB region, to ensure rapid operationalization of the products and services (sub-activity 4.2.1.2). Other technical assistance will include support for improving risk analysis methodologies for loans in the 6 value chains climate resilient, low-carbon pathways to ensure that climate risks do not compound financial risks; improving accessibility conditions such as collateral requirements, repayment terms, interest rates; and analyzing social barriers to access to any of the various products and developing internal ESG mitigation procedures.
178. The features of the new or revised financial products will include the following:
- Targets application of climate resilient, low-carbon technologies as identified in the project and the county green lists.
 - Ease of access or revision of eligibility conditions, including size of lender, collateral requirements, to ensure FO and cooperatives, vulnerable groups have equitable access.
 - Value chain specificity
 - Increasing concessionality, including through the mobilization of guarantees ⁸⁵, interest rate buy-downs, first loss guarantees, modified repayment schedules, etc. These mechanisms will be identified during the project's first year and the project will support Banks in their negotiations or in leveraging agreement.
 - Contain or propose risk reduction mechanisms for farmers and farmer organizations (e.g., insurance)
 - Leverage digital technologies (e.g., digital payment; remote credit appraisal based on data analytics)
 - Be complementary to the current Bank offer in support of climate resilient, low-carbon practices in the agriculture sector.
179. This technical assistance will be executed by FAO with funding from GCF.

Activity 4.2.2. Support smallholders and their business units in the development of bankable business plans, with particular focus on social inclusion and gender-based access

180. In parallel to activity 4.2.1, the project will also support farmer organizations, value chain actors (e.g processors) and cooperatives in accessing the financial services offered by the partner financial institutions. This will take on two forms: first, the project will deliver training on financial literacy and financial management to farmers, farmer organizations and cooperatives (sub-activity 4.2.2.1).
181. Second, cooperatives who operate on the more "mature" end of the spectrum will also benefit from assistance in the development of bankable business plans and finance requests that will be submitted to participating financial institutions (sub-activity 4.2.2.2). Kenya Commercial Bank (KCB), Co-operative Bank and Equity Bank have expressed their strong interest in providing up to USD 10 million in loans as leveraged financing. The purpose of this assistance is to ensure that cooperatives have, in the long-term, the autonomous capacity to mobilize their own financing. This will bridge the gap or the mismatch between supply and demand of climate resilient, low-carbon finance in the 6 value chains.

⁸⁵ For example, the project will explore opportunities for FI to link with agriculture lending incentive providers such as ACELI. ACELI Africa Trust is currently offering risk-reduction incentives to member financial institutions in Kenya. These can reach 10% of the lending portfolio, provided the loans meet several criteria, which are similar to the project's eligibility criteria. .

182. Activity 4.2.2 will be executed by Agriterra using Danish MFA grant funding, in continuity with activities under Output 3.2.

Activity 4.2.3 Facilitate smallholders access to financial incentives schemes for agroforestry

183. As part of the innovations introduced in this project, and to support continuous mobilization of finance and investment into CRLCSA value chains, the project will assist select cooperatives in the 6 value chains in accessing cooperatives in the 6 value chains in accessing carbon markets, payments for ecosystem services, biodiversity offsets or conservation finance. Cooperatives selected for this activity will be those who show the most success in implementing CRLCSA practices at farm level. In year 3, the project will begin by undertaking an analysis of the most promising avenues based on the rate of adoption and success of the technologies promoted in the project (sub-activity 4.2.3.1). Options considered may include, inter alia: the ACORN facility, Net Zero Adaptation Facility, Conservation Finance, Payments for Ecosystem Services or Biodiversity Offsets, or private sector impact investment funds.

184. Support provided to cooperatives and their members will include raising awareness on the availability and opportunity for the various financial mechanisms (sub-activity 4.2.3.2) and developing training for participating cooperatives on the requirements of each selected mechanism (sub-activity 4.2.3.3). This may include training on conducting farm-level carbon accounting, implementing traceability requirements, monitoring of carbon stocks in plantations and forests and monitoring biodiversity at various scales, and managing reporting requirements for finance.

185. This activity will be executed by Agriterra using GCF grant for sub-activity 4.2.3.1 and 4.2.3.3 and Danish MFA grant for sub-activity 4.2.3.2.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Accredited Entity (AE)

186. FAO will serve as the Accredited Entity (AE) for this project. As such, FAO will be responsible for the overall management of the project, including: (i) all aspects of project appraisal; (ii) administrative, financial and technical oversight and supervision throughout project implementation; (iii) ensuring funds are effectively managed to deliver results and achieve objectives; (iv) ensuring the quality of project monitoring, as well as the timeliness and quality of reporting to the GCF; and (v) project closure and evaluation. FAO will ensure these responsibilities in accordance with the detailed provisions outlined in the Accreditation Master Agreement (AMA) between FAO and GCF.

187. FAO role as AE will be attributed to the relevant offices and divisions in FAO Headquarters located in Rome, Italy (HQ), Sub-Regional Office for Eastern Africa located in Addis Ababa, Ethiopia, and the Country Representation Office for Kenya (FAO- Kenya).

188. To fulfil the AE functions, FAO will set up a dedicated Project Task Force (PTF) in line with FAO project cycle guidelines. The PTF will be composed by the Budget Holder (BH), the Lead Technical Officer (LTO), Funding Liaison Officer (FLO), HQ Technical Officer and other officers, as appropriate.

189. The PTF will remain independent from the Executing Entity functions also performed by FAO (see Project execution section below). In line with the GCF policy on fees adopted through GCF Board Decision B.19/09, the above-mentioned segregation of responsibilities within FAO will ensure that the Organization can independently and effectively perform the AE functions listed in the GCF General principles and indicative list of eligible costs covered under GCF fees and project management costs.

Project co-financing and leveraged financing

190. The Ministry of Foreign Affairs of the Government of Denmark (DMFA) will provide co-financing of DKK 40 million in the form of Grant to the project. The host country represented by the National Treasury of Kenya will be providing co-financing of USD 14 million in the form of grants. FAO will provide co-financing to the project in the form of cash (grant) through the GEF-7 funded Integrated Landscape Management for conservation and restoration of the Mt. Elgon in Western Kenya and the FAO's Technical Cooperation Programme, for a total amount of USD 985,910.

191. The co-financiers are responsible for reporting to the AE in accordance with the detailed provisions outlined in the GCF policies as well as AMA, Funded Activity Agreement (FAA) between FAO and GCF and the co-financing agreement signed between the co-financier and FAO in its capacity of AE, on co-financing activities execution, and the disbursed and allocated co-financing amount. Each co-financier will be responsible for executing and managing and their co-financing funds under the coordination of the Project Management Unit (PMU) and through the Project Steering Committee (see Project execution section below).

192. In addition, the project will include up to USD 10 million in leveraged financing in Loans from the Kenya Commercial Bank, Equity Bank and the Cooperative Bank of Kenya⁸⁶ which will be channeled to farmers, cooperatives and value chain actors on the basis of bankable proposals and business plans developed under Component 4.

Other project partners

193. The partners will be engaged during project implementation as a service provider in accordance with relevant FAO Manual Sections.

Project execution

Executing Entities (EE)

194. The project will be executed by FAO and Agriterra in a co-execution modality to deliver the project activities funded by GCF proceeds and Danish MFA (DMFA) funding. A description of each EE is indicated in section 7 of the FS, Part C, including track record and previous similar initiatives.

195. FAO will act as co-EE and will ensure strong country-driven execution of project activities and will oversee the execution of selected activities funded by GCF proceeds (see Table B.4.1) based on its comparative advantages.

196. A FAO Project Execution Team (PET) will be set up in FAO-Kenya, comprising relevant FAO country personnel (Assistant FAO Representative Programme, Administrative and operational staff) and will be responsible for FAO EE function.

197. Agriterra is a global farmer advisory organization that takes its roots in the agricultural sector and farmer associations in the Netherlands. It focuses on strengthening the position of cooperatives and farmers' organizations in counties including Kenya. As a member of the Cooperative Partnership for Climate Smart Food and Forestry, a consortium of Danish, Dutch, and Finnish agriculture development agencies and experts, Agriterra has been implementing technology transfer programs in Kenya leveraging expertise of Dutch agriculture experts on climate resilient agriculture. Decades of work with cooperatives has led to evolving a comprehensive model of technology transfer that leverages the experience of farmers to benefit other farmers. This enables learning by doing, creating rich relationships and business partnerships, and developing locally adapted solutions creatively.

198. The Government of Kenya, acting through National Treasury, will execute the activities co-financed by the Government of Kenya. The National Treasury will be responsible to carry out the Kenya Government co-financed activities 2.1.2. The National Treasury will exercise decision making power and retain overall responsibility over the use of the co-finance to support the activities in the targeted counties. The National Treasury will have the final authority in selecting the sites and interventions and the implementation of the activities in accordance with G-FLLoCA procedures. The Government of Kenya will not administer GCF proceeds as EE.

199. The co-EEs will conduct the procurement following their own procurement policy and rules.

Table 7: Responsibilities of Executing Entities at sub- activity level.

Activity	Sub-activity	Executing Entity	Funding source
1.1.1 Develop and deploy innovative and efficient extension methods for disseminating and demonstrating CRLCSA knowledge, technologies, and practices in gender-responsive and socially inclusive ways	1.1.1.1	FAO	GCF
	1.1.1.2	FAO	GCF
	1.1.1.3	FAO	FAO
	1.1.1.4	FAO	FAO
	1.1.1.5	FAO	GCF
1.1.2 Strengthen the dissemination of climate information services to last-mile users including women, youth and PLWD through cooperatives and Farmer Organizations.	1.1.2.1	FAO	GCF
	1.1.2.2	FAO	GCF
	1.1.2.3	FAO	GCF
	1.1.2.4	FAO	GCF
	1.1.2.5	FAO	GCF
	1.1.2.6	FAO	GCF
	1.1.2.7	FAO	GCF
1.1.3 Develop and test methodologies for decentralized carbon accounting	1.1.3.1	FAO	FAO

⁸⁶ The specific amount leveraged from the Kenya Commercial Bank is currently under negotiation and will be confirmed during the project's review process.

	1.1.3.2	FAO	FAO
	1.1.3.3	FAO	FAO
	1.1.3.4	FAO	FAO
1.1.4 Upgrade and update agricultural databases, crop and productivity datasets, cooperative census	1.1.4.1	FAO	FAO
	1.1.4.2	FAO	FAO
	1.1.4.3	FAO	GCF
	1.1.4.4	FAO	GCF
1.1.5 Assess local climate change impacts and eligible climate solutions for the ag sector	1.1.5.1	FAO	FAO
	1.1.5.2	FAO	GCF
1.1.6 Share knowledge and lessons learned through existing platforms	1.1.6.1	FAO	GCF
	1.1.6.2	FAO	GCF
	1.1.6.3	FAO	GCF
2.1.1 Develop a county climate-resilient and low-carbon agricultural landscape management strategy and implementation plan, including improved watershed management, land use planning, reforestation, and natural regeneration	2.1.1.1	FAO	GCF
	2.1.1.2	FAO	GCF
	2.1.1.3	FAO	GCF
	2.1.1.4	FAO	GCF
	2.1.1.5	FAO	GCF
2.1.2 Implement and monitor climate-resilient and low-carbon landscape management plans.	2.1.2.1	GoK	GoK
	2.1.2.2	GoK	GoK
	2.1.2.3	GoK	GoK
	2.1.2.4	FAO	FAO
	2.1.2.5	GoK	GoK
3.1.1 Deploy CRLCSA production/ processing assets and training to smallholder farmers, farmer organizations and associations	3.1.1.1	FAO	GCF
	3.1.1.2	FAO	GCF
	3.1.1.3	FAO	GCF
	3.1.1.4	FAO	GCF
	3.1.1.5	FAO	GCF
	3.1.1.6	FAO	GCF
3.1.2 Disseminate CRLCSA technology, knowledge, and assets to cooperative members through peer-to-peer networks and exchanges	3.1.2.1	AGT	DMFA
	3.1.2.2	AGT	GCF
	3.1.2.3	AGT	DMFA
	3.1.2.4	AGT	DMFA
3.1.3 Support smallholder farmer aggregation into cooperatives and other business units as climate risk reduction and risk sharing mechanisms	3.1.3.1	AGT	DMFA
	3.1.3.2	AGT	GCF
	3.1.3.3	AGT	DMFA
3.1.4 Support improvements in social inclusion and women's meaningful participation in CRLC value chains	3.1.4.1	FAO	GCF
	3.1.4.2	FAO	GCF
	3.1.4.3	FAO	GCF
4.1.1 Work with buyers and aggregators to increase demand and market opportunities for CRLCSA commodities	4.1.1.1	FAO	GCF
	4.1.1.2	FAO	GCF
	4.1.1.3	FAO	GCF
	4.1.1.4	FAO	GCF
	4.1.1.5	FAO	FAO
	4.1.1.6	FAO	GCF
	4.1.1.7	FAO	GCF
4.1.2 Increase access to various certification and labeling schemes	4.1.2.1	AGT	GCF
	4.1.2.2	AGT	GCF
4.2.1 Develop gender-responsive and socially inclusive private finance tools, procedures, and products to promote the upscale of CRLCSA value chains	4.2.1.1	FAO	GCF
	4.2.1.2	FAO	GCF
4.2.2 Support smallholders and their business units in the development of bankable business plans, with particular focus on social inclusion and gender-based access	4.2.2.1	AGT	DMFA
	4.2.2.2	AGT	DMFA
4.2.3 Facilitate smallholders access to financial incentives schemes for agroforestry	4.2.3.1	AGT	GCF

	4.2.3.2	AGT	DMFA
	4.2.3.3	AGT	GCF

200. A dedicated Project Management Unit (PMU) will be established and hosted by FAO in Nairobi, Kenya with a regional antenna located in Kisumu. The PMU will be functional for the entire duration of the project. The PMU will coordinate directly with all relevant government departments, county administrations, external partners, service providers, NGOs and communities, and will be responsible for providing support to the execution of day-to-day activities.

201. The PMU will be led by the assistant FAO Representative (program) and a National Project Coordinator (NPC) who will be recruited by FAO. The NPC will be responsible for overall project management and coordination with project stakeholders. The PMU will also include finance, procurement and administrative officers, Human resources, legal and contracting agents; Regional project coordinators, program assistants, a gender and social inclusion specialist and ESS specialist, as well as M&E officers.

202. Please refer to Annex 2, FS, Part C, Section 8 for more detail.

Project Governance

203. A Project Steering Committee (PSC) will be established to provide strategic guidance for the project. The PSC will be co-chaired by the PS of the Treasury (NDA) and the Principal Secretary of the MoA. As indicated in section B.4, FAO is part of the PSC and as such is a party to the final decision-making process. Members will also include:

- The Principal Secretary (Representative) from the Ministry of Cooperative Development & Marketing (Ministry of Industry, Trade And Cooperatives),
- The Principal Secretary (Representative) of Ministry of Environment
- The Chair of Agriculture Committee Council of Governors
- The Chair of Climate Change Committee Chair of the CoG
- The Chief Executive of the Lake Region Economic Bloc and
- The Representative of FAO in Kenya

204. The role of the PSC will be to: (i) provide overall guidance and direction to the project, ensuring it remains within any specified constraints; (ii) address project issues as raised by the national project coordinator; (iii) monitor project risks and the effectiveness of mitigation measures, and provide guidance on new project risks, and agree on possible countermeasures and management actions to address specific risks; (iv) review the project progress, and provide direction and recommendations to ensure that the agreed deliverables are produced satisfactorily according to plans; (v) review and agree with annual work plan and provide necessary strategic guidance for its implementation; (vi) appraise the annual project implementation report, including the quality assessment rating report; (vii) make recommendations for subsequent work plans to build on achievements and address any shortcomings; and (viii) provide ad hoc direction and advice for exceptional situations when the project coordinator's tolerances are exceeded.

205. The PSC will be expected to meet formally at least once every 12 months. Formal meetings will be scheduled and arranged by the National Project Coordinator in consultation with, and at the request of PSC members (with tentative dates for the following meeting being agreed under Any Other Business). Extraordinary meetings of the PSC can be requested by any of its members.

206. The project will also be supported by a Project Technical Coordinating Committee (PTCC). The PTCC will be chaired by FAO Kenya, and supported by representatives from the Ministry of Environment, Ministry of Agriculture, and the Ministry of Cooperatives, Agriterria, the Cooperative Partnership, and representatives from private sector and civil society. A representative of Indigenous Peoples will also be invited to participate in the PTCC, and attention will be paid to ensuring equal gender participation in the Committee. The PTCC will meet every semi-annually using in-person and online platforms and its function will be to conduct overall Monitoring, learning, and reporting on project delivery, conduct regular risk management, and address any implementation issue prior to and in between high-level PSC meetings. The PTCC will provide technical guidance to PMU and to the PSC.

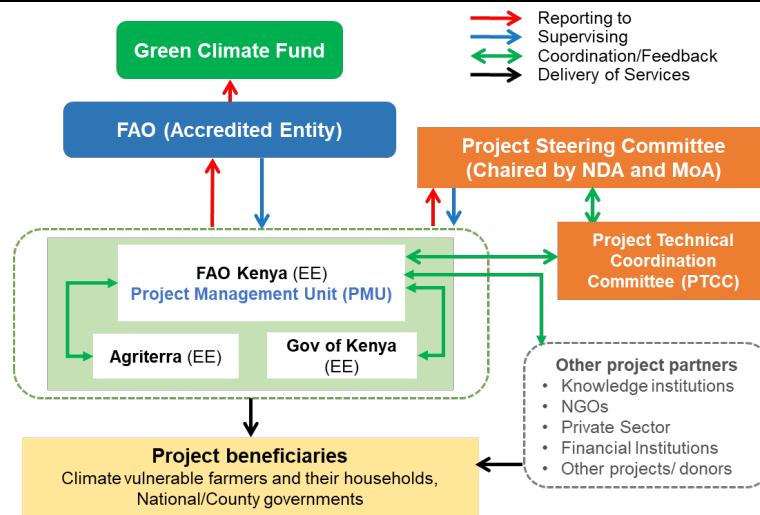


Figure 3: Implementation arrangements

Flow of funds and legal arrangements

207. In line with the project implementation arrangements outlined above, GCF proceeds received by FAO in its capacity as Accredited Entity will flow to the co-Executing Entities, namely FAO-Kenya and Agriterra for the implementation of project activities.
208. The contractual arrangements include; 1) the Project Agreement between FAO and the Government of Kenya, which also include relevant clauses related to co-financing (and execution of the cofinancing) from the Government (Subsidiary Agreement), 2) the execution agreement between FAO and Agriterra (Subsidiary Agreement), and 3) the co-financing agreement between FAO and the DMFA.

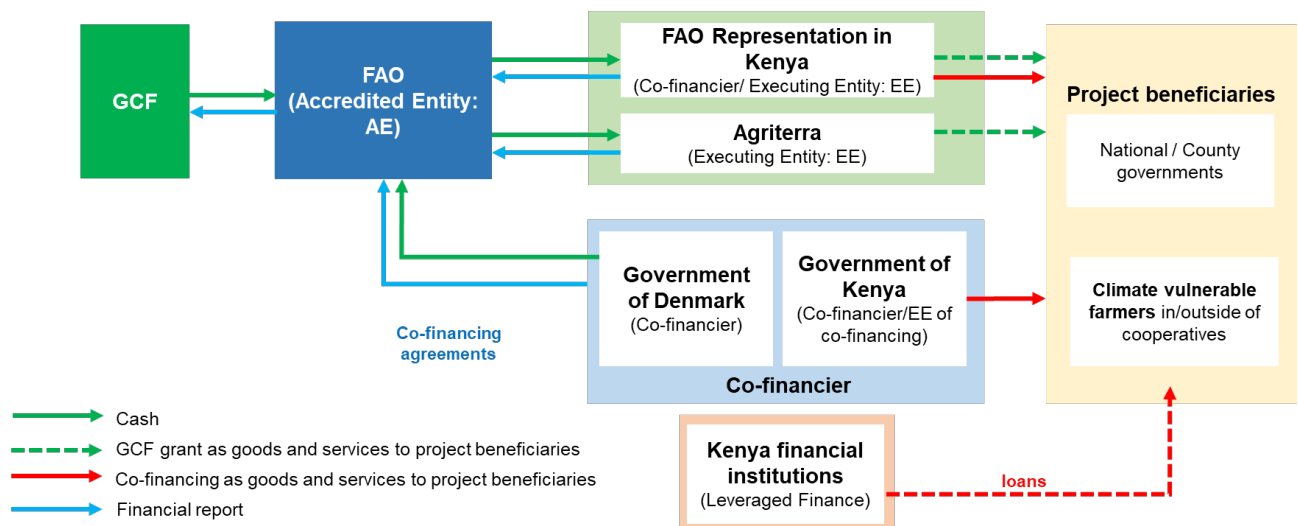


Figure 4: Flow of funds

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

209. From 2015 to 2019, Kenya's economic growth averaged 5.7%, making it one of the fastest growing economies in Sub-Saharan Africa. The public debt to GDP ratio has risen considerably over the past few years to 62% at the end of 2020. Half of public debt is external. This limits the GoK's ability to finance projects and activities to deal with the climate change crisis. Without GCF support for ongoing efforts to address climate change resilience barriers, the GoK cannot take adequate steps on its own. In addition, GCF investments enable existing efforts and transformative projects to scale-up by enabling private sector, cooperatives, and financial institutions to set the stage for large scale autonomous adaptation. The GoK has explored alternative modes of financing for this project but has limited the taking of large loans to investment programs for infrastructure programs, such as those delivered by the World Bank under the National irrigation program. These provide a solid baseline on using GCF resources to support this project.

210. In 2018, the Kenyan government was able to disburse only 752.4 million USD in climate-related development expenditures. The same report indicates that international partners would need to sustain 87% of the NDC costs and that development partners were providing less than 1/3 of all finance tracked, concluding that “there is an urgent need for international investors to scale up their investments in adaptation sectors, which requires more innovative financing models”. GCF support can help meet this financing gap and foster the emergence of a healthy agriculture private sector, while taking pressure off the National Treasury in the face of competing priorities. The GCF resources will enhance the implementation of actions that bring positive impacts to the most vulnerable communities and livelihoods in the LREB. Thus, GCF financing will contribute to faster economic growth and development in the targeted counties leading to the realization of the aspirations of national climate change policies as detailed in the NDC.
211. Smallholders are, by definition, highly vulnerable on their own. Currently there are limited mechanisms whereby farmers can access financing to transform their agricultural practices into climate resilient low carbon modalities. Access to assets is low, adaptive capacity is low, and aggregation among farmers is seen here as a way of sharing risk, pooling resources, and accelerating access to technology. All direct beneficiaries in the project are climate vulnerable smallholders. Some are reached through informal groupings that either exist or will need to be formed for the project (e.g., associations, self-help groups, FFS). Some are reached through existing groupings, like cooperatives. In terms of exposure and sensitivity to climate hazards, all are the same.
212. All smallholders (regardless of their groupings) are vulnerable to climate variability and climate change: small farms in size and number of animals, generally subsistence production or limited surpluses, low productivity, little or no diversification. The main difference between coop members and non-members is that coop members have access to additional adaptive/risk reduction mechanisms (slightly higher incomes, the ability to access knowledge and markets more effectively through the cooperative social network). While they are a little more resilient than others, coop members are still vulnerable. That is why we want to encourage non-members become members at the end of the project (activity 3.1.3).
213. Cooperatives vary in size, climate capacity and maturity. The number of members in a coop (coop size) is not linearly related to coop maturity, to climate capacity, or to the climate resilience of individual members - a large cooperative does not necessarily provide more climate services to its members; a small coop can be more mature than a large one. Each cooperative has its own challenges and in particular challenges related to climate change adaptation and mitigation.
214. We defined a resilience index based on the Resilience Index Measurement and Assessment Framework (RIMA) and the IPCC6 definitions of climate risk, adaptive capacity, sensitivity etc. This index is measured at household or individual level. It will allow us to differentiate between members and non-members. By our simulation, using high-level aggregated data, coop members appear to be between 10 and 15% more resilient than non-members. Non-members rank in the 40% range, whereas members rank in the less than 55% range. Because our target is a % improvement on the initial resilience score, we will also be able to measure increases for all types of vulnerable farmers, and also understand attribution to various variables on which the project has acted.
215. GCF grants will be used to procure technical assistance, goods, and services to support smallholders directly. One group of beneficiaries will receive support through FFS and semi-formal organizations while the other will receive their support through their cooperative, building on the existing adaptive mechanisms and social networks. GCF grants are not used to develop the commercial capacity of coops but to increase their capacity to change the behaviour of individual members - through knowledge/training, processes, institutional climate servicing capacity, access to markets, acquisition of inputs, etc. All according to the specific barriers each coop faces in addressing climate risks.
216. While this project mobilizes grants from GCF and other partners, the concessionality provided to different beneficiaries will vary: individual farmer beneficiaries on the lower end of the resilience spectrum are expected to benefit from technical support and training, assets and other investments designed to reduce their vulnerability, funded entirely from grants with no financial counterpart contribution required. Cooperatives on the relatively higher end of the resilience spectrum will be required to contribute counterpart funding, reducing the use of GCF grants and leveraging private finance from Banks and other donors. The exit strategy for this project entails “graduating” beneficiaries from one level to another on the resilience spectrum – at the final stage, cooperatives who can access loans from financial institutions will be enabled to independently and continuously acquire and transfer climate resilient, low-carbon technologies to their members.
217. No other initiative targets cooperatives or integrates business models aimed at making the economic case for climate smart farming in the targeted area of the project -- the Lake Region Economic Bloc. Coffee, tea, and dairy value chains are critical for Kenya’s agricultural economy and farmers’ livelihood, while there is a severe shortage of investment from both public and private sectors to assist the key players, especially the smallholders, to transit

to a low carbon and climate resilient pathway. The investment in fruit trees, poultry, and indigenous vegetable value chains have been long neglected in the targeted region. Although there are increasing public investment in these value chains in the past decades, partially because of these value chains' climate resilience potential, the majority of the producers and producer groups are in a small scale, and the organization of the value chains is fragmented. Furthermore, the GCF financing would also serve to create a model for technology transfer that is not currently present in Kenya, without which individual farmers would be left to make ill-informed choices. Increasing and accelerating access to technology that can unlock significant climate and economic benefits is one aspect of the project that requires upfront support, but whose benefits are also expected to outlast project execution.

218. Given the difficult socio-economic context, and the introduction of fiscal adjustment measures to control debt (exacerbated by climate change, socio-economic, and environmental challenges), the GoK and the participating financial institutions seek the highest level of concessionality from the GCF that will unlock non-grant financing, including from direct access entities in the near future.

B.6. Exit strategy (max. 500 words, approximately 1 page)

219. The exit strategy of this project has multiple aspects: the first part is an "internal" exit strategy that develops the capacity of smallholders to practice climate resilient, low-carbon agriculture and that, at the end of the project, bring them in to existing cooperatives so that they too can benefit from the added resilience and risk sharing mechanisms that a cooperative provides to their members. In this part of the project, vulnerable smallholders will access climate technologies and practices that will enable them to increase their resilience, increase their productivity and production and increase their income. These farmers can then join cooperatives as a means of continually accessing climate extension services and also increasing their market access.
220. The second part of the project's exit strategy is making sure that cooperatives themselves become self-sufficient in the financing of climate solutions. This is achieved by building their capacity to integrate the cost of implementing climate resilient, low-carbon practices in the business plan and financial management of the cooperative itself. Ensuring that the costs of implementing new practices are reflected in the price of commodities sold by the cooperative will ensure that farmers do not face excessive risks in adapting to climate change. The project also helps cooperatives develop their own climate extension services, making them more cost-effective and ensuring that all members meet the new climate standards. These new services would be covered by existing membership fees and dues, as an added benefit of membership. Because the project also supports the creation of market linkages for climate resilient, low-carbon commodities (including labeling and certification), it ensures that farmers receive fair prices, and that market demand is met.
221. A third part of the project's exit strategy lies in the link established with financial institutions who are ready to assist. By bridging the gap between demand and supply of finance for climate resilient, low-carbon agriculture, the project contributes to creating a lasting, replicable, and easily scalable mechanism that can be applied to any cooperative, farmer organization, value chain or region. This part of the exit strategy builds on the capacity building provided to cooperatives to articulate better business plans and funding requests and on the institutional capacity strengthened within FI for developing and rolling out more appropriate, accessible financial products for climate resilient, low-carbon agriculture (under outcome 4).

Operations & Maintenance

222. The project strengthens capacity, disseminates knowledge and assets to smallholder farmers and their organizations, county governments and financial institutions. Some of the project activities create requirements for in-project and post-project operations and maintenance which are detailed as follows:
223. Under **Component 1**, the county governments will continue to deliver and disseminate climate information services, extension, data and other support to farmers and cooperatives on the basis of improved capacity built by the project.
224. Under **Component 2**, the counties implement their Landscape Management Strategies using their own budgets. The cost of maintenance of rehabilitated, restored, sustainably managed agricultural landscapes are estimated at a maximum of 5% of the costs of initial implementation, annually (USD 225 per hectare per year). These costs include monitoring of land degradation and ecological trends, but also continued maintenance such as clearing paths, creating fire buffers, ensuring drainage and other regular land maintenance activities. Total costs of maintaining all project-supported lands (i.e., not counting any possible upscaling) are USD 45,000 per county per year.
225. Under **Component 3**, the project supports the delivery of technologies, knowledge, and assets to smallholder farmers through Farmer Field Schools and Cooperatives. Some maintenance costs may be incurred by farmer groups and farmer organizations for the project-supported assets that may include greenhouses, irrigation kits,

earth ponds and water cisterns, post-harvest storage, or processing machinery and equipment. Costs of Operations and maintenance will be factored into the business plans developed under **Component 4**, facilitating the integration of all costs into the pricing of commodities. Selection of technologies to be implemented by cooperatives and farmers will also be based on a careful consideration and integration of ongoing O&M costs. Costs have been calculated based on 3% of project implementation costs, equivalent to USD 2.66 per participating farmer per year (total of USD 35,508 per year). These costs are to be met from cooperatives and/or farmers own resources after the project. Further details are available in Annex 3 (Economic and Financial Analysis) and the Annex 2 FS, Part C, Section 7.3.

C. FINANCING INFORMATION							
C.1. Total financing							
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount				Currency		
	29,215,309				USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing			
(i) Senior loans	Enter amount	Enter years	Enter years	Enter %			
(ii) Subordinated loans	Enter amount	Enter years	Enter years	Enter %			
(iii) Equity	Enter amount			Enter % equity return			
(iv) Guarantees	Enter amount	Enter years					
(v) Reimbursable grants	Enter amount						
(vi) Grants	29,215,309						
(vii) Results-based payments	Enter amount						
(b) Co-financing information	Total amount				Currency		
	20,772,529				USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority	
Government of Kenya	Grant	14,000,000	USD (\$)	Enter years Enter years	Enter%	Options	
Government of Denmark	Grant	5,791,019 ⁸⁷	USD (\$)	Enter years Enter years	Enter%	Options	
FAO	Grant	985,910	USD (\$)	Enter years Enter years	Enter%	Options	
(c) Total financing (c) = (a)+(b)	Amount				Currency		
	49,997,957				USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Other in-kind contributions include material assets, land, and current operational expenditures of the cooperatives themselves. These will be reported on separately as leveraged financing (an estimated amount of 1,387,500 USD). The project also leverages up to 10 million USD in Loans from Cooperative Bank, Kenya Commercial Bank (KCB) and Equity Bank in support of farmers, value chain actors and cooperatives. Parallel financing includes the other initiatives supported by IFAD particularly ARCAFIM and RK-FINFA, which will become operational in 2023-2024.						
C.2. Financing by component							
Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions

⁸⁷ This is an estimated equivalent amount of DKK 40,000,000 which is stated in the co-financing letter from the Government of Denmark.

Component 1 – Enabling local government support for adaptation and mitigation	Output 1.1 Local administrations deploy improved climate knowledge, extension, and methodologies to support producers and value chain actors	2.44	2.00	Grants	0.44	Grants	FAO
Component 2 – Sustainable Resilient Agricultural Landscapes	Output 2.1 Agricultural landscapes are managed under strategies that conserve, restore, and sustainably manage community forest and agriculture land, and reduce emissions	13.61	0.32	Grants	12.99	Grants Grants	Governme nt of Kenya FAO
Component 3 – Resilient livelihoods	Output 3.1 Vulnerable smallholders adopt gender-responsive and socially inclusive climate resilient and low carbon production and processing practices, technologies, assets, and risk reduction mechanisms	25.64	22.04	Grants	3.60	Grants	Governme nt of Denmark
Component 4 – Scaling through CRLCSA market and finance	Output 4.1 Increased access to markets and profitability of climate smart, low carbon sustainable agricultural products	1.32	1.07	Grants	0.25	Grants	FAO
	Output 4.2 Vulnerable smallholders and their organizations have increased access to gender-responsive and socially inclusive financial products that support climate resilient, low carbon growth	2.84	0.9	Grants	1.94	Grants	Governme nt of Denmark
Monitoring and Evaluation		1.64	1.55	Grants	0.08	Grants	Governme nt of Denmark
Project Management		2.50	1.33	Grants	1.17	Grants Grants	Governme nt of Denmark Governme nt of Kenya
Indicative total cost (USD)		49.99 million	29.22 million		20.78 million		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☒ No ☐

Capacity Building and Technology transfer are two central aspects of this project, and they are built into all outputs as seen in the table below:

OUTPUT	TOTAL AMOUNT	CAPACITY BUILDING / TECHNOLOGY TRANSFER	COMMENTS

Output 1.1 Enhanced public agro-climate services support farmer-led proactive adaptation and mitigation actions.	2,575,733	Capacity building	All activities in this output are designed to build the institutional and technical capacity of county governments to provide better, more tailored services to smallholder farmers and their organizations in order to support climate resilient, low-carbon value chains. This will include deployment of technical advisory services, research, training, access to information and knowledge, and the facilitation of collaborative or participatory processes at county and LREB levels.
Output 2.1 Agricultural landscapes are managed under strategies that conserve, restore, and sustainably manage community forest and agriculture land, and reduce emissions	13,612,383	Capacity Building	All activities in this output are designed to build the technical capacity of counties to deploy climate resilient, low-carbon landscape management strategies and approaches that are supportive of agriculture. Activities include training, learning by doing, access to data and information, and the facilitation of collaborative processes.
Output 3.1 Vulnerable smallholders adopt gender-responsive and socially-inclusive climate resilient and low carbon production and processing practices, technologies, assets, and risk reduction mechanisms	25,861,980	Technology Transfer	The largest part of the output focuses on deploying technology transfer (including knowledge, know-how, approaches, methods, processes, and goods) related to climate resilient, low-carbon practices in the 6 value chains. Mechanisms used for transferring technologies include farmer field schools to reach less aggregated farmers, and farmer-to-farmer extension through cooperatives to reach farmers in existing business groupings. Technologies will be sourced nationally and internationally, and the list of technologies will be regularly vetted and validated by county governments using the capacity developed under Output 1.1. Some capacity building is also provided to farmers and their organizations to support further farmer aggregation.
Output 4.1 Increased access to markets and profitability of climate smart, low carbon sustainable agricultural products	1,322,688	Capacity building	In this output the project builds the capacity of farmers and cooperatives, buyers, and other value chain actors on the market side. This includes building the awareness of buyers to identify climate resilient, low-carbon products and their benefits, and building the capacity of farmers to access certification and labelling schemes designed to give them a market advantage. Methods used for capacity building will include training, technical advisory, advertising, and public relations campaigns.
Output 4.2 Vulnerable smallholders and their organizations have increased access to gender-responsive and socially inclusive financial products that support climate resilient, low carbon growth	2,501,131	Capacity building	Under this output some activities are designed to build the institutional and technical capacity of financial institutions to develop and deliver more accessible financial products to farmers and cooperatives. Methods used for capacity building will include the mobilization of expertise, training, and technical advisory, as well as facilitation of participatory processes.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

226. **Direct beneficiaries:** The project reaches 572,000 **vulnerable smallholder farmers** whose livelihoods depend on any of the 6 VC in the targeted region. Of these, the project expects to reach at least 80,000 individual cooperative members (project participants), with direct benefits accruing to their households (a total of 320,000 people based on the conservative estimate of 4 people per household). Other beneficiaries are smallholder farmers who are not part of cooperatives (63,000 participating farmers with direct beneficiaries accruing to their households, 252,000 people).
227. Additional beneficiaries not included in the figure above include government officials and county administration representatives, who would add a total of approximately 50 people per county (700 persons); buyers, who will be targeted under outcome 3 (12 buyers); and other value chain actors.
228. For **indirect beneficiaries**, we have assumed that benefits would accrue to members on non-participating cooperatives in the 6 value chains (1,342,140.00⁸⁸), and that participants in the FFS would also transfer indirect benefits to extended family members and households in surrounding villages (756,000 people)⁸⁹. This brings the total of immediate indirect beneficiaries to 2,098,140 people, 50% of which are women, and which represents nearly 15% of the Lake Victoria Economic Block population. However, our estimate is that, in the long-term, thanks to the upscaling strategy and the presence of Kenyan Banks willing to support eligible cooperatives through loans, indirect benefits may be accrued by a much larger number of people in the Lake region.
229. The Gender Action Plan details how the goal of intersectional equity will be mainstreamed in two ways: participation in activities and the content of activities. Women, female youth, female-headed households, and females living with disabilities all have specific quotas that are required for each activity. Mainstreaming in terms of participation will be ensured by setting the target proportion of participants from each of these intersections to be roughly equal to the relative population sizes at the national level: 50% women, 16.5% FHH, 1% for women LWD, and 25% female youth (aged 18-34 years). The male youth (aged 18-34) participation quota is 25% and 1% for men LWD. Mainstreaming in terms of content will be ensured by integrating Gender Action Learning Systems (GALS) and gender transformative concepts into 100% of training materials.
230. Please refer to FS, Part C, section 6 for additional detail on beneficiaries and benefits. The benefits that are expected from this project are as follows:

Improving the Health, well-being, food and water security (ARA2) of the Most vulnerable people and communities (ARA1)

- 1) Improved adaptive capacity through increased access to climate resilient and low carbon technologies, market, finance, social safety nets and risk sharing mechanisms by participating in organized farmer groups/cooperatives.
- 2) Improved levels of food and nutrition security through increased production of key crops and commodities that can either be directly consumed or sold.
- 3) Increased incomes through improved production, productivity, value addition, and access to markets
- 4) Reduced exposure to climate hazards such as droughts and floods from improved watershed management and maintenance of soil cover, and from increased access to climate information services including Early warning.
- 5) Improved access to water through measures designed to promote water conservation, harvesting and erosion control.
- 6) Increased gender equality and social inclusion in access to/control over agricultural inputs, agricultural finance, and agro-climate information and services and agricultural extension.
- 7) Increased women's, PLWD, and youth representation and participation in decision-making in cooperative societies and value chains.
- 8) Increased access to gender-responsive and socially inclusive adaptation and mitigation technologies, markets, and labour practices.

⁸⁸ As per Cooperative census, FAO and Agriterria (2022)

⁸⁹ <https://link.springer.com/article/10.1007/s10457-006-9007-8> and <https://link.springer.com/article/10.1057/s41287-020-00323-7>

- 9) Increased access to gender and social safeguards to reduce climate risks for the most vulnerable, as well as risks to women's health, wellbeing, and livelihoods from increased engagement in cooperative organizations and leadership.
231. *Resilience in this project is a direct result of adaptive capacity and sensitivity.* To measure progress in resilience and to ensure the inclusion of all project-influenced variables, the project has created a specific Resilience Index that draws heavily on the Resilience Index Measurement and Analysis Framework (RIMA-II⁹⁰) which was developed by FAO. Originally conceived as a tool to measure socio-economic resilience to food-related shocks, the RIMA-II contains sub-indicators that can also inform climate resilience. The RIMA index classifies data according to 5 groups of elements of resilience. For this project, we have retained those indicators that are specifically impacted by climate change and that align with our theory of change and projected impact pathways. Furthermore, we cross-referenced the RIMA framework with the IPCC 6 definitions of climate risk and resilience, to include measures of sensitivity and adaptive capacity specifically related to climate change (as opposed to food-related). The sub-indicators include:
- Access to Basic Services (ABS): This includes access to water/sanitation, energy, markets and roads, health services. The reason these indicators are included is because they facilitate or hinder access to agricultural assets (e.g., inputs) and therefore determine the ability to generate income in the face of climate change; they also facilitate or hinder access to public services and aid in the case of severe climate events, along with the transmission of climate information services.
 - Access to Assets (AST): This includes assets directly related to the production of income from agriculture (an indicator that the project will seek to influence) but also protective assets, meaning those that can serve as buffers to climate impacts. For this project we have retained the following production assets: access to land, access to crop and livestock for production, access to savings and finance and access to processing and value addition facilities, as well as the direct measure of income from the main VC commodity - and the following non-productive assets: access to climate resilient, low-carbon technology, access to post harvest storage and access to shelter. These sub-are all sub-indicators that the project intends to directly influence. Change in the measurement of these is expected as a direct result of project intervention.
 - Adaptive Capacity (AC): This category also includes sub-indicators that will be directly influenced by the project and are limited to adaptive capacity in relation to climate change. Though all the other indicators also contribute to adaptive capacity, we have retained the following three dedicated sub-indicators: Access to climate resilient, low-carbon technology, access to education and training, and access to climate information and early warning.
 - Access to Social Safety Nets (SSN): In this category we included only one indicator which is related to membership in a farming group, business unit or cooperative (to be disaggregated). In conformity with the project theory of change, membership in a farming organization serves as a mechanism to reduce risk and accelerate knowledge and technology acquisition. This sub-indicator is not included in the RIMA framework but are added in for the purpose of this project as means of sharing climate risk or reducing climate risk through social networks, public or private. Furthermore, the sub-indicator will help us understand differences in resilience scores among the different types of beneficiaries (members of FFS vs members of COOPs).
 - Sensitivity (S): In the context of the project sensitivity relates to exposure to climate risk as well as to the ability to resist climate shocks, which may be facilitated or undermined by some underlying socio-economic factors. As noted earlier, overall "geographic" sensitivity and the climate risks and hazards to which beneficiaries are exposed, is established in section 2. In this category we have included socio-economic drivers of sensitivity, such as gender and age, disability, current level of food security or malnutrition, and the dependence on rainfed agriculture, given that precipitation is one of the key factors to be influenced by climate variability and change.
232. The measurement will be made at household level and the measure will be made through a Resilience Survey of a representative sample of project participants at inception, mid-term, and final evaluation). To facilitate tracking of project indicators and deliverables, the survey will be designed at inception, and will

⁹⁰ The RIMA framework was created using the following definition of resilience: "The capacity of a household to bounce back to a previous level of well-being (for instance food security) after a shock." While this definition (and the indicators that support it) do not exclude climate shocks, they do not consider slow-onset climate trends and individual sensitivity/exposure to those, as a "shock." Furthermore, the RIMA II resilience index contains two parts: one direct, descriptive measure and the other indirect and inferential that looks at determinants of resilience. For the purposes of this project, it was felt that the second part would be superfluous since we are only considering the climate drivers of resilience and not the full spectrum of underlying causes. Please refer to <https://www.fao.org/3/i5665e/i5665E.pdf> for further reading on RIMA II.

also include additional questions in order to minimize the reporting burden on farmers and the project. Please refer to Part C, section 6 of the FS for further detail on indicators and sub-indicators of the resilience index.

Reducing Emissions from Forest and Land use (MRA4)

233. The calculation of mitigation benefits for this project was made using two different methodologies. The FAO Ex-ACT methodology and model was used to calculate emissions from AFOLU activities in all value chains except dairy and poultry, and the GLEAM model was used to calculate emissions in the livestock value chains (dairy/poultry). Overall, while the dairy and poultry value chains remain net emitters, the project achieves significant reductions in emissions intensity per unit of output. Globally, the project achieves a net emissions reduction of 4,268,492.04 tCo₂ eq by 2044.

234. Mitigation benefits are expected from the following activities and technologies:

- Reforestation, regeneration, rehabilitation, and conservation of land under the Landscape Management Strategies (Output 2.1)
- Integration of agro-forestry in existing value chains (Output 3.1)
- Improvements in land preparation, tillage, soil cover (Output 3.1)
- Improvements in water and energy use efficiency (Output 3.1)
- Increased use of renewable energy, waste-to-energy, and reduction of deforestation (Output 3.1)
- Improvements in emissions reduction per liter of milk, Kg of meat and kg of egg through improved productivity and feed/management changes (Output 3.1)
- Reductions in emissions from food loss and wastage (Output 3.1)

235. Key assumptions for the calculation of mitigation for each model are included in FS, Part C, section 6.4 and Annex 22.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

236. The fundamental **paradigm shift** that this project fosters is the transformation of value chains from low productivity and low resilience to high performing, financially profitable, climate resilient, low-carbon pathways through locally-led approach. The project leverages cooperatives as a climate risk reduction mechanism – not only a business strategy that enables farmers to pool resources and leverage better negotiating power on the market. This is done by strengthening the ability of farmer organization and cooperatives to self-identify climate problems and solutions and to easily access relevant adaptation and mitigation technologies to enhance proactively climate resilience in production and business. Ensuring that the technologies and practices promoted fit into the business plans and models of cooperatives means that farmers will implement climate solutions that make both climate and financial sense to them.

237. Having cooperatives include climate technologies and extension as part of their business plans will ensure that climate costs are integrated as part of the regular cost of farming. Prices of commodities will therefore reflect the full cost of doing business under a climate change scenario, reducing the risk borne by individual farmers in adapting to climate change. The increased market opportunities fostered under Outcome 4 will further facilitate transformational changes to the value chains as farmers will be able to fetch better prices, stronger bargaining positions and increased market shares, because both production and quality of produce under climate resilient, low-carbon practices will be increased.

238. This project dovetails with local governance and devolution processes, recognizing that while climate capacity is stronger at national level, it is imperative to strengthen it at local level. The project generates a paradigm shift in the way in which county governments relate and provide services to farmers and agri-businesses. Recognizing the limits and constraints faced by conventional extension services, the project builds the capacity of county governments to exercise normative influence on markets and producers by setting norms, providing enabling climate services, and assuring the maintenance of public goods, including ecological services that underpin agriculture. Conventional extension services will therefore be complemented by farmer-to-farmer climate extension that will be integrated into the business practices of any cooperative – as climate related services that cooperatives offer their members to sustain climate resilience, decarbonization and profitability at the same time.

Potential for scaling up and replication.

239. Climate change adaptation measures introduced in the project will support resilient, low carbon, sustainable, and competitive agri-food value chains in the long-term that can create a model for further upscaling to other beneficiaries in the same value chains or extended to other value chains and other sectors and regions in the country. Activity 1.1.6 is specifically designed to foster such upscaling, replication and knowledge exchanges. Under Outcome 1, innovative extension methods and climate information

services developed and delivered by county governments can readily be made accessible to other counties through the action of the Council of Governors or the Maarifa Center. The decentralized carbon accounting methods designed under outcome 1 as well as the rapid climate risk assessment processes delivered under Outcome 2 can also be extended to other counties. Under Outcome 2, the integrated climate resilient, low-carbon landscape management strategies can also be replicated in other counties, especially once the lessons are drawn regarding the impact of such on agricultural landscapes and agricultural production. All climate technologies promoted under Outcome 3 – whether through FFS or cooperatives – are replicable and can be scaled up. All climate technologies promoted under Outcome 3 – whether through FFS or cooperatives – are replicable and can be scaled up. The premise in this project is that participating farmers will see and document the economic benefits of applying these technologies as well as the climate benefits – in fact, the economic benefits may accrue earlier than the climate benefits. Farmers in the same value chains who are not participating in the project will be able to see – through the awareness raising and research activities of the project – the benefits of climate resilient, low-carbon technologies; because the technology transfer mechanism created under outcome 3 will continue to be available after the project, cooperatives and farmer organizations will be able to access the technologies and knowledge independently (albeit for a price).

240. All climate technologies promoted under Outcome 3 – whether through FFS or cooperatives – are replicable and can be scaled up. The premise in this project is that participating farmers will see and document the economic benefits of applying these technologies as well as the climate benefits – in fact, the economic benefits may accrue earlier than the climate benefits. Farmers in the same or different value chains who are not participating in the project will be able to see – through the awareness raising and research activities of the project – the benefits of climate resilient, low-carbon technologies; because the technology transfer mechanism created under outcome 3 will continue to be available after the project, cooperatives and farmer organizations will be able to access the technologies and knowledge independently (albeit for a price).
241. The project also creates an internal scaling process whereby smallholder farmers can aggregate into more mature farmer organizations, culminating in high-performing climate resilient, low carbon cooperatives. Visible economic and climate benefits experienced by cooperative members are expected to create an attraction effect which combined with project efforts to remove barriers to aggregation, will support the scalability of project outputs. The project also creates a push effect, increasing the supply of in-demand climate smart commodities.
242. Under outcome 4, all activities are naturally scalable and replicable. The project creates capacity in financial institutions to develop and deliver concessional finance to farmers designed specifically for climate resilient, low-carbon agriculture. This capacity and these products, which can immediately be deployed by the financial institutions to eligible beneficiary cooperatives (whose capacity to develop bankable proposals will also be strengthened), can be upscaled and replicated by financial institutions to other areas and other value chains easily and at very low cost.

Potential for knowledge sharing and learning.

243. Knowledge sharing and learning are fundamental elements of this project's strategy. Under Outcome 1, county administrations, particularly those departments concerned with climate change, agriculture, extension and cooperatives, will be capacitated to develop new information and knowledge services with the participation of farmers and farmer's organizations – including climate information services – that will be deployed to last mile users. In particular, the project leverages farmers' own organizations as vehicles of climate knowledge dissemination vs the traditional model of on-demand, government-to-farmer extension. Under Outcome 2, the project also generates and share knowledge and learning around the benefits of climate resilient, low-carbon landscape management in support of agriculture. In this regard, the project leverages the LREB Multi-stakeholder climate change knowledge platform and other mechanisms to promote knowledge sharing and learning.
244. The most important part of the knowledge sharing and learning strategy lies in the development and acquisition of knowledge by farmers themselves, by leveraging cooperatives and providers of technology in Kenya and elsewhere under Outcome 3. This includes generating knowledge about specific climate issues and solutions, empowering farmers to undertake their own diagnosis and prioritization of climate problems in the context of agriculture as a business. Methods used for knowledge sharing and learning include trainings, demonstrations, exchange visits, community meetings, market fairs and county level forums as well as LREB-wide meetings and workshops, the Multi-sectoral Knowledge platforms, Council of Governors and Maarifa center.

245. The project also generates knowledge and learning for other value chain actors, and particularly buyers, aggregators and finance institutions. New knowledge products such as research and comparative analysis of quality, grading and pricing of climate resilient, low-carbon products, market insights, agricultural databases, will also be generated by this project to support future similar endeavours.
246. Finally, the project includes sub-activities related to the evaluation and documentation of the effectiveness of major strategies promoted, generating significant evidence and lessons learned that will be disseminated and shared to all stakeholders through activity 1.1.6. Lessons learned, case studies, information products and other reports will be disseminated at county and national level and at international levels through the project website, print and mass media.
247. Further details are found in the Monitoring, Evaluation and Learning plan in Annex 11.

Contribution to the creation of an enabling environment

248. The project supports the improvement of the enabling environment for farmers and cooperatives. First, by strengthening the capacity of county administration to provide climate-related services and market-access services, farmers will be enabled to produce more and better-quality products that will reach the market more readily. The services provided that support this enabling environment include improved climate information services and warnings; agrometeorological advisories; council-approved lists of climate resilient, low-carbon technologies, improved extension services; market studies, market information and sales data.
249. The approaches taken for transferring technology under Outcome 3 are intended to reduce the costs and risks faced by farmers when attempting to adapt or to adopt low carbon practices. First, all practices are tested and approved by county administrations in relation to local vulnerabilities (outcome 1). Farmer to farmer extension and networking reduce the costs of disseminating knowledge and practices, while ensuring upscaling to other areas and value chains. Because the costs of adaptation and mitigation will be fully considered in farmers business cases when proposed, farmers will not face any surprises or untold risks. Demonstrated profitability of climate resilient, low-carbon methods and improved access to markets further reduce costs and risks.
250. The project also supports the improvement of the enabling environment for framers and their organizations to access finance that supports climate resilient, low-carbon value chains, by improving the capacity of lenders to develop suitable products, and by improving the capacity of business entities to develop bankable business cases. This portion of the project, under Outcomes 3 and 4, is a significant pillar of the project's exit strategy that – together with documenting the profitability of climate resilient, low-carbon technologies – will ensure long-term sustainability of project outcomes. By also working with buyers, the project ensures that increased market space is made for the commodities produced under this project.

Contribution to the regulatory framework and policies

251. The project contributes to improving regulatory frameworks and policies at county level in each county where it will operate. Specific sub-activities are included to support the linkages of the project's activities with the county's CIDP and investment plans (e.g., the landscape management strategies) and Council Assemblies (e.g. approval of technology green-lists). The project also supports the implementation of each county's climate change acts and action plans (refer to FS, Part B, section 3.3). Through the Council of Governors and the Climate Multi-stakeholder Platforms, it is expected that lessons learned with policy or regulatory application will also be extended to other counties and escalated to national level policy making. The project also makes an indirect contrition to the implementation of existing national and county policies and regulations related to biodiversity, water, energy and pollution.
252. For more details on how this project contributes to national policies, strategies and plans, please refer to section D.6 and section 3 of the feasibility study.

D.3. Sustainable development (max. 500 words, approximately 1 page)

253. Direct benefits accrued by project beneficiaries include improved climate resilience through improved food security, increased levels of income, improved access to assets, improved access to food, increased access to markets, social networks, and social safety nets. Combined, the improved access to nutritious food locally, the increased access to markets and increased incomes, will result in improved food security outcomes for project beneficiaries. Resilience will be measured though a composite index specifically composed for this project, comprised of many sub-indicators that directly relate to the SDGs including food and nutrition security, among others. Please refer to FS, Part C, section 6 for details on how the Resilience

Index is composed and measured. Project activities will have direct linkage to SDGs 1, 2, 3, 5, 8, 13 and 15.

254. The measurable co-benefit of this project is job creation in the 6 value chains, while indirect benefits include increased economic activity in the supply chains for agricultural inputs and feed value chains, and increased employment in the transporting and bulking aspects of the value chain. In particular, the project expects to create meaningful employment for women and youth within cooperatives, farming groups and processing industries.
255. Additionally, the project expects to generate benefits to the broader population of the LREB through the implementation of climate-sensitive landscape management strategies, and the overall improvement of the economic outlook in the region.
256. Other indirect socio-economic benefits include:
- Improved access to education for children and especially girls (whose working tasks may be reduced, or through improved ability to pay school fees)
 - Improved health through access to increased, more diverse and better quality food (either produced or purchased)
 - Improved access to health services (increased ability to pay)
 - Increased gender equality
 - Reduced rates of gender-based violence and child labour
 - Increased access to meaningful employment for youth, women, persons with disabilities

Environmental Benefits

257. At the farm level, environmental benefits arise from the implementation of sustainable land management techniques, anti-erosion work, increase in soil cover and the combination of crops or crop rotation, as well as the judicious application of fertilizer, reductions in use of noxious agro-chemicals and herbicides. Benefits include:
- improved land productivity and increased biomass production,
 - improved soil fertility,
 - soil moisture retention and micro-organism content,
 - increases in agro-biodiversity at farm and landscape levels.
258. At the landscape level the project also expects to deliver, through the implementation of the landscape management strategies (output 2.1) combined with other, non-project interventions, improved forest cover and forest density, reductions in rates of run-off and erosion, improvements in biodiversity status, and improved water conservation and aquifer recharge.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

259. **The vulnerability in LREB is due to agriculture primarily being rainfed and small-scale production.** Therefore, crop production is extremely vulnerable to precipitation changes, causing crop failures in the short-term and production declines in the long-term⁹¹. Most smallholder farmers are particularly vulnerable to interannual rainfall variability and oscillating crop yield and livestock production trends, resulting in unstable and unreliable incomes. For example, a survey of smallholder farmers in Kakamega and Vihiga counties found that smallholder farmers strongly rely on rainwater to feed crops and do not have access to stable financial resources and credits from public or private supporting institutions⁹². Overall, increasing rainfall variability has led 10% of the population in the Lake Victoria Basin to suffer chronic food insecurity, requiring short-term emergency food relief and long-term development programs⁹³.
260. **Economic and social development level of the smallholders and the value chain actors hinder their actions to responds to climate hazard and require the project supports to reduce costs of climate actions and increase efficiency and coordination between governments and private actors in delivering climate services and easing access to market.** Smallholders working small plots (0.2 ha on average), and/or owning 3-4 cows or approximately 10 heads of poultry. They are characterized by low

⁹¹ World Bank Group. WBG. 2021. Climate Risk Profile: Kenya (2021): The World Bank Group.

https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15724-WB_Kenya%20Country%20Profile-WEB.pdf

⁹² Karugia et al. 2005. The Role of Rural Factor Markets in Reducing Poverty, Risks and Vulnerability in Rural Kenya: The Case of Kakamega and Vihiga Districts. Strategies and Analysis for Growth and Access.

⁹³ USAID. 2018. Lake Victoria Basin: Climate Change Adaptation Strategy and Action Plan, 2018 - 2023. Retrieved from: https://www.climate-links.org/sites/default/files/asset/document/2018_USAID-PREPARED_Lake-Victoria-Basin-Adaptation-Strategy-Action-Plan.pdf

access to basic services (e.g., sanitation, education, health, markets). They also have limited access to climate-proof inputs (e.g., seeds, varieties, breeds, etc.). Approximately 75% of their income depends on a single commodity which is usually rainfall dependent or climate sensitive.

261. In the LREB, yields are symptomatic of low productivity aligned to climate variability, high levels of loss and wastage in the post-harvest and processing cycles. Many farmers are producing with limited technical means, low levels of mechanization, with limited access to inputs. Most practices can be characterized as unsustainable mono-cropping land and water use, although many farmers are becoming aware of land degradation issues and its impacts on their productivity.
262. Smallholders who produce surpluses (i.e. above household consumption) can pool production and access processing and value addition through business groups (e.g., Business units, coops). However, individual farmers have very little access to financing on their own. Cooperatives and farmers organizations generally also have limited access due to high requirements, high costs of lending, and the general low capacity in articulating bankable business propositions for financial institutions. Women, youth and persons living with disabilities face increased barriers to access to inputs, land, income, finance and technical support. Please refer to the Gender Assessment for further detail.
263. This project therefore creates a system where governments and private sector come together to remove the barriers to autonomous financing of value chains through cooperatives. At one end of the spectrum, vulnerable smallholders are trained and capacitated in the best available climate resilient, low-carbon practices for their value chain. They derive short and long-term adaptation benefits from these, and also have the opportunity to further reduce their costs and risks by joining cooperatives. At the other end of the spectrum, mature cooperatives exit the project with members who are all practicing climate resilient, low-carbon agriculture, have access to improved assets, increase their sales and market presence, and are able to include in the regular costs of doing agri-business the provision of climate services to their members. This gives them the leading edge and additional capacity to access private finance, and thus perpetuate the model autonomously.
264. County budgets are limited and even though counties are legally mandated to dedicate 2% of their budgets to climate action, priorities are many, and the linkages to private agri-businesses are not enough to direct public resources to cooperatives. This is why the project designed to strengthen the capacity of county governments to serve farmers and farmer organizations in a supportive, enabling, normative manner, by providing guidelines, regulations, climate information, and advisory.

D.5. Country ownership (max. 500 words, approximately 1 page)

265. The project supports the implementation of Kenya's national climate change policies, strategies and plans as follows:
 1. **The National Climate Change Act of 2016**⁹⁴: Paragraph 2 of Section 3⁹⁵ (which lays out specifically how the Act will affect Kenya's climate change response), mandates the formulation of programs and projects to enhance resilience that are intergenerationally, gender equity mainstreamed and community led, while also contributing to disaster risk reduction. This paragraph also stipulates the need to promote low carbon technologies. All points are supported by the project's activities. The Act of 2016 also follows the decentralization framework. The Act establishes the Climate Change Directorate, which coordinates and facilitates climate change adaptation policies and action plans government agencies between counties, and county level climate change units, which will be supported by the project
 2. **Kenya's Updated NDC**⁹⁶: Kenya's 2020 Updated Nationally Determined Contribution (NDC) sets an abatement target of 32% of the business-as-usual (BAU) scenario. Under the BAU scenario, Kenya's projected emissions are projected to increase to approximately 143 MtCO₂ by 2030, which the NDC wants to limit to 32% relative to the BAU. Over the course of the project lifespan, the project would tentatively mitigate a projected 4.2MtCO₂.
 3. **Kenya's National Adaptation Plan 2015 – 2030**⁹⁷: The National Action Plan 2015 – 2030, coordinates climate change adaptation at the national level while also contributing to economic development to boost resilience. Outcome 1 of the project supports the mainstreaming of climate

⁹⁴ Retrieved from: <https://faolex.fao.org/docs/pdf/ken160982.pdf>

⁹⁵ Section 3 of the Climate Change Act of 2016, Paragraph 1 states "This Act shall be applied for the development, management, implementation and regulation of mechanism to enhance climate change resilience and low carbon development for the sustainable development of Kenya."

⁹⁶ To be confirmed.

⁹⁷ /https://www4.unfccc.int/sites/NAPC/Documents%20NAP/Kenya_NAP_Final.pdf

change adaptation into all public sector offices with the trainings provided. Land management reforms that will be carried out in Outcome 2 support the land management reforms of the NAP. Outcome 3 further develops the bullet point #5 via the farmer field schools, cooperatives development, and gender-, generation- streamed climate change adaptation policies and actions plans that the cooperatives will develop.

4. **National Climate Change Action Plan 2018 - 2022⁹⁸** : The National Climate Change Action Plan (NCCAP) 2018-2022 sets out several priorities and actions that needs to be accomplished including link climate change adaptation with development, and improve water conservation by reversing main water towers degradation and rehabilitating all catchments. Activities the project will undertake in Outcomes 1, 2 and 3 will contribute to implementing the NCCAP. Specifically, the transfer of technologies to farmers, the development and implementation of landscape management strategies, the dissemination of climate information services (CIS), will support the NCCAP priorities.

266. For an analysis of how the project contributes to sectoral policies (energy, water, environment, forests and others) please refer to section 3 of the FS, Part B.

267. The National Designated Authority (NDA) has been closely involved in the design of this proposal. The NDA has participated in all major meetings and consultations and provided input at regular intervals to ensure that the proposal meshed with the government's priorities. And the NDA will play a key role in continuing coordination with county governments, linking with the Council of Governors.

268. This project is the culmination of a long consultative process that involved the active participation of many partners at national and county levels. At national level, consultation involved all key sectoral ministries, central government agencies and development partners to ensure consistency of intervention with major policy orientation and synergies with other initiatives, in particular G-FLLOCA and the IFAD RK-FINFA program. The lessons from past projects and coordination with other ongoing or pipelined initiative have been integrated into this design. Please refer to Annex 7 Stakeholder Engagement Plan for more details.

269. The project is deigned based on the needs of targeted population. County level consultations took place on multiple occasions, starting with the initial consultative meeting, which brought together over 90 participants (men and women) from all 14 counties (cooperatives, environment, agriculture and climate departments) which enabled the prioritization of value chains and the initial design of the project's theory of change and proposed interventions. Consultations continued with the County Capacity Needs Assessment, in which representatives (men and women) from all 14 counties were interviewed to determine climate related capacity gaps. Further consultations took place as the design advanced and county representatives provided input into the design of various activities. Several validation meetings were held after the finalization of the first draft of the proposal package.

270. At local level consultations also took place with smallholder farmers, men and women. Individual farmers were contacted and interviewed regarding application of climate resilient, low-carbon technologies or potential barriers faced when adapting to climate change. A Climate Change Survey was distributed to famers who were members of cooperatives and others who were not (women included). Gender-specific questions were included in the survey as well. A comprehensive Cooperative Census was also undertaken. This included a detailed questionnaire designed to assess and screen potential beneficiary cooperatives in the 6 value chains, understand challenges regarding all aspects of climate resilient, low-carbon agriculture. This census allowed the project design team to access local data, understand beneficiaries and design the best possible mechanism to enable cooperatives to accede to higher levels of climate performance.

D.6. Efficiency and effectiveness (max` . 500 words, approximately 1 page)

271. The financial structure proposed for this project uses grants to support technical assistance, technology transfer, capacity building for vulnerable smallholders, along with nationally-funded targeted investments in public service goods and services (e.g., ecosystem restoration and management, extension and information services). Considering the high level of needs for climate finance in Kenya (Refer to section B5) and the limited investment in adaptation to climate change and in AFOLU-related mitigation in the LREB region, this proposed initiative makes optimal use of the GCF contribution to generate a public-private partnership towards transformation of key value chains. The expected direct and indirect benefits of the project for the LREB and the national economy justify the level of financing requested. The project design team has prepared the project cost. Total project costs were estimated at USD 49.992 million. The total cost comprises of a GCF grant of USD 29.215 million (58% of total project cost); Government of Kenya

⁹⁸ <https://napglobalnetwork.org/wp-content/uploads/2022/01/napgn-en-2022-kenya-NCCAP-2018-2022-Implementantation-Status-Report.pdf>

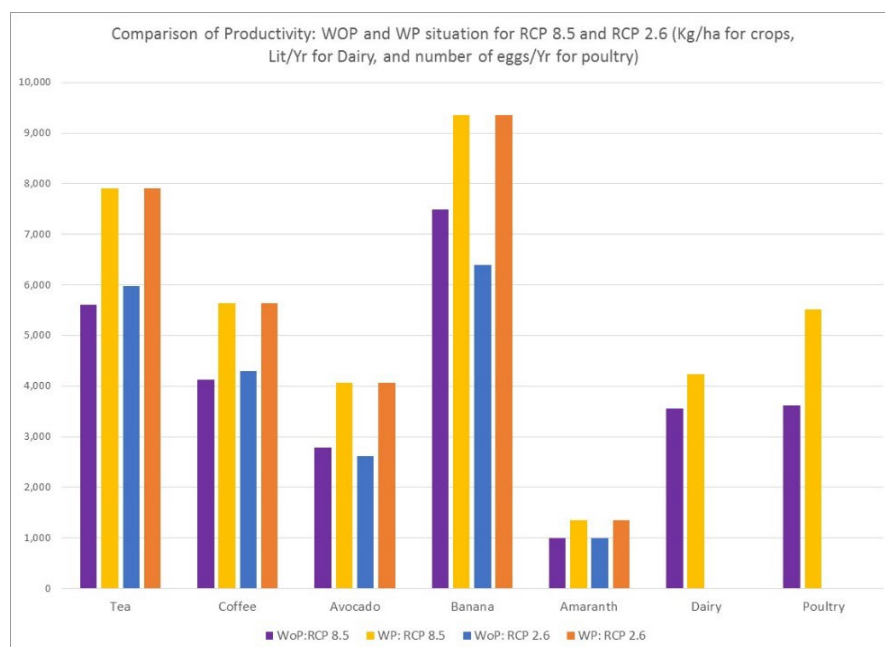
through National Treasury of Kenya contributing USD 14.0 million (28%); a Grant of USD 5.791 million (11.6%) from the Ministry of Foreign Affairs of the Government of Denmark (DMFA); and USD 0.981 million (2%) from Food and Agriculture Organization (FAO).

272. The EFA has explored the financial and economic performance of the activities considered in the Funding Proposal of the project “with” and “without” GCF support. The project focuses on the top 7 priority value chains, three of which have more of a market orientation (dairy, coffee and tea) and four that have more of a household food security orientation (African leafy vegetables, poultry and fruit tree). The total project beneficiaries, 143,000 households, will be reached during the project period with various project assistance. Out of the total beneficiaries the project assumes that, 35,750 will have tea; another 35,750, coffee; 42,900 will have African leafy vegetables; 14,300, avocado, and another 14,300, banana. In addition, 42,900 and 28,600 beneficiaries will have dairy and poultry respectively. These beneficiaries will also have crops that are listed above.

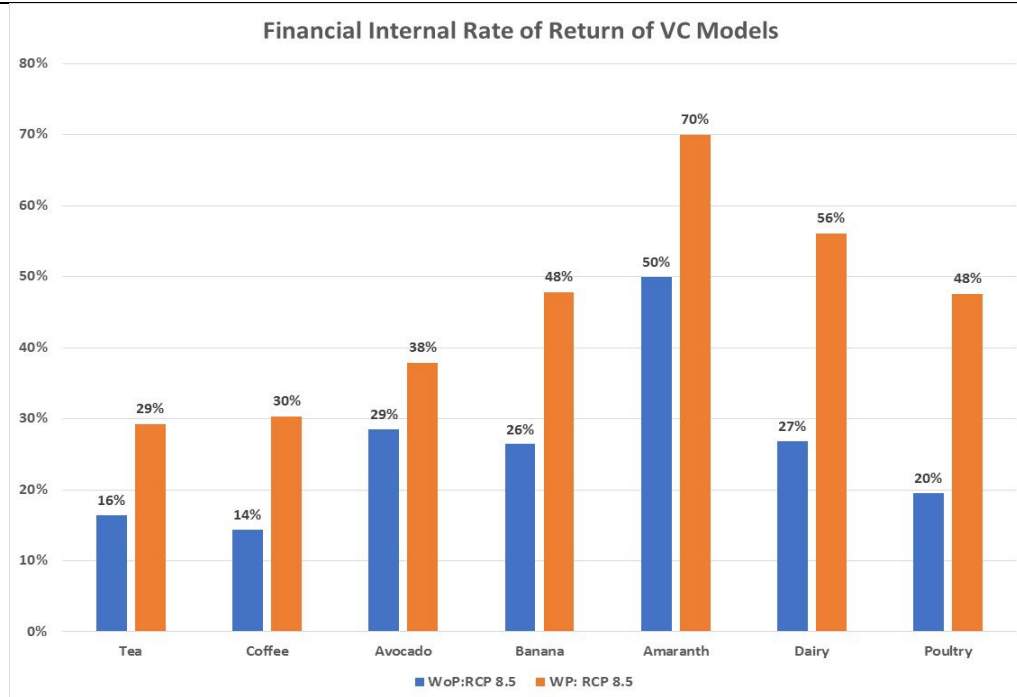
273. The incremental production of the targeted value chain products provided the benefits for the EFA. The increment was estimated by taking the difference between the baseline productivity (without project case) and estimated increased productivity under climate sensitive and low carbon technologies (with project case). The profitability indicators that were estimated for each value chain product at the beneficiary level include the Net Present Value (NPV, economic and financial), the Internal Rate of Return (IRR, economic and financial), the Benefit-costs ratio (B/C), the pay-back period and increase in returns to family labor (for the financial analysis). Using the aggregated cost and benefits flows of the VC product models and the project cost, the same profitability indicators were estimated for the project. The financial and economic viabilities were estimated for two climate change scenarios - RCP 2.6 and RCP 8.5 - for both

the beneficiary level and the project level. A 20-year timespan was considered for the stream of benefits from climate change adaptation activities involving agricultural and livestock techniques and practices.

274. The bar chart on the left shows the productivity level of 7 products from 6 value chains for the WOP and WP cases under RCP 8.5 and RCP 2.6 scenarios. For livestock, the changes were not analysed as the projection of yield decline was not available. Both type of yields was based on the published literature and the Value Chain studies that FAO has conducted for this project.



275. On the basis of the yield levels and cost of production under WOP and WP cases, the profitability indicators were estimated. Considering the Financial Internal Rate of Return (IRR) for 20-year period as a summary indicator to demonstrate the contribution of the project in increasing income for the beneficiaries, the second bar chart below compares the IRRs of WOP and WP cases for all VC models under RCP 8.5 scenario. The same trend of results was obtained for the RCP 2.6 scenario as well.



276. As the bar chart shows, the farmers who participate in these VCs can generate enhanced profits starting with relatively low profitability and decreasing yield due to climate change under the WoP or baseline situation. The project intervention will allow them to increase their profitability from cultivating the same amount of land with new climate sensitive and low carbon technologies under the WP scenario, which also contributes to reducing the carbon emission.
277. Based on individual VC commodity model analyses, the project level financial and economic profitability indicators were analysed. The financial internal rate of return of the project is 23% with a benefit-cost ratio of 1.11; and 25% and 1.13 respectively for RCP 8.5 and RCP 2.6 scenarios for a 20-years period at 18% discount rate. The project will be able to earn USD 10.7 million as a net present value under the same discount rate for 20 years under RCP 8.5 (USD 12.9 for RCP 2.6). The project can breakeven the project expenditure plus the beneficiaries' investments after 9 years (pay-back period) of project commencement for both climate change scenarios.
278. The project level economic internal rate of return is 21% with a benefit-cost ratio of 1.24; and 22% and 1.25 respectively for RCP 8.5 and RCP 2.6 scenarios under 10% economic discount rate. The project is also viable under many sensitivity scenarios. The EFA analyses indicate that the project is therefore adequately viable in financial, economic, and social (environmental) terms and also has the capacity to face many risk factors while being viable. The project also has the potential reduce GHG emission to a significant level with social net present value of USD 113 million and USD 116 million for RCP 8.5 and RCP 2.6 scenarios. The project therefore is suitable for receiving public funds for investments.

Further details on the economic and financial analyses undertaken for the project can be found in Annex 3.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

☒ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Farmers in the LREB are vulnerable to climate variability and climate change impacts. There has to date been limited dissemination and penetration of best practices in terms of low carbon, climate resilient agriculture among local farmers in the LREB. This is due to limits in the availability of climate informed extension service, low availability of data on landscape degradation, and low capacity among local governments to identify and plan adaptation solutions. Cooperatives have not been targeted from previous interventions as key agents of change in the LREB landscape.	<u>Low</u>	The paradigm shift will involve transition towards climate resilient, low-carbon practices in the 6 targeted value chains. Technical support will be provided through FFS and Cooperatives to scale up climate resilient, low-carbon agriculture in the LREB; technical assistance will also be provided to enable access to financial services and markets that can further support resilient VC crop production and sale. As productivity increases, and marketing of key VC products is boosted, farmers' behavior will change towards sustainable, climate-resilient and low- emission agriculture in the region, leading to a broader scale of application of CRLCSA practices in the 6 value chains and beyond.	The project targets 143,000 participating farmers (men and women) with direct benefits in terms improved resilience and increased income accruing to their households or 572,000 direct beneficiaries. Component 2 will improve landscape management for all residents of the LREB region, generating emissions reductions. Farmers practicing climate resilient, low-carbon agriculture will also generate emissions reductions for crop VC and reductions in emissions intensity for livestock value chains. The technical support provided to cooperative will be leveraged for current and future members and can be extended at scale to reach farmers in other value chains and counties.

Replicability	This project builds on the replication and upscaling of proven, tested climate resilient, low-carbon practices in Kenya and abroad. However, not many of these practices are being implemented by local farmers due to the costs and risks involved.	<u>Medium</u>	The project's strategy for replication builds on various components: Building climate change responses into cooperatives' business model and demonstrating their profitability will ensure that farmers joining cooperatives have permanent access to the best available climate information, extension and services. leveraging the county government's participation in the project to identify and disseminate best practices to farmers beyond project beneficiaries, after project implementation. Deploying an active learning and knowledge sharing strategy that builds on institutions such as Council of Governors, Climate change Multisectoral platforms and national government.	<i>All proposed technologies are applicable to similar crop or livestock value chains. Replication can be foreseen at various levels: to beneficiaries not targeted in the project in the same value chains; to other value chains in the region, and to other regions.</i> <i>The technology transfer mechanisms set up by the project can continue to be available to farmers after project completion.</i>
Sustainability	There is a good level of commitment among county governments to address climate change. However, the conventional approaches to delivering climate services to farmers (information, extension, inputs) has not been sustainable so far. Most of it is costly and does not reach all last mile users and beneficiaries.	<u>Medium</u>	The project transforms the way in which government support farmers and their organization. By focusing on providing guidance, enabling investment, normative frameworks, and establishing rules, counties can partner with farmer organizations and businesses to create better conditions for agriculture in the region. As a result of the project, farmers will see both the economic and climate benefit of technologies proposed and will face less risk in adopting them.	<i>The project will support the creation of technology transfer mechanisms, knowledge and capacity that will outlast the duration of the project. Linkages to CIDPs and other county level strategies and plans are established. Once climate resilient, low-carbon technology becomes mainstreamed in cooperatives business models, all farmers entering the market will have access to this knowledge and will benefit from climate related services provided directly by cooperatives.</i>
E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)				

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term ⁹⁹	Final ¹⁰⁰	
MRA4 Forestry and land use	Core 1: GHG emissions reduced, avoided or removed/sequestered	EX-ACT, GLEAM, Decentralized carbon accounting in each county, supported by remote sensing imaging. Secondary source of information through Global Forest Watch and Kenya Forest Service annual reports.	0	-643,794 tCO ₂ eq	-1,287,589 tCO ₂ eq by project end (2030) -4,268,492 tCO ₂ Eq over project lifetime	A: External shocks do not damage or encourage encroachment on forest lands A: Population growth and migratory patterns remain unchanged; A: No wholesale of land occurs and land tenure regimes remain stable A: Counties continue to implement their climate laws and programs according to plans Data sources: Baseline emissions from Global Forest Watch. AFOLU emissions reductions estimates were obtained using EXACT (see Annex 22). Livestock related emissions reductions were obtained using a downscaled version of the GLEAM model (please see Annex 22). Both figures calculated to project lifetime of 20 years (6 years of project implementation + 14 years of

⁹⁹ Note that this figure does not include the 10,000 ha of gazetted forests to be restored/rehabilitated under the FAO-co-financing, since these are already accounted for under the GEF Mount Elgon project.

¹⁰⁰ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

						capitalization). Monitoring of carbon stocks in each county will be conducted using the decentralized carbon accounting methods developed under Outcome 1.
<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Cooperative census and Resilience survey of project beneficiaries	0	286,000 people, of which 143,000 women (direct beneficiaries) 1,049,070 people, of which 524,535 women (indirect beneficiaries)	572,000 people, of which 286,000 women (direct beneficiaries) 2,098,140 People, of which 1,049,070 women (indirect beneficiaries)	<i>A: No external shocks (political or exogenous) occur with impacts on the targeted beneficiaries and sites</i> <i>A: Population growth and migratory patterns remain unchanged;</i> <i>A: No wholesale of land occurs and land tenure regimes remain stable</i> <i>Project expects to reach 63,000 smallholders and their households through FFS and 80,000 smallholders and their households through cooperatives. Methods for targeting beneficiaries are detailed in section 6, Part C, of the FS.</i>
<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Cooperative census and Resilience survey of project beneficiaries	0	286,000 people, of which 143,000 women (direct beneficiaries) 1,049,070 people, of which 524,535 women (indirect beneficiaries)	572,000 people, of which 286,000 women (direct beneficiaries) 2,098,140 People, of which 1,049,070 women (indirect beneficiaries)	<i>Data sources: Project Annual Reports, FPIC and group formation reports (activity 1.1.1), Stakeholder Engagement reports, Cooperative Census and Resilience Survey</i>

<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Cooperative census and Resilience survey of project beneficiaries	0	286,000 people, of which 143,000 women (direct beneficiaries) 1,049,070 people, of which 524,535 women (indirect beneficiaries)	572,000 people, of which 286,000 women (direct beneficiaries) 2,098,140 people, of which 1,049,070 women (indirect beneficiaries)	
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary Indicator 2.5 Beneficiaries (female/male) adopting innovations that strengthen climate change resilience</u>	Resilience Household survey of project beneficiaries	0	25% of the beneficiaries (143,000) adopt at least one of the innovations that strengthen climate change resilience. 50% of the uptakers are women	60% of the beneficiaries (343,200) beneficiaries adopted at least one of the innovations that strengthen climate change resilience. 50% of the uptakers are women	
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	<i>County Landscape Management Strategies' Implementation Reports</i> <i>Physical Observation</i>	0	15,000 ha	30,540 ha	<i>A: No major forest fire or natural disaster occurs that would destroy large areas of land in LREB.</i> <i>A: Population growth and migratory patterns remain unchanged;</i> <i>A: Counties continue to implement their climate laws</i>

						and programs according to plans Target number of hectares is based hectares to be brought under low-emission/ climate resilient management under output 2.1 and output 3.1.
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E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	While there exists knowledge, practices, know how related to climate resilient low carbon agriculture in the 6 value chains, this knowledge is dispersed; not always adapted to the Kenyan context; difficult to source; and difficult to adopt by individual smallholder farmers. There is currently no systematic mechanism for identifying and disseminating CRLCSA	<u>low</u>	In the target scenario, smallholders can identify climate challenges as they materialize, and access and adopt appropriate knowledge, methods, processes, inputs, services, advice, or assets from a variety of recognized sources in a cost-efficient manner.	The project brings together governments and smallholder farmers (private sector) in the creation and deployment of a comprehensive system of technology transfer that helps smallholders to identify and understand current and future climate challenges, develop or co-develop climate solutions leveraging public and private extension services and technologies.	<u>Multiple sub-national areas within a country</u>

	technology at scale in western Kenya.				
Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	In the baseline situation, the 6 commodities targeted in the project are not produced in a climate resilient, low carbon manner; demand for these commodities is strong and continues to increase, creating a pull effect on the value chains.	low	<i>In the target scenario, all commodities are produced and processing using climate resilient, low-carbon practices and the value chains can respond to demand without generating emissions and despite the climate risks faced by the region.</i>	The project will work with private sector entities to reorient the 6 value chains towards more productive, climate resilient low carbon pathways. The project will also support increased market access for such commodities, and will also work with financial institutions to support the long-term continued investment in climate resilient, low-carbon practices. As a result, the markets for the 6 commodities will be transformed.	Multiple sub-national areas within a country
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	A few knowledge exchange mechanisms already exist in Kenya to share climate knowledge, on which this project will build.	medium	<i>In the target scenario, stakeholders and value chain actors build on the good practices, methodologies and networks created by this project to generate further change across the agriculture sector.</i>	The project will generate knowledge, best practice and lessons learned that will be applicable to all value chains and the entire agriculture sector in the longer term. This will be done by working through county government extension services, cooperatives, the Multi-stakeholder climate change knowledge platform regionally and nationally as well as through the	National level (one country)

				Council of Governors and national government.	
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E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1 Enhanced public agro- climate services support farmer-led proactive adaptation and mitigation actions.	# of men and women benefiting from improved knowledge, extension and climate services, disaggregated by gender, age, and type of actor	Resilience Household Survey (primary) Government of Kenya extension service data (secondary)	Less than half of smallholders receive extension services, which are limited in scope, not easily understandable, or not timely, and not suited to climate risk management	71,500 people (50% women) receive improved or new extension and climate services	143,000 people (50% women) receive improved or new extension and climate services	<i>A: The Kenya meteorology Department continues to decentralize its services.</i> <i>A: Climate data is available free of charge from government.</i> <i>A: Cell phone and internet connectivity rates increase at annual rate of 2%¹⁰¹</i> <i>Data sources and methodologies applied for estimating baseline and targets: Cooperative census, Climate Survey, and Survey of county extension departments (2022).</i>
Output 1.1 Local administrations deploy improved climate	# of methods, tools, databases and climate knowledge	County government climate change	The government uses traditional extension public	A climate responsive extension method	At least 1 new/ upgraded	A: County governments continue to prioritize climate change in local budgets and to

¹⁰¹ GSM Association, <https://www.gsma.com/newsroom/press-release/reform-policy-to-boost-kenyas-mobile-connectivity-says-gsma/>

knowledge, extension and methodologies to support producers and value chain actors	products developed and implemented	directorates annual reports (secondary) Information products and studies (Primary)	services which have little focus on building climate resilience; The methods used are not gender-responsive; the agricultural databases are not complete or regularly updated. No methods for decentralized carbon accounting or climate risk assessment are being used;	for delivering gender-responsive agro-climate services is developed/ upgraded and deployed by mid-term.	method for delivering gender-responsive agro-climate services is approved and deployed by end of project; the agricultural databases for the 6 VC are filled and updated on a regular basis by county administrations	provide operational budgets for extension services; A: Government of Kenya disbursements to counties are conducted in accordance with laws and regulations. <i>Data sources and methodologies applied for estimating baseline and targets: County capacity assessment (2022)</i>
Outcome 2.1 Reduced emissions from the AFOLU sector	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	EX-ACT (Primary); GLEAM (Primary); decentralized carbon accounting methodology (activity 1.1.1) (Secondary).	0	-643,794 tons Co2 removed through improved agricultural practices in crop and livestock production, land use management and landscape restoration	-1,287,589 TCo2 eq by project end (2030) -4,268,492 tCo2Eq over project lifetime through improved agricultural practices in crop and livestock production,	A: there are continued budgets for the improved management of public lands. <i>Data sources and methodologies applied for estimating baseline and targets: EXACT</i>

					land use management and landscape restoration	
Outcome 2.2 Increased ecosystem resilience to climate change	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	County Landscape Management Strategies' Implementation Reports Physical Observation	0	15,000 ha	30,540 ha	<i>A: No major forest fire or natural disaster occurs that would destroy large areas of land in LREB.</i> <i>A: Population growth and migratory patterns remain unchanged;</i> <i>A: Counties continue to implement their climate laws and programs according to plans</i> <i>Target number of hectares is based hectares to be brought under low-emission/ climate resilient management under output 2.1 and output 3.1.</i>
Output 2.1 Agricultural landscapes are managed under strategies that conserve, restore, and sustainably manage community forest and agriculture land, and reduce emissions	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under resoration and/or improved ecosystems</u>	County Landscape Management Strategies' Implementation Reports, Physical observation (Primary), county forestry department reports (Secondary).	No improved climate-based landscape management strategies exist.	The landscape management strategies are approved, and implementation is initiated in each county by year 3	2,800 ha of land are under improved landscape management by end of project.	<i>A: Climate risks do not cause excessive tree mortality or loss of vegetative cover.</i> <i>Data sources and methodologies applied for estimating baseline and targets: physical observation and consultations with county forestry departments</i>

Outcome 3. Increased climate resilience of smallholders' livelihoods using climate-resilient, low carbon technologies.	# of beneficiaries showing an improvement of the resilience score against the Resilience Score Card (disaggregated by gender, age and type of beneficiary)	Resilience Household Survey. (Primary)	0 people showing improvement on Resilience Survey score, (disaggregated by gender, age and type of beneficiary);	At least 100,000 beneficiaries show an improvement of 30% against the resilience scorecard by end of project;	At least 295,000 beneficiaries (60% of total project beneficiaries) show an improvement of more than 30% against the resilience scorecard by end of project;	A: Land tenure regimes do not change and there is no influx of migrants. A: Climate resilient, low carbon technologies are made available without infringement on intellectual property rights <i>Data sources and methodologies applied for estimating baseline and targets: Resilience Score Card; Ex-ACT</i>
Output 3.1 Vulnerable smallholders adopt gender-responsive and socially-inclusive climate resilient and low carbon production and processing practices, technologies, assets, and risk reduction mechanisms	# of people participating in trainings provided by the project (disaggregated by type of training (FFS vs Coop) and gender/age)	Resilience Household Survey	0	75,000 people (of which 40% are women) have participated in at least one project supported training by end of project;	143,000 people (40% women) have participated in at least 3 trainings by end of project;	A: Cultural factors encourage participation in trainings. A: No external shocks (political or exogenous) with impacts on the targeted beneficiaries and sites
	<u>Supplementary Indicator 2.5 Beneficiaries (female/male) adopting innovations that strengthen climate change resilience</u>	Resilience Household Survey	0	25% of the participants (35,750) adopt at least one of the innovations that strengthen climate change resilience. 50% of the uptakers are women	60% of the participants (35,750) adopt at least one of the innovations that strengthen climate change resilience. 50% of the uptakers are women	A: Supply chains operate to ensure continued availability of inputs Data sources and methodologies applied for estimating baseline and targets: Coop Census 2022

Outcome 4 Climate resilient, low carbon value chains are sustained financially and upscaled by the adoption of new business models and practices among farmers and value chain actors.	<i>% income increase among project beneficiaries (men and women) after implementation of project activities</i>	Resilience Household Survey, Cooperative accounting books;	0 people report an increase	75,000 people are reporting (40% women) an increase in income generated from the target value chains with the supports of the project interventions by year three	143,000 people (40% women) are reporting a minimum 15% increase in income generated from the target value chains with the supports of the project interventions by year six	A: Sales are recorded by farmer groups and project activities do not feed into the informal or grey market. A: No sudden price changes, drastic price fluctuations or demand of the targeted products occur during project implementation Data sources and methodologies for baseline data: Coop census 2022.
Output 4.1 Increased access to markets and profitability of climate smart, low carbon sustainable agricultural products	% increase in sales in the targeted value chains of the FO and Cooperatives	Resilience Household Survey, Cooperative accounting books;	0 increase	8% increase in sales of climate resilient, low-carbon commodities by project beneficiaries at the end of year 3 of the project	15% increase in sales of climate resilient, low-carbon commodities by project beneficiaries at the end of the project	A: local and national food demand continues to grow. A: No major disruptions to roads or infrastructures that would prevent physical access to markets Data sources and methodologies for baseline: Coop census
Output 4.2 Vulnerable smallholders and their organizations have increased access to gender-responsive and socially inclusive financial products that support climate	Number of beneficiaries, farmer organizations and Cooperatives with improved capacity to access private sector finance-disaggregated by	Resilience Survey; Cooperative Survey (Primary), Bankable proposals and financial requests	15% of surveyed farmer organizations and cooperatives were able to access loans from financial institutions	25% of beneficiary farmer organizations and cooperatives have improved capacity to access loans from financial institutions by end of year 3.	At least 50% of beneficiary farmer organizations and cooperatives have improved capacity to access loans	A: There is no major change in interest rates and lending conditions or legal frameworks. Data sources and methodologies: Data provided

resilient, low carbon growth	gender, age and type of financial product.	submitted by cooperatives	(Cooperative Survey).		from financial institutions by end of project.	through the cooperative census.
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Project/programme co-benefit indicators

Co-benefit 1: Increased socially inclusive rural employment	<i>Number of jobs created and sustained for men, women, PLWD, and youth,</i>	County government data (employment department); project reports	0	1,000 jobs created (to be disaggregated by group)	3,000 jobs created (to be disaggregated by each group) by end of project.	A: there is no major change in labor laws and minimum wages. A: Data sources and methodologies: Government of Kenya employment records
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E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
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Component 1 – Enabling local government support for adaptation and mitigation

Output 1.1 Local administrations deploy improved climate knowledge, extension and methodologies to producers and value chain actors

Activity 1.1.1 Develop and deploy innovative and efficient extension methods for disseminating and demonstrating CRLCSA knowledge, technologies and practices in gender-responsive and socially inclusive ways	The purpose of this activity is to assist county governments in developing and deploying improved agro-climate extension services, leveraging national and regional best practice, innovative and efficiency-boosting technologies, and participatory processes. Key stakeholders involved in this activity are county governments extension, meteorology, agriculture and cooperative departments.	1.1.1.1 Stocktaking of extension methods and public and private extension service providers and actors and gaps/needs analysis 1.1.1.2 Participatory design of innovative and efficient CRLCSA extension methods including the feedback mechanisms between the services recipients and the providers	1.1.1.1 One stocktaking report on the state of CRLCSA extension in the LREB 1.1.1.2 One regional extension design workshop (4-days) 1.1.1.3. One Set of SOP for each new extension method adopted in each county 1.1.1.4. Cluster-level training on the new SOP, three cluster, one training
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	Methods of delivery for the activity include training, mobilization of external expertise and advisory, learning by doing and technical backstopping from FAO.	1.1.1.3 Develop/improve the Standard Operating procedures for the deployment of extension methods in each county 1.1.1.4 Regional and county level training for extension and cooperative departments on climate sensitive extension services 1.1.1.5 Monitor the effectiveness and efficiency of CRLCSA extension service	per cluster, 30 participants per workshop, 5 days 1.1.1.5 One report on extension services in each county every 2 years.
Activity 1.1.2 Strengthen the dissemination of climate information services to last-mile users including women, youth and PLWD through cooperatives and Farmer Organizations.	This activity supports county governments in deploying climate information services, advisories, early warnings and data to last mile users, with a particular focus on farmer organizations and cooperatives as key knowledge intermediaries, making sure vulnerable groups receive and can act upon climate information within the agricultural production context. Key stakeholders in this activity are the county governments meteorological, agriculture, extension, disaster risk management and cooperative departments. Methods for delivery of this activity include training, workshops, co-planning, participatory scenario planning and co-design workshops, as well as the mobilization of technical advisory and expertise.	1.1.2.1 Conduct a stock take and gap analysis of climate information services in each county for the 6 value chains. 1.1.2.2 Develop new climate information services or products for the 6 value chains in cooperation with National and County Meteorological Department 1.1.2.3 Training for agro-meteorological departments (county level) in the development and delivery of climate information services to FO and cooperatives in the 6 value chains 1.1.2.4 Seasonal planning at the ward level to develop locally relevant and tailored seasonal advisories in the targeted value chains and natural resources management 1.1.2.5 Disseminate timely information on climate impacts and weather forecasts through selected community radio, social media, print channels and extension 1.1.2.6 Monitor reach and level of use for each new product. 1.1.2.7 Develop registry of intermediaries of climate information	1.1.2.1 One climate information services status report in each county 1.1.2.2 One new climate information service or product in each county 1.1.2.3. Two 5-day training workshops for 140 people (in Year 1 and in Year 3) 1.1.2.4 Bi-annual workshops in each county for co-development of agrometeorological advisories, 2 workshop per year in each county. in y1, each workshop is 4 days and in y2-6, each workshop is 2 days 1.1.2.5. 4 seasonal bulletins, 2 early warnings, 30 rainfall forecasts, 100 3-day weather forecasts disseminated annually through various channels in each county 1.1.2.6 Three bi-annual climate information services report in each county for the 6 value chains. 1.1.2.7 One registry in each county

		and weather forecast targeting disadvantage groups	
Activity 1.1.3 Develop and test methodologies for decentralized carbon accounting	Under this activity, county governments will acquire technical capacity to conduct decentralized carbon accounting, through training as well as through learning by doing. The methodologies will be applied in linkages to the development of landscape management strategies (Outcome 2). The methods used for delivering this activity include training, development of processes and institutional capacity, acquisition of material capacity, and the mobilization of technical expertise to backstop emissions reporting in each county.	1.1.3.1 Conduct a participatory design workshop for selecting and developing a LREB-specific decentralized carbon accounting methodology. 1.1.3.2 Set up the process for data gathering and computation 1.1.3.3 Develop operating manuals and guidance documents 1.1.3.4. Conduct learning by doing accounting exercises in each county and prepare emissions reports.	1.1.3.1 Cluster-level workshop bringing together county-level Meteorological Department, agriculture/agro-meteorology and extension services, forest and environment departments and climate change units, three clusters, 5 days 1.1.3.2 Software and data collection protocol 1.1.3.3 One set of operating manuals and guidance documents 1.1.3.4. One emissions report per county at year 2, and one at year 5.
Activity 1.1.4 Upgrade and update agricultural databases, crop and productivity datasets, cooperative census	Under this activity, the project supports county governments in developing more refined data and information services that can serve cooperatives, business and private sector development, while also supporting the documentation of climate benefits from application of climate resilient, low-carbon technologies. This activity builds on ongoing efforts to digitize agricultural extension and services.	1.1.4.1 Conduct systems needs assessment and gap analysis for database in each county 1.1.4.2. Upgrade and update agricultural databases with technical support, draft technical and training guides and reporting schedule for dissemination purposes 1.1.4.3. Collect and analyse crop, climate, productivity, market and economic data annually through county extension services and integrate into databases. 1.1.4.4. Deliver technical assistance to conduct biennial census of cooperatives	1.1.4.1. One stock-take and gap analysis report on agricultural databases 1.1.4.2. 14 upgraded databases 1.1.4.3. Value chain level report highlighting productivity vs climate trends 1.1.4.4 Three Coop Census reports
Activity 1.1.5 Assess eligible climate solutions for the agriculture sector in relation to climate impacts	In this activity, the county governments build their capacity to understand local climate risk to agricultural production and to select the most suitable climate resilient, low-carbon technologies in which to invest. The selection of technologies forms a blueprint or a menu from which cooperatives and financial institutions can select when making decisions about investment and lending.	1.1.5.1 Training for county climate change units, meteorological departments and agriculture departments on selection of climate solutions. 1.1.5.2 Conduct a participatory screening and pre-selection of climate resilient, low carbon solutions and technologies for the 6 Value Chains	1.1.5.1 Cluster based training in climate change impact assessment and climate solutions identification, three clusters, 3 days 1.1.5.2 A county council-endorsed list of county-specific climate resilient, low carbon solutions and technologies in year 2 and updated in year 5.

	The activity is designed to assist county governments in delivering better normative guidance to private sector. The activity is mainly supported through training and the mobilization of technical expertise.		
Activity 1.1.6 Share knowledge and lessons learned through existing platforms	This activity, which also supports the project's M&E plan, includes measures to identify and share best practice, lessons learned and case illustrations from the project to stakeholders within and outside project boundaries. This supports the broader replication and upscaling of project outcomes and outputs to other value chains, counties and regions.	1.1.6.1 Identify lessons learned and best practices arising from project implementation 1.1.6.2 Develop a project knowledge and learning strategy to be implemented through a suite of knowledge products, fact sheets, studies and reports 1.1.6.3 Hold regular knowledge sharing events on project activities, outputs and outcomes in each county and jointly with the Multi-Stakeholder Climate Change Knowledge Platform and the Council of Governors.	1.1.6.1. 24 Lessons learned documents (one per year per outcome starting year 3) and one synthesis report at the end of the project 1.1.6.2. A communication strategy and 15 knowledge products 1.1.6.3. 15 knowledge sharing events and venues (twice a year + 3 at the end of project)
Component 2 – Sustainable Resilient Agricultural Landscapes			
Output 2.1 Agricultural landscapes are managed under strategies that conserve, restore, and sustainably manage community forest and agriculture land, and reduce emissions			
Activity 2.1.1 Develop a county climate-resilient and low-carbon agricultural landscape management strategy and implementation plan, including improved watershed management, land use planning, reforestation and natural regeneration	Under this activity, FAO will provide technical assistance support to each county government towards the development of a climate resilient, low-carbon landscape management strategy and action plan, which will then be brought together in a LREB-wide integrated landscape management strategy. This activity builds on the outcome of The Restoration Initiative ¹⁰² , which supported implementation of the National Forest and Landscape Restoration Plan (FOLAREP), which sets the ambition to restore 5.1 million ha by 2030, and generates significant emissions reductions. The purpose of the activity is to ensure that	2.1.1.1 Conduct an assessment and participatory mapping of landscape degradation in each county 2.1.1.2 Deliver training on climate resilient, low-carbon landscape management for county administrations (forestry, land, agriculture and water departments) 2.1.1.3 Develop, with technical assistance, a county climate resilient, low-carbon landscape management strategy 2.1.1.4 Raise community awareness of the benefits of ensuring restoration, conservation and sustainable	2.1.1.1 One landscape report in each county 2.1.1.2 Three 2-day regional trainings on climate resilient, low-carbon landscape management (60 people trained) 2.1.1.3. 14 county landscape management strategies. 2.1.1.4. Landscape management workshops and awareness events (3 per county for 2 years) 2.1.1.5. Two one-day regional workshop

¹⁰² Restoration of arid and semi-arid lands (ASAL) of Kenya through bio-enterprise development and other incentives under The Restoration Initiative, a GEF-supported initiative.

	agricultural landscapes are managed in a way that enhances productivity and ensures long-term resilience.	management of their agriculture landscape 2.1.1.5 Hold a regional workshop bringing together counties for lessons sharing and development of a regional LMS	
Activity 2.1.2 Implement and monitor climate-resilient and low-carbon landscape management plans.	This activity is financed with a combination of GCF funding and co-financing from the Government of Kenya and FAO-GEF through county governments, leveraging resources from the G-FLLOCA program and the GEF-7 Mount Elgon Project ¹⁰³ . Local consultations will be deployed in each county to ensure local community buy-in and participation in the LMS implementation plan. Each county will implement selected reforestation, conservation, restoration, afforestation, watershed management, forest landscape restoration and/or improved land management actions in degraded hotspots of 200 ha around the agricultural landscapes targeted by this project and in gazetted forests and reserves over the 6 years of project implementation.	2.1.2.1 Draft implementation and M&E plan and budget for each climate resilient, low-carbon Landscape Management plan (LMP). 2.1.2.2. Conduct local consultations to develop LMS implementation plans those related to restoration and management of agriculture landscape 2.1.2.3 Implement the LMS with national budgets (incl. FLLOCA) through fencing/live-fencing, reforestation, afforestation and watershed management or rehabilitation in priority areas in each county. 2.1.2.4 Implement the LMS with support from county governments in gazetted forests and water towers 2.1.2.5. Monitor and survey state of landscape management after 5 years (along with the decentralized carbon accounting under 1.1.1.4)	2.1.2.1 One landscape management strategy implementation plan in each county. 2.1.2.2. 28 community consultations targeting restoration and management of the agriculture landscape 2.1.2.3 200 ha under climate resilient landscape management in each county 2.1.2.4 10,000 ha of forests and water towers rehabilitated 2.1.2.4 14 State of agricultural landscape reports and recommendations for upscaling.
Component 3 - Resilient livelihoods			
Output 3.1 Vulnerable smallholders adopt gender responsive and socially inclusive climate resilient and low –carbon production and processing practices, technologies, assets and risk reduction mechanisms			
Activity 3.1.1 Deploy CRLCSA production/processing assets and training to smallholder farmers, farmer organizations and associations	This activity supports the transfer of technologies to smallholder farmers who are not in cooperatives, using the Farmer Field School model. Trainings will include technologies as listed in the Feasibility Study and be linked to government	3.1.1.1 Technology needs and climate vulnerability assessment and a Free Prior Informed Consent campaign 3.1.1.2 Targeting and group formation 3.1.1.3 Curriculum Development for the FFS	3.1.1.1 One TA Assessment and one FPIC report 3.1.1.2 2520 FFS formed 3.1.1.3 A Complete FFS curriculum for each VC in relation to climate change

¹⁰³ Integrated Landscape Management for conservation and restoration of the Mt. Elgon Ecosystem in Western Kenya – GEF ID 10598

	extension services developed and strengthened under Outcome 1. This will also include gender-related training, social inclusion measures.	3.1.1.4 Training of trainers and facilitators 3.1.1.5 Deploy FFS in 14 counties 3.1.1.6 Equipping participating counties with inputs and tools (seeds, post-harvest, equipment)	3.1.1.4 One 2-week refresher training for existing trainers + 4 week-training of 315 trainers and subsequent refresher trainings up to year 3 3.1.1.5. 2520 FFS implemented (63,000 people trained) 3.1.1.6 Inputs and tools procured
Activity 3.1.2 Disseminate CRLCSA technology, knowledge and assets to cooperative members through peer-to-peer networks and exchanges	This activity supports the transfer of technologies to smallholder farmers members of cooperatives, using the Agriterra model for technology transfer that leverages peer-to-peer networks and North-South/South-South technology transfer. Cooperatives are also supported in strengthening governance and financial management and business planning under outcome 4. The activity is implemented by Agriterra.	3.1.2.1 Conduct Climate Clever Check and Cooperative assessment 3.1.2.2 Transfer CRLCSA technologies through peer-to-peer training, capacity development, 3.1.2.3 Procure and distribute inputs and tools (seeds, post-harvest, equipment) 3.1.2.4 Disseminate and share knowledge within and among cooperatives	3.1.2.1. 130 Climate Clever Checks and Cooperative Audits. 3.1.2.2. 130 cooperatives receive trainings (80,000 people) 3.1.2.3 Inputs and Tools procured 3.1.2.4. 112 knowledge sharing workshops within and among cooperatives.
Activity 3.1.3 Support smallholder farmer aggregation into cooperatives and other business units as climate risk reduction and risk sharing mechanisms	In this activity, the project strengthens the ability of cooperatives to manage and integrate new members. At the end of each FFS Cohort, farmers participating in FFS will be invited to join existing cooperatives. This strengthening of cooperatives helps farmers pool resources, reduce costs, improve revenues and reduce climate risks more efficiently. This activity is implemented by Agriterra.	3.1.3.1 Deliver technical assistance to increase business and management capacities of farmers' organizations 3.1.3.2 Increase farmers' awareness on benefits to join cooperatives through workshops, site visits and information products 3.1.3.3 Increase existing cooperative's capacities to manage a larger amount of memberships through training and material support	3.1.3.1 300 people and 130 organizations trained in business management 3.1.3.2 100 demo days 3.1.3.3 Computers, phones, office supplies, membership management software and other inputs procured to apply the MyCoop Tool.
Activity 3.1.4 Support improvements in social inclusion and women's meaningful participation in climate resilient, low-carbon value chains	This dedicated activity ensures mobilization of stakeholders and budgets for the achievement of more meaningful outcomes for women, youth and persons living with disabilities within the 6 value chains. The goal is at least 10,000 women, youth and PLWD acceded to roles of more meaningful participation in value chains. The project will also support the maintenance of gender and social inclusion safeguards and standards through this activity.	3.1.4.1 Training for SEAH, GBV, etc. (GAP) 3.1.4.2 Deliver Women's and Youth Leadership trainings and GALS awareness raising to Smallholders, FFS and FO including cooperatives 3.1.4.3 Monitor gender objectives through surveys and studies (WEAI)	3.1.4.1. At least 280 people trained in SEAH, GBV and GAP in project 3.1.4.2 50,000 farmers (M/F) trained in gender related approaches 3.1.4.3. One gender impact assessment at end of project

Component 4 – Scaling through CRLCSA market and finance

Output 4.1 Increased access to markets and profitability of climate smart, low carbon sustainable agricultural products

Activity 4.1.1 Work with buyers and aggregators to increase demand and market opportunities for CRLCSA commodities	The project will work on the demand and market side of the value chains to increase uptake of CRLCSA commodities produced by project beneficiaries. Profitability of adopting climate resilient, low-carbon technologies will be key to encouraging continued sustainability, replication, and upscaling. To create further market opportunities for targeted commodities, the project will team up with relevant value chain actors as well as county governments, to develop and deliver marketing campaigns highlighting the benefits and attractiveness of the products under climate resilient, low-carbon technologies. The project will provide technical assistance and business mentoring to the most mature cooperatives to negotiate sales agreements and contracts with identified buyers.	4.1.1.1 Conduct an assessment of markets and buyers demand of the targeted products including the needs in quantity and quantity of the target products and buyer's affordability 4.1.1.2 Research and comparative analysis of quality, grading and pricing of products supported under the project will be conducted 4.1.1.3 Prepare and deliver an advertising campaign on the six targeted commodities 4.1.1.4 Provide technical assistance and business mentoring to mature cooperatives in signing purchasing agreements and contracts with identified buyers. 4.1.1.5 Develop and Deploy innovative business hubs to promote market access and services, including business incubators for smallholder farmers pursuing nature-based livelihoods 4.1.1.6 Organize marketing events and awareness campaigns 4.1.1.7 Develop seasonal market insights and share with the FOs and Cooperatives (link to agricultural databases under 1.1.1)	4.1.1.1 One Market and buyer assessment report (year 1) including identification and analysis of 12 buyers for the 6 commodities. 4.1.1.2 6 sets of product sheets, pamphlets, brochures and presentations 4.1.1.3 One advertising campaign running from year 3 to 6. 4.1.1.4. 12 new signed agreements. 4.1.1.5 15 new smallholder businesses supported 4.1.1.6. 14 annual county-led Marketing events (starting year 3) 4.1.1.7. 4 seasonal market insight reports for each value chain annually, distributed by county governments to cooperatives.
Activity 4.1.2 Increase access to various certification and labeling schemes	This activity, which will be delivered jointly with the above, will provide technical assistance to farmer organizations and cooperatives in accessing and complying with relevant certification schemes. These organizations will be supported in identifying the most suitable certification scheme– i.e., the one that would bring the most added	4.1.2.1 Select FOs and Cooperatives and other value chain actors, such as aggregators to receive the certifications based on assessments 4.1.2.2 Provide Technical Assistance, to selected cooperatives for the identification of and compliance with suitable certification or labels.	4.1.2.1 30 beneficiary cooperatives and FOs are selected to receive TA for certification 4.1.2.2 30 trainings delivered to cooperatives on certification and compliance.

	value both economically and from a climate resilience or decarbonization point of view. Each organization will then receive targeted technical assistance, training and expert advisory services to understand and apply the requirements.		
Output 4.2 Vulnerable smallholders and their organizations have increased access to gender-responsive and socially inclusive financial products that support climate resilient, low carbon growth			
Activity 4.2.1 Develop gender-responsive and socially inclusive private finance tools, procedures and products to promote the upscale of CRLCSA value chains	The project will support the development of capacity within financial institutions to support climate resilient, low-carbon value chains. This activity will help strengthen and increase the supply and accessibility of financial products available to support climate resilient, low-carbon agriculture in the region. The project will support financial institutions, through consultancies and expert advice, in the development and roll out of new financial products or in the revision of existing products and services to access concessional finance.	4.2.1.1 Develop new financial or revise the existing products that invest in climate resilience and low carbon value chains with financial institutions 4.2.1.2 Deliver training for financial institutions on the financial products supporting climate resilient, low-carbon agriculture	4.2.1.2 6 new financial products developed supporting climate resilient, low-carbon value chains. 4.2.1.3 6 trainings for financial institutions (200 people)
Activity 4.2.2 Support smallholders and their business units in the development of bankable business plans, with particular focus on social inclusion and gender-based access	The project will support farmer organizations and cooperatives in accessing the financial services offered by the partner financial institutions. The purpose of this assistance is to ensure that cooperatives have, in the long-term, the autonomous capacity to mobilize their own financing, through the development or revision of their own business plan, and through the articulation of solid, bankable financial proposals. This will bridge the gap or the mismatch between supply and demand of climate resilient, low-carbon finance in the 6 value chains	4.2.2.1 Deliver training on financial literacy and financial management to farmers, FOs and Cooperatives 4.2.2.2 Assist the FOs, cooperatives and other rural enterprises to develop bankable business plans to invest in climate resilience and low carbon practices	4.2.2.1 50 cooperatives trained in financial literacy and financial management for agriculture value chains 4.2.2.2 130 new or revised business plans.
Activity 4.2.3 Facilitate smallholders access to financial incentives schemes for agroforestry	The project will assist select cooperatives in the 6 value chains in accessing cooperatives in the 6 value chains in accessing carbon markets, payments for ecosystem services, biodiversity offsets or	4.2.3.1 Conduct assessment of technical modalities of implementing carbon and conservation finance incentives in targeted cooperatives	4.2.3.1 One modality report and a set of locally specific agroforestry designs 4.2.3.2 5 information products 4.2.3.3 2 trainings (20 cooperatives each)

	conservation finance. Support provided to farmers and cooperatives will include raising awareness on the availability and opportunity for the various financial mechanisms and developing training for participating cooperatives on the requirements of each selected mechanism.	(including coop selection, land and scheme selection) 4.2.3.2 Raise awareness of availability of carbon-based mechanisms and conservation finance (biodiversity offset) for agroforestry 4.2.3.3 Deliver training to selected cooperatives in relevant schemes, including all compliance and monitoring requirements.	
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E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

279. MEL will be undertaken in compliance with FAO and GCF policies. As AE, FAO will ensure that all EE are making use of a common monitoring and measurement system to analyse and quantify the causal pathways and attributable changes, the contribution of project activities and the overall results of the project. This includes the implementation of: (i) a *monitoring system* to regularly measure progress and results; (ii) a Risk monitoring and management plan; (iii) an *evaluation plan* that helps understand efficacy, pathways to impact and to verify the assumptions that the program is making and (iv) a *learning strategy* through which the results of the project can feed back into project implementation and exit strategy.

Monitoring System

280. Project management, supervision and day to day monitoring will be carried out by the Project Management Unit (PMU), with the support of FAO-Kenya and information flows from each EE. Monitoring will ensure that: (i) project deliverables and outputs are produced in accordance with the project results framework and leading to the achievement of project outcomes; (ii) project outcomes are leading to the achievement of the project objective; (iii) risks are continuously identified and monitored and appropriate mitigation strategies are applied; and (iv) agreed project climate-related benefits are being delivered. The data collection tools will be made on the basis of the indicators from the project's Results Framework (RF), and the Annual Work Plan and Budgets (AWPBs).

281. Project outputs will be monitored separately as well as in relation to the achievement of higher level project results and overall GCF goals. Reporting on results will be required as follows:

Project Inception Report

282. A project inception workshop will be conducted in the first month of implementation in order to: a) orient project stakeholders to the project strategy and discuss any change in the overall context that might influence implementation; b) discuss the roles and responsibilities of the project team, including reporting and communication lines; c) review the results framework and discussion, present in detail the Theory of Change of the project, reporting, monitoring and evaluation roles and responsibilities, and to finalize the M&E plans; d) review financial reporting requirements; and e) planning and scheduling of M&E meetings; and f) finalize the first year work plan. The workshop will generate an inception report that will be presented and approved by FAO within the 4 weeks following the workshop.

Project Progress Reports

283. **Semi Annual Progress Reports.** The PMU will prepare 6-months progress reports which will be sent to the Budget Holder. The progress reports will include an overall assessment of achievements against targets and deliverables. Executing Entities will also report on their financial and operational progress on a six-month basis.

284. **Annual Progress Reports.** The M&E Officer and EE will provide inputs to the Annual Report for each year of implementation. The M&E Officer will ensure that the indicators in the results framework are monitored annually through the online tools and other procedures. The annual performance reports will be due

at GCF 60 days after the end of each calendar year. The final project annual report and the terminal evaluation report will serve as the final project report package.

285. **Technical Reports** will be prepared by consultants as part of project outputs (deliverables) and to document and share project outcomes and lessons learned. The drafts of any technical reports will be submitted by the PMU. The Lead Technical Officer (LTO) will be responsible for ensuring appropriate technical review and clearance of the report. Copies of the technical reports will be distributed to project partners and the PTCC and PSC as appropriate.

286. Annual Financial and Co –financing Reports: containing information compiled from all EEs on expenditures against targets, including expenditures from all sources of co-financing.

287. **Project Completion report (Final APR):** A final APR report will be submitted by the AE, with input from all EE, to the GCF, no later than 6 months after technical completion of activities.

288. For field-level activities, the project will make use of georeferenced monitoring. The project places particular attention on ensuring the monitoring of differential impacts by sex, age and vulnerability, especially monitoring regarding activities relevant to women and more vulnerable populations.

Monitoring of Core GCF Results - GHG emissions reductions and Adaptation Benefits

289. The project will conduct EXACT and GLEAM simulation exercises in the final year of the project, leveraging updated data on project execution and output delivery regarding specifically:

- Number of heads of cattle and productivity levels by end of project
- Extent of land under improved management, improved cover, reforestation, afforestation and conservation
- Using these simulations the project's final evaluation will compare projected achieved emissions reductions with anticipated reductions at inception. (please refer to Annex 22 for details on the methodologies). A report analysing changes and factors of success or challenges will be compiled at the end of the project.

Resilience Survey

290. As indicated in FS, Part C section 6.3, a resilience survey will be deployed at inception, mid-term and end point of the project. The survey will be designed at inception, to facilitate tracking of project indicators and deliverables as well as all sub-indices contained in the resilience index. It will be tailored to include additional questions to minimize the reporting burden on farmers and the project and to maximize data collection. The measurement will be made at household level with a representative sample of project participants.

Cooperative Census

291. The cooperative census, which was crafted with climate, gender equality and social inclusion intent will be carried out by counties on a biennial basis, assessing evolution in membership and participation, social inclusion, sales and income, environmental and climate challenges and access to assets and inputs (land, water, energy).

Mid-Term and Terminal Evaluations

292. To provide an external viewpoint on the progress of the project and the achievement of its objectives, and in line with the AMA signed with the GCF, two independent project evaluations will be conducted by FAO– mid-term and final evaluations. In line with the FAO policy on evaluations, the interim (midterm) and final evaluations may be either managed by FAO Office of Evaluation or decentralized (under the responsibility of decentralized offices). In both cases, they will be carried out by a team of independent external consultants. The evaluations will also be conducted in line with the GCF Evaluation Policy, and will follow the GCF's Integrated Results Management Framework, namely containing an assessment of the paradigm shift potential and its enabling environment. The project's beneficiary surveys will constitute important inputs for the interim and final evaluations. Please refer to Annex 11 for the detailed M&E plan.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

293. A risk assessment been conducted for this project, including an evaluation of financial, technical, operational and institutional risks that may prevent the project from achieving its objectives. An environmental and social risk assessment has also been conducted, the findings of which are presented in Annex X. FAO as AE has conducted due legal diligence assessment on all project partners and executing entities. FAO confirms that none of the activities, counterparties, beneficiaries, executing entities, co-financiers or implementing partners are subject to or impacted by any United Nations Security Council Sanctions. No entity or individual in the project is subject to or listed on any UN Security Council sanctions list, including the UN Security Council Consolidated Sanctions list. This will be reviewed annually on the basis of updated lists of sanctioned entities, during the Annual Project Performance Review.

294. There is a **low risk of reversal and no risk of leakage** the carbon sink creating activities of this project. All activities designed to reduce GHG emissions from agricultural production and processing are also designed to ensure that activities are not displaced, or delegated to other value chain actors. There is strong will and potential for continued agricultural production in the area. Regarding the risk of reversal, this applies to the implementation of the Landscape Management Strategies executed by county governments under Outcome 2 and to agroforestry or any other increases in land cover achieved by the project. The risk of reversal arises from unlikely risks not controlled by the project, such as major natural disasters (earthquake, landslides, fires), sudden population changes (rapid influx of migrants or refugees, pandemic). Measures to prevent reversal are in-built into the project's implementation strategy: by tying increases in land cover with economic benefits for farmers, by ensuring that the economic benefits of improved landscape management are documented, seen and integrated by communities, county governments and private sector, and by ensuring that commodities produced under climate resilient, low-carbon practices fetch fair prices and attract more financing than others. The decentralized carbon accounting methodologies and exercises deployed by counties under Outcome 1 will also serve as a means to assess any risk of reversal or leakage during project implementation because it is repeated at 2 intervals and can inform a redress strategy.

295. Main risks identified for this project are illustrated in the table below.

Selected Risk Factor 1 Land tenure, land ownership or land conflict

Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>Medium</u>
Description		
Outdated land registries, limited access to land registries and information, poor dispute resolution, poor implementation of land reforms and overlapping mandates between governing institutions are some of the challenges associated with land governance in Kenya. ¹⁰⁴ Weak land use planning and insecure land tenure have proven to be a barrier to the adoption of practices and technologies (e.g., agroforestry, irrigation infrastructure and soil conservation) that can reduce vulnerability to climate change. It also discourages long-term planning in favour of maximizing short-term profits and complicates the implementation of effective climate change adaptation and mitigation plans. Regarding land use/ access issues for Indigenous Peoples, it is not foreseen that Indigenous Peoples will be directly targeted as Indigenous Peoples in the region (such as pastoralists and hunter-gathers) are not engaged in the project target VCs. However, the project will develop and implement county's landscape management strategies which could cause restrictions on land use and access to natural resources subject to traditional ownership or under customary use, if the proposed strategies and its implementation overlap with Indigenous Peoples's territories and their livelihood.		
Mitigation Measure(s)		
Identification of existing tenure rights and right holders, including the identification of customary use of land and resources, such as cultural, ceremonial or spiritual use, and any ad hoc, seasonal or cyclical use of land and natural resources (Component 2). In case the project confirms the presence of Indigenous Peoples in the project target areas, the Indigenous Peoples Planning Framework (IPPF) and related plans will be applied to these Indigenous Peoples who might be impacted by the project intervention.		

¹⁰⁴ IGAD, 2018. Kenya - Land Governance Country Profile. Available at: <https://land.igad.int/index.php/documents-1/countries/kenya/profiles-3/785-land-governance-in-igad-region-kenya-country-profile/file>

Selected Risk Factor 2 Market reorientation and supply chain constraints in selected value chains

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>
Description		
There is a very low risk that the national food market would see a change in demand in the commodities targeted by this project. For coffee and tea, demand projections at national and global levels have shown steady increasing trends, even despite market shocks and the pandemic. However, supply chains may be constrained due to unforeseen international events, such as war, natural disasters, or sudden trade policy changes, leading to input and output barriers (barriers to acquisition of inputs, and barriers to delivery of outputs). This is highly unlikely, as even during the pandemic, the agricultural sector in Kenya barely contracted, and trade continued unhindered.		
Mitigation Measure(s)		
Measures to mitigate this risk are embedded into Activity 4.1.1, where the project will work with and for buyers to demonstrate the quality and economic value of outputs produced using climate resilient, low-carbon technologies. Support to cooperatives to reach buyers, negotiate trade agreements, navigate supply chains, and conduct appropriate price-setting as part of activity 4.2.2 will also assist cooperatives in understanding and assessing the impacts of value chain risks to their own production. Furthermore, the project also works with counties to monitor trends for value chains as a service given to farmers and cooperatives under activity 1.1.4. This will serve the dual purpose of obtaining advance warning of any changing trends in demand or supply in the 6 value chains, but also of informing county-level trade and development policy to better support farmers in their responses to market trends.		

Selected Risk Factor 3 Decreasing national budgets for management of natural resources

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>
Description		
There is a small risk, given the increasing demands on national budgets and given the fact that some of the sectors such as Forestry have not yet undergone full devolution in Kenya, that national and decentralized budgets for the improved management of natural resources and public lands could decrease. This would mean the national and county governments would progressively divest themselves from the conservation and management of natural reserves, parks, forests or public lands, leaving private sector interest to fill the vacuum in ways that might be contrary to the project's objectives and the country's best interest. This risk is very low – given the historical high level commitment of the Government of Kenya to the protection of natural resources (regardless of political interests and orientation), the presence of a solid corpus of national laws related to conservation and natural resources management, and the increasing trends of public investment in the management and conservation of the country's natural capital. National and county budgets and priorities are set annually and enshrined in the Budget Policy Statement." The BPS includes (a) an assessment of the current state of the economy and the financial outlook over the medium term, including macro-economic forecasts; (b) the financial outlook with respect to Government revenues, expenditures and borrowing for the next financial year and over the medium term;(c) the proposed expenditure limits for the national government, and indicative transfers to county governments; (d) the fiscal responsibility principles and financial objectives over the medium term including limits on total annual debt; (e) fiscal risk statement".		
Mitigation Measure(s)		
The project includes measures to build the county's capacity to understand the risks to natural resources posed by climate change and unsustainable use, and to also understand the economic and development value of the maintenance of ecosystem services. This will be done through the development of the integrated landscape management strategies, which are underpinned by rapid climate risk assessments (Output 2.2). Awareness raising of the ecological, social and economic benefits of healthy ecosystems and agricultural landscapes is also built into the project's strategy, further strengthening the local stewardship of ecosystems. Measures to integrate activities related to the project (e.g., extension, climate services, etc.) into the County Integrated Development Plans (CIDP) are also in place. In addition, FAO will review, in collaboration with the National Treasury and the NDA, any impacts of the BPS on the project annually.		

Selected Risk Factor 4 Changes in financial policy hinder financial institutions' ability to lend to beneficiaries

Category	Probability	Impact
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Technical and operational		Low	Low
Description			
There is a possibility that changes in monetary policy or rates will impact financial institutions' ability to lend to project beneficiaries, which will make finance more difficult to access or could constrain the availability of supply despite project efforts to build FI and farmer organization capacity. Currently, Commercial banks in Kenya are required by law to keep a specified proportion of their total deposits at the Central Bank (5.25%). The Central Bank sets rates that determine other bank's lending rates: currently the central bank rate is high, at 8.75%. Inflation rate in Kenya in March 2023 is also high, at 9.6%. Changes in these conditions will impact the participating banks' lending appetite, ability and conditions. However, the risk of significant rapid change is low, as the central policy principle for the Central Bank is to maintain annual price stability. Monetary policy is guided by a monetary programme, which is premised on the economic growth and inflation targets provided by the National Treasury. Monetary policy decisions are made by the Monetary Policy Committee (MPC). The MPC meets at least once every two months and reviews data and analysis from various sources including the Central Bank Departments enabling it to decide on any action to maintain or vary its stance. Information from the monetary policy and the general macro-economic indicators of the country are also integrated into the annual budget policy statement.			
Mitigation Measure(s)			
The project will monitor, together with the participating financial institutions, any changes in monetary policy and its potential impact on the ability of the institutions to deliver financing. Capacity development for financial institutions provided under outcome 4 will also include capacity to conserve lending appetite and increase concessionality within the parameters of monetary policy evolution. FAO will also monitor changes in the country's monetary policy, macro-economic indicators and results of annual budget policy statements.			
Selected Risk Factor 5: Political (collaboration between Government stakeholders)			
Category	Probability	Impact	
Technical and operational	Medium	Medium	
Description			
Limited collaboration between stakeholders from different ministries (e.g. Ministry of Agriculture - MoA, Ministry of Environment and the National Treasury as well as the county governments) on land restoration activities and the coordination of the project implementation. This low collaboration may negatively affect project implementation, resulting in limited information and knowledge-sharing on climate-resilience practices and technologies, and ultimately impair the overall long-term project impact.			
Mitigation Measure(s)			
All actors will be engaged in the Project Steering Committee (PSC) and Project Technical Coordination Committee (PTCC) to ensure that government organisations work hand-in hand. At local (county) level, all activities will be coordinated under the county level PSC and PTCC to ensure collaboration between all government stakeholders. These actions will lower the probability of risk to low.			
Selected Risk Factor 6: Resistance on women's participation and empowerment			
Category	Probability	Impact	
Other	Low	Low	
Description			
The project may encounter passive or active resistance to women's participation as well as when working to strengthen women's empowerment in communities.			
Mitigation Measure(s)			
The project targeting mechanisms including direct, community-based and self-targeting mechanisms will promote women's participation including vulnerable groups such as female-head of household, female youth and girls, and persons living with HIV/AIDS. The project will rely on participative community approach and on gender-transformative approach: Women's Leadership-Youth leadership trainings developed by Agriterra as well as the Gender Action Learning System (GALS). These trainings will focus on all aspects of the value chain, accelerating the inclusion of women/ vulnerable groups in production, processing and value addition, and trade. During the Women's Leadership Workshops, the role of women as members, entrepreneurs and leaders will be discussed and together with male and female farmers an action plan is developed to improve the participation of women in the cooperative. All along the project lifecycle, the project will refer to the GAP developed during the design process to monitor the progress made on GEWE and take necessary corrective actions as appropriate.			
Selected Risk Factor 7: Money laundering and countering the financing of terrorism (ML/TF)			
Category	Probability	Impact	
ML/TF	Low	Low	
Description			
The risk of money laundering and countering the financing of terrorism			

Mitigation Measure(s)

FAO includes in the project agreement signed between FAO and the Government of Kenya clauses related to AML/CFT, as follows:

- a) The Government shall comply, and shall require all persons and entities engaged in its activities under the Project to comply, with all internal anti-money laundering, counter-terrorism financing laws, rules, and regulations;
- b) The Government confirms it has obtained sufficient undertakings from all persons and entities involved in its activities under the Project that they shall not engage in any prohibited practices; the Government undertakes and confirm that it shall comply with the substantive objectives of the GCF's Policy on Prohibited Practices;
- c) Consistent with numerous United Nations Security Council resolutions adopted under Chapter VII of the UN Charter, the Government and FAO are firmly committed to the international fight against terrorism and, in particular, against the financing of terrorism. It is the policy of the Government and FAO to seek to ensure that none of their funds are used, directly or indirectly, to provide support to individuals or entities:
 - i) associated with terrorism, as included in the list maintained by the Security Council Committee established pursuant to its Resolutions 1267 (1999), 1989 (2011) and 2253 (2015); or ii) that are the subject of sanctions or other enforcement measures promulgated by the United Nations Security Council. This provision must be included in all agreements that may be concluded with third parties for the implementation of activities under the Project.

During project implementation FAO, as AE, will ensure close monitoring and supervision through its offices in the regional office and HQ in order to ensure that the activities are implemented in full compliance with the signed project.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

296. The project is aligned with the FAO Environmental and Social Standards (2022), and GCF policies including the revised GCF Environmental and Social Policy (2021), GCF Indigenous Peoples Policy (IPP) and the GCF Information Disclosure Policy (2016), GCF Policy on the Prevention and Protection from Sexual Exploitation, Sexual Abuse and Sexual Harassment (GCF SEAH Policy), among others. Stakeholder consultations process was carried planned at the community level in the project area through period 2021-2023, considering GCF (2022) guidance on designing and ensuring meaningful stakeholder engagement.
297. The project is expected to predominantly generate substantial positive environmental and social benefits, including, but not limited to enhance resilience of vulnerable smallholder and key value chains to the impacts of climate change, reduce GHG emissions from Agriculture, Forestry and Land Use sectors (AFOLU), strengthen land use planning, among others. Nonetheless, the project has the potential to generate adverse social and environmental impacts that need to be carefully managed and monitored.
298. The FAO Project Environmental and Social Screening Checklist was prepared, and the project was categorized as a Moderate Risk Project (Category B project), where:
299. There are identified potential adverse environmental and social impacts requiring the need for environmental and social management plans, including: proximity to protected areas, potential generation of new environmental waste from agro-industrial activities, indirect trigger for increased pesticides use (even when IPM is being promoted) and construction of small scale of agro-facilities. It is expected that the project activities, as described in Chapter 2, will trigger the following Environmental and Social Safeguard Policies: ESS1, ESS2, ESS4, ESS5, ESS6, ESS7, ESS8 (see Table below).
300. Potential impacts are limited to the project's footprint, not unprecedented in the project area, neither irreversible nor cumulative. Potential adverse impacts can be addressed using recognized good management or pollution abatement practices that include FAO guidelines as well as World Bank Group Environmental, Health and Safety Guidelines (known as EHS Guidelines¹⁰⁵)
301. Project sub-activities will be subject to project screening process (using FAO Screening checklist) to identify potential E&S risks and impacts. Depending on the classification and level of risk identified, Environmental and Social Management Plans (ESMPs) will then be prepared. Given that the project is considered medium risk overall, it is expected that only some of the project activities will require ESMPs following the screening phase. Additionally, all project activities will have to ensure compliance with Kenya's environmental and social regulation and obtain necessary construction and or operation permits and licenses.
302. The Government of Kenya has not yet adopted the United Nations Declaration on the Rights of Indigenous Peoples and has yet to ratify Convention 169 of the International Labor Organization, the country is yet to ratify the ILO convention on Indigenous and Tribal Peoples. Though, Kenya has no specific law on Indigenous Peoples, the country has ratified the: i) International Convention on the Elimination of All Forms of Racial Discrimination (ICERD); ii) the Convention on the Elimination of Discrimination against Women (CEDAW); and iii) the Convention on the Rights of the Child (CRC). Kenya does not have specific legislation on Indigenous Peoples, and does not identify with the term indigenous, rather considers those segments of the population who have continued to live by their traditional customs and belief systems and way of life, as marginalized (these communities include mostly hunter-gatherers and pastoralists).
303. Participants to the project will be invited to self-identify as members of a vulnerable, indigenous, ethnic or religious minority and specific measures are included to support meaningful participation of Indigenous People in the project, including in the Project Committees. Before initiating project activities which could impact Indigenous Peoples, a Free Prior and Informed Consent (FPIC) process will be implemented following FAO guidelines, to ensure an effective and inclusive participation approach of potential affected indigenous communities. An Indigenous Peoples Planning Framework (IPPF) has been included within the project Environmental and Social Management Framework (ESMF - Annex 6) of the FP.
304. The overall ESMF budget (Annex 6) is USD 697,356 for a blend of human resources to respond to ad-hoc requirements for safeguards activities. Gender activities in the Gender Action Plan are included in the project activities budget. ESS compliance and GAP monitoring are also included in the baseline, mid and end-line surveys,

¹⁰⁵ www.ifc.org/ehsguidelines

project Monitoring and Evaluation and Management Information System (MIS) for which separate budget are included.

305. Further information on the risks identified and assessed, as well as mitigation measures, is provided in the table below, which follows FAO Environmental and Social Standards (ESS). The project has also developed/ will develop the following environmental and social assessment and management instruments:

Safeguard Policy	Triggered	Safeguard Instruments & Mitigation Measures
ESS 1. Biodiversity conservation, and sustainable management of natural resources	Yes	Non-eligible activities (ESMF Appendix 1 Exclusion List) Elaboration of Landscape Management Strategy Biodiversity Management Planning Framework
ESS2. Resource efficiency and pollution prevention and management	Yes	Non-eligible activities (ESMF Appendix 1 Exclusion List) ESMF/ESMP include Integrated Pest Management Approach; FAO/WHO International Code of Conduct on Pesticide Management; International Code of Conduct for the sustainable use and management of fertilizers.
ESS3. Climate change and disaster risk reduction	Yes	The Project already incorporates a Climate Risk Assessment and findings from this assessment have been fully considered in the project design.
ESS4. Decent work	Yes	ESMF/ESMP. Labour Management Plan (Appendix 12 of ESMF) The project promotes compliance with national employment and labor laws; as well as with safe and healthy working conditions. Training for farmers and sensitization sessions on Occupational, Health, and Safety (OHS) will take place, project will utilize the Voluntary Guidelines on the Responsible Governance of Tenure of Land, Fisheries, and Forests.
ESS5. Community health, safety, and security	Yes	ESMF/ESMP Zero tolerance of SEAH. The project includes training for all project staff on SEAH. Labour Management Plan (Appendix 12 of ESMF)
ESS6. Gender equality and prevention of gender-based violence	Yes	Project design and implementation integrates gender as a cross-cutting element; a Gender Analysis and Action Plan, with specific gender-targeted activities built into the project design to address main gender barriers and challenges (FP Annex 8). The project will have zero tolerance for all forms of SEAH. The project's GRM will be accessible for all project-related complaints, including SEAH-specific complaints. The GRM will be survivor-centered and gender responsive, and will have specific procedures for SEAH, including confidential reporting and safe and ethical documenting (see Chapter 5 of ESMF). ESMF/ ESMP also includes measures to facilitate social inclusion and enhance gender equality, and safeguard against SEAH.
ESS7. Land tenure, displacement, and resettlement	Yes	Non-eligible activities (ESMF Appendix 1 Exclusion List) Use of Voluntary Guidelines on the Responsible Governance of Tenure of Land, Fisheries, and Forests
ESS8. Indigenous Peoples	Yes	Indigenous Peoples Planning Framework (IPPF) in ESMF and subsequent ESMP and Indigenous Peoples Plan (IPP) Due to the nature of the proposed project and its underlying activities, it is not anticipated that the ethnic minority groups will be exposed to any additional negative impacts or risks arising from the project beyond ESMF (see Chapter 6&7 of ESMF). Ethnic minority groups living in the targeted districts and selected sites could, however, experience particular difficulties in accessing project support. A careful targeting strategy to avoid exclusion of minority groups will be implemented. FPIC following FAO's guidelines and manuals will be implemented.
ESS9. Cultural heritage:	No	As per the Non-eligible activities (ESMF Appendix 1 Exclusion List). The project will not invest in areas identified as cultural heritage sites. The project specific ESMP will ensure that the scoping and screening exercises capture these kinds of issues. Though the risks are considered not very likely. However, for any activities that involve earthwork Chance find procedure (see Appendix 5 of ESMF) will be made available and communicated to all involved actors.

Stakeholder Engagement Plan (SEP)

306. This project has adopted an active stakeholder engagement strategy since its inception and throughout development and design following a bottom-up approach that is aligned to the devolution process in Kenya. Particular attention has been paid to identifying key stakeholders and involving them meaningfully and at the right

time, to ensure their advice, knowledge and inputs are duly considered in all aspects of this project. Stakeholder engagement is fully mainstreamed in all project activities.

307. The identification of stakeholders was done using a collaborative approach, mobilizing Kenyan government authorities at national and county level as well as FAO-Kenya's experience and expertise. Analysis of each stakeholders' potential influence on, benefits from, and contribution to the project was done combining desk review and interviews or discussions.
308. The initial identification of stakeholders was done through exchanges between The National Treasury who is the NDA, and FAO on the basis of the project's proposed regional and thematic scope. Subsequent discussions expanded to include relevant government departments, agencies and development partners at national and sub-national levels who have helped with refining the concept note and ensured inclusivity of relevant actors. Stakeholders in this project fall in the following groups and categories, as detailed in Annex 7.
- *Farmers*: individual smallholder farmers involved in the six value chains namely coffee, tea, dairy, African leafy vegetables, fruit trees and poultry. Farmers are direct and indirect beneficiaries of the project. The project also identified vulnerable groups whose participation in project activities will be the object of particular attention: women, youth, persons living with disabilities, and persons from indigenous peoples or ethnic and religious minorities.
 - *Farmer organizations*: this includes informal self-help groups, business units, farming groups, community forest associations, farmer field schools and cooperatives in the six value chains, as well as value chain actors not involved in production, such as buyers, transporters, wholesalers, input and service providers. Farmer organizations are also direct and indirect beneficiaries of the project.
 - *County governments*: This includes county Council Assemblies, Governors, and public administration in all the devolved sectors. This also includes ward-level decentralized administration and services. Counties are direct beneficiaries of the project and implementing partners.
 - *Regional (LREB) institutions*: including the LREB secretariat and relevant caucuses such as agriculture, water and environment; LREB Centre of excellence in Agriculture and Water; the Lake Region Development Authority; Lake Victoria North and South Water Service Boards, as well as knowledge sharing platforms on various sectors. Regional institutions play a key role in upscaling and knowledge sharing.
 - *National Government Departments*: The National Treasury (NT); State department of Agriculture; State Department of Cooperatives; Ministry of Environment, Climate Change and Forestry; Kenya Meteorological Department, Kenya Agricultural and Livestock Research Organization (KALRO); Department of Resource Survey and Resource Survey, among others.
 - *Non-state actors*: such as Non-Governmental Organizations, Faith Based Organizations, community leadership, associations, academic and research institutions, and media.
 - *International organizations*: This includes development partners working or implementing projects in the LREB (e.g., World Bank, IFAD, DANIDA, Netherlands, GEF, etc).
 - *Financial institutions*: This includes community based credit and savings cooperatives, savings groups, as well as formal lending and grant-making institutions (e.g., Banks, micro-finance institutions).
309. A participatory process for the design of this project was conducted starting in late 2021, culminating in April 2023. This process involved gathering ideas, suggestions, inputs, and contributions from stakeholders identified above, including farmers, county governments, development partners and others. Details of meetings, consultations and feedback received is included in Annex 7.
310. A series of analytical consultations, designed to assess needs and gaps, understand barriers, and refine the technical components of the project were also deployed. These included in-person and virtual surveys, interviews and workshops with all categories of stakeholders.
311. Throughout implementation, the project will make every effort to provide relevant information to stakeholders in appropriate and timely manner.
312. Dissemination of information will be done using a variety of media such as print (newspapers, posters), radios, public meetings and workshops to ensure maximum reach to stakeholders. Particular attention will be paid to the adequate communication of project criteria, targeting approaches, activities and benefits to vulnerable and traditionally excluded groups, including women, elderly people, indigenous/ethnic groups and persons living with disabilities. Provisions will be made for the translation in local language of key project documentary outputs, the Grievance Redress Mechanism and FPIC documents, and for the use of pictorial or image-based communication in order to ensure communication about the project is not limited by literacy rates.
313. Details of how each stakeholder and stakeholder group is to be engaged are included in Annex 7.

Grievance Redress Mechanism

314. The project will establish one grievance mechanisms at the field level to file complaints, sensitive to the location wherein the project is being implemented. Both (i) contact information and (ii) information on the process one must follow to file a complaint will be disclosed in all meetings, workshops and other related events throughout the life of the project. It is also expected that all awareness raising material to be distributed will include the necessary information regarding the process for filing grievances and key contacts. The project will be responsible for documenting and reporting, as part of the safeguard's performance monitoring, on any grievances received and how they were addressed.
315. The Project Management Unit (PMU) will be responsible for addressing incoming grievances regarding environmental and social standards; as part of the safeguard's performance monitoring, the Project Coordinator of the PMU will be responsible for documenting and reporting on any grievances received and how they were addressed. FAO as well as other executing entities (EEs) will inform communities about the GRM through culturally appropriate mechanisms, ensuring information on the mechanisms at different levels through preferred communication channels, which are to be agreed and discussed as part of the project FPIC.
316. The GRM will include methods/process to (i) receive and register external communications from the public; (ii) screen and assess the issues raised and determine how to address them; (iii) provide, track, and document responses, if any; and (iv) adjust the management program, as appropriate. Please refer to Annex 7 for further detail.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

317. **The Gender Assessment** documented the current state of women in Kenya compared to men, as well as how gender intersects with age, household headship status, and disability status to influence participation and benefit from engaging in agricultural value chains in the context of a changing climate. The assessment revealed that female youth, female-headed households (including widows), and females living with disabilities, are the most under-represented and vulnerable groups in CSA value chains, based on how their gender interacts with other axes of social differentiation and inequality. Thus, the project's gender mainstreaming strategy makes a concerted effort to build in intersectional equity principles into its targeting metrics, to ensure that female youth, female-headed households, and females living with disabilities are equitably represented in all project activities. Given this intersectional gender approach, gender data is expected to be collected beyond sex-disaggregation (i.e., number and % of female farmers disaggregated by youth status, household headship status, and disability status vs. number and % of male farmers disaggregated by youth status, household headship status, and disability status).
318. **The Gender Action Plan (GAP)** tackles gender inequalities across several priority areas in climate resilient and low-carbon sustainable agriculture across the 6 priority value chains. These include closing gender gaps and existing inequalities in participation (at intra-household level and within producer organizations/cooperatives), workloads (prioritizing agricultural technologies and practices for adaptation and mitigation that reduce workloads and negative impacts on women), access and use of productive resources (such as agro-climate information, technologies, livelihood incomes, credit), and collective action (working with women's groups as platforms for enhancing access, agency, women's leadership, and voice in climate-smart agriculture). In closing these gender gaps, the project will address the underlying socio-cultural norms that perpetuate gender inequality and constrain women's capabilities. This requires a culturally sensitive, multi-level approach that includes women and men in all their diversity.
319. The Gender Action Plan details how the goal of intersectional equity will be mainstreamed in two ways: participation in activities and the content of activities. Women, female youth, female-headed households, and females living with disabilities all have specific quotas that are required for each activity. Mainstreaming in terms of participation will be ensured by setting the target proportion of participants from each of these intersections to be roughly equal to the relative population sizes at the national level: 50% women, 16.5% FHH, 1% for women LWD, and 25% (12.5% of total project population) are female youth (aged 18-35 years). The male youth participation quota is 25% (25% of total project population) and 1% for men LWD (0.5% of total project population). Mainstreaming in terms of content will be ensured by integrating Gender Action Learning Systems (GALS) and gender transformative concepts into 100% of training materials. These training materials include the "gender- and youth- specific" trainings, such as the Specialized Agriterra Training Programmes in Women's Leadership and Youth Leadership, as well as more "general" training materials on agricultural production, markets, finance, etc. Gender Action Plan claims half of the total project budget. The relationships among the project objectives, actions, indicators, targets, and baselines are as shown in Annex 8.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

320. As Accredited Entity for this project, FAO will ensure that financial management and procurement of goods and services using GCF resources adheres to relevant FAO rules and regulations, as well as relevant provisions in the Accreditation Master Agreement (AMA) signed between FAO and GCF. These rules and regulations were reviewed and deemed satisfactory by the GCF Secretariat and Accreditation Panel as part of FAO's accreditation process.
321. FAO has deployed an Oracle-based Enterprise Resource Planning (ERP) system, the 'Global Resources Management System' (GRMS), which provides all FAO employees in all locations globally with travel, human resources, procurement, and finance functionalities. Using GRMS improves the flow of financial information, supports financial monitoring and reporting, increases transparency and visibility, and strengthens internal control. FAO maintains a Chart of Accounts which is used by the whole Organization and that allows for a separation of income and expenditure by donor and project, and it provides a standardized coding structure that enables data to be recorded, classified, and summarized to facilitate internal management and external reporting requirements.
322. Direct procurement by FAO is done in accordance with its Manual Section 502 on "Procurement of Goods, Works, and Services". To sub-contract the delivery of specific activities using Letters of Agreement, FAO operates in accordance with its Manual Section 507 on "Letters of Agreement". Such services are managed by the FAO Procurement Service, which provides policy and operational support to ensure that the Organization procures goods, works, and services based on "Best Value for Money" principles.
323. Financial management and procurement executed by Agriterra as Executing Entity of GCF proceeds will be overseen and supervised by the FAO-GCF project supervision team. The FAO-GCF project supervision team will undertake regular supervision missions and recruit a qualified, internationally recognized auditing firm to perform frequent spot checks and audits to ensure financial management and procurement by the PMU and executing entities are conducted in line with agreed standards and practices. This will be governed by the agreements to be signed between FAO and Agriterra before the project becomes operational. Audits will be performed by a qualified, internationally recognized auditing firm, competitively selected by FAO.
324. Controls against money laundering, terrorist financing, and prohibited practices will be duly exercised by FAO according to its Policy Against Fraud and other Corrupt Practices. The FAO office of the inspector General handles complaint related to:
- Unlawful acts related to FAO activities (for example theft, fraud, embezzlement, solicitation/acceptance of bribes or kickbacks, extortion);
 - Misrepresentation, forgery, or false certification in connection with any official business;
 - Fraud committed to obtain undue financial benefits or entitlements (e.g., fraudulent claims for rental subsidy, education grant, travel, medical insurance);
 - Fraud, favoritism, disclosure of confidential bidding information, or misconduct related to contract bids, performance of contract obligations or evaluation;
 - Retaliation as defined under the Whistleblower Protection Policy;
 - Staff violations of the Standards of Conduct for International Civil Servants
 - Complaints of workplace harassment pursuant to the Policy on the Prevention of Harassment, Sexual Harassment and Abuse of Authority.
325. The project will transfer the ownership of the procured item(s) to the beneficiary group immediately upon the attainment of technical installation clearance, the project will withhold ownership documents and prepare a full handover schedule to prevent any fraudulent activities or illegal disposition of the asset(s) not according to the business plan. The beneficiary group will assume the full duty of care upon completion of ownership transfer. All steps of the asset management will be guided by FAO relevant rules.

G.4. Disclosure of funding proposal

- ☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
- ☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:
- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
 - redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☒ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☒ Others: Indigenous Peoples Planning Framework
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☒ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☒ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)
- ☒ Annex 23 Market Study and Value Chain Analyses



REPUBLIC OF KENYA
THE NATIONAL TREASURY AND ECONOMIC PLANNING

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THE NATIONAL TREASURY
P O BOX 30007 – 00100
NAIROBI

When Replying Please Quote

Ref: TNT/CONF/36/021/ C (41)

Date: July 7, 2023

Ms. Mafalda Duarte
Executive Director
Green Climate Fund
Songdo Business District
175 Art center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Dear

Mafalda

Re: Funding proposal for the GCF by Food and Agriculture Organization regarding 'Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya'

We refer to the project titled *'Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc'* in Kenya as included in the funding proposal submitted by the Food and Agriculture Organization (FAO) to us on 26th June 2023.

The undersigned is the duly authorized representative of The National Treasury, the National Designated Authority of Kenya.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The Government of Kenya has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Kenya;

(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours

Sincerely


DR. CHRIS K. KIPTOO, CBS
PRINCIPAL SECRETARY/NATIONAL TREASURY

Copy to: **Njuguna Ndung'u, CBS**
Cabinet Secretary
The National Treasury and Economic Planning
NAIROBI

Ms. Carla Mucavi
FAO Representative in Kenya
NAIROBI

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	Food and Agriculture Organization of the United Nations (FAO)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Bomet, Bungoma, Busia, Home Bay, Kakamega, Kericho, Kisii, Kisumu, Migori, Nandi, Nyamira, Siaya, Trans Nzoia, and Vihiga counties of the Republic of Kenya
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, January 15, 2025
Language(s) of disclosure	English and Swahili
Explanation on language	English and Swahili are the official languages of Kenya, which are language understandable by affected peoples/stakeholders.
Link to disclosure	English: https://openknowledge.fao.org/items/c66bd532-fa2e-4a08-8fde-3d54789c04e6 Swahili: https://openknowledge.fao.org/items/587da971-a5c8-4cf5-9af1-bfe70ebfb712
Other link(s)	FAO disclosure portal (English and Swahili): https://www.fao.org/environmental-social-safeguards/disclosure/en FAO Representation in Kenya website (English and Swahili): https://www.fao.org/kenya/resources/en/
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the “Annex 6: Environmental and Social Management Framework (ESMF)”.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, January 15, 2025
Language(s) of disclosure	English and Swahili
Explanation on language	English and Swahili are the official languages of Kenya, which are language understandable by affected peoples/stakeholders.

Link to disclosure	English: https://openknowledge.fao.org/items/c66bd532-fa2e-4a08-8fde-3d54789c04e6 Swahili: https://openknowledge.fao.org/items/587da971-a5c8-4cf5-9af1-bfe70ebfb712
Other link(s)	FAO disclosure portal (English and Swahili): https://www.fao.org/environmental-social-safeguards/disclosure/en FAO Representation in Kenya website (English and Swahili): https://www.fao.org/kenya/resources/en/
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the “Annex 6: Environmental and Social Management Framework (ESMF)”.
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report	Indigenous Peoples Plan Framework (IPPF)
Date of disclosure on accredited entity’s website	Wednesday, January 15, 2025
Language(s) of disclosure	English and Swahili
Explanation on language	English: https://openknowledge.fao.org/items/c66bd532-fa2e-4a08-8fde-3d54789c04e6 Swahili: https://openknowledge.fao.org/items/587da971-a5c8-4cf5-9af1-bfe70ebfb712
Link to disclosure	FAO disclosure portal (English and Swahili): https://www.fao.org/environmental-social-safeguards/disclosure/en FAO Representation in Kenya website (English and Swahili): https://www.fao.org/kenya/resources/en/
Other link(s)	N/A
Remarks	An IPP Framework consistent with the requirements for a Category B project is contained in the “Annex 6:

	Environmental and Social Management Framework (ESMF)".
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, January 17, 2025
Place	The hard copies will be made available in English and Swahili in the following places; FAO Representation in Kenya: UN Complex, Gigiri, Nairobi, Kenya National Treasury: Old Treasury Building, Harambee Ave, Nairobi, Kenya Ministry of Agriculture and Livestock Development: Kilimo House, Nairobi, Kenya Ministry of Agriculture of each county office, total 14 counties Lake Region Economic Bloc office: Prosperity House
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, February 17, 2025

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP255

Proposal name:	Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya
Accredited entity:	Food and Agriculture Organization of the United Nations
Country:	Kenya
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
This cross-cutting project is building on demonstrated approaches of working with cooperatives as key agents of change in the Kenyan agriculture sector, while also supporting individual smallholder farmers to build resilience and organize better.	
The project proposes a value-chain approach for multiple commodities – both staple crops and cash crops – providing end-to-end support to climate-affected producers and value-chain actors.	

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity (AE) and, if considered appropriate, subject to the conditions set out in annex II to document GCF/B.41/02.

II. Summary of the Secretariat's assessment

2.1 Project background

3. This project, "Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya (CRLCSA)", aims to transform agricultural value chains towards low-carbon, climate-resilient pathways using a locally-led approach. It leverages the private sector, smallholder farming networks, and the cooperative movement's strengths in Kenya to establish a framework for technology transfer and scaling up, drawing on the expertise and knowledge within national and international farming networks. With a focus on six priority value chains (dairy, poultry, coffee, tea, fruit trees, and African leafy vegetables) the project seeks to introduce and expand both adaptation and mitigation technologies at every stage, from production and harvesting to processing and marketing.

4. The project is organized around four main outcomes/components. Outcome 1 focuses on strengthening county-level government services to support the expansion of climate-resilient, low-carbon and sustainable value chains, emphasizing how governments can assist cooperatives and vulnerable farmers. Outcome 2 collaborates with county administrations to invest in community-led, sustainable agricultural landscape management and restoration, laying a foundation for climate resilience within the agricultural sector. Outcome 3 focuses on transferring production, harvest and processing technologies directly to farmers and their organizations to reduce emissions, enhance resilience and optimize natural resource management in response to climate variability and impacts.

5. Under Outcome 4, the project aims to improve farmers' market access by partnering with value-chain actors, leveraging private sector financing to support scalability, and providing risk mitigation through Kenya Commercial Bank (a direct access entity in Kenya), Co-operative Bank of Kenya, and Equity Bank. Additional grant co-financing is provided by the Government of Denmark, Food and Agriculture Organization of the United Nations (FAO), and the Government of Kenya. The project will be implemented through FAO as the AE, with FAO and Agriterria serving as co-executing entities (EEs), while the Government of Kenya will execute the activities it co-finances. The project aims to directly benefit 572,000 individuals, primarily smallholder farmers, including both men and women, and will also indirectly benefit 2.1 million individuals.

2.2 Component-by-component analysis

Component 1: Enabling public support of adaptation and mitigation (total cost: USD 2.44 million; GCF cost: USD 2.00 million)

6. Component 1 of this project focuses on enhancing climate-resilient and socially inclusive agricultural practices by developing innovative, gender-responsive extension methods to disseminate CRLCSA knowledge (activity 1.1.1). It aims to strengthen climate information services for underserved communities, including women, youth and persons with disabilities, through cooperative networks (activity 1.1.2).

7. Additionally, the project will develop decentralized carbon accounting methodologies, update agricultural databases, and assess climate solutions tailored for the agriculture sector (activities 1.1.3, 1.1.4 and 1.1.5). Knowledge-sharing through existing platforms will also support broader adaptation and learning (activity 1.1.6).

Component 2: Sustainable resilient agricultural landscapes (total cost: USD 13.61 million; GCF cost: 0.32 million)

8. Under Component 2, the project aims to create county and regional strategies for climate-resilient, low-carbon agricultural landscape management, focusing on enhanced watershed management, sustainable land use, reforestation and natural regeneration (activity 2.1.1). These strategies will be implemented and monitored to ensure effective climate adaptation and landscape sustainability (activity 2.1.1).

Component 3: Resilient livelihoods (total cost: USD 25.64 million; GCF cost: USD 22.04 million)

9. Component 3 will provide CRLCSA production and processing assets, along with training, to smallholder farmers and farmer organizations, enhancing their climate resilience (activity 3.1.1). It aims to spread CRLCSA technology and knowledge through cooperative networks and peer exchanges, encouraging smallholder farmers to form cooperatives as a means to mitigate and share climate risks (activities 3.1.2 and 3.1.3). Additionally, the project emphasizes improving social inclusion and fostering women's active participation within climate-resilient value chains (activity 3.1.4).

Component 4: Scaling through CRLCSA market and finance (total cost: USD 4.16 million; GCF cost: USD 1.97 million)

10. The project aims to boost demand and market access for CRLCSA commodities by collaborating with buyers and aggregators while expanding access to third-party certification and labelling schemes (activities 4.1.1 and 4.1.2). It will also develop inclusive finance tools and products that promote CRLCSA value chains, emphasizing gender-responsiveness and social inclusion (activities 4.2.1 and 4.2.2). Additionally, support will be provided to smallholders and their business units in creating viable business plans, enhancing access to financial incentives for agroforestry, and fostering equitable opportunities across the sector (activity 4.2.3).

Additional allocation in parallel provided by other commercial banks in the form of loans (total cost: USD 10 million)

11. The project will secure up to USD 10 million in leveraged financing through loans from Kenya Commercial Bank, Equity Bank, and the Cooperative Bank of Kenya, which will be directed to farmers, cooperatives and value-chain actors based on bankable proposals and business plans developed under Component 4.

Additional allocation: Project management (total cost: USD 4.14 million; GCF cost: USD 2.88 million)

12. This project will be managed by the AE, FAO, who is also the main EE of the project. FAO will be responsible for the overall management of the project, including (i) all aspects of project appraisal; (ii) administrative, financial and technical oversight and supervision throughout project implementation; (iii) ensuring funds are effectively managed to deliver results and achieve objectives; (iv) ensuring the quality of project monitoring, as well as the timeliness and quality of reporting to GCF; and (v) project closure and evaluation.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: N/A

13. The project will support 572,000 vulnerable smallholder farmers whose livelihoods rely on one of the six targeted value chains in the region. This includes an expected reach of at least 80,000 cooperative members (project participants), benefiting approximately 320,000 individuals based on a conservative estimate of 4 people per household. Additional beneficiaries include 63,000 smallholder farmers outside the cooperatives, reaching an estimated 252,000 individuals within their households.

14. The indirect beneficiaries include the non-participating cooperatives within the six value chains (1,342,140 individuals), with additional benefits reaching extended family members and households in nearby villages through participants in the farmer field school programme (756,000 people). This brings the total immediate indirect beneficiaries to 2,098,140, with women comprising 50 per cent of this group – nearly 15 per cent of the Lake Region Economic Block (LREB) population. In the long term, even broader indirect benefits across the Lake region are anticipated, supported by an upscaling strategy and the willingness of Kenyan banks to provide loans to eligible cooperatives.

15. The project also aims to enhance health, well-being, and food and water security for the most vulnerable communities by improving adaptive capacities through climate-resilient and low-carbon technologies, access to markets, finance, social safety nets and risk-sharing mechanisms within organized farmer groups and cooperatives. It will boost food and nutrition security by increasing the production of essential crops and commodities for both consumption and sale, thereby improving incomes through enhanced productivity, value addition, and market access.

16. The project also seeks to reduce exposure to climate risks like droughts and floods through watershed management, soil conservation and increased access to climate information

services, including early warnings. Improved water access will be achieved by promoting water conservation, harvesting and erosion control measures. Gender equality and social inclusion are central to the project, with an emphasis on equitable access to agricultural inputs, finance, agroclimate information, and extension services, as well as increased representation of women, youth and persons living with disabilities in decision-making roles within cooperative societies and value chains. Additionally, access to gender-responsive and socially inclusive adaptation and mitigation technologies, markets and labour practices will be expanded.

17. The project is estimated to achieve a net emissions reduction of 4,268,492.04 tonnes of carbon dioxide equivalent (t CO₂ eq) by 2044. The mitigation benefits for this project have been calculated using two distinct methodologies: the FAO Ex-ACT model assessed emissions from agriculture, forestry and other land use activities across all value chains except dairy and poultry, while the Global Livestock Environmental Assessment Model estimated emissions specifically for these livestock value chains.

3.2 Paradigm shift potential

Scale: N/A

18. The project promotes a fundamental paradigm shift by transforming value chains from low productivity and resilience to high-performing, financially profitable and climate-resilient low-carbon pathways through a locally-led approach. It utilizes cooperatives as a mechanism for reducing climate risk, functioning not only as a business strategy that allows farmers to pool resources but also to enhance their negotiating power in the market. This is achieved by empowering farmer organizations and cooperatives to identify climate challenges and solutions while facilitating access to relevant adaptation and mitigation technologies, thereby proactively enhancing climate resilience in their production and business practices. By ensuring that the promoted technologies and practices align with the business plans and models of cooperatives, farmers can adopt climate solutions that are both environmentally beneficial and financially viable.

19. Overall, the project has significant potential for scaling up and replication, optimizing knowledge-sharing and learning through training sessions, demonstrations, community meetings and other methods. It also enhances the enabling environment for farmers and cooperatives by strengthening the capacity of county administrations to deliver climate-related and market access services. This support will empower farmers to produce more and higher-quality products that can reach the market more efficiently. Key services that contribute to this enabling environment include improved climate information services and warnings, agrometeorological advisories, council-approved lists of climate-resilient, low-carbon technologies, enhanced extension services, as well as market studies, information and sales data.

3.3 Sustainable development potential

Scale: N/A

20. The project activities will have direct linkages to the following Sustainable Development Goals (SDGs): SDG 1: no poverty; SDG 2: zero hunger; SDG 3: good health and well-being; SDG 5: gender equality; SDG 8: decent work and economic growth; SDG 13: climate action; and SDG 15: life on land. These goals will guide the project's implementation and ensure that it contributes effectively to addressing critical global challenges related to poverty reduction, food security, health, gender equity, sustainable economic growth, climate change and biodiversity conservation.

21. The project has a measurable co-benefit in job creation across the six value chains, while indirect benefits include increased economic activity in the supply chains for agricultural inputs and feed, as well as heightened employment in the transportation and bulking segments of the value chain. Notably, the project aims to provide meaningful employment opportunities for

women and youth within cooperatives, farming groups and processing industries. Furthermore, it is anticipated to benefit the broader population of the LREB through the implementation of climate-sensitive landscape management strategies and overall enhancements in the region's economic outlook.

22. Additional indirect socioeconomic benefits encompass improved access to education for children, particularly for girls, whose workloads may be reduced or whose families may have greater ability to pay school fees. Health improvements are expected due to increased access to a more diverse and higher quality food supply, whether produced or purchased, along with better access to health services facilitated by greater financial capacity. The project also aims to promote greater gender equality, decrease rates of gender-based violence and child labour, and enhance access to meaningful employment for youth, women and persons with disabilities.

23. At the farm level, environmental benefits arise from implementing sustainable land management practices, anti-erosion initiatives, increased soil cover, crop diversification or rotation, and the careful application of fertilizers, alongside reductions in harmful agro-chemicals and herbicides. These efforts lead to improved land productivity, increased biomass production, enhanced soil fertility, better soil moisture retention and microorganism content, and increased agrobiodiversity at both farm and landscape levels.

3.4 Needs of the recipient

Scale: N/A

24. Smallholder farmers, in particular in the LREB faces significant challenges due to inter-annual rainfall variability and fluctuating trends in crop yields and livestock production, leading to unstable and unreliable incomes. The increasing variability in rainfall has exacerbated these issues, resulting in chronic food insecurity for 10 per cent of the population in the LREB. This situation necessitates both short-term emergency food relief efforts and comprehensive long-term development programmes to address the underlying vulnerabilities in the agricultural sector and improve resilience among affected communities.

25. Furthermore, in the LREB, agricultural yields reflect low productivity due to climate variability and substantial post-harvest and processing losses. Many farmers operate with minimal technical support, low mechanization levels, and limited access to necessary inputs. Current practices largely consist of unsustainable monocropping with significant land and water use impacts.

26. On the other hand, smallholder farmers have very limited financial capacity to invest in their crop production. Cooperatives and farmers' organizations typically face limited access to financing due to high eligibility requirements, elevated lending costs and challenges in presenting viable business proposals to financial institutions.

27. In order to overcome the challenges of climate vulnerability and limited financial availability, this project establishes a system where governments and the private sector work together to remove barriers to self-financing value chains through cooperatives. On one side, vulnerable smallholders receive training and resources for adopting the best climate-resilient, low-carbon practices tailored to their value chain. This approach provides both immediate and long-term benefits for adapting to climate challenges, while also allowing smallholders to further reduce costs and risks by joining cooperatives. On the other side, established cooperatives exit the project with members who all implement climate-resilient, low-carbon agriculture, have access to better assets, increase their market presence, and can integrate climate service provision into their regular business expenses. This enables them to remain competitive and gain additional capacity to access private financing, supporting the model's long-term sustainability.

28. Hence, the CRLCSA project is needed to reduce climate action costs and to improve efficiency and coordination among governments and private actors in delivering climate services and facilitating market access.

3.5 Country ownership

Scale: N/A

29. The country ownership of this project is very high for several reasons: (i) it strongly supports the implementation of national climate change policies and strategies such as the National Climate Change Act of 2016, updated nationally determined contribution and the National Adaptation Plan (2015–2030) of Kenya; (ii) the national designated authority (NDA) is involved in the project from the inception and project design stage. The NDA actively participated in all major meetings and consultations, providing regular input to align the project proposal with government priorities. Once the project is approved, the NDA will play a crucial role in coordinating with county governments and liaising with the Council of Governors; (iii) the Government of Kenya is also one of the EEs and through the National Treasury of Kenya has co-financed USD 14 million in the project and will execute activities that are co-financed by the Treasury; (iv) involvement of several partners at the regional and national level throughout the project design; several discussions were held at the county level where the consultations took place on multiple occasions, starting with the initial consultative meeting which brought together over 90 participants (men and women) including cooperatives, environment, agriculture and climate departments from all 14 counties; and (v) numerous consultations were held at the local level with smallholder farmers, including climate change surveys, individual interviews with regard to applications for the climate resilient and low-carbon technologies.

3.6 Efficiency and effectiveness

Scale: N/A

30. The proposal effectively demonstrates the project's efficiency and effectiveness through a comprehensive economic and financial analysis. Both analyses have been conducted for six priority value chains – coffee, dairy, fruit trees, vegetables, poultry, and tea – under two climate scenarios, Representative Concentration Pathway (RCP) 2.6 and RCP 8.5.

31. In the context of the macroeconomic environment in Kenya, inflation has risen steadily, fluctuating moderately between 5 and 7 per cent over the past five years—a relatively stable trend compared to broader African trends. However, food prices remain pressured by ongoing droughts impacting agricultural productivity. The AE identifies access to finance as a key barrier preventing local farmers from further enhancing the agricultural value chains.

32. Economic viability in the long run is substantiated by the positive results of economic indicators across the six value chains. Economic benefits were valued based on gross revenue for each value chain, adjusted to economic prices using calculated conversion factors, along with reduced and sequestered greenhouse gas (GHG) emissions. The model projects a positive economic net present value (NPV) of USD 120 million and an internal rate of return (IRR) of 40 per cent under RCP 8.5, and an NPV of USD 123 million with an IRR of 43 per cent under RCP 2.6, applying a social discount rate of 10 per cent over a 20-year period. Additional economic benefits, such as nutritional and health improvements and enhanced natural resources (e.g. soil quality), were not included in the model due to data limitations.

33. Financial viability over the long term is supported by the financial indicators across the same value chains. At the project level, incremental financial NPV is USD 10.7 million with a financial IRR of 23.4 per cent and a 12-year payback period under RCP 8.5. Under RCP 2.6, the financial NPV is USD 13 million with a financial IRR of 24.9 per cent and an 11-year payback period, indicating the project's financial feasibility using an 18 per cent financial discount rate over 20 years.

34. The financial model projects an average income increase of 20 per cent for farmers, with a conservative target of 15 per cent incorporated into the logical framework under Outcome 4. The model also indicates the financial viability of leveraged financing from Equity Bank and Cooperative Bank, amounting to USD 10 million. Even with an 18 per cent interest rate, a 3-year tenor, and no grace period for these agricultural loans, the investments are shown to be marginally viable, with a financial IRR slightly above the financial discount rate and a debt service coverage ratio exceeding 3, ensuring sufficient cash flows for debt servicing post-project completion.

35. The adaptation cost per direct beneficiary is USD 50.6, and USD 13.8 per total beneficiary, which are moderate within the GCF agricultural and adaptation portfolio. The mitigation cost is USD 0.07 per t CO₂ eq, positioning it at the very lower end of the GCF portfolio. The co-financing ratio stands at 0.71. The proposal addresses the limited dissemination of best practices within the LREB and underscores the project's potential to expand and reinforce best practices in low-emission, climate-resilient agriculture among farmers in the region.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

36. **Environmental and social risk category.** The AE has assigned the project an environmental and social (E&S) risk category of Category B based on its screening of the proposed project activities. The Secretariat agrees with this categorization and confirms that the risk category is within the AE's E&S risk accreditation level.

37. The project is expected to predominantly generate substantial positive E&S benefits through the transformation of key agricultural value chains by the adoption of climate resilient, low-carbon sustainable agricultural practices and technologies. Some of main benefits include strengthening the resilience of farmers (reducing their sensitivity to current and projected climate risks and simultaneously improving their adaptive and coping capacity); and reducing GHG emissions (promotion of renewable energy, improvements in land and water management, for example, agroforestry, reforestation, watershed management). Besides all positive impacts, the project also has the potential to generate adverse social and environmental impacts that need to be carefully managed and monitored.

38. **Performance Standard (PS) 1: Assessment and management of environmental and social risks and impacts.** The potential impacts from project activities are envisaged not to be unprecedented in the project area but limited to the project footprint and are neither irreversible nor cumulative. The management and mitigation of identified environmental and social risks/impacts will require the development of Environmental and Social Management Plans (ESMPs). The AE has developed an environmental and social management framework (ESMF), the umbrella instrument that will guide in the preparation of site-specific ESMPs. Final consultations with regard to the validation of the ESMF are still pending.

39. **PS2: Labour and working conditions.** The AE has identified risks related to labour and working conditions. Such risks are likely to include the risk of employing children in project activities, discrimination in hiring and selection processes, abusive working conditions, and a high degree of informality and vulnerability (e.g. lack of social protection). The project will work mostly with smallholders and family farmers characterized by family-focused motives such as favouring the stability of the farm household system, using mainly family labour for production and using part of the produce for family consumption. Additionally, due to the country context, youth and children often assist with the farming work of their respective families, and there is a risk that the work they undertake might be beyond what is age-appropriate (unless closely

monitored). Additionally, decent work standards and occupational, health and safety guidelines will be promoted.

40. **PS3: Resource efficiency and pollution prevention.** The project will promote practices and technologies that are aimed at improving efficiency in the use of natural resources (water, land, soil, energy) through an inclusive and participatory approach that includes women, men, youth, elders, persons with disabilities and Indigenous Peoples. The project will incorporate an integrated water management approach, including water conservation measures, restoration of water catchment areas and watershed management; conservation and restoration of riparian zones; development of several small-scale irrigation schemes rather than one large system; promoting water-saving irrigation technologies (drip irrigation, water-saving equipment, etc.); rehabilitation of existing irrigation schemes; promoting the rational use of industrial water (reuse, cascade use, water-saving processes and equipment); promoting water and rainwater harvesting technologies/structure; and raising public awareness on efficient water use. In terms of energy, the project will promote the use of solar energy and energy-efficient machinery and equipment (e.g. tractors, ventilation systems, drying and storage systems, cooling devices).

41. **PS4: Community health, safety and security.** Community health and safety risks are likely to arise from exposure to agro-chemicals, water and vector-borne diseases, zoonotic diseases, and pollution resulting from some project interventions. Risks are likely to include the marginalization of women, youth and persons with disabilities. Proposed mitigation measures include: integrated pesticides management plans, awareness creation and sensitization on social distancing for potential diseases, adoption of good industry practices and waste management plans. The project proposes to establish a strong quota and target with a dedicated budget line for women, youth and vulnerable groups in selected value chains for each training, grant and investment opportunity; additionally, the project will adopt a participatory and demand-driven approach to assess community needs and develop support packages tailored to attend their needs, challenges and priorities. Education and awareness-raising with the local end users regarding sanitation and health issues related to water-borne and/or water-related diseases, as well as intermittent water quality monitoring of the main water reservoirs, will also be embraced and adopted.

42. **PS5: Land acquisition and involuntary resettlement.** Based on the Secretariat assessment of the AE's submission, project activities are unlikely to lead to involuntary resettlement or displacement of people or communities; resources from the project will be used for land acquisition and this is further reinforced by an exclusion list. Proposed activities will imply the involvement of communities on a purely voluntary and demand-driven basis. The country reports historic difficulties related to land access and tenure security rights that could compromise the large-scale adoption of CRLCSA practices and increase competition over natural resources among marginalized communities. Existing land uses and land ownership are considered as factors that might influence the adoption/selection of specific production technologies and practices, for example private landowners are more likely to conserve land as they are assured of retaining the long-term gains from investments in conservation. Security of tenure might limit the participation of women, youth and persons with disabilities in the project activities, and can influence socioeconomic activities such as a change in the range of income generating activities, lifestyles, etc.

43. **PS6: Biodiversity conservation and sustainable management of living natural resources.** The activities promoted by the project are expected to generate substantial positive environmental benefits from the scaling up of climate-resilient land use practices, increasing not only the resilience of ecosystems and local communities to climate change, but also reducing land degradation and maintaining/enhancing biodiversity and ecosystem functionality through an ecosystem-based approach. Considering that some agricultural and forestry activities could take place near or in the buffer zones of critical habitat or natural protected areas, there are

restrictions regarding the type of activities that can be financed/promoted by the project to avoid any potential significant adverse impact on biodiversity or habitats (refer to the exclusion list in appendix 1).

44. **PS7: Indigenous Peoples.** Pastoralist Indigenous Peoples are not engaged in the project activities. However, the project will develop and implement the county landscape management strategies which could have potential impact on Indigenous Peoples. The AE has provided an Indigenous Peoples Plan to comply with the Indigenous Peoples Policy, to be applied in case the project confirms the presence of Indigenous Peoples in the project target areas. The AE is recommended to engage with the GCF Indigenous Peoples Advisory Group on relevant Indigenous Peoples' issues that may be encountered.

45. **PS8: Cultural heritage.** The project will preserve and protect the cultural heritage (tangible and intangible), avoiding and reducing the adverse impacts that could potentially affect the cultural heritage. Based on the submission of the AE, the project does not aim to make use of intangible forms of culture such as cultural knowledge, innovations and practices of communities embodying traditional lifestyles for commercial purposes. Adequate consultation regarding activities to be carried out on Indigenous Peoples' territories will include free, prior and informed consent and an exclusion list has been developed. Activities related to planning process (Landscape Management Strategies of the counties) shall map or delimit these areas. Additionally, a Chance Find Procedure has been developed.

46. **Grievance redress mechanism (GRM):** The project will establish one or more grievance mechanisms at the field level to file complaints in the location where the project is being implemented. The GRM will serve as a mechanism for timely resolution of an issue directly linked to project activities and prevent escalation of problems into social conflict; and as an accountability mechanism where people who are directly or indirectly affected by the project can seek remedy. However, the GRM will also serve as a proactive tool to facilitate dialogue with affected communities at all stages of the project cycle, including to determination of information the communities require and the best means for them to receive it.

47. **Sexual exploitation, abuse and harassment (SEAH).** The project will adopt a zero-tolerance approach to all forms of SEAH. Potential incidents may arise from interaction of project staff or partners with community members and lack of awareness about SEAH and how to address it. Specific SEAH risks and their level of significance will be determined during subactivity screening prior to implementation. Proposed mitigation measures to manage risks include regular SEAH sensitization campaigns and training to all project-related personnel, such as those from the EEs, and putting in place a code of conduct for personnel. The farmer field schools curricula activities will be utilized as an opportunity to raise awareness and incorporate good practices to address SEAH in specific value chains. In addition, SEAH provisions, such as training and disciplinary actions in case of incidents, will be included in agreements with EEs. The project-level grievance redress mechanism (GRM) will be accessible for reporting and responding to SEAH complaints. It will be survivor-centred and gender-responsive and will be used for monitoring SEAH incidents. In addition, the GRM will be complemented by referral pathways that may be used for survivors and victims. Project personnel will be trained on the GRM to handle such incidents and ensure safe working conditions. The AE is advised to pay particular attention to tea-related activities that will be supported by the project given the prevalence of sexual exploitation of women in this value chain as indicated in the ESMF.

4.2 Gender policy

48. The AE has provided a gender analysis and action plan and therefore complies with the GCF gender policy. The gender assessment identifies several key challenges faced by women in the LREB of Kenya, in the agricultural sector. Women make up a significant portion of the agricultural labour force but face substantial barriers in accessing land, finance and agricultural

inputs. Given that climate change amplifies existing socioeconomic inequalities, such as control over or access to resources, women smallholder farmers in Kenya lack secure land rights, livestock assets and technologies, constraining them from adapting to and coping with climate shocks. Added vulnerabilities include increased risk for gender-based violence due to conditions such as longer distances to collect water and loss of livelihoods, creating tensions within households. Sociocultural norms limit their participation in decision-making and high-value agricultural value chains, which are often dominated by men. Additionally, women face a disproportionate work burden, restricted access to agricultural extension services, and increased vulnerability to climate change impacts, all of which affect their ability to improve livelihoods and contribute to sustainable agriculture. Women also have limited autonomy in health-related decision-making, further hindering their socioeconomic progress.

49. The Gender Action Plan aims to address these barriers through targeted interventions and defines activities, indicators and targets for women's participation and benefits with corresponding timelines, budget and gender expertise. It addresses gender inequalities across several priority areas in agri-climate adaptation and mitigation. These include closing gender gaps and existing inequalities in participation at the intra-household level and within producer organizations or cooperatives; improving access to and use of productive resources such as climate information, technologies and credit; and collective action by working with women's groups as platforms for enhancing access, agency and voice in climate-smart agriculture. In addition, the project will provide access to extension services for women. The plan also includes measures for increasing access to markets and developing agricultural landscape management strategies in consultation with women during the development stages.

4.3 Risks

4.3.1. Overall project assessment (medium risk)

50. The proposed cross-cutting project will help address climate change impacts in 14 counties in Kenya (the LREB) by investing in climate-resilient low-carbon, sustainable agricultural production and will assist local government in supporting producers and value-chain participants in agriculture. Target GCF result areas are: forestry and land use; most vulnerable people and communities; health and well-being; and food and water security. GCF is requested to provide a grant of USD 29.2 million. Together with grant contributions of USD 14.0 million from the Government of Kenya, USD 5.8 million from the Government of Denmark, and USD 0.98 million from FAO, the total project cost is USD 50 million. The co-financing ratio is 1:0.71 and will account for 42 per cent of total financing.

4.3.2. Accredited entity/executing entity capability to execute the current project (medium risk)

51. FAO and Agritererra will jointly share project implementation responsibility for this funding proposal. In addition to overall project management, FAO will serve as EE and handle the enabling of local government support, agricultural landscape management, social inclusion, and market opportunities and inclusive finance. On the other hand, Agritererra will be responsible for dissemination of climate technology, support to smallholder farmers, business plan development and access to financial incentives for farmers.

52. FAO was accredited by GCF as an AE in 2016. FAO is a specialized agency established to fight hunger and achieve food security with operations in over 130 countries worldwide. Agritererra is a non-profit agricultural agency that works on farmer organization and cooperative development. Agritererra provides training and sharing of experiences to farmers' cooperatives by experts to strengthen governance, leadership, operational management and sustainable

financial services. Since its founding 25 years ago, Agriterra has worked with over 320 farmers' organizations supporting over 1 million farmers across Africa, Asia and South America.

4.3.3. Project-specific execution risks (medium risk)

53. The proposed project aims to reduce climate emissions via conservation of the ecosystem, dissemination of technologies and assets, and reorientation of business practices towards long-term sustainability. Grant funding will also be used by local government units for related climate support activities such as knowledge dissemination and upgrade of agricultural databases. At least 28 per cent of the USD 50 million project funding is expected to be provided by the Government of Kenya. However, the credit rating of the Government of Kenya was recently downgraded to B- (S&P and Fitch) and Caa1 (Moody's) over issues related to wider fiscal deficit, rising interest payments, high debt burden and heightened sociopolitical risk. The government reported larger budget deficit compared to that originally planned due to higher spending and tax revenue shortfall. High external debt level, tight monetary policy, and revenue shortfalls put pressure on the national budget to support publicly financed projects. This concern is partly mitigated by the government's historically high level of commitment to natural resources conservation, good access to concessional external financing, and no immediate external liquidity pressure; 74 per cent of total external debt is on concessional terms with long tenors. In relation to weaker credit standing, the Kenyan Shilling has been depreciating against the United States dollar in the last five years and is projected to further depreciate by 4 per cent per annum in 2025–27 in the S&P base case scenario. During the six-year implementation period, the project will be exposed to foreign currency depreciation and inflation risk which can lead to cost overrun.

4.3.4. Project viability and concessionality

54. The proposed project has outlined numerous activities for a sustainable agricultural landscape and resilient livelihoods. These include activities in communities to strengthen data management, manage community forest and agricultural land, climate resilient crop production and technologies, and strengthening the cooperative movement. Both FAO as AE and EE and Agriterra as co-EE have extensive experience implementing these types of activities and should raise no concerns regarding project viability. The proposed GCF grant will represent 58 per cent of total programme size in a wholly grant-funded project. The proposed concessionality level is justified given the low country rating of Kenya (B- by S&P and Fitch and Caa1 by Moody's) and limited public resources. The project is expected to generate significant mitigation and adaptation benefits for Kenya.

4.3.5. Compliance risk (low risk)

55. The AE and Agriterra, a Netherlands-based cooperative development-focused international non-governmental organization, will be the EEs for the activities funded by GCF proceeds, as well as those co-financed by FAO and the Government of Denmark. The Government of Kenya, through its NDA, the National Treasury, will execute the activities that it co-finances (activity 2.1.2) and will not administer GCF proceeds as an EE. The proposed project activities generally present relatively lower risks of money laundering and terrorist financing (ML/TF).

56. The AE has assessed the ML/TF risk as low due to mitigants that will be put in place such as anti-money-laundering/countering the financing of terrorism (AML/CFT) clauses in project subsidiary agreements, as well as its own AML/CFT controls. The AE also highlights that close monitoring and oversight during project implementation will be made possible through its Eastern Africa sub-regional office in Ethiopia and country representation office in Kenya.

57. Given the nature of the proposed project activities, the AE's established AML/CFT controls, and the additional on-the-ground oversight during implementation, the overall compliance risk is determined to be low.

4.3.6. GCF portfolio concentration risk (low risk)

58. In case of approval, the impact of this proposal on the GCF concentration risk remains within the monitoring thresholds of the Risk Appetite Statement in terms of results areas, single proposal or AE concentration.

4.3.7. Recommended risk rating

59. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall project	Medium
Accredited entity/executing entity capability to implement this project	Medium
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Low

4.4 Fiduciary

60. The Food and Agriculture Organization (FAO) will serve as the accredited entity (AE) for this project, responsible for overall management, including project appraisal, administrative oversight, financial supervision, and technical supervision throughout implementation. FAO will establish a Project Task Force to fulfil its AE functions, maintaining independence from executing entity functions.

61. The project will be co-executed by FAO and Agriterra as executing entities (EEs). FAO-Kenya will establish a Project Execution Team for its EE functions, while Agriterra, a global farmer advisory organization, will leverage its expertise in strengthening cooperatives and farmers' organizations. The Government of Kenya, through the National Treasury, will execute activities co-financed by the government but will not administer GCF proceeds.

62. A Project Management Unit will be established in Nairobi with a regional antenna in Kisumu, led by the assistant FAO Representative and a National Project Coordinator. The Project Management Unit will coordinate with relevant stakeholders and support day-to-day activities.

63. The Project Steering Committee, co-chaired by the Principal Secretary of the Treasury and the Principal Secretary of the Ministry of Agriculture, will provide strategic guidance. The Project Technical Coordinating Committee, chaired by FAO-Kenya, will conduct monitoring, learning, reporting and risk management.

64. Regarding fund flow, GCF proceeds received by FAO as AE will flow to the co-EEs (FAO-Kenya and Agriterra) for project implementation. The arrangement is governed by various agreements, including the project agreement between FAO and the Government of Kenya, execution agreement between FAO and Agriterra, and co-financing agreement between FAO and Ministry of Foreign Affairs of the Government of Denmark (DMFA).

65. Financial management and procurement will follow FAO rules and regulations, utilizing its Global Resources Management System. FAO will oversee Agriterra's financial management and procurement through regular supervision missions. Independent audits will be performed by an internationally recognized auditing firm, selected competitively by FAO, to ensure compliance with agreed standards and practices.

4.5 Results monitoring and reporting

66. The main causal pathway that the funding proposal sets out to achieve as contained in the theory of change is based on the premise that empowering cooperatives, improving access of smallholder farmers and value-chain actors to improved climate information, technologies, markets and finance to support production, processing and marketing of targeted value chains will lead to improved resilience and reduced emissions from the agriculture, forestry and other land use sector. This pathway of change was clearly set out in the theory of change.

67. The project aims to contribute to three GCF results areas: forestry and land use (mitigation), most vulnerable people and communities (adaptation), and health and well-being and food and water security (adaptation). As such, the project will report against GCF Core Indicator 1: GHG emissions reduced, avoided or removed/sequestered; Core Indicator 2: Direct and indirect beneficiaries reached; Core Indicator 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice to capture mitigation and adaptation results respectively; and Supplementary Indicator 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience. Appropriate sources of information and means of verification have been specified in the logical framework and the monitoring and evaluation plan for the tracking and reporting of these results areas.

68. The proposal also allocated around USD 1.5 million for all monitoring, evaluation and learning activities and the Secretariat deems this sufficient to enable compliance with the requirements of the Integrated Results Management Framework and the GCF Evaluation Policy.

4.6 Legal assessment

69. The Accreditation Master Agreement was signed with the accredited entity on 8 June 2018 (the “AMA”), and it became effective on 4 October 2018. The accredited entity’s five-year accreditation term lapsed on 3 October 2023 but was extended by decision B.37/18, paragraph (q) until the earlier of three years from the date of lapse of the five-year accreditation term or the date on which a revised accreditation framework is adopted by the Board. The Board approved the re-accreditation of the accredited entity pursuant to decision B.37/18, and negotiation of the Amended and Restated Accreditation Master Agreement is ongoing.

70. The accredited entity has provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

71. The proposed project will be implemented in the Republic of Kenya in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in such countries may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

72. Therefore, it is recommended that the Board considers whether disbursements of GCF proceeds should only be made after GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities for GCF and its personnel.

73. To address the matters raised in this section it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP255

Proposal name:	Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya
Accredited entity:	Food and Agriculture Organization of the United Nations
Country:	Kenya
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1. This is the first submission of the funding proposal "Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya" submitted to the independent Technical Advisory Panel (iTAP) under the standard proposal approval process. The accredited entity (AE) is the Food and Agriculture Organization of the United Nations (FAO). FAO and Agriterro will be executing entities (EE) for GCF proceeds and for the co-financing from FAO and the Government of Denmark. The host country, the Government of Kenya, acting through the National Treasury, will be the EE for the sub-activities co-funded by the Government of Kenya.
2. This cross-cutting funding proposal is aimed at fostering the emergence of climate-resilient, low-carbon, environmentally sustainable, and financially viable agricultural value chains by accelerating the transfer of technology, knowledge, assets and services with a focus on agrifood cooperatives as key agents to leverage rural change.
3. The expected mitigation outcome is 4,268,492 tonnes of carbon dioxide equivalent (t CO₂ eq) over the 20-year lifespan of the project. The proposal focuses on the mitigation area of forestry and land use to achieve this outcome.
4. The expected adaptation outcome includes 572,000 direct and 2,098,140 indirect beneficiaries, for which the project focuses on three adaptation result areas (ARAs): most vulnerable people and communities (ARA1); health and well-being, and food and water security (ARA2); and ecosystems and ecosystem services (ARA4).
5. The project will achieve its mitigation and adaptation outcomes through activities organized in four components:
 - (a) Enabling support for adaptation and mitigation;
 - (b) Sustainable resilient agricultural landscapes;
 - (c) Resilient livelihoods; and
 - (d) Scaling through climate-resilient, low-carbon, sustainable agriculture (CRLCSA) market and finance.
6. The overall budget for the six years implementation period is USD 49,992,238, with a funding request to GCF of USD 29,215,309 as a grant. The Government of Kenya will contribute USD 14 million, the Government of Denmark will provide USD 5,791,910 and FAO will provide USD 985,910 as co-financing to this funding proposal, all in the form of grants.

7. In its assessment, the iTAP considered all documents as submitted on 4 November 2024, the answers to the iTAP as received on 25 November 2024 and the additional information provided during the call between the iTAP, the AE and the GCF Secretariat on 27 November 2024.

1.1 Impact potential

Scale: Medium to high

8. The funding proposal addresses climate change mitigation and adaptation in 14 counties of the Lake Region Economic Bloc in Kenya (LREB) from different perspectives.

9. Component 1 focuses on improving the enabling conditions for CRLCSA across all 14 counties and enhancing the public agro-climate services support services. The project will develop innovative and efficient extension methods for demonstrating and disseminating CRLCSA practices, secure the access of tailored climate information to last-mile users, develop and test methodologies for carbon accounting at a decentralized level, improve databases related to agricultural production and the cooperative system, prepare Climate Technology Green Lists (Green Lists) that include technologies approaches or practices that can face climate risks and reduce greenhouse gas (GHG) emissions, and facilitate knowledge and experience exchange.

10. Component 2 aims at reducing emissions from and increasing adaptive capacity of the sector on agriculture, forestry and other land uses (AFOLU), as well as at increasing ecosystem resilience to climate change. To achieve these outcomes the project will develop, implement and monitor country and regional landscape management strategies (LMS) using the Green Lists developed in Component 1.

11. The third component focuses on increasing climate resilience of the smallholders' livelihoods. The project will facilitate the adoption of gender-responsive and socially inclusive CRLCSA by providing training and assets to farmers and farmers' organizations, disseminating knowledge to cooperatives and through peer-to-peer exchanges, supporting smallholder farmers' aggregation into cooperatives and supporting women's participation.

12. The fourth component prepares and tests upscaling mechanisms through market development and the development of gender-responsive and socially inclusive private finance tools. Activities under this component focus on identifying mechanisms to enhance market access for CRLCSA commodities, such as certification schemes. Furthermore, the project aims to support smallholders and cooperatives in creating bankable business models. The project will, jointly with the Kenya Commercial Bank (KCB), the Cooperative Bank and Equity Bank, support the capacity development within financial institutions to support CRLCSA activities and will facilitate access to carbon markets, payments for ecosystem services, biodiversity offsets, or to conservation finance.

1.1.1. Adaptation potential

13. The funding proposal includes detailed and robust information about the three key factors determining climate risks for the agricultural sector in LREB: climate hazards, exposure and adaptive capacity (annex 2, part A).

14. The funding proposal documents historical trends in temperature, precipitation and extreme weather events from 1981 to 2010, along with their observed impacts on the agricultural sector based on the level of exposure and the vulnerability of crops, livestock and associated livelihoods. It also includes an analysis of projected climate impacts on agricultural systems under Representative Concentration Pathway (RCP) 2.6 and RCP 8.5 by differentiating impacts on the six value chains selected for this project: tea, coffee, fruit trees (avocado and banana), African leafy vegetables, dairy and poultry, as well as an analysis of the climate impact

on water resources. The information explains how despite previous interventions, the local adaptive capacity remains low due to prevalent, unsustainable management practices and lack of mechanisms that facilitate the transition to climate-responsive production.

15. As complementary information, the funding proposal includes a diagnostic of the national framework on climate services in agriculture, explaining the challenges, barriers and opportunities for securing last-mile climate information in the 14 counties, and a list of recommended climate resilient practices for each value chain.

16. The funding proposal outlines a coherent set of activities that respond to the climate risks from different angles. The project aims to mainstream technologies, approaches and management practices that respond to the identified climate risks, referred to as CRLCSA. It will support public actors on including CRLCSA into county-level landscape planning, on providing climate information and the extension services necessary to promote the implementation of CRLCSA practices, and on monitoring progress on climate change adaptation. Further, the proposal will start the implementation of CRLCSA activities, promote peer-to-peer learning and exchange for smallholder farmers, strengthen farmers' cooperatives and improve market access and conditions for climate resilient commodities.

17. The funding proposal includes a monitoring, evaluation and learning plan (annex 11) that clarifies the methods, collection tools, frequency and indicators for assessing, quantifying, documenting and reporting progress on achieving the adaptation outcomes. In addition to measuring the GCF indicators for adaptation, the project includes in its monitoring the measurement of changes in adaptive capacity through a resilience index developed by FAO.¹ The iTAP considers this as a good practice that can contribute to a better understanding of the adaptation impact over time.

18. Annex 11 also clarifies the roles and responsibilities of the different actors involved in monitoring, evaluation and reporting activities and includes a learning and knowledge plan as well. The project's budget (annex 4) includes information about the yearly costs for monitoring, evaluation and learning, with a total of USD 1,635,646 to be jointly funded by GCF (95 per cent) and the Government of Denmark (5 per cent).

19. The iTAP assesses that the funding proposal is well aligned with the national plans and climate strategies, including the Kenya National Adaptation Plan 2015–2030² and the National Climate Change Action Plan (NCCAP) II 2018–2022³ and III 2023–2027⁴ as well as with the priorities included in the updated nationally determined contribution (NDC) of Kenya (2020)⁵. More information about coherence with existing policies in Kenya is discussed in section 1.5 on country ownership.

1.1.2. Mitigation potential

20. The updated NDC of Kenya (2020) states that the country seeks to abate its GHG emissions by 32 per cent and that, subject to national circumstances, Kenya intends to bear 21 per cent of the mitigation cost from domestic sources⁶. The iTAP acknowledges that the Government of Kenya is co-funding this project with 28 per cent of the total budget and that over 90 per cent of this contribution will be invested in field activities under Component 2.

¹ See FAO (2016). [RIMA II. Resilience Index Measurement and Analysis – II.](#)

² [https://www.unfccc.int/sites/NAPC/Documents%20NAP/Kenya NAP Final.pdf](https://www.unfccc.int/sites/NAPC/Documents%20NAP/Kenya%20NAP%20Final.pdf)

³ <https://faolex.fao.org/docs/pdf/ken190169.pdf>

⁴ <https://emsi.co.ke/other-publications/national-climate-change-action-plan-nccap-iii-2023-2027/>

⁵ <https://unfccc.int/sites/default/files/NDC/2022-06/Kenya%27s%20First%20NDC%20%28updated%20version%29.pdf>

⁶ Ibid

21. Furthermore, the low carbon technologies and practices included in this proposal are aligned with the priority mitigation activities listed in the updated NDC 2020, especially those regarding making progress towards the achievement of at least 10 per cent of the land area with tree cover, reducing deforestation and degradation, scaling up nature-based solutions (NbS) for mitigation and promoting climate-smart agriculture.
22. The funding proposal uses a comprehensive set of tools for quantifying the mitigation impact potential before and during implementation of the project: on one side, the EX-Ante Carbon Tool (EX-ACT) for crop-related activities and on the other the Global Livestock Environmental Assessment Model (GLEAM) for livestock activities. Both tools, and the methodologies behind them, have been developed by FAO, widely reviewed, open sourced and based on updated scientific information.
23. Annex 22a of the funding proposal explains the assumptions for quantifying the GHG emissions in the baseline and in the project scenarios, differentiating by assumptions relevant to the specific activities: reduced deforestation, cropland management, grassland management (to be calculated with EX-ACT) and management of dairy cattle and management of poultry (to be calculated with GLEAM). Annex 22b documents the calculation of the mitigation impact potential with EX-ACT and annex 22c the calculation with GLEAM. The iTAP acknowledges the completeness, clarity and coherence of the methodological approach and the well documented calculations.
24. Annex 11 explains how the project will monitor the mitigation impact of the project at the county level. When asked about the link to the national monitoring, reporting and verification (MRV) system of Kenya, the AE stated that “the MRV system at the project level will align with Kenya’s integrated national MRV+ framework by adopting the same shorter reference and monitoring periods of five years, ensuring consistency with the country’s reporting timelines. Emission factors used in the project will mirror those applied in the national framework, maintaining coherence and comparability. The project will identify emission factors that are specific for the LREB and not captured by the existing national MRV+ default values through literature review and participatory design workshops. This identification process will contribute to the national MRV+ system in Kenya. This approach ensures the project’s MRV system integrates seamlessly with Kenya’s national system, enhancing the accuracy and reliability of emissions reporting while supporting continuous improvement and alignment with Kenya’s climate action goals.”⁷ The project will collaborate with the county administrations to improve their capacity to undertake monitoring activities that feed into the national-level MRV system. The iTAP recognizes that this approach can serve not only as a tool for monitoring the mitigation impact potential within this project but also as a pilot initiative that could inform the use of decentralized carbon accounting methods and the possible use in MRV national systems in other regions or countries.
25. The iTAP is concerned about how the project will secure the timely distribution of seeds and other planting material essential for ensuring the adoption of CRLCSA practices outlined in the Green Lists, which are critical for achieving the adaptation and mitigation impact potential. The proposal does not include the establishment of nurseries or the provision of seeds in its activities or budget. When asked by iTAP, the AE responded that the project would work with community forest associations and cooperatives, but this answer does not clarify the availability of seeds and other planting material necessary for the wide range of agricultural activities mentioned in the funding proposal. Reduced availability of seeds and other vegetal material can cause delays in achieving the adaptation and mitigation impact potential of this funding proposal.

⁷ Text from the written answers from the AE to the iTAP questions, received on 25 November 2024.

26. Considering the above, the iTAP assesses the climate impact potential as medium to high, contingent on the project's ability to ensure the timely availability of seeds and planting materials throughout its implementation.

1.2 Paradigm shift potential

Scale: High

27. Several aspects of this project can contribute to a shift towards sustainable, climate-resilient and low-carbon commodity production LREB. The funding proposal is designed to overcome the barriers that, despite the efforts made by Kenya over the past decade, are still delaying the transition to a climate-resilient low-carbon development pathway. These barriers include insufficient public services to farmers and cooperatives to enable value chain transformation, limited integration of climate resilient and low carbon strategies into land-use planning and landscape management at county level, limited capacity of smallholder farmers to access proactive assets and face risks related to the adoption of climate-resilient, low-carbon technologies and the mismatch between supply and demand of finance to support smallholder farmers' transition. The four components into which the funding proposal is structured propose clear pathways to address and overcome these barriers.

28. Further, the funding proposal leverages Kenya's devolution process by strengthening the administration in 14 counties to provide improved climate knowledge, extension services and methodologies that support value chain actors in adopting CRLCSA. This way the county administrations will be able to plan and facilitate the implementation and monitoring of CRLCSA in the project area.

29. The project will also support smallholder farmers in increasing their competences, implementing and monitoring climate resilient, low-carbon landscape management, and sustainable agricultural technologies proposed in Components 2 and 3, and will strengthen their organization in farmer groups and cooperatives. Component 4 will facilitate access to new markets and mobilize new financing mechanisms that facilitate upscaling the adoption of new business models.

30. The funding proposal includes various learning and exchange mechanisms, both as part of the implementation and as part of the monitoring, evaluation and learning.

31. Considering the above, the iTAP assesses the paradigm impact potential of this funding proposal as high.

1.3 Sustainable development potential

Scale: High

32. The funding proposal can contribute to multiple Sustainable Development Goals (SDGs), including SDG 1– no poverty, SDG2 – zero hunger, SDG 5 – gender equality, SDG 8 – decent work and economic growth, SDG 13 – climate action and SDG 15 – life on land.

33. Potential socioeconomic co-benefits of this project include better and more sustainable income sources from CRLCSA commodities, improved food security due to both improvement of food produced for own consumption and access to better food due to increased income.

34. In addition, the proposal introduces a gender-responsive and socially inclusive approach in most activities. The gender assessment and project-level plan (annex 8) present a detailed diagnostic of gender challenges at the national and county levels as well as along the six selected value chains. This diagnostic includes the gender dynamics of cooperative membership, leadership and capacity by county and value chain. The proposal includes a Gender Action Plan developed using a tailored strategy specifically designed to integrate gender considerations into CRLCSA. The activities mentioned in the Gender Action Plan are included in both the project description in section B of the main document and in the project budget (annex 4). Annex 8 also

clarifies the targets, the roles and responsibilities, the specific budget and the monitoring regarding the gender-responsive and socially inclusive approach.

35. The project can provide environmental co-benefits at both the landscape and farm levels: at the landscape level the reforestation activities (Component 2) can improve forest cover, reduce run-off and land degradation, and contribute to biodiversity conservation; and at the farm level, activities included in Component 3 will increase biological diversity and soil health.

36. Considering the above, the iTAP assesses the sustainable impact potential of this funding proposal as high.

1.4 Needs of the recipient

Scale: High

37. The International Monetary Fund estimates that the population of Kenya in 2024 is 52.4 million⁸ with a rural population over 70 per cent of the total and a Human Development Index of 0.6 in 2022⁹.

38. Kenya, a lower-middle income leading economy in East Africa, improved its economic performance thanks to both market reforms and a successful translation of economic growth into poverty alleviation^{10,11}. The sharpest gains in poverty reduction occurred prior to 2015/16, but this trend was reversed initially after 2015/16 with a strong setback due to national circumstances and even more after 202 as a consequence of the coronavirus disease 2019 pandemic, especially in urban areas. According to the funding proposal, major development challenges for Kenya include the increasing debt, inflation, climate variability and climate change impacts on rain-fed agriculture (annex 2, part B-C).

39. Over the last decade, Kenya has prioritized activities to mitigate and adapt to climate change. Despite progress, Kenya still has both a great need for investment and innovations to improve readiness and a great urgency for action. According to the country index of the Notre Dame Global Adaptation Initiative, over 187 countries, Kenya is ranked 52nd most vulnerable and 146th in the readiness index¹².

40. The funding proposal focuses on 14 counties of the LREB, where agricultural activities play a major role for food security and income generation for local farmers. The purpose of the LREB is to coordinate and facilitate service delivery; promote trade linkages, investment, and economic activity; enhance public investment and promote developmental research, innovation in social planning and coordination of activities. Importantly, the 14 counties of the LREB share a common vision of sustainable development and are mobilized to take joint action to tackle environmental degradation, poverty and climate change (annex 2, part B-C).

41. The funding proposal provides clear and well documented information about the current institutional and socioeconomic context in LREB, one of the most densely populated regions of Kenya, and includes key development indicators for each county covered by the project.

42. In terms of institutional development, the funding proposal builds on the efforts by Kenya regarding the devolution process and the attempts at decentralizing the climate governance systems. However, the proposal explains that while climate capacity is stronger at national level, it is imperative to strengthen it at local level. An example of this need is the

⁸ <https://www.imf.org/en/Publications>

⁹ <https://hdr.undp.org/data-center/human-development-index#/indicies/HDI>

¹⁰ World Bank (2023). Kenya Poverty Assessment 2023 – From Poverty to Prosperity: Making Growth More Inclusive. World Bank.

¹¹ <https://www.usaid.gov/kenya/economic-growth-and-trade>

¹² <https://gain.nd.edu/our-work/country-index/rankings/>

current limited integration of climate resilient and low carbon strategies into landscape management at the county level. As a response, the project proposes activities aimed at strengthening the institutional and implementation capacities of the local governments and extension services.

43. Agriculture is central for both income and food security of the population in LREB and occupies over 70 per cent of the total working population. The main challenges faced by the agricultural sector in LREB are declining yields, small-size land holdings, increasing population that requires more land for housing and services, inadequate farming techniques and mismanagement as well as lack of title holding among smallholder farmers. Additionally, smallholder farmers have limited or no financing sources for climate resilient agriculture, especially if they are not engaged in a cooperative. The activities proposed in Components 2, 3 and 4 of this funding proposal are a response to these needs.
44. Considering the above, the iTAP assesses the needs of the recipient as high.

1.5 Country ownership

Scale: High

45. The funding proposal includes a no-objection letter dated 7 July 2023, signed by Dr. Chris K. Kiptoo, CBS, Principal Secretary, National Treasury of the Republic of Kenya (annex 1).
46. In its annex 2 (part B-C) the funding proposal clarifies the alignment of the project with policies of Kenya at different levels. At the national level, the annex discusses the alignment with the national constitution (2010), the national gender and equality commission act, as well as forest, energy, landscape management and environmental management laws. It includes a detailed analysis of the alignment of the proposal with the climate change policy (see section 1.1), as well as information about its alignment with the national strategy on climate smart agriculture and the national policy on disaster risk management. Further, the annex introduces how the project will serve the objectives, help to achieve climate change policies, and improve institutional capacity at the county level.
47. The funding proposal mentions three EEs: FAO, Agritererra and the Government of Kenya. The iTAP acknowledges the capacity of FAO, accredited in October 2016, as AE and as EE for this proposal. To date FAO has been the AE and has co-financed 23 projects funded by GCF, many of them implemented in countries in Africa¹³. Agritererra seeks to develop and strengthen cooperatives to make them bankable and with sustained access to finance.
48. Agritererra, is an international organization with headquarters in the Netherlands, which provides expert advice and training to cooperatives and farmer organizations in emerging economies. Agritererra's approach to addressing climate change is based upon climate smart agriculture (CSA). Furthermore, its previous experiences and partnerships, including cooperation with the International Center for Tropical Agriculture (CIAT) and Bioversity International on "Climate Clever" coffee production in Viet Nam, or the Climate Resilient Agribusiness in Kenya, Tanzania and Uganda, demonstrate an appropriate level of expertise and competencies as an EE for this funding proposal.
49. The Government of Kenya, acting through the National Treasury, will execute the activities co-financed by the Government of Kenya. The National Treasury plays an important role in defining the national policy on climate change and has proven experience on complementary activities to this project such as the "Financing locally-led climate action programme" (FLLOCA).

¹³ FAO has a longstanding experience in dealing with climate change mitigation and adaptation, both at the international level and in Kenya.

50. The funding proposal provides clear information regarding the engagement with civil society organizations and other relevant stakeholders during both the project design and implementation (annex 6 on environmental and social management framework and annex 7 on stakeholder engagement plan). These annexes document a wide range of consultations with farmers, farmers' organizations and value chain actors, county governments, regional institutions, national government departments, non-state actors and financial institutions. In addition, a consultative meeting with representatives from Indigenous Peoples' (IP) groups was convened in November 2023. The inputs, concerns and suggestions from this participative process are well documented in the funding proposal as well as how these were considered in the project design. The stakeholder engagement plan (annex 7) clarifies the expected level of engagement of multiple actors from the public and private sectors as well as from research organizations and civil society. It also clarifies the engagement methodology to be used by each actor. The overall environmental and social management budget is USD 697,356 for activities during the implementation phase.

51. The funding proposal includes important safeguards for securing a participative approach during the project implementation, including obtaining free prior and informed consent (FPIC) when an activity could affect Indigenous communities, and annual stakeholder consultations with fair access by vulnerable actors including people living with disabilities (PLWD), women, Indigenous Peoples' representatives or youth. A detailed description of the safeguards instruments and mitigation measures is included in the funding proposal.

52. Altogether the iTAP recognizes the alignment of this funding proposal with key GCF policies, including the Environmental and Social Policy, the Indigenous Peoples Policy, the Information and Disclosure Policy as well as with the FAO Environmental and Social Standards of 2022.

53. Considering the above, the iTAP assesses the country ownership of this funding proposal as high.

1.6 Efficiency and effectiveness

Scale: Medium to high

54. The overall budget for the six-year implementation period is USD 49,992,238, with a funding request to GCF of USD 29,215,309 as a grant. The Government of Kenya will contribute USD 14 million, the Government of Denmark will provide USD 5,791,910 and FAO will provide USD 985,910 as co-financing to this funding proposal, all in the form of grants. This means a co-funding ratio of 1:0.71.

55. The funding proposal includes an economic and financial analysis (EFA) prepared following the guidelines prepared by the GCF Secretariat. The results show that the financial internal rate of return of the project is 23 per cent with a benefit-cost ratio of 1.11; and 25 per cent with a benefit-cost ratio of 1.13 respectively for RCP 8.5 and RCP 2.6 scenarios for a 20-year period at 18 per cent discount rate. The project will be able to earn USD 10.7 million as a net present value using the same discount rate for 20 years under RCP 8.5 (USD 12.9 for RCP 2.6). For both climate change scenarios, the project can break even the project expenditure plus the beneficiaries' investments nine years (payback period) after the project begins.

56. The economic and financial analysis also states that the project-level economic internal rate of return is 21 per cent with a benefit-cost ratio of 1.24; and 22 per cent and 1.25 respectively for RCP 8.5 and RCP 2.6 scenarios using a 10 per cent economic discount rate. The sensitivity analysis shows that the project is viable under many scenarios. Furthermore, the project has the potential to reduce GHG emissions to a significant level with social net present value of USD 113 million and USD 116 million for RCP 8.5 and RCP 2.6 scenarios.

57. The iTAP considers the high concessionality of the GCF investment to be justified based on the arguments presented in section 1.5, "Needs of the Recipient". Additionally, the iTAP emphasizes the significant co-funding contribution by the Government of Kenya and underscores the importance of activities under Component 4 in ensuring sustainable financing for CRLCSA beyond project implementation. In this regard, the proposed capacity-building efforts to enhance financial literacy among cooperatives and farmers' organizations, coupled with collaboration with local financial institutions, are assessed as a promising and strategic approach.

58. However, much of the long-term success of the project depends on the improved income due to better or premium prices of CRLCSA commodities. This is challenging for two reasons. First, improving income for smallholder farmers requires that cooperatives are adequately prepared and able to respond promptly to short-term price fluctuations. While this can be achieved through tools such as a digital electronic market price platform that displays "offer" and "sell" prices alongside matching volumes for various commodities, the current funding proposal does not include such a tool.

59. Second, achieving premium prices for sustainably produced commodities requires the implementation of a labeling or certification system that enables consumers to identify these products in the market. The iTAP recognizes that in some cases, such as certified coffee, there are premium prices and that studies showing willingness to pay for more sustainable commodities are available for Kenya. However, the evidence on the actual economic benefits that certification means for the farmers is contested. Furthermore, there is no clear and proven avenue securing higher prices for all the CRLCSA commodities. Even more, there is concern about the risk of turning certification into one additional request to farmers instead of an (economic) recognition for sustainable production. In other words, the concern is that a situation might arise in which participation in the certification scheme comes at a higher cost (for inputs, certification cost, etc.) compared with production for non-certified markets, and is not fully offset by the price differential obtained by selling into this scheme, in comparison with the other markets.

60. The challenges, or lack of concrete evidence for securing a premium price, are highlighted in the analysis of the various value chains included in this proposal (annex 23).

61. The iTAP acknowledges both the risks and the potential value of promoting CRLCSA commodities and recognizes their likelihood of achieving better productivity under future climate scenarios. It also appreciates the project's efforts to enhance market conditions for these commodities. Thus, the iTAP evaluates the efficiency and effectiveness of this proposal as medium to high, contingent on the project's dedicated strategy to secure better incomes for farmers producing CRLCSA commodities.

II. Overall remarks from the independent Technical Advisory Panel

62. This cross-cutting funding proposal aims to achieve a mitigation outcome of over 4 million tonnes of carbon dioxide equivalent (t CO₂ eq) over the 20-year lifespan of the project. The expected adaptation outcome includes 572,000 direct and over 2 million indirect beneficiaries. The funding proposal is clear, complete and coherent. It combines a strong scientific foundation with local knowledge and institutional capacity within the country and the project region. It also addresses the priorities, opportunities and challenges of multiple stakeholders who are essential for climate-resilient, low-carbon development in the project region and offers a transformative pathway forward.

63. However, the iTAP identified two concerns that could affect how the project achieves the criteria on climate impact potential, and efficiency and effectiveness. First, the project's activities do not include provision for seed production or the establishment of nurseries, which

are crucial for ensuring the timely availability of planting materials required to implement the climate-resilient, low-carbon sustainable agricultural (CRLCSA) practices outlined in the Green Lists. This can lead to significant delays in scaling up these practices. Second, there is a potential risk to local farmers if the market does not offer premium prices for CRLCSA commodities or if these premiums do not directly benefit the farmers and outweigh the increased production costs.

64. To address these concerns, the iTAP recommends the following to the AE:
 - (a) During the inception phase, the AE should assess whether the existing seed producers and nurseries in the region can provide sufficient planting material for all CRLCSA activities in a timely manner, and during the implementation phase, the AE should monitor progress on the provision of seeds and other planting materials.
 - (b) Throughout the implementation phase, the AE should track price developments of the CRLCSA commodities and monitor whether the economic benefits to farmers outweigh the production costs.
65. The iTAP recommends that the Board approve this funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP255)

Proposal name:	Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya
Accredited entity:	Food and Agriculture Organization of the United Nations
Country:	Kenya
Project/programme size:	Small

Impact potential
The AE acknowledges that the overall impact potential is assessed as medium to high by ITAP. The AE would like to confirm that the project will assess and ensure the availability of planting material for CRLCSA activities. Activity 3.1.1 will contribute to the establishment or strengthening of community nurseries.
Paradigm shift potential
The AE acknowledges that the overall paradigm shift potential is assessed as high by ITAP.
Sustainable development potential
The AE acknowledges that the overall sustainable development potential is assessed as high by ITAP.
Needs of the recipient
The AE acknowledges that the overall needs of the recipient is assessed as high by ITAP.
Country ownership
The AE acknowledges that the overall country ownership is assessed as high by ITAP.
Efficiency and effectiveness

The AE acknowledges that the overall efficiency and effectiveness is assessed as **medium to high** by ITAP.

The AE confirms that the Economic and Financial Analysis (Annex 3), which considered the prevailing prices, demonstrated positive return for farmers. During project implementation, the AE will monitor the price developmemnts of the commodities and the resulting economic benefits for farmers.

Overall remarks from the independent Technical Advisory Panel:

The AE acknowledges iTAP overall assessment and recommendation for the Board approval.



Food and Agriculture Organization
of the United Nations

Annex 8

Gender Assessment and project-level action plan

For the GCF-FAO Project “Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya”

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ACRONYMS AND ABBREVIATIONS

ADPs	Annual Development Plans
AFOLU	Agriculture, Forestry, and Other Land Use
AI	Artificial Insemination
CC	Climate Change
CCOs	County Coordinating Officers
CECMs	County Executive Committee Members
CEDAW	Convention on the Elimination of All Forms of Discrimination against Women
CIDPs	County Integrated Development Plans
CRA	Climate Resilient Agriculture
CRLCSA	Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya
CSA	Climate Smart Agriculture
DHS	Demographic and Health Survey
EADD	East African Dairy Development
FAO	Food and Agriculture Organization of the United Nations
GAP	Gender Action Plan
GBV	Gender Based Violence
GCF	Green Climate Fund
GDI	Gender Development Index
GESI	Gender Equality and Social Inclusion
GGGI	Global Gender Gap Index
GII	Gender Inequality Index
GoK	Government of Kenya
HDI	Human Development Index
KALRO	Kenya Agricultural and Livestock Research Organization
KBNS	Kenyan Bureau of National Statistics
LDC	Least Developed Countries
LVB	Lake Victoria Basin
LREB	Lake Region Economic Bloc
MCA	Members of the County Assembly
MoALFC	Ministry of Agriculture, Livestock, Fisheries and Cooperatives
MDG	Millenium Development Goal
NAP	National Adaptation Plan
NAPA	National Adaptation Programme of Action
NCCAP	National Climate Change Action Plans
NCCSR	National Climate Change Strategy Response
NDCs	Nationally Determined Contributions
PLWD	People Living With Disability
Pos	Producer Organizations
SACCO	Savings and Credit Cooperative Organisation
SRHR	Sexual Reproductive Health and Rights
STI	Sexually Transmitted Infection
UNFCCC	United Nations Framework Convention on Climate Change
VCs	Value Chains

EXECUTIVE SUMMARY

Gender Challenges in Kenya

Kenya has made significant strides in enhancing gender equality and social inclusion (GESI) over the last decade. The Government of Kenya (GoK)'s introduction of Kenya's Constitution of 2010 and its development program, Kenya's Vision 2030 Blueprint, have generated policy and organizational reforms that have formally enabled greater gender equality across political, economic, and social spheres.¹ The process of gender mainstreaming in the devolution of government authority, resources, and functions to Kenya's 47 counties has laid a foundation for community-responsive development and created new opportunities for women and men, boys and girls, to participate as leaders and decision-makers at county level.²

Despite these political commitments, on-the-ground realities at county level and across different sectors indicate that gender gaps and inequalities persist in Kenya. These gender gaps are evidenced in various equality-based indexes. The Gender Development Index (GDI)³ and Kenya's Gender Inequality Index (GII)⁴ highlight gender gaps in education and living standards between women and men. According to the most recent Global Gender Gap Index (GGGI) report, Kenya ranks 57th out of 146 countries, demonstrating gender gaps in economic opportunities, educational attainment, health, and political empowerment.⁵

Women in Kenya are incredibly diverse, with differences in age, marital status, socio-economic status, disability status, religion, and ethnic community influencing their levels of inequality and vulnerability relative to men. In the 14 counties targeted by the CRLCSA project in the Lake Victoria Basin (LVB)/Lake Region Economic Block (LREB), there are four ethnic communities that form the majority of inhabitants: In Bomet, Kericho, and Nandi (Kalenjin), In Kisii and Nyamira (Kisii), In Bungoma, Busia, Kakamega, Trans-Nzoia, and Vihiga (Luhya), and in Homa Bay, Kisumu, Migori, and Siaya (Luo).⁶ Specific local and historical dynamics and socio-cultural and gender norms mediate diverse women's access to and control over resources, influencing their engagement across different sectors, including agriculture.⁷ Where possible, this Gender Assessment includes discussion of how gender interacts with age status (i.e., young women and men) and disability status⁸, as they represent important dimensions of social differentiation that influence participation, benefits, and use of adaptation and mitigation strategies in agricultural value chains. In this way, the assessment views both GESI as core strategies towards the achievement of diverse women, People Living With Disability (PLWD), and youth to equitably benefit from project activities.

¹ UN Women. Global Gender Equality Constitutional Database. Constitution of the Republic of Kenya. <https://constitutions.unwomen.org/en/countries/africa/kenya?provisioncategory=d91f71586bb54610baa13236037086c1>

² Hyun, M., Okolo, W., and Munene, A. 2020. USAID/Kenya Gender Analysis Report. Prepared by Banyan Global. 2020. <https://banyanglobal.com/wp-content/uploads/2020/05/USAID-Kenya-Final-Gender-Analysis-Report.pdf>

³ <https://hdr.undp.org/data-center/specific-country-data#/countries/KEN>

⁴ <https://hdr.undp.org/data-center/thematic-composite-indices/gender-inequality-index#/indicies/GII>

⁵ Global Gender Gap Report. 2022. <https://www.weforum.org/reports/global-gender-gap-report-2022/in-full/economy-profiles-5b89d90ea5>

⁶ Nyabira, B.C. and Ayele, Z.A., 2016. The state of political inclusion of ethnic communities under Kenya's devolved system. *Law, Democracy & Development*, 20, pp.131-153.

⁷ Brisebois, A., Hallstrøm Eriksen, S. E., & Crane, T. A. The politics of governing resilience: Gendered Dimensions of Climate-Smart Agriculture in Kenya. *Frontiers in Climate*, 86. <https://www.frontiersin.org/articles/10.3389/fclim.2022.864292/full>

⁸ Veronica Wanjiku N'gang'a. PWD in agriculture in Kenya: access to resources and training. Presented at Promoting equity: cross-cutting disability in international development research. Cross Cutting Disability Research Programme (CCDRP) Final Dissemination Conference, University College London, London, UK, 17 June 2013. (2013) 14 pp.

Women's representation and meaningful participation in the decision-making process trails men at national and county level, despite Kenya's Constitution of 2010 requiring a minimum of one-third representation of either gender in all elected and public posts. While the political participation rate for voting at national level is 47% women, and 52% men,⁹ the proportion of seats held by women in national parliaments and local governments are lower. As of 2018, less than one-third of the National Parliament were women (31.8% of National Assembly (MPs) and 31.3% of Senators).¹⁰ For local government at the county assembly level, women's representation rates were significantly lower, with only 6.4% of governors and 14.9% of Deputy Governors.¹¹ Thus, while gender parity quotas are enshrined in Kenyan legislation, patriarchal customs and discriminatory socio-cultural and gender norms hamper women's ability to meaningfully and equally participate in policy-making and top decision-making and leadership roles.¹² Women who do enter politics and leadership positions at national level, as well as within local and community-based organizations (including agricultural cooperatives and producer organizations) often face backlash from men, including Gender-Based Violence (GBV) and intimidation, highlighting the continuing challenge in challenging discriminatory attitudes and practices towards women's role outside the home.¹³ Devastatingly, 39.4% of women in Kenya experience GBV in their lifetime.¹⁴

Knowledge and resource gaps in women's healthcare and sexual reproductive health and rights (SRHR) persist in Kenya. While women have a greater life expectancy than men (65 to 60 years), women and girls have greater malnutrition-wasting rates, obesity rates, and HIV infection rates than men.¹⁵ Anemia is also a significant health issue, with 29% of women of reproductive age (15-49 years old) reported as anemic in the 2019 national census.¹⁶ While the maternal mortality and infant mortality rates have decreased in recent years, 30% of births are not attended by skilled health personnel, and only 56% of the population has universal health coverage.¹⁷ There are also clear gaps between women's needs and the reality of SRHR healthcare provisioning. Almost 1/3 of women of reproductive age do not have their need for family planning satisfied with modern methods, and women and girls under 25 years of age had the lowest level of understanding of reproductive rights.¹⁸ A 2021 study by the African Women's Development and Communication Network found that married women in Kenya expressed that they experienced a lack of autonomy, choice, or decision-making in negotiating for safe sex or not to have sex. Moreover, women and girls lack access to information about their basic reproductive health, with only 12% of girls aged 12-19 and 38% of women ages 21-30 are knowledgeable about menstruation.¹⁹

⁹ Nasimiyu, M.L. and E. Mariano. (2022). Sociocultural factors influencing women's participation in political leadership in Kakamega County, Kenya. *International Journal of Research and Innovation in Social Science*, 5(5) ISSN 2454-6186 <https://www.rsisinternational.org/journals/ijriss/Digital-Library/volume-6-issue-6/650-656.pdf>

¹⁰ <https://data.unwomen.org/country/kenya>

¹¹ *ibid.*

¹² Nasimiyu, M.L. and E. Mariano. (2022). Sociocultural factors influencing women's participation in political leadership in Kakamega County, Kenya. *International Journal of Research and Innovation in Social Science*, 5(5) ISSN 2454-6186 <https://www.rsisinternational.org/journals/ijriss/Digital-Library/volume-6-issue-6/650-656.pdf>

¹³ Mwambi, M., Bijman, J., & Galie, A. (2021, July). The effect of membership in producer organizations on women's empowerment: Evidence from Kenya. In *Women's Studies International Forum* (Vol. 87, p. 102492). Pergamon.

¹⁴ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

¹⁵ <https://data.unwomen.org/country/kenya>

¹⁶ <https://data.worldbank.org/indicator/SH.ANM.ALLW.ZS?end=2019&locations=KE&start=2000&view=chart>

¹⁷ The African Women's Development and Communication Network. 2022. Sexual Reproductive Health and Rights at a Glance: Fact Sheet for Kenya.

¹⁸ *ibid.*

¹⁹ *ibid.*

Women in Kenya have higher rates of poverty than men, and rural women have the highest levels of poverty and food insecurity.²⁰ Of the female population living below the national poverty line, young girls aged 0-17 years and elderly women 70 years and older are the most vulnerable and have the highest poverty levels (41% young girls and 39.1% elderly women are below the poverty line).²¹ Kenya has a 5% gender gap in adult literacy rates, with only 80% of women over 15 years old able to read compared to 85% of men.²² Although the gender gap in youth literacy rates and reading comprehension has closed between boys and girls²³, a 3.1% gender gap remains in mathematics comprehension at the Grade 2/3 level.²⁴ Gender gaps also exist in Technical and Vocational Education Training (TVET), with only 43.2% (210,795) women compared to 56.8% (276,906) men enrolled.²⁵ According to the 2022 GGI index, women had near-equal rights for access to financial services, but restricted rights for widows and daughters, and uneven rights for access to land assets and access to non-land assets.²⁶ For economic participation, women have higher rates of unemployment (9.6%) compared to men (5.3%)²⁷ and women earn 32% less than their male counterparts.²⁸ The 2019 Kenya census estimates a national labor force of 22.3 million, with women accounting for more than 50% of the total working population. According to the World Bank (2020) Report, Kenya's informal sector accounts for at least 87% of employment opportunities. The informal sector in Kenya offers employment to approximately 15 million Kenyans, according to 2018 estimates, compared to the 2.9 million who work in the formal sector.²⁹ These 15 million Kenyans are domestic workers, cleaners, beauticians, mechanics, and street vendors, among many more. Besides, the informal sector in Kenya is highly unregulated, with workers therein having limited or no social or labor protections.

Women in Agriculture

Recent estimates suggest that women account for approximately 75% of the agricultural labour force in Kenya and manage 40% of its small-scale farms.³⁰ While agriculture is central to the Kenyan economy accounting for 34.2% of its GDP and employing over 60% of the population, 70% who are living in rural areas³¹, women's participation as producers and sellers in formal agricultural value chain markets has lagged behind men. This is largely due to discriminatory social and gender norms that hinder women's involvement in agricultural value chains by shaping gender roles and responsibilities, and directing which types of crops, livestock species, marketing opportunities, and networks are appropriate for women to access.³² For example, traditional cash crops including coffee and tea, and the emergent formal dairy

²⁰ <https://data.unwomen.org/country/kenya>

²¹ *ibid.*

²² UNESCO Institute for Statistics (UIS). UIS Stat Bulk Data Download Service. Accessed November 24, 2022. apiportal.uis.unesco.org/bdd.

²³ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

²⁴ <https://data.unwomen.org/country/kenya>

²⁵ <https://data.unwomen.org/country/kenya>

²⁶ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

²⁷ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

²⁸ Equileap. 2019. Gender Equality in Kenya: Assessing 60 Leading Companies on Workplace Equality. Special Report. 24pp. https://equileap.com/wp-content/uploads/2019/11/Gender-equality-in-Kenya_Special-report-by-Equileap.pdf

²⁹ <https://www.genderandcovid-19.org/uncategorized/informal-women-workers-and-missed-opportunities-the-pandemic-economic-stimulus-package-in-kenya/>

³⁰ The Kenya Institute for Public Policy Research and Analysis (KIPPRA). 2020. Women's Access to Agricultural Finance in Kenya. Policy Brief No. 03 of 2020-2021. <https://repository.kippira.or.ke/handle/123456789/2782>

³¹ *ibid.*

³² David, S. 2021. Women in agribusiness value chains in Africa: A white paper on constraints and opportunities for developing a gender-responsive agribusiness sector.

sector, are associated with men or are masculinized commodities, whereas informal food security crops and small livestock (e.g., poultry, rabbits) are feminized commodities.³³

Barriers to women's participation and benefit in agricultural value chains include limited access to affordable agricultural finance, affordable and appropriate technology, limited access to extension services, limited access to markets, and lack of access to inputs. Structural inequalities continue to prevent women from accessing higher levels of value chains and trading opportunities both at national, intra-regional and global level and in different areas of production.³⁴ For example, unequal land rights for rural women make it more difficult for them to engage in value chains that require large tracts of land to produce at scale (e.g., coffee, tea).³⁵

Intersectional gender differences also influence aspirations and opportunities for engaging in agriculture and cooperative societies in Kenya.³⁶ For example, young women are disincentivized to pursue agribusiness activities when they are the least likely to have access to productive resources and assets, access to credit and financial services, or business support services compared to men, women, or boys.³⁷ Young women and men usually do not hold positions of leadership or decision-making within cooperative organizations due to power dynamics that prioritize the opinions of elder men – this can also act as a disincentive.³⁸ PLWD are marginalized from participating in agricultural production more broadly, and the belief that they are unable to contribute as producers, marketers, or leaders.³⁹

Women and Climate Change in Agriculture

Women are disproportionately vulnerable to climate impacts based on existing inequalities in their roles, rights, and opportunities, which are defined by gender norms and socio-economic status.⁴⁰ For example, women tend to be more reliant on the natural resource base for securing their daily livelihoods (e.g., the collection of fuel wood for household energy, water collection for domestic and productive use), so the reduction in these resources due to climate change affects them more directly.⁴¹ Given that climate change amplifies existing socio-economic inequalities (e.g., control over or access rights to resources), women smallholder farmers in Kenya's lack of secure land rights, livestock assets, and technologies

³³ For more details on the gendered power embedded in agricultural commodities, see Tavenner, K., & Crane, T. A. (2018). Gender power in Kenyan dairy: cows, commodities, and commercialization. *Agriculture and Human Values*, 35(3), 701-715.

³⁴ Tavenner, K. and Crane, T., 2016. Best practice guide to socially and gender-inclusive development in the Kenyan intensive dairy sector. ILRI (aka ILCA and ILRAD).

³⁵ Rubin, D., Boonabaana, B., & Manfre, C. (2019). Building an inclusive agriculture: Strengthening gender equality in agricultural value chains. Annual Trends and Outlook Report: Gender Equality in Rural Africa: From Commitments to Outcomes, 83-96. <https://rtbfoods.cirad.fr/content/download/4186/31890/version/1/file/Building+an+inclusive+agriculture+-+Strengthening+gender+equality+in+agricultural+value+chains.pdf>

³⁶ Bullock, R. and Crane, T., 2021. Young Women's and Men's Opportunity Spaces in Dairy Intensification in Kenya. *Rural Sociology*, 86(4), pp.777-808. <https://onlinelibrary.wiley.com/doi/pdfdirect/10.1111/ruso.12385>

³⁷ *ibid.*

³⁸ Bullock, R. and Crane, T., 2021. Young Women's and Men's Opportunity Spaces in Dairy Intensification in Kenya. *Rural Sociology*, 86(4), pp.777-808. <https://onlinelibrary.wiley.com/doi/pdfdirect/10.1111/ruso.12385>

³⁹ Veronica Wanjiku N'gang'a. 2013. PWD in agriculture in Kenya: access to resources and training. Presented at Promoting equity: cross-cutting disability in international development research. Cross Cutting Disability Research Programme (CCDRP) Final Dissemination Conference, University College London, London, UK, 17 June 2013. (2013) 14 pp.

⁴⁰ Awiti, A. 2022. Climate change and gender in Africa: A review of impact and gender-responsive solutions. *Frontiers in Climate*. 4:895950. <https://doi.org/10.3389/fclim.2022.895950>

⁴¹ Caroli G, Tavenner K, Huyer S, Sarzana C, Belli A, Elias M, Pacillo G, Läderach P. 2022. The Gender-Climate-Security Nexus: Conceptual Framework, CGIAR Portfolio Review, and Recommendations towards an Agenda for One CGIAR. Position Paper No. 2022/1. CGIAR FOCUS Climate Security. <https://hdl.handle.net/10568/117590>

constrain them from adapting and coping with climate shocks.⁴² Added vulnerabilities include increased risk for GBV due to conditions such as longer distances to collect water and loss of livelihoods creating tensions within households.

While commercialization has been identified as a key strategy in assisting farmers adapt to climate change in Kenya⁴³, current business-as-usual practices in the way commercialization is performed tends to weaken women's control over decision-making and previously controlled incomes.⁴⁴ In Kenya, and specifically the 14 counties targeted by the CRLCSA project, this dynamic is underpinned by socio-cultural norms that assign gendered responsibility to certain agricultural activities and value chains as being "for men" (e.g., cash crops, coffee, tea, dairy) and others as "for women" (poultry, vegetables and fruits for homestead consumption).⁴⁵ Moreover, even in these activities that are customarily for women, as these activities become more profitable, men usually control the economic benefits gained from these activities.⁴⁶

Thus, value chain upgrading, and agricultural intensification in the context of a changing climate must be gender-responsive and acknowledge the gendered trade-offs that guide women's decisions in whether to engage in increasing marketization and/or formal marketing of agricultural and livestock products.⁴⁷ Gender-responsive solutions to these trade-offs would start with interventions working with local women's groups and cooperative societies to identify appropriate strategies to ensure women are equitably benefitting from participation in the value chain. For example, adding agricultural diversification or nutrition-based programming and actively investing in value chain activities that are shown to benefit women more directly (e.g., poultry production) in addition to planned commercialization activities.⁴⁸

The Gender Action Plan (GAP) for the proposed project tackles gender inequalities across several priority areas in agri-climate adaptation and mitigation⁴⁹. These include closing gender gaps and existing inequalities in participation (at intra-household level and within producer organizations/cooperatives), workloads (prioritizing agricultural technologies and practices for adaptation and mitigation that reduce workloads and negative impacts on women), access and use of productive resources (such as agri-climate information, technologies, livelihood incomes, credit), and collective action (working with women's groups as platforms for enhancing access, agency, and voice in climate-smart agriculture). In closing these

⁴² FAO and ARC. 2021. Women's leadership and gender equality in climate action and disaster risk reduction in Africa – A call for action. Accra, FAO. <https://doi.org/10.4060/cb7431en>

⁴³ GoK. Agricultural Sector Development Strategy 2010-2020. <https://faolex.fao.org/docs/pdf/ken140935.pdf>

⁴⁴ Tavenner, K., Van Wijk, M., Fraval, S., Hammond, J., Baltenweck, I., Teufel, N., ... & Manda, L. 2019. Intensifying inequality? Gendered trends in commercializing and diversifying smallholder farming systems in East Africa. *Frontiers in Sustainable Food Systems*, 3, 10.

⁴⁵ Ihalainen, M., Shaikh, S., Mujawamariya, G., Mayanja, S., Adetonah, S., Tavenner, K. and Elias, M., 2021. Promise and contradiction: value chain participation and women's empowerment. *Advancing gender equality through agricultural and environmental research: past, present and future*, pp.147-188.

⁴⁶ Tavenner, K. and Crane, T.A., 2018. Gender power in Kenyan dairy: cows, commodities, and commercialization. *Agriculture and Human Values*, 35(3), pp.701-715.

⁴⁷ Safa Barraza, A. and Berthelin, L. 2022. *Climate resilience and disaster risk analysis for gender-sensitive value chains: A guidance note*. Rome, FAO. <https://doi.org/10.4060/cc0051en>

⁴⁸ Tavenner, K., Van Wijk, M., Fraval, S., Hammond, J., Baltenweck, I., Teufel, N., ... & Manda, L. (2019). Intensifying inequality? Gendered trends in commercializing and diversifying smallholder farming systems in East Africa. *Frontiers in Sustainable Food Systems*, 3, 10.

⁴⁹ Adaptation requires adopting specific practices to lessen climate change impacts, while mitigation deals with addressing the root causes of climate change (i.e., Greenhouse Gas Emissions).

gender gaps, the project will use Gender Transformative Approaches (GTA),⁵⁰ and more specifically, Gender Action Learning Systems (GALS) to address the underlying discriminatory socio-cultural and gender norms that perpetuate gender inequality and constrain women's capabilities. This requires a culturally sensitive, multi-level approach that includes women and men in all their diversity.

The proposed project will use eight core strategies to achieve its gender and social inclusion (GESI) objectives. These are: (1) Supporting and strengthening Kenya's existing gender-responsive legal and institutional frameworks related to gender equality, climate change, and agriculture at national and community level (including within producer organizations, county agencies, and private sector); (2) Supporting gender-responsive and socially inclusive agri-climate information and services for vulnerable smallholders, both women and men, in adopting climate-resilient and low-carbon production and processing practices, technologies, assets, and risk reduction mechanisms; (3) Strengthening women, PLWD, and youth representation and participation in meaningful decision-making and leadership in cooperative societies and agrifood value chains; (4) Supporting agricultural extension to disseminate and demonstrate CRLCSA knowledge, technologies and practices in ways that are gender-responsive and socially-inclusive; (5) Requiring sex- and age-disaggregated data and relevant gender, agriculture, and climate indicators be collected, analyzed, and fed back into project activities iteratively and in a participatory manner; (6) Support GESI agri-climate finance for vulnerable smallholders and their organizations by increasing access to gender-responsive and socially inclusive financial products that support climate-resilient and low-carbon growth/Supporting gender-responsive and socially inclusive financial services, climate finance, and bundled services for enhancing women, PLWD, and youth actions towards climate adaptation and mitigation; (7) identifying and promoting gender-responsive and socially inclusive adaptation and mitigation technologies, markets, and labour practices; and (8) Promote and monitor gender and social safeguards to reduce climate risks for the most vulnerable, as well as risks to women's health, wellbeing, and livelihoods from increased engagement in cooperative organizations and leadership (for example, the risk of GBV or men's cooption of benefits from women's value chain labour).

The Gender Action Plan (GAP) should be further developed with the participation of women, men, girls, boys, youth, PLWD, and other vulnerable people who are most affected by the climate crisis within the agricultural sector to ensure all planned activities are gendered and meet the rights, needs, and experiences of women, men, girls, boys, youth, PLWD, and other vulnerable people.⁵¹

PART I: GENDER ASSESSMENT

1. Introduction

This Gender Assessment and accompanying Gender Action Plan (GAP) are presented in relation to the proposed project, "Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya (CRLCSA)." The project mainstreams the achievement of gender equality and demonstrates its importance as a cross-cutting theme and prioritized climate and development goal. As such, the proposal has integrated gender mainstreaming into all project outputs and activities, as evidenced in the Gender Action Plan.

⁵⁰ For a full list of publications by the FAO Joint Programme on Gender Transformative Approaches for Food Security and Nutrition see: <https://www.fao.org/joint-programme-gender-transformative-approaches/resources/publications/en>

⁵¹ In alignment with the key recommendations from the 2020 report "Review of the National and County Planning and Budgeting Processes with a Gender Responsiveness and Social Inclusion Lens in Agriculture Sector, Kenya"

To illustrate that the project has meaningfully considered gender,⁵² and will commit actions towards the achievement of gender equality⁵³ through gender-responsive⁵⁴ programming, this assessment provides a situational analysis of the relevant gender dynamics in the proposed project sites in the Lake Victoria Basin (LVB), (also referred to as the Lake Region Economic Bloc, or LREB). Targeting is at county level⁵⁵, and includes the following 14 counties: Bomet, Bungoma, Busia, Homa Bay, Kakamega, Kericho, Kisii, Kisumu, Migori, Nandi, Nyamira, Siaya, Trans-Nzoia, and Vihiga, and across six value chains: coffee, tea, fruit trees, African leafy vegetables, livestock, and poultry.⁵⁶ Climate change impacts these value chains in significant ways – limited water supply coupled with high temperatures puts additional stress on crops such as coffee, tea, and African leafy vegetables, as well as dairy cows and poultry due to inadequate feed and fodder, water resources, and pest and disease outbreaks. There is generally a lack of sex and gender-disaggregated data at sub-national level⁵⁷, but county-level data on gender gaps in agricultural value chains and climate risks are presented using primary qualitative and quantitative data sources. Recommendations regarding where the project can have gender-responsive and gender-transformative⁵⁸ programming and results are discussed in Section 6.

This Gender Assessment recognizes the ways that gender intersects with other factors such as social differentiation, such as age, assets base, marital status, ethnic community, religion and class, within specific historical and cultural contingent contexts to influence people's ability to benefit from agricultural development and their vulnerability and resilience to climate change.⁵⁹ However, sex- and age-disaggregated data (qualitative or quantitative data that is collected and presented separately on men and women and by age⁶⁰) that goes beyond gender binaries are rarely collected. In this assessment, sex- and age-disaggregated data is presented from the 2019 Kenyan Census's national and county-level population statistics, as well as two quantitative surveys conducted in the Feasibility Study (i.e., climate risk survey and cooperative survey). The results of the qualitative research conducted in the Feasibility study are presented in section 5.2.7.

⁵² 'Gender' refers to the socially constructed system of classification that ascribes qualities of masculinity and femininity to people, often based on their biological sex. Gender characteristics can change over time and are different between cultures. We also recognize that gender interacts with other social categories (e.g., age, caste, class, ethnic community, disability, etc.) in unique ways that mediate people's engagement with agriculture and climate change adaptation based on relative privilege and sociocultural norms.

⁵³ 'Gender equality' refers to equal rights, responsibilities and opportunities of women and men and girls and boys. European Institute for Gender Equality (EIGE). 2022. Glossary and thesaurus. <https://eige.europa.eu/thesaurus/terms/1168>

⁵⁴ 'Gender responsive' refers to approaches that reflect an understanding of and response to, socially constructed gender relations and roles in ways that try to address gender inequalities including encouraging equal, active participation equal opportunities, and fair distribution of benefits. Gender responsiveness is accomplished through gender analysis and gender inclusiveness. (Source: Nelson, G. (2015). Gender Responsive National Communications Toolkit. United Nations Development Programme).

⁵⁵ Following the Kenya Constitution 2010, the country is divided into 47 counties with devolved governments.

⁵⁶ Value chains were selected using ten criteria and a participatory approach (refer to FS Section 6 for detail on targeting and selection of VC). After consultation in the LREB, the value chains selected were three commercial value chains (dairy, tea, coffee) and three value chains that offer more direct food security benefits (fruit tree, African leafy vegetables, and poultry)

⁵⁷ <https://www.genderinkkenya.org/wp-content/uploads/2020/12/Gender-Sector-Statistics-Plan.pdf>

⁵⁸ 'Gender transformative' refers to approaches that seek to tackle the structured root causes of entrenched gender inequalities at multiple scales, including gender norms and roles, rather than merely responding to the symptoms of gender inequality such structures produce. Farhall, K. and Rickards, L., 2021. The "gender agenda" in agriculture for development and its (lack of) alignment with feminist scholarship. *Frontiers in Sustainable Food Systems*, 5, p.573424. <https://doi.org/10.3389/fsufs.2021.573424>

⁵⁹ Ravera, F., Martín-López, B., Pascual, U. et al. The diversity of gendered adaptation strategies to climate change of Indian farmers: A feminist intersectional approach. *Ambio* 45 (Suppl 3), 335–351 (2016). <https://doi.org/10.1007/s13280-016-0833-2>

⁶⁰ Nelson, S. & Hill, C., 2019. *Gender in adaptation planning for the agriculture sectors: Guide for trainers*. Rome.

In Kenya, men's and women's gender roles, relations, and responsibilities are intimately connected to their fragility and resilience to climate change. On the one hand, existing gender inequalities exacerbate the negative impacts of climate change on women – their lack of access to resources, information, and technologies makes them more vulnerable to both rapid and slow-onset climate events⁶¹. On the other hand, women's specialized agricultural and environmental knowledge strengthens their adaptive capacity to climate impacts⁶². Thus, the gender-differentiated impacts and opportunities posed by climate change underscore women's vulnerability and disadvantage them in the face of a changing climate, as well as their resilience and power as agents of change in their households and communities in adapting to and mitigating climate change. In line with these observations, approaches to climate change adaptation have recently identified the co-benefit of women's empowerment.⁶³

In general, women face disproportionate climate change impacts given existing social inequalities. Even though women make up 75% of agricultural labor force⁶⁴, LVB/LREB communities are patriarchal and undervalue, exclude, and marginalize women's contribution to agriculture across sectors and value chain nodes. While women and children spend significant amounts of time laboring in production, harvesting, and post-harvesting activities, and may have decision-making power over food security crops, most women and youth do not own the land or other farm assets and do not have decision-making power over cash crops, or food security crops or livestock activities that become commercialized. Thus, women in the LVB/LREB region cannot use land for collateral, affecting their ability to access investment and loans to strengthen or expand their farming activities. Having fewer physical, capital, and financial resources to draw upon, women in the LVB/LREB are expected to face greater vulnerability as climate change impacts agricultural production.

While gender equality is part of Kenya's 2015 – 2030 climate change adaptation plan, in the LVB/LREB women are facing social and gender inequalities that climate change is exacerbating. These gender-based challenges are also evident in agricultural value chains in the context of a changing climate. Women's work is often invisible in agriculture – they frequently receive no payment for agricultural labor and are often not considered farmers by agricultural extension staff. Since women produce a large share of agricultural output and supply a large share of the labor (which has been increasing over time), any successful agricultural intervention requires raising women's empowerment and ensuring that gender-specific strategies are at the core of development programs.⁶⁵

To identify and create actionable steps towards addressing these challenges, gender analysis was used to explore the gender-based constraints and opportunities that influence women's engagement in agricultural value chains in the context of a changing climate. Gender considerations are crucial to ensure the gender gap in agriculture is addressed and closed. The gender gap refers to “the underperformance of the agriculture sectors in many developing countries, partly caused by women lacking equal access to

⁶¹ Ngigi, M. W., Mueller, U., & Birner, R. (2017). Gender differences in climate change adaptation strategies and participation in group-based approaches: An intra-household analysis from rural Kenya. *Ecological Economics*, 138, 99-108.

⁶² Bryan, E., Ringler, C., Okoba, B., Roncoli, C., Silvestri, S., & Herrero, M. (2013). Adapting agriculture to climate change in Kenya: Household strategies and determinants. *Journal of environmental management*, 114, 26-35.

⁶³ Women's empowerment refers to “the process by which women gain power and control over their own lives and acquire the ability to make strategic choices.” European Institute for Gender Equality (EIGE). 2022. Glossary and thesaurus. <https://eige.europa.eu/thesaurus/terms/1102>

⁶⁴ The Kenya Institute for Public Policy Research and Analysis (KIPPRA). 2020. Women's Access to Agricultural Finance in Kenya. Policy Brief No. 03 of 2020-2021. <https://repository.kippira.or.ke/handle/123456789/2782>

⁶⁵ Kristjanson, P., Bryan, E., Bernier, Q., Twyman, J., Meinzen-Dick, R., Kieran, C., Ringler, C., Jost, C. and Doss, C., 2017. Addressing gender in agricultural research for development in the face of a changing climate: where are we and where should we be going?. *International Journal of Agricultural Sustainability*, 15(5), pp.482-500.

the resources, decision making, and opportunities that they need to be productive. The gender gap harms society due to lost agricultural output, decreased food security, and stunted economic growth” (FAO, 2011)⁶⁶. There are also documented differences in women’s and men’s perception of climate change in Kenya, with gender-differentiated adaptation strategies evident at the intra-household level.⁶⁷ Their climate change adaptation actions suggest women and men pursue adaptation practices based on intra-household and community gender roles, responsibilities, and norms.⁶⁸

In creating action-based recommendations towards strengthening gender equality in the intervention sites, the CRLCSA project is integrated with the local agri-climate information ecosystem and seeks to leverage the potential for agricultural cooperatives and producer organizations as sites of gender-responsive programming and transformation. Indeed, agricultural cooperatives have been proven sites of resilience for women, who have leveraged their power in cooperatives to emerge as key agents for rural change.⁶⁹ The role of cooperatives in enhancing women’s participation, performance, benefits, and empowerment from engaging in agricultural value chains is well-documented, including in Kenya.⁷⁰ For example, by providing informational and technological resources, access to individual and communal credit schemes, access to farming inputs, and collective action for bargaining, cooperatives serve to build women’s confidence and provide them with the tools needed to increase their adaptive capacity and resilience to climate change.⁷¹

2. Methodology

This gender assessment presents a situational gender analysis of Kenya and the 14 targeted counties in the CRLCSA project. Gender analysis is defined as, “a systematic analytical process for organizing, collecting, analyzing, and interpreting qualitative and quantitative information that examines gender relations in a particular context, ranging from households to communities to nations.”⁷² The goal of gender analysis is to understand the specific roles of men and women, the relationships between men and women, their access to resources, their activities, and the constraints they face relative to each other. In identifying the different roles, needs, interests and opportunities for women and men, boys, and girls, a gender analysis helps identify relevant entry points, policies, and opportunities for enhancing gender equality (and social inclusion) in a particular intervention. In the context of agriculture and climate change interventions, a gender assessment also helps identify multiple causes of vulnerability, including gender and other social inequalities, and to build on the diverse knowledge and capacities within communities/households that can be used to make them more resilient to climate-related shocks and risks.⁷³ In alignment with GTA, the gender assessment sees that key to understanding gender power and

66 <https://www.fao.org/3/i2050e/i2050e00.htm>

⁶⁷ Ngigi, M., Mueller, U., & Birner, R. (2016). Gender differences in climate change perceptions and adaptation strategies: an intra-household analysis from rural Kenya. *Available at SSRN 2747856*.

⁶⁸ Nunow, A., Muthama, N. J., & Kinama, J. M. (2020). Analysis of gender parity in climate change adaptation actions within Kajiado and Kiambu counties, Kenya. *East African Journal of Science, Technology and Innovation*, 1(2).

⁶⁹ Lecoutere, E. (2017). The impact of agricultural co-operatives on women’s empowerment: Evidence from Uganda. *Journal of Co-operative Organization and Management*, 5(1), 14-27. <https://doi.org/10.1016/j.jcom.2017.03.001>

⁷⁰ Lodiaga, M.D. 2020. The Cooperative Movement in Kenya: Women Only Cooperatives Their Potential for Women’s Empowerment and Enhancement of Gender-Just Peace. 7(4). *AJBSR*.MS.ID.001177. DOI:10.34297/AJBSR.2020.07.001177

⁷¹ *ibid*.

⁷² Mehar, M. and McDougall, C. (2017). Methods and tools for gender analysis in FISH: A preliminary consolidation and reference guide. Penang, Malaysia: CGIAR Research Program on Fish Agri-Food Systems. Internal document.

⁷³ Caroli G, Tavenner K, Huyer S, Sarzana C, Belli A, Elias M, Pacillo G, Läderach P. 2022. The Gender-ClimateSecurity Nexus: Conceptual Framework, CGIAR Portfolio Review, and Recommendations towards an Agenda for One CGIAR. Position Paper No. 2022/1. CGIAR FOCUS Climate Security.

relationships that mediate access, control, and decision-making, as well as assets, resources, and workloads, is to investigate the underlying socio-cultural norms, values, and beliefs that are expressed in the construction of gender identities and inequalities.⁷⁴ For additional guidance in performing gender-sensitive value chain studies, FAO has several key resources.⁷⁵

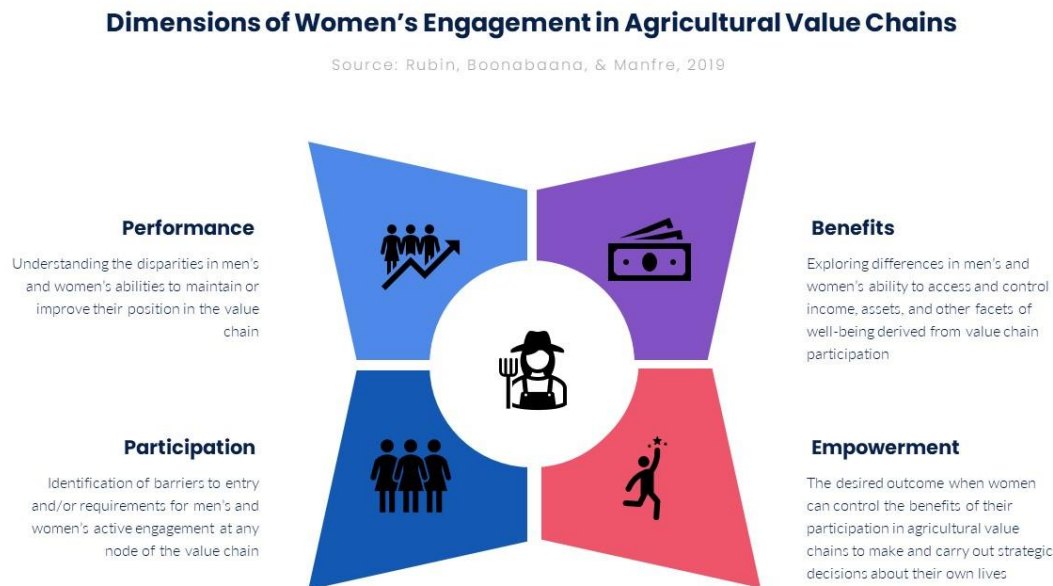
The Gender Assessment focuses on the current situation of rural women in the counties targeted by the proposed project: Bomet, Bungoma, Busia, Homa Bay, Kakamega, Kericho, Kisii, Kisumu, Migori, Nandi, Nyamira, Siaya, Trans-Nzoia, and Vihiga. The Gender Assessment is based on a desk review of recent literature on gender, climate change, cooperatives, and agricultural value chains in Kenya. Academic and grey literature from international institutions and NGOs, as well as national and county-level data and statistics are used to provide an empirical evidence base for the inclusion of gender in the proposal. Primary data was also collected in the counties targeted via three fieldwork activities. The first quantitative assessment presents data related to gender, climate information services, and agricultural decision-making in crop and livestock activities. The second assessment is qualitative, and explores the gender barriers, opportunities, and dynamics regarding women's participation, performance, and empowerment in the six value chains targeted. The third assessment is quantitative and explores gendered membership in cooperatives and value chains targeted in each county. The tools used for qualitative data collection are presented in the Appendix, while the quantitative data collection tools are annexed in the main report.

Qualitative interviews with cooperative members were designed to better understand the gender-based barriers and opportunities for women's engagement in the agricultural value chain, that influence their participation, performance, benefits, and empowerment. Interviews with cooperative leadership were designed to identify the key gender issues and inequalities within specific agricultural cooperatives, and discuss solutions towards improving women's participation, performance, benefits, and empowerment. These dimensions were chosen based on the gender analytic approach of Rubin et al. (2019) in investigating relevant dimensions of women's engagement in agricultural value chains (see Figure 1).

⁷⁴ Mehar, M. and McDougall, C. (2017). Methods and tools for gender analysis in FISH: A preliminary consolidation and reference guide. Penang, Malaysia: CGIAR Research Program on Fish Agri-Food Systems. Internal document.

⁷⁵ See: FAO. (2016). Developing gender-sensitive value chains – A guiding framework. Rome. [Developing gender-sensitive value chains \(fao.org\)](#); FAO. (2018). Developing gender-sensitive value chains – Guidelines for practitioners. Rome. [Developing gender-sensitive value chains \(fao.org\)](#); and Safa Barraza, A. and Berthelin, L. (2022). Climate resilience and disaster risk analysis for gender-sensitive value chains: A guidance note. Rome, FAO. <https://doi.org/10.4060/cc0051en>

Figure 1. Dimensions of Women's Engagement in Agricultural Value Chains



These questionnaires were used to address the 'big questions' emergent in gender analysis of agricultural value chains in the context of a changing climate across the four dimensions presented in Figure 1. For cooperative members, these included: What are the barriers to entry and/or requirements for men's and women's active engagement at any node of the value chain? What are the disparities in men's and women's ability to maintain or improve their position in the value chain? What are the differences in men's and women's ability to access and control income, assets, or other facets of well-being derived from value chain participation? What steps/changes are needed so that women can control the benefits of their participation in agricultural value chains to make and carry out strategic decisions about their own lives? For cooperative leadership, these included: What are the barriers to entry and/or requirements for men's and women's active engagement at any node of the value chain? What are the disparities in men's and women's ability to maintain or improve their position in the value chain? What are the differences in men's and women's ability to access and control income, assets, or other facets of well-being derived from value chain participation? What steps/changes are needed so that women can control the benefits of their participation in agricultural value chains to make and carry out strategic decisions about their own lives?

2.1 Organization of the Report

The report is intended as a 'living document' whereby the assessment data herein and Gender Action Plan (GAP) should be updated periodically based on new information obtained during the inception and implementation phases. It is expected that the project's Monitoring, Evaluation, and Learning (MEL) system and the mid-term evaluation will consider progress and gaps in the implementation of the GAP and provide feedback to further adjust and refine it. The gender expert assigned to the project will be responsible for a review of this document on a bi-annual basis.

The remainder of the Gender Assessment is organized as follows: Section 3 provides an overview of the legal and policy framework for the promotion of gender equality in Kenya. Section 4 presents a general

overview of the status of women and gender equality in Kenya. Section 5 consists of a gender analysis in the context of the project implementation sites. Section 6 introduces project strategies to integrate gender equality and social inclusion, which are further elaborated in the GAP.

3. Legal and policy framework for promotion of gender equality in Kenya

This section maps and assesses the existing institutional, policy, and legal frameworks pertaining to gender equality in Kenya. These include international frameworks and conventions on gender equality, national legal frameworks for the promotion of gender equality, and national climate-related policies. A brief analysis of the level of gender integration of the national climate-related policies is presented, before an overview of sectoral policies relevant to this proposal – namely, agriculture, livestock, and natural resources.

3.1 International Frameworks and Conventions on Gender Equality

There are multiple international instruments, policies and declarations that require states to develop and implement programmes and policies contributing to gender equality and women's rights. Article 2(6) of the Constitution of Kenya (2010) compels the Government to implement the obligations of the international treaties it has ratified. These include: the 1994 International Conference on Population and Development (ICPD) Programme of Action; the Programme of Action of the World Summit on Social Development (1995) and its review held in 2009; the BPfA (1995); the United Nations Commission on the Status of Women; the Joint United Nations Programme on HIV/AIDS (UNAIDS) Action Framework on Women, Girls, Gender Equality and HIV (2009); and Committee on the Elimination of Discrimination Against Women (CEDAW).⁷⁶ The latest CEDAW Committee concluding observations on the eight periodic report of Kenya gives specific recommendations for rural women under Article 42. The Committee calls upon the GoK to: (a) To promote the participation of rural women in decision-making processes and their access to high-quality health care, education and adequate water and sanitation; (b) To facilitate the access of rural women to land, eliminate all customs and traditional practices that impede their equal access to land and establish a clear legislative framework to protect their rights to inheritance and land ownership; (c) To develop and implement a national gender policy on agricultural development as set forth in the agricultural sector development strategy covering the period 2010–2020; (d) To ensure access to high-quality health care for rural women, including through increased training of midwives; (e) To ensure the equal participation of rural women and girls in policymaking processes on disaster mitigation and climate change; and (f) To implement the Climate Change Act of 2016 in a manner that prioritizes women's rights. While progress has been made in domesticating international treaties and conventions, the implementation and monitoring of some of these remains weak.

At the regional level, Kenya has ratified the following commitments: the AU Charter and its Protocol on Human and People's Rights on the Rights of Women in Africa (2003); and as a member of the Intergovernmental Authority on Development (IGAD), it recognizes that there are sharp gender inequalities in access to key productive assets including: land, labour, financial services, technology, and inputs; coupled with education and health care.⁷⁷ Despite these commitments, challenges for gender statistics persist in the country, as producers and users are not adequately coordinated. There is a need to harmonize methods and standards across producers and users of gender statistics in Kenya to address the gaps in sex-disaggregated and gender-specific statistics.

⁷⁶https://tbinternet.ohchr.org/_layouts/15/treatybodyexternal/Download.aspx?symbolno=CEDAW%2FC%2FKEN%2FCO%2F8&Lang=en

⁷⁷ <https://www.genderinkenya.org/wp-content/uploads/2020/12/Gender-Sector-Statistics-Plan.pdf>

A recent review of gender in climate change and agriculture policies in the East African region found that while there is increasing gender responsiveness in the region, (i) gender issues are still interpreted as “women issues,” (ii) there is disharmony in gender mainstreaming across governance levels, (iii) budgeting for gender is not yet fully embraced by governments, (iii) allocations to gender at sub-national level remain inconsistently low with sharp differences between estimated and actual budgets, and (iv) gender activities do not address any structural inequalities.⁷⁸

3.2 National Legal Frameworks for the Promotion of Gender Equality

There are several legal frameworks to promote, enforce and monitor equality and non-discrimination based on sex. These include: The Constitution of Kenya, 2010, Article 23; National Gender and Equality Commission Act, No. 15 of 2011, Section 8; The Prohibition of FGM Act, 2011; Prevention against Domestic Violence Act, 2015.⁷⁹

The 2010 Constitution is the cornerstone of Kenya's legal framework on gender equality and women's empowerment. Key provisions include:

- **Article 27:** Guarantees equality and freedom from discrimination. It prohibits discrimination based on gender, ensuring that men and women have equal opportunities in political, economic, cultural, and social spheres.
- **Two-thirds Gender Rule (Article 81(b)):** The Constitution mandates that not more than two-thirds of members of elective and appointive bodies should be of the same gender, promoting gender parity in political representation.
- **Affirmative Action (Article 27(8)):** This allows for measures to be taken to promote women's representation and participation in various spheres, including political, economic, and social.
- **Bill of Rights (Chapter 4):** Guarantees equal protection under the law for all citizens, including women, and prohibits any form of discrimination.

Several statutory laws have been enacted to implement these constitutional principles and promote gender equality:

- **The Sexual Offences Act (2006):** Provides comprehensive legal protection against sexual violence, aiming to safeguard women's bodily integrity and prosecute perpetrators of sexual offenses.
- **The Matrimonial Property Act (2013):** Protects women's property rights in marriage and after divorce, recognizing both monetary and non-monetary contributions made during the marriage.
- **The Prohibition of Female Genital Mutilation (FGM) Act (2011):** Criminalizes female genital mutilation and provides for the protection of women and girls from this harmful practice.
- **The Employment Act (2007):** Prohibits discrimination in employment on grounds of gender and mandates equal pay for work of equal value, ensuring that women are treated fairly in the workplace.
- **The Protection Against Domestic Violence Act (2015):** Provides legal protection for victims of domestic violence, who are predominantly women, and facilitates prosecution of offenders.

⁷⁸ Ampaire, E. L., Acosta, M., Huyer, S., Kigonya, R., Muchunguzi, P., Muna, R., et al. (2020). Gender in climate change, agriculture, and natural resource policies: insights from East Africa. *Clim. Change* 158, 43–60. doi: 10.1007/s10584-019-02447-0

⁷⁹ KNBS and UN Women. SDG Fact Sheet 2021 Kenya.

<https://data.unwomen.org/sites/default/files/documents/Publications/KNBS-Kenya-factsheet.pdf>

The **National Gender and Equality Commission**⁸⁰ (NGEC) was established by the National Gender and Equality Commission Act, 2011 pursuant to Article 59 (4) of the Constitution of Kenya. It is one of the three (3) successor commissions, NGEC, Commission on Administrative Justice and Kenya National Commission on Human Rights), to the Kenya National Human Rights and Equality Commission (KNHREC) established in Article 59 of the Constitution of Kenya 2010. NGEC's mandate is informed by Section 8 of the National Gender and Equality Commission Act 2011. Its mandate is to promote and ensure gender equality, principles of equality and non-discrimination for all persons in Kenya with a focus on women, persons with disability, children, youth, older members of society, minority and marginalised groups.

It monitors, facilitates and advises on the integration of the principles of equality and freedom from discrimination in all national and county policies, laws, and administrative regulations in all public and private institutions; and it also investigates and ensures compliance with legal frameworks and strategic plans.

3.3 National Climate-related Policies and Gender

Several government policies seek to improve gender equality and women's empowerment:

- **The National Gender and Equality Commission (NGEC):** Established under the National Gender and Equality Commission Act (2011), NGEC is a constitutional commission mandated to promote gender equality and freedom from discrimination. It monitors the implementation of laws and policies related to gender equality, advises the government, and conducts public education.
- **The Kenya Vision 2030:** Kenya's long-term development blueprint emphasizes gender equity in economic, social, and political spheres. It seeks to empower women through increased participation in leadership and decision-making, as well as enhancing women's access to education, healthcare, and economic opportunities.
- **The National Policy on Gender and Development (2019):** This policy provides a framework for mainstreaming gender issues into all aspects of government policy and program implementation. It focuses on addressing gender inequalities in areas such as education, health, and political participation.
- **Kenya Women's Economic Empowerment Strategy (2019–2023):** This strategy focuses on creating opportunities for women's economic participation by improving access to financial resources, land, employment, and entrepreneurial opportunities.

There are five recent national climate-related policies that are particularly relevant to the proposed project. Each of these is discussed in turn, followed by a table analyzing the level of gender integration of each of these policies.

Central Bank of Kenya - Guidance on Climate-Related Risk Management.⁸¹ **Passed 2019.** In October 2021, the Central Bank of Kenya issued this guidance under section 33(4) of the Banking Act, which empowers the Central Bank of Kenya to guide institutions to maintain a stable and efficient banking and financial system. The guidance recognizes that climate change poses a substantial risk and can pose an opportunity for the financial sector and requires banks to embed the consideration of the financial risks from climate change in their governance arrangements; incorporate the financial risks from climate change into their existing financial risk management practice; and develop an approach to disclosure on the financial risks from climate change.

⁸⁰ <https://www.ngeckenya.org/about/15/mandate>

⁸¹ <https://www.centralbank.go.ke/wp-content/uploads/2021/10/Guidance-on-Climate-Related-Risk-Management.pdf>

National Climate Change Action Plan 2018-2022 (NCCAP).⁸² (2018). This plan aims to strengthen the country's path towards sustainable, climate-resilient development while achieving low carbon climate resilient development. It builds on the previous Action Plan spanning the period 2013-2017. The NCCAP consists of three documents, including an Adaptation Technical Analysis Report (volume II), and a Mitigation Technical Analysis Report (volume III). The Plan seeks in particular to: 1) reduce risks to communities and infrastructure resulting from climate-related disasters such as droughts and floods, 2) Increase food and nutrition security through enhanced productivity and resilience of the agricultural sector in as low- carbon manner as possible, 3) Enhance resilience of the Blue Economy and water sector by ensuring access to and efficient use of water for agriculture, manufacturing, domestic, wildlife and other uses, 4) Increase forest cover to 10% of total land area; rehabilitate degraded lands, including rangelands; increase resilience of the wildlife and tourism sector, 5) Mainstream climate change adaptation into the health sector; and increase the resilience of human settlements, including improved solid waste management in urban areas, 6) Improve energy and resource efficiency in the manufacturing sector, and 7) Climate-proof energy and transport infrastructure; encourage electricity supply based on renewable energy; encourage the transition to clean cooking; and develop sustainable transport systems.

Kenya Climate Smart Agriculture Strategy 2017-2026.⁸³ (2017). The broad objective of the Kenya Climate Smart Agriculture Strategy 2017-2026 (KCSAS) is to adapt to climate change, build the resilience of agricultural systems, and minimize emissions for enhanced food and nutritional security and improved livelihoods. The specific objectives of the KCSAS are to (i) enhance the adaptive capacity and resilience of farmers, pastoralists, and fisher-folk to the adverse impacts of climate change; (ii) develop mechanisms that minimize greenhouse gas emissions from agricultural production systems; (iii) create an enabling regulatory and institutional framework; and (iv) address cross-cutting issues that adversely impact climate smart agriculture. Four broad strategic areas have been identified for KCSAS: (i) Adaptation and building resilience by addressing vulnerability to changes in rainfall and temperature, extreme weather events, and unsustainable land/water management and utilization; (ii) Mitigation of greenhouse gas emissions from key and minor sources in the agriculture sector; (iii) Establishment of an enabling policy, legal, and institutional framework for effective implementation of climate smart agriculture; and (iv) Minimizing the effects of underlying cross-cutting issues, such as human resource capacity and finance, which would potentially constrain the realization of climate smart agriculture objectives.

National Policy on Climate Finance.⁸⁴ (2016). This policy accomplishes several goals. First, it describes the current legal and policy framework for climate financing that is relevant for Kenya, focusing on both domestic and international sources. Second, it outlines the role that climate financing could play in each of Kenya's most important economic sectors (agriculture, forestry, energy, transport, trade, tourism, manufacturing, water and sanitation, disaster risk management, and research and innovation). Third, it describes the policy interventions the Kenyan government intends to make with respect to climate financing, including to establish a national Climate Change Fund, identify climate financing sources and create a national system for tracking them, enhancing Kenya's carbon trading system, and exploring the possibility of green bonds.

⁸² Government of Kenya. 2018. National Climate Change Action Plan (Kenya) 2018-2022. Ministry of Environment and Forestry, Nairobi, Kenya.

⁸³ Government of Kenya. 2017. Kenya Climate Smart Agriculture Strategy-2017-2026

⁸⁴ The National Treasury of Kenya. 2016. National Policy on Climate Finance.

National Adaptation Plan 2015-2030.⁸⁵ Passed in 2016. This document identifies Kenya's vulnerabilities to the effect of climate change, adaptation actions and implementation strategies. **National adaptation plan (NAP):** A NAP is the process developed by the UNFCCC to facilitate adaptation planning in LDCs and other developing countries as a means of identifying medium- and long-term adaptation needs. NAPs help countries to develop strategies and programmes to address those needs.⁸⁶

Table 1: Gender integration in climate-related policies in Kenya

Year Passed	Title of Climate Policy	Level of Gender Integration
Passed 2019.	Central Bank of Kenya - Guidance on Climate-Related Risk Management.	Gender-blind [no mention of gender/women].
Passed 2018.	National Climate Change Action Plan 2018-2022 (NCCAP).	Gender-responsive. Gender/women-related issues clearly integrated into the policy.
Passed 2017.	Kenya Climate Smart Agriculture Strategy 2017-2026.	Gender-sensitive. Gender/women-related issues framed as a cross-cutting theme.
Passed 2016.	National Policy on Climate Finance.	Minimally gender-sensitive – 1 mention of women as a vulnerable group to be prioritized.
Passed 2016.	National Adaptation Plan 2015-2030.	Gender-sensitive. Gender/women-related issues framed as a cross-cutting theme.

3.4 Sectoral Policies

3.4.1 Agriculture and Livestock

A recent review⁸⁷ on guidelines to mainstreaming gender and social equality and social inclusion during planning and budgeting processes in the agriculture sector in Kenya outlines the current policy environment for gender and social inclusion in the agriculture sector. The review found that specific gender and social inclusion policies exist that call for mainstreaming GESI into agriculture sectors. The Agricultural Sector Gender Policy (2013) aims to ensure gender equality in agriculture for enhanced and equitable productivity, food security, growth, and national development.' Moreover, the policy calls for gender-responsive programming and institutional transformation in the agricultural sector, strengthening institutional capacity to mainstream gender in the agricultural sector, promoting support and accountability for gender mainstreaming in the agricultural sector, and harnessing and coordinating sector efforts in gender mainstreaming for greater impact (GoK, 2013).⁸⁸ As a recent example of how this policy has materialized into development planning, Kenya's National Agriculture Investment Plan (NAIP) (2019-2024)⁸⁹ has several specific performance indicators tied to enhancing gender equality and women's empowerment, including the proportion of rural women that are empowered by agriculture, the proportion of women and men engaged in agriculture with access to financial services, and the growth rate of minimum dietary diversity of women. NAIP Flagships 1,2,3,4, and 7 commit to promote women as agricultural producers and decision makers, thereby enhancing their food security.

⁸⁵ Kenya National Adaptation Plan: 2015-2030, Government of Kenya, July 2016

https://www4.unfccc.int/sites/NAPC/Documents%20NAP/Kenya_NAP_Final.pdf

⁸⁶ Nelson, S. & Hill, C, 2019. *Gender in adaptation planning for the agriculture sectors: Guide for trainers*. Rome.

⁸⁷ Nyasimi, M., Ndetu, V., and Kidera, S. 2021. Guidelines to mainstreaming gender and social equality and social inclusion during planning and budgeting processes in the agriculture sector in Kenya.

⁸⁸ Nyasimi, M., Ndetu, V., and Kidera S. 2020. Review of the National and County Planning and Budgeting Processes with a Gender Responsiveness and Social Inclusion Lens in Agriculture Sector, Kenya.

⁸⁹ National Agriculture Investment Plan (NAIP). 2019-2024. TOWARDS SUSTAINABLE AGRICULTURAL TRANSFORMATION and FOOD SECURITY IN KENYA. <https://faolex.fao.org/docs/pdf/ken189052.pdf>

More recently, The Ministry of Agriculture, Livestock, Fisheries, and Cooperatives (MoALFC) has developed agricultural policies and strategies that address the inclusion of women, youth, and other special interest groups. For example, The Agricultural Sector Transformation and Growth Strategy (ASTGS, 2019-2029) is addressing the ‘unique challenges and opportunities for women and youth in the sector by incorporating tailored opportunities for these groups as an integral part of delivering the ASTGS’. The ASTGS further acknowledges that women and youth are underrepresented in agriculture and therefore, do not receive full benefits of the sector.⁹⁰ Policies and recommendations related to livestock are generally embedded in agricultural policies – however, Kenya is in the process of developing a NAMA for the dairy sector that is considering the role of gender to deliver benefits to women and other marginalized groups at large scale. The Kenyan Dairy NAMA includes “increasing on-farm dairy productivity through private sector investment in gender-inclusive extension services and fodder supply” as one of four components in its finalization.⁹¹

4. Overview of the Status of Women in Kenya

This section provides an overview of the status of women in Kenya. Information on specific sub-groups that are particularly vulnerable (e.g., girls, women-headed households, elderly women, and widows⁹²) are provided where possible.

4.1 Gender Parity

Gender parity in Kenya can be assessed using data from indexes and indicators that demonstrate women’s status in the country. These include the Human Development Index (HDI), Gender Development Index (GDI), Global Gender Gap Index (GGGI), and Gender Inequality Index (GII). Each of these indexes and their related indicators are described below.

Human Development Index (HDI), 0.575 HDI, 152 HDI ranking (2021)⁹³

The HDI is a summary measure for assessing long-term progress in three basic dimensions of human development: a long and healthy life, access to knowledge and a decent standard of living. Kenya’s HDI value for 2021 is 0.575— which puts the country in the Medium human development category— positioning it at 152 out of 191 countries and territories. Between 1990 and 2021, Kenya’s HDI value changed from 0.474 to 0.575, a change of 21.3 %. Between 1990 and 2021, Kenya’s life expectancy at birth changed by 2.8 years, mean years of schooling changed by 2.9 years and expected years of schooling changed by 3.1 years. Kenya’s GNI per capita changed by about 29.7 % between 1990 and 2021.

Kenya Human Development Index (HDI)⁹⁴

2021 HDI Value	0.575
HDI change from 2020	-0.003

⁹⁰ [ASTGS]. Agricultural Sector Transformation and Growth Strategy. 2019. <https://kilimo.go.ke/wp-content/uploads/2022/03/ASTGS-Abridged-version.pdf>

⁹¹ CGIAR Research Program on Climate Change, Agriculture and Food Security. 2017. CCAFS GSI-sponsored gender research informs Kenya Dairy NAMA to increase on-farm dairy productivity through private sector investment in gender-inclusive extension. Reported in Climate Change, Agriculture and Food Security Annual Report 2017. Outcome Impact Case Report. <https://hdl.handle.net/10568/121866>

⁹² Birech, J. K. (2019). The Contribution of the Government and other Stakeholders in enhancing the Socioeconomic Status of the Widows in Kenya. *Advances in Social Sciences Research Journal*, 6(10), 20–38. <https://doi.org/10.14738/assrj.610.7249>

⁹³ <https://hdr.undp.org/data-center/specific-country-data#/countries/KEN>

⁹⁴ <https://hdr.undp.org/data-center/human-development-index#/indicies/HDI>

Life expectancy at birth	61.4 years
Expected years of schooling	10.7 years
Mean years of schooling	6.7 years
Gross national income per capital (2017 PPP\$) ⁹⁵	4,474

Gender Development Index (GDI)

The GDI measures gender gaps in achievements in three basic dimensions of human development: health (measured by female and male life expectancy at birth), knowledge (measured by female and male expected years of schooling for children and mean years of schooling for adults aged 25 years and older) and living standards (measured by female and male estimated GNI per capita). It is a ratio of the female to the male HDI. The 2021 female HDI value for Kenya is 0.557 in contrast with 0.592 for males, resulting in a GDI value of 0.941, placing it into Group 3.⁹⁶

Kenya GDI⁹⁷

2021 GDI Value 0.941

GDI change from 2020 +0.004

	Female	Male	Gender Gap
HDI Value	0.557	0.592	-0.035
Life expectancy at birth	64.1 years	58.9 years	5.2 years
Expected years of schooling	10.3 years	11.1 years	-0.7 years
Mean years of schooling	6.1 years	7.3 years	-1.2 years
Gross national income per capital (2017 PPP\$) ⁹⁸	3,873	5,084	-1,211

Global Gender Gap Index (GGGI)⁹⁹ Report.¹⁰⁰ According to the 2022 edition of the GGGI report, Kenya ranks 57th out of 146 countries, with a score of 0.729 (where imparity=0 and parity=1). The GGGI uses four aggregated index indicators – Economic Participation and Opportunity, Educational Attainment, Health and Survival, and Political Empowerment to calculate Kenya’s GGGI score. The table below describes the disaggregated indicators that comprise each aggregated score.

GGGI Indicator	Rank	Score
Economic Participation and Opportunity	6 th	0.811
Labour-force participation rate %	9 th	0.939
Wage equality for similar work 1-7 (best)	56 th	0.685
Estimated earned income int’l \$1,000	10 th	0.826
Legislators, senior officials, and managers	12 th	0.985
Professional and technical workers	101 st	0.677
Educational Attainment	118 th	0.939
Literacy rate %	103 rd	0.920
Enrollment in primary education %	1 st	1.000

⁹⁵ Purchasing Power Parity

⁹⁶ <https://hdr.undp.org/data-center/specific-country-data#/countries/KEN>

⁹⁷ <https://hdr.undp.org/gender-development-index#/indices/GDI>

⁹⁸ Purchasing Power Parity

⁹⁹ The gender gap is defined as the discrepancy between men and women in terms of opportunities, status, attitudes, and other variables. The Gender Gap Index ranges between 0 and 1. Hence, a score of 1 reflects equality between men and women in the variables considered in the index, while a score of 0 shows significant inequality.

¹⁰⁰ <https://www.weforum.org/reports/global-gender-gap-report-2022/in-full/economy-profiles-5b89d90ea5>

Enrollment in tertiary education	122 nd	0.737
Health and survival	57 th	0.975
Sex ratio at birth	1 st	0.944
Healthy life expectancy	63 rd	1.045
Political empowerment	81 st	0.192
Women in parliament %	89 th	0.272

Indicators showing gender inequalities, such as the **Gender Inequality Index (GII)**. The GII measures gender inequalities (the loss in human development due to inequality between female and male achievements) in three key dimensions – reproductive health, empowerment, and labour market. A low GII value indicates low inequality between women and men, and vice-versa. Reproductive health is measured by maternal mortality ratio and adolescent birth rates; empowerment is measured by the shares of parliamentary seats held and population with at least some secondary education by each gender; and labour market participation is measured by the labour force participation rates for women and men. Kenya has a GII value of 0.506, ranking it 128 out of 170 countries in 2021.¹⁰¹

Kenya GII¹⁰²

2021 GII Value 0.506

GII change from 2020 -0.001

Maternal mortality rate 342 deaths/100,000 live births

Adolescent birth rate 64.2 births/100,000 women ages 15-19

	Female	Male	Gender Gap
Share of seats in parliament	23.2%	76.8%	-53.5%
Population with at least some secondary education (age 25 and older)	31.1%	37.7%	-6.6%
Labor force participation rate (age 15 and older)	71%	75.6%	-4.6%

Kenya – Gender Inequality Index by County¹⁰³

The GII scores for the 14 target counties are all higher than the national average, with the lowest county score being 0.62 (Bomet, Bungoma, Kericho, Trans Nzoia) and the highest county score being Migori (0.69).

County	Gender Inequality Index Score
Bomet	0.62
Bungoma	0.62
Busia	0.65
Homa Bay	0.67
Kakamega	0.63
Kericho	0.62
Kisii	0.63
Kisumu	0.63
Migori	0.69
Nandi	0.65
Nyamire	0.65
Siaya	0.65
Trans Nzoia	0.62
Vihiga	0.63

¹⁰¹ <https://hdr.undp.org/data-center/specific-country-data#/countries/KEN>

¹⁰² <https://hdr.undp.org/data-center/thematic-composite-indices/gender-inequality-index#/indicies/GII>

¹⁰³ The Kenya National Inequality Index per county report. 2015.

4.2 Population

As per the last census (2019), the Kenyan population was 23,548,056 males, 24,014,716 females, 1,524 intersex for a total of 47,564,296.¹⁰⁴ A 2021 World Bank population model estimated Kenya has a total female population of 26,726,429 (50.4%) and a total male population of 26,279,184 (50%). The population sex ratio (female/male) is 1.01, with a population growth rate of 2.25%.¹⁰⁵ The gender and youth populations for the 14 counties targeted in the CRLCSA project are listed below.

Gender population and percentage by county¹⁰⁶

National/County	Men		Women		Intersex		Total
	N	%	N	%	N	%	N (100%)
Kenya	23,548,056	49.5	24,014,716	50.5	1,524	0.003	47,564,296
Bomet	434,287	49.59	441,379	50.40	23	0.003	875,689
Bungoma	812,146	48.6	858,389	51.4	35	0.002	1,670,570
Busia	426,252	47.7	467,401	52.3	28	0.003	893,681
Homa Bay	539,560	47.7	592,367	52.3	23	0.002	1,131,950
Kakamega	897,133	48	970,406	52	40	0.002	1,867,579
Kericho	450,741	49.98	451,008	50.01	28	0.003	901,777
Kisii	605,784	47.82	661,038	52.18	38	0.003	1,266,860
Kisumu	560,942	48.54	594,609	51.46	23	0.002	1,155,574
Migori	536,187	48.03	580,214	51.97	35	0.003	1,116,436
Nandi	441,259	48.82	444,430	50.18	22	0.002	885,711
Nyamire	290,907	48.04	314,656	51.96	13	0.002	605,576
Siaya	471,669	47.49	521,496	52.51	18	0.002	993,183
Trans Nzoia	489,107	49.39	501,206	50.61	28	0.003	990,341
Vihiga	283,678	48.08	306,323	51.92	12	0.002	590,013

National Youth Population and Percentage by Gender¹⁰⁷

Youth Age Bracket	Male Youth		Female Youth		Intersex Youth		Total Youth
	N	%	N	%	N	%	N (100%)
15-19	2,686,264	50.8	2,599,442	49.17	151	0.002	5,285,857
20-24	2,112,690	47.2	2,334,778	52.5	206	0.004	4,447,674
25-29	1,839,543	47.8	2,014,859	52.27	153	0.004	3,854,555
30-34	1,698,678	47.57	1,871,887	52.42	154	0.004	3,570,719

Bomet County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	1,104	1,347	2,451
20-24	1,835	2,330	4,165

¹⁰⁴ Kenya Census 2019 Population by County and sub-County. <https://dc.sourceafrica.net/documents/119530-Kenya-Census-2019-Population-by-County-and-Sub.html>

¹⁰⁵ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

¹⁰⁶ Kenya Census 2019 Population by County and sub-County. <https://dc.sourceafrica.net/documents/119530-Kenya-Census-2019-Population-by-County-and-Sub.html>

¹⁰⁷ The 2019 Kenya Population and Housing Census. Volume III: Distribution of Population by Age and Sex. <https://www.knbs.or.ke/?wpdmpo=2019-kenya-population-and-housing-census-volume-iii-distribution-of-population-by-age-sex-and-administrative-units>

25-29	1,945	2,029	3,974
30-34	1,625	1,465	3,090

Bungoma County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	10,775	11,647	22,422
20-24	8,629	10,619	19,248
25-29	7,919	9,509	17,428
30-34	7,399	8,472	15,871

Busia County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	5,709	6,728	12,437
20-24	5,274	6,762	12,036
25-29	5,177	6,295	11,472
30-34	5,101	5,303	10,404

Homa Bay County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	5,365	6,652	12,017
20-24	5,344	7,745	13,089
25-29	5,425	6,758	12,183
30-34	5,080	5,668	10,748

Kakamega County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	9,872	10,602	20,474
20-24	10,070	11,283	21,353
25-29	8,544	9,295	17,839
30-34	7,547	8,294	15,841

Kericho County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	2,545	2,672	5,217
20-24	1,951	2,570	4,521
25-29	1,957	2,586	4,543
30-34	1,961	2,282	4,243

Kisii County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	7,112	8,249	15,361
20-24	8,220	10,558	18,778
25-29	8,277	9,654	17,931
30-34	7,130	7,645	14,775

Kisumu County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	21,593	25,274	46,867
20-24	23,455	30,061	53,516
25-29	23,382	26,765	50,147
30-34	21,442	21,760	43,202

Migori County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	8,699	10,225	18,924
20-24	7,800	11,047	18,847
25-29	7,602	9,124	16,726
30-34	6,777	7,286	14,163

Nandi County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	52,592	50,073	102,665
20-24	35,899	37,502	73,401
25-29	27,630	30,924	58,554
30-34	27,607	31,543	59,150

Nyamira County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	32,881	32,295	65,176
20-24	17,981	21,804	39,785
25-29	15,605	21,550	37,155
30-34	16,537	24,766	41,303

Siaya County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	4,246	5,262	9,508
20-24	4,246	5,758	10,004
25-29	4,127	5,037	9,164
30-34	3,705	4,105	7,810

Trans Nzoia County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	9,505	10,312	19,817
20-24	8,729	10,427	19,156
25-29	8,392	9,334	17,726
30-34	7,760	8,121	15,881

Vihiga County Youth Population by Gender

Youth Age Bracket	Male Youth	Female Youth	Total Youth
	N	N	N (100%)
15-19	3,193	3,284	6,477
20-24	2,577	2,860	5,437
25-29	2,221	2,579	4,800
30-34	2,027	2,462	4,489

Women-headed households in Kenya make up 31% of all households as per the 2021 reporting of the Demographic and Health Survey (DHS).¹⁰⁸

The counties targeted by CRLCSA all have majority ethnic communities. Bomet, Kericho, and Nandi are predominately Kalenjin, Kisii and Nyamira are predominately Kisii, Bungoma, Busia, Kakamega, Trans-Nzoia, and Vihiga are predominately Luhya, and Homa Bay, Kisumu, Migori, and Siaya are predominately Luo.

Counties where various ethnic communities form the majority¹⁰⁹

Ethnic communities	Counties targeted by CRLCSA
Kalenjin	Bomet, Kericho, Nandi
Kisii	Kisii, Nyamira
Luhya	Bungoma, Busia, Kakamega, Trans-Nzoia, Vihiga
Luo	Homa Bay, Kisumu, Migori, Siaya

4.3 Poverty and Food Insecurity

Wealth inequalities in Kenya are extreme, with 15.2% of all people living below 50% of median income, with a .1 % gender gap between women and men (15.2% and 15.1%, respectively).¹¹⁰ As of 2018, the proportion of men, women, and children of all ages living in poverty in all its dimensions according to national definitions was 53% national average, with 54% of women and 52% of men living in poverty.¹¹¹ In assessing multidimensional poverty, the national average was 38.9%, with higher rural rates of poverty (48.4%) compared to urban rates (20.3%).¹¹² Girls aged 0-17 years and elderly women over the age of 70 experience the highest levels of poverty at national level (41.5% and 39.1%, respectively).

Proportion of female population below the national poverty line %¹¹³

Age	% Below national poverty line
0-17 years	41.5%
18-35 years	29.1%
36-59 years	32.5%
60-69 years	36.2%
70+ years	39.1%

¹⁰⁸ <https://data.worldbank.org/indicator/SP.HOU.FEMA.ZS?locations=KE>

¹⁰⁹ Nyabira, B.C. and Ayele, Z.A., 2016. The state of political inclusion of ethnic communities under Kenya's devolved system. Law, Democracy & Development, 20, pp.131-153.

¹¹⁰ <https://data.unwomen.org/country/kenya>

¹¹¹ *ibid.*

¹¹² *ibid.*

¹¹³ *ibid.*

Over half of Kenya's population are food insecure, with the 2022 State of Food Security and Nutrition in the World Report (SOFI) categorizing Kenya as a low-income food deficit country¹¹⁴. The Food Insecurity Experience Scale (FIES), which rates the prevalence of moderate to severe food insecurity in the population, rates 56.5% of the population as experiencing moderate food insecurity and 19.1% of the population with severe food insecurity.¹¹⁵ National census data from 2019 reported that the prevalence of anemia among women of reproductive age (18-49) is 29%.¹¹⁶

4.4 Health

Women have a higher life expectancy compared to men (65 years for women compared to 60 years for men).

Women's life expectancy compared to men.

Women	65 years (2020 most recent year) ¹¹⁷
Men	60 (2020 most recent year) ¹¹⁸

Additional women's health status statistics, including maternal mortality rate, infant mortality rate, malnutrition (stunting and wasting) rate, and obesity rate is listed in the table below.

Women's Health Statistics

Maternal mortality rate	377 per 100,000 live births (2014 most recent year) ¹¹⁹
Infant mortality rate	31 per 1,000 live births (2020 most recent year) ¹²⁰
Malnutrition - stunting rate	Prevalence of stunting among children under the 5 years of age is 26.9% female and 32.8% male. The national average is 29.9%. ¹²¹
Malnutrition-wasting rate	National average is 6.7%, female 6.1% and male 7.2% (2016 most recent year) ¹²²
Obesity rate	National average is 4.9%, 6.5% female and 4.7% male (2016 most recent year) ¹²³
Proportion of births attended by a skilled health personnel	70.2% (2016 most recent year) – an 8.4% increase from 61.8% in 2014. ¹²⁴
Universal health coverage	56% as of 2022. ¹²⁵

¹¹⁴ FAO, IFAD, UNICEF, WFP and WHO. 2022. The State of Food Security and Nutrition in the World 2022. Repurposing food and agricultural policies to make healthy diets more affordable. Rome, FAO. <https://doi.org/10.4060/cc0639en>

¹¹⁵ *ibid.*

¹¹⁶ <https://data.worldbank.org/indicator/SH.ANM.ALLW.ZS?end=2019&locations=KE&start=2000&view=chart>

¹¹⁷ <https://data.worldbank.org/indicator/SP.DYN.LE00.FE.IN?locations=KE>

¹¹⁸ <https://data.worldbank.org/indicator/SP.DYN.LE00.MA.IN?locations=KE>

¹¹⁹ <https://data.worldbank.org/indicator/SH.STA.MMRT.NE?locations=KE>

¹²⁰ <https://data.worldbank.org/indicator/SP.DYN.IMRT.IN?locations=KE>

¹²¹ <https://data.unwomen.org/country/kenya>

¹²² *ibid.*

¹²³ *ibid.*

¹²⁴ *ibid.*

¹²⁵ The African Women's Development and Communication Network. (2022). Sexual Reproductive Health and Rights at a Glance: Fact Sheet for Kenya.

Number of new HIV infections per 1,000 uninfected population	1.4 total average (1.3 male and 1.5 female) in 2018 – a decrease from an average of 5 (5 male and 5 female) in 2012. ¹²⁶
Proportion of women of reproductive age (aged 15-49 years) who have their need for family planning satisfied with modern methods	70.7% (2014 most recent year) ¹²⁷
Mean age of women at birth of first child (years)	28.7 years (2022 most recent year) ¹²⁸

A 2021 report by the Commission for Sexual and Reproductive rights¹²⁹ found that 36% of women and girls under 25 years of age had the lowest level of understanding of reproductive rights. The majority of women and girls, irrespective of age, are unaware of the constitutional provisions on sexual reproductive health rights, including abortion. Only 12% of girls aged 12-19 and 38% of women ages 21-30 are knowledgeable about menstruation. Girls aged 15 and below are least knowledgeable on how to prevent unintended pregnancies. Women aged 21–30 are most knowledgeable on how to prevent unintended pregnancies. 80% of these women use contraception, compared with 20% who reported abstaining from sex. Most married women expressed that they experienced a lack of autonomy, choice, or decision-making in negotiating for safe sex or not to have sex. Respondents reported that it is up to women, and not men, to take the necessary precautions to prevent pregnancies and STIs.

4.5 Gender-Based Violence

Twenty-one percent (21%) of girls and women aged 15-49 years have undergone female genital mutilation/cutting. Of these, 11.4% were between 15-19 years of age, 14.7% were between 20-24 years of age, 18% were between 25-29 years of age, 22.9% were between 30-34 years of age, 27.8% were between 35-39 years of age, 32.1% were between 40-44 years of age, and 40.9% were between 45-49 years of age, as per 2014 reporting.¹³⁰ At national level, 4.4% of women aged 20-24 years of age who were married or in a union were married before age 15, while 22.9% were married before age 18, as per 2014 reporting.¹³¹ The proportion of women and girls aged 15 years and older subjected to sexual violence by persons other than an intimate partner in the previous 12 months is 22.7% national average (2014 reporting year).¹³² GBV is a critical issue in Kenya, across its many forms. For example, 47.1% of ever-partnered women and girls aged 15 years and older were subjected to physical, sexual, or psychological violence by a current or former intimate partner in the previous 12 months, with 36.9% physical violence, 13.3% sexual violence, 32.4% emotional/psychological violence reported as per 2014.¹³³ According to the 2022 GGI, 39.4% of women will experience GBV in their lifetime.¹³⁴

¹²⁶ <https://data.unwomen.org/country/kenya>

¹²⁷ *ibid.*

¹²⁸ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

¹²⁹ <https://reproductiverights.org/new-report-women-and-girls-in-kenya-lack-access-to-comprehensive-information-on-sexual-and-reproductive-health-rights-despite-constitutional-framework/>

¹³⁰ <https://data.unwomen.org/country/kenya>

¹³¹ *ibid.*

¹³² *ibid.*

¹³³ *ibid.*

¹³⁴ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

4.6 Literacy and Education

Nationally, Kenya has a 5% gender gap in adult literacy rates, with only 80% of women over 15 years old able to read compared to 85% of men.¹³⁵ Although the gender gap in youth literacy rates and reading comprehension has closed between boys and girls¹³⁶, a 3.1% gender gap remains in mathematics comprehension at the Grade 2/3 level.¹³⁷ Gender gaps also exist in Technical and Vocational Education Training (TVET), with only 43.2% (210,795) young women compared to 56.8% (276,906) young men enrolled. Interestingly, these trends are reversed for older adults, with women making up almost 70% (143,585) of those enrolled compared to 31.3% (65,497) of men.¹³⁸

Adult Literacy Rate (% females and males aged 15 years and older)¹³⁹

Women%	80% (2021 most recent year)
Men%	85% (2021 most recent year)

Adult illiteracy Rate (%females and males aged 15 years and older)

Women%	20% (based on 2021 estimate)
Men%	15% (based on 2021 estimate)

Educational Status of Girls and Boys

Youth Literacy Rates (% females and males aged 15-24)

Girls %	89% (2021 most recent year)
Boys %	88% (2021 most recent year) ¹⁴⁰

Quality Education¹⁴¹

Level	National Average	Girls	Boys
Grade 2/3 mathematics	42.1%	43.6%	40.7%
Grade 2/3 reading	53.1%	57.1%	49.4%
Primary reading	44.3%	47.1%	41.6%

Sex disaggregated literacy in target counties¹⁴²

County Name	Women%	Men%	% Difference
<i>Bomet</i>	-	-	-
<i>Bungoma</i>	80	88	-8
<i>Busia</i>	66	85	-19
<i>Homa Bay</i>	74	91	-17
<i>Kakamega</i>	77	87	-10
<i>Kericho</i>	-	-	-
<i>Kisii</i>	79	92	-13
<i>Kisumu</i>	84	94	-10

¹³⁵ UNESCO Institute for Statistics (UIS). UIS Stat Bulk Data Download Service. Accessed November 24, 2022. apiportal.uis.unesco.org/bdd.

¹³⁶ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

¹³⁷ <https://data.unwomen.org/country/kenya>

¹³⁸ <https://data.unwomen.org/country/kenya>

¹³⁹ UNESCO Institute for Statistics (UIS). UIS.Stat Bulk Data Download Service. Accessed October 24, 2022. apiportal.uis.unesco.org/bdd.

¹⁴⁰ <https://data.worldbank.org/indicator/SE.ADT.1524.LT.MA.ZS?locations=KE>

¹⁴¹ <https://data.unwomen.org/country/kenya>

¹⁴² Source: PFS document, missing data for Bomet, Kericho, Nandi, and Trans Nzoia.

<i>Nandi</i>	-	-	-
<i>Migori</i>	77	92	-15
<i>Nyamira</i>	83	91	-8
<i>Siaya</i>	69	91	-22
<i>Trans Nzoia</i>	-	-	-
<i>Vihiga</i>	79	90	-11

4.7 Employment and Economic Participation of Women

The proportion of youth (aged 15-24 years) not in education, employment or training was 18.3% national average, with female 21.6% and male 15.1%, as per 2019 most recent year.¹⁴³ The proportion of informal employment in non-agriculture employment, by sex, is 83.4% national average, with 43.5% female and 56.5% male, as per 2019 reporting year.¹⁴⁴ The proportion of adults (15 years and older) with an account at a bank or other financial institution or with a mobile-money service provider is 81% national average (2018 reporting year).¹⁴⁵ As per 2016 reporting year, child labor exists in Kenya, with a national average of 13.1% of children aged 5-17 years engaged in child labor.¹⁴⁶ According to the 2022 GGGI index, for unemployed adults (% of labour force aged 15-64), females made up 5.85% and males made up 5.21%, with a national average of 5.53%.¹⁴⁷

Employment Information

Labour force participation rate %	National average for labour force participation rate is 71% for females, 75.6% for males, with a gender gap of -4.6% (2021 most recent year) ¹⁴⁸
Unemployment rate %	National average for unemployment is 7.4%, with 9.6% female and 5.3% male (2016 most recent year) ¹⁴⁹

4.8 Women's Representation and Political Participation

Women received the right to vote in 1963, and as of 2023 there has never been a female head of state, and the seats held in the upper house is currently 30.9%.¹⁵⁰ Women's representation and meaningful participation in the decision-making process trails men at national and county level, despite Kenya's Constitution of 2010 requiring a minimum of one-third representation of either gender in all elected and public posts. While the political participation rate for voting at national level is 47% women, and 52% men,¹⁵¹ the proportion of seats held by women in national parliaments and local governments are lower. As of 2018, less than one-third of the National Parliament were women (31.8% of National Assembly (MPs) and 31.3% of Senators).¹⁵² For local government at the county assembly level, women's representation

¹⁴³ *ibid.*

¹⁴⁴ *ibid.*

¹⁴⁵ *ibid.*

¹⁴⁶ *ibid.*

¹⁴⁷ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

¹⁴⁸ <https://hdr.undp.org/data-center/thematic-composite-indices/gender-inequality-index#/indicies/GII>

¹⁴⁹ <https://data.unwomen.org/country/kenya>

¹⁵⁰ *ibid.*

¹⁵¹ Nasimiyu, M.L. and E. Mariano. (2022). Sociocultural factors influencing women's participation in political leadership in Kakamega County, Kenya. International Journal of Research and Innovation in Social Science, 5(5) ISSN 2454-6186 <https://www.rsisinternational.org/journals/ijriss/Digital-Library/volume-6-issue-6/650-656.pdf>

¹⁵² <https://data.unwomen.org/country/kenya>

rates were significantly lower, with only 6.4% of governors and 14.9% of Deputy Governors.¹⁵³ Only Members of the County Assembly (MCA) representation had slightly over one-third (34.2%) of women MCA representatives.¹⁵⁴

Thus while gender equity quotas are enshrined in Kenyan legislation, patriarchal customs and socio-cultural norms hamper women's ability to meaningfully and equally participate in political culture and top decision-making roles.¹⁵⁵ Women who do enter politics and leadership positions at national level, as well as within local and community-based organizations (including agricultural cooperatives and producer organizations) often face backlash from men, including Gender-Based Violence (GBV) and intimidation, highlighting the continuing challenge in challenging discriminatory attitudes and practices towards women's role outside the home.¹⁵⁶ These dynamics also exist in the private sector, as according to the 2022 GGI Index, 13.2% of firms had female majority ownership, and 18.1% of firms had females as top managers.¹⁵⁷

5. Gender issues in the project implementation sites

This section explores the gender dynamics and differential needs of women, men, youth, and other vulnerable groups in the 14 counties targeted by the CRLCSA project. The gender equality and social inclusion issues presented are focused on agricultural production in the LVEB in the context of a changing climate. The existing norms, societal expectations, and stereotypes that underpin gender roles and relations and influence multi-dimensional vulnerability are discussed in each sub-section related to the constraints that women, youth, PLWD, and other marginalized groups face. The capacities of these groups as sources of resilience to climate change and catalysts for change are also discussed, where possible.

The first sub-section discusses the overall gender dynamics and inequalities in crop and livestock systems, followed by more specific discussion on issues related to women's access and control over resources in the project implementation sites – these include land use and ownership, agricultural finance, access to trainings, technology, and information, and time as a limited resource considering women's domestic, productive, and community roles and caretaking responsibilities. Primary data from the quantitative Climate Risk and Value Chains Consultation Survey collected for the CRLCSA feasibility study is presented in this section to highlight gender equality in access to climate information, gender differentiation in types of climate information needed, and intra-household decision-making in crop, livestock, and poultry production.

The second sub-section details more in-depth discussion on the specific value chains targeted in the CRLCSA project and presents the findings from the qualitative study on gendered participation, performance, and empowerment in the six value chains targeted for the project.

The third sub-section discusses gender and social dynamics in producer organizations and cooperatives. Primary data from the quantitative cooperative survey is presented and discussed in this section. This

¹⁵³ *ibid.*

¹⁵⁴ *ibid.*

¹⁵⁵ Nasimiyu, M.L. and E. Mariano. (2022). Sociocultural factors influencing women's participation in political leadership in Kakamega County, Kenya. *International Journal of Research and Innovation in Social Science*, 5(5) ISSN 2454-6186 <https://www.rsisinternational.org/journals/ijriss/Digital-Library/volume-6-issue-6/650-656.pdf>

¹⁵⁶ Mwambi, M., Bijman, J., & Galie, A. (2021, July). The effect of membership in producer organizations on women's empowerment: Evidence from Kenya. In *Women's Studies International Forum* (Vol. 87, p. 102492). Pergamon.

¹⁵⁷ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

includes data on the types of producer organizations and associated value chains women, men, girls, and boys are members of, in each target county.

5.1 Gender and social inequalities in crop and livestock systems in the LREB

5.1.1 Crop production

Crop production in the LREB is generally separated into crops that are produced and consumed at home versus crops that are produced and marketed/sold for profit. While these could potentially be the same species (e.g., vegetables, African leafy vegetables, fruit trees), crops that are sold for profit are generally clustered under high value commodity crops, such as maize, coffee, and tea. These crops require a significant amount of land to cultivate at scale, and given women's lack of land rights in Kenya, they generally do not have ownership or decision-making ability for these crops. However, it is quite common for women to work as laborers across all types of crop production. For some crops, there is a more rigid gender division of labor in production that determines who does what, and when. For example, post-harvesting and marketing activities are usually the purview of men, particularly for higher value commodity crops. Women are generally tasked with the cultivation of 'domestic or food security crops' for household consumption, which are often grown on small plots of land that have been allocated to them by male members of their household. However, because the process of sub-division of land is through patrilineal descent and is a limited resource, arable land is reserved for son's wives; young girls almost never have land allocated to them for crop production for either home-based or market-based plant species.

The gender dimensions of crop production in the LREB have been documented in the context of Climate-Smart Agriculture (CSA) practices.¹⁵⁸ A review of key issues found that gendered differences in crop adaptation strategies were closely linked to husbands' and wives' roles and responsibilities, social norms, risk perceptions, and access to resources. Due to a lack of access to climate information, income, land, access to financial resources, access to digital technologies, women are less likely than men to act in response to climate shocks, including adapting CSA practices in crop production. In the context of crop production in a changing climate, the prioritization of gender and agriculture issues should consider the following dimensions of gender inequalities in developing inclusive interventions: (1) participation in decision-making at all levels; (2) work burden; (3) access to and use of productive resources such as agroclimatic information, technology, livelihood incomes, and credit; and (4) collective action to address and mitigate climate impacts.¹⁵⁹

5.1.2 Livestock

In Kenya, livestock is reared under different agricultural systems (i.e., pastoral, and mixed-crop farming). In the LREB, mixed-crop farming is the most common. Investment in livestock value chains, including intensified dairy and poultry production in LREB, is ongoing, with a growing body of research in Kenya showing that empowering women in livestock value chains leads to healthier communities, animals, and

¹⁵⁸ Brisebois, A., Hallstrøm Eriksen, S. E., & Crane, T. A. The politics of governing resilience: Gendered Dimensions of Climate-Smart Agriculture in Kenya. *Frontiers in Climate*, 86. <https://www.frontiersin.org/articles/10.3389/fclim.2022.864292/full>

¹⁵⁹ Huyer, S. and Gumucio, T., 2020. Going Back to the Well: Women, Agency, and Climate Adaptation. *World Journal of Agriculture and Soil Science*, 5(3), pp.1-3.

environments.¹⁶⁰ This aligns with global analyses that show investing in gender-equitable livestock production has the potential to build more economically and climate-resilient agricultural systems, increase the availability of nutritionally rich animal-source foods in vulnerable communities, and restore ecosystems globally.¹⁶¹

In the LREB, cows, their related commodities (i.e., bulked milk and value-added products such as yogurt), and the commercialization process in collecting and bulking milk at cooperative societies are all influenced by sociocultural norms that grant ownership and authority to elder men.¹⁶² While women and youth participate in the dairy value chain, it is more common for them to participate in informal nodes outside of structured and formal marketplaces and export markets.¹⁶³ For example, male youth may use motorbikes to transport milk to local markets, and women may sell their milk to neighbors or local intermediaries. While participating in informal markets generally helps women better control the proceeds from their dairy labor, they also face high social culpability and danger from engaging in these types of sales, as there has been a ‘crackdown’ by the Kenyan Dairy Board (KDP) in recent years to outlaw such practices of selling unpasteurized milk in the name of public health and safety.¹⁶⁴

Poultry and small ruminant production tend to be women’s preferred value chains, as these species require less intensive labour and resource inputs, and they face a lower barrier to entry regarding capital and discriminatory socio-cultural norms that discourage women and youth engagement.¹⁶⁵ A recent study found that livestock raising across a range of species (chickens, goats, sheep, rabbits, cows) increased the overall wellbeing and resilience of Kenyan households through increased incomes, food security, social benefits, and time and labor savings. However, these benefits largely promoted long-term household resilience rather than immediate gains. Livestock ownership also had major time and labor costs, such as the general daily care of livestock and provision of water and fodder, which were overwhelmingly borne by women and children. Climate shocks in the LREB, including drought, would further burden women and youth by making them travel further distances for water, and may affect their ability to produce fodder crops, or purchase feed inputs at reasonable costs (assuming crop scarcity would drive the price of feeds, which are already expensive for smallholders to afford). Despite this investment, women had limited livestock ownership rights, decision-making power, control over income, or access to meat.¹⁶⁶

Livestock ownership requires significant investments of household time and labor, which disproportionately burden women. Prevailing gender inequalities may therefore constrain the net benefit of livestock ownership for many women and their households in some contexts. Livestock development programs must assess both program benefits and costs at multiple levels to ensure that women’s participation in livestock production leads to improved individual and household outcomes.¹⁶⁷

¹⁶⁰ Walingo, M.K., 2009. Role of livestock projects in empowering women smallholder farmers for sustainable food security in rural Kenya. *African Journal of Food, Agriculture, Nutrition and Development*, 9(7).

¹⁶¹ FAO. 2013. Understanding and integrating gender issues into livestock projects and programmes: A checklist for practitioners. <https://www.fao.org/3/i3216e/i3216e.pdf>

¹⁶² Tavenner, K., & Crane, T. A. (2018). Gender power in Kenyan dairy: cows, commodities, and commercialization. *Agriculture and Human Values*, 35(3), 701-715.

¹⁶³ Tavenner, K., Crane, T.A. and Saxena, T., 2021. “Breaking Even” under Intensification? Gendered Trade-Offs for Women Milk Marketers in Kenya. *Rural Sociology*, 86(1), pp.110-138.

¹⁶⁴ *ibid.*

¹⁶⁵ Dumas, S. E., Maranga, A., Mbullo, P., Collins, S., Wekesa, P., Onono, M., & Young, S. L. (2018). “Men are in front at eating time, but not when it comes to rearing the chicken”: unpacking the gendered benefits and costs of livestock ownership in Kenya. *Food and Nutrition Bulletin*, 39(1), 3-27. <https://doi.org/10.1177/0379572117737428>

¹⁶⁶ *ibid.*

¹⁶⁷ *ibid.*

5.1.3 Women's work burden and time as a resource

In Kenya, employment and national accounts data do not capture non-market activities (where women predominate) and therefore fail to demonstrate the full contribution of women to the care economy and the extent of the female work burden. Women in Kenya are time-poor because of their dual roles in the care economy and the labour market. On average, women in Kenya work longer hours (12.9 hours) each day than men (8.2 hours), yet they earn less because more of the hours they work are not remunerated. In Kenya girls spend more time on non-remunerated work in the form of household work compared to boys. Women in rural areas of Kenya are burdened with household tasks such as pounding grain, collecting firewood, fetching water, tilling land, planting, weeding, and harvesting, looking after livestock, caring for children, and cooking for the family. Only 30 % of households in Kenya have access to potable water and fetching water alone can account for up to 40 % of a woman's day, taking from 3 to 5.25 hours. Men in Kenya spend 258 minutes per day doing agricultural work compared to 372 minutes for women.¹⁶⁸

Childcare is also an important time burden for women in Kenya. Women's labour time and flexibility are therefore more constrained than men. The disproportionate cost borne by women in terms of child-rearing and family responsibilities also limits the time that they can devote to economic activities, which means they may have less time to develop and grow their businesses. In addition, because women tend to be time-poor (combining family duties with running their businesses) and have limited access to financial resources, they are less likely to register their businesses or insure them and also to access business and trade opportunities. As a result, they are not able to access loans from formal financial institutions or even recover after a fire, theft, or any other setback, which limits the growth of their enterprises and their participation in national and intra-regional markets.¹⁶⁹

5.1.4 Land use and ownership

Land can be used in many ways, including for subsistence farming, cash crop farming, or as collateral for credit to finance other businesses. Access to, control over, and ownership of land is influenced by diverse factors that include gender, age, and marital status. According to the 2022 GGGI index, women have restricted rights for widows and daughters, and uneven rights for access to land assets and access to non-land assets.¹⁷⁰ Land in Kenya is mainly controlled by male household heads on the assumption that the rights are held in trust for all in the household. Women hold only about 1% of registered land titles in Kenya, with around 5 to 6% of registered titles held in joint names, meaning that 99 % of the land is in men's hands.¹⁷¹ Therefore, even though women have access to use the land, most of them do not have freedom to make decisions about its use. These decisions are made by the husband or male relatives, or by the community if the land is communally owned. In most cases, these decisions are made in the absence of women.

Even under the new Constitution, which allows women to inherit land, customary land law still prevails because it is hard to change people's mindsets, and women still must fight for the same land rights as men, especially at the household level. It is even worse for unmarried or divorced women who have returned to their parents' home, since women are expected to be married and inherit land from their

¹⁶⁸ Kiriti-Nganga, T. W. (2015). Gender and trade liberalization in Kenya: The case of women retail traders.

¹⁶⁹ Kiriti-Nganga, T. W. (2015). Gender and trade liberalization in Kenya: The case of women retail traders.

¹⁷⁰ https://widgets.weforum.org/GGGR/edition-22-ranking/pdf/2022/gggr_index_2022_072_KEN.pdf

¹⁷¹ Kenya Land Alliance and FIDA, 2006.

husbands. In such cases, the brothers or male relatives may disinherit them. Although some women try to fight for their rights, they find it difficult because legal procedures are costly and in most instances the disputes lead to family feuds that most women want to avoid. According to customary land law, women in Kenya have usufruct rights to land only when their husbands are alive, but after that, male relatives have the right to take the land from them.¹⁷²

5.1.5 Agricultural finance

According to a 2019 financial access (FinAccess) household survey supported by The Kenya Institute for Public Policy Research and Analysis (KIPPRA), access to agricultural finance is generally low for both women and men in Kenya.¹⁷³ Across the country only 14.66% of the agricultural population had access to agricultural finance (both formal and informal sources). Of these, 9.61% accesses agricultural finance through formal prudential sources, and 5.3% access finance from “excluded sources”, comprising social networks and individual arrangements, while a whopping 84.81% of the agricultural population does not use any form of agricultural finance. Women mainly source finance for agricultural operations from non-prudential sources and informal sources such as family and friends. This could be explained by lack of control over assets that could be used as collateral in accessing credit from formal sources. In assessing who has access to agricultural finance by gender and age cohorts, the study found that young women (16-34 years of age) had the lowest levels of access, while men 65 years and older had the highest rate of access. Women in rural areas were also more likely to access loans from informal sources compared to women in urban areas. Compared to men, women had lower levels of financial literacy and access to agricultural finance information, with women aged 16-34 years of age having the lowest levels of financial literacy and access to agricultural finance information. Female youth are often denied access to agricultural finance due to low savings, existing debts, lack of collateral, and bad credit history.

5.1.6 Access to agricultural training, climate information, and technologies

Degree of access to agricultural training, climate adaptation information, and technologies are a function of multiple intersecting factors.¹⁷⁴ Evidence from across crop and livestock sectors suggests that there are gender differences in access to agricultural training and extension services, with women and youth less likely to receive visits from extension staff or attend trainings. There are several explanations for this trend: For one, a male bias often appears to exist in extension service provision based on the belief that men are decision makers and female farmers only marginal producers. Larger farms, which tend to be operated by men, are more likely to be targeted by extension agents due to economies of scale and higher efficiency in service provision. Furthermore, time constraints related to the double burden of household tasks and farm work, as well as social norms affecting their mobility, may negatively affect women’s ability to participate in farmer training.¹⁷⁵

Climate change is a significant threat to agriculture-related livelihoods, and its impacts amplify prevailing gender inequalities. Climate information services (CIS) are crucial enablers in adapting to climate change

¹⁷² Kiriti-Nganga, T. W. (2015). Gender and trade liberalization in Kenya: The case of women retail traders.

¹⁷³ KIPPRA. 2021. Women’s Access to Agricultural Finance in Kenya. Policy Brief No.03/2020-2021.

¹⁷⁴ <https://data.unwomen.org/country/kenya>

¹⁷⁵ International Coffee Organization. 2018. Gender Equality in the Coffee Sector: An Insight Report from the International Coffee Organization. 45pp. <https://www.ico.org/documents/cv2017-18/icc-122-11e-gender-equality.pdf>

and managing climate-related risks for smallholder farmers.¹⁷⁶ However, initiatives aimed at improving access to climate information services appear not to recognize women as an information market that requires different outreach strategies. For example, a study in Bungoma County in Western Kenya similarly shows that critical sectoral environmental policies do not effectively address women's climate change adaptation needs, potentially due to the disengagement of governmental services.¹⁷⁷ Moreover, given that there are distinct gendered preferences for how to receive CIS information (see table below), it is crucial that interventions design dissemination pathways informed by gender analysis – preferably at county level.

Gender differences in preferences for CIS dissemination pathways¹⁷⁸

Dissemination pathway	Overall (% yes)	Husbands (% yes)	Wives (% yes)	Difference in % point
Radio	75.00	68.59	81.41	– 12.82**
Extension officers	31.41	42.31	20.51	21.79***
Television	19.73	22.15	17.31	4.84*
Social groups	16.03	12.18	19.87	– 7.69**
Other farmers	6.73	7.05	6.41	0.64
Local leaders	3.21	5.13	1.28	3.85**
Printed media—news paper	2.89	5.13	0.64	4.49*
Field days	1.28	1.92	0.64	1.28
NGOs	0.32	0.64	0.00	0.64
Number of sources of CIS (mean)	1.88	1.91	1.85	0.06
Trust score on sources of CIS (mean)	0.68	0.65	0.70	– 0.05**
N	312	156	156	

*p < 0.1, **p < 0.05, ***p < 0.01, following a z-test for equality of proportions and t-test for difference in means

5.1.7 Enabling Environment for Gender Mainstreaming in Building the Climate Resilient Value Chain

176 Ngigi, M.W., Muange, E.N. Access to climate information services and climate-smart agriculture in Kenya: a gender-based analysis. *Climatic Change* 174, 21 (2022). <https://doi.org/10.1007/s10584-022-03445-5>

177 Lupao, C. W. 2016. The efficacy of Kenya's critical sectoral environmental policies in meeting women's climate change mitigation and adaptation needs: Bungoma County, Kenya (Doctoral dissertation), University of Nairobi, Nairobi, Kenya.

178 Ngigi, M.W., Muange, E.N. Access to climate information services and climate-smart agriculture in Kenya: a gender-based analysis. *Climatic Change* 174, 21 (2022). <https://doi.org/10.1007/s10584-022-03445-5>

While efforts are being made to create an enabling environment for gender mainstreaming in building climate resilient value chains, many challenges remain. There's a lack of data that specifically examines the gendered impacts of climate change on agriculture. This makes it difficult to understand the unique challenges women farmers face and tailor interventions accordingly.

A significant challenge is the absence of comprehensive gender-disaggregated data in agricultural policies and planning. Without this data, it is difficult to track the specific needs and contributions of women in agriculture, which often leads to their exclusion from program benefits. While gender-disaggregated data is more readily available at national level, counties have not yet fully implemented disaggregation in their decentralized programs and data collection protocols. Beyond the collection of basic gender-disaggregated data on social indicators, full gender mainstreaming in agricultural extension and climate information services would require the monitoring of a larger suite of indicators¹⁷⁹. The Kenya Climate Smart Agriculture Strategy also notes that there continues to be “Misrepresentation and misunderstanding of gender issues” among civil society and other actors in the agriculture space, which could be in part fueled by this lack of data. This is also recognized in the CSA implementation plan (2023-2027) which advocates for the establishment of “gender sensitive CSA knowledge at the community, county and national levels”.¹⁸⁰

Additionally, there is limited representation from women within the agriculture services themselves, hindering county level administrations’ ability to define and provide gender-tailored services. The Climate Smart Agriculture Strategy notes that while there is an objective to establish “Social Protection and Safety net programmes to cushion women, youth and vulnerable groups from the impacts of climate change”, these are prevented by a lack of participation by these same vulnerable groups in the definition of services that are destined for them.

Although many counties in the Lake Region Economic Bloc (LREB) have developed climate change policies and action plans, these policies often focus primarily on environmental and climate resilience without explicitly addressing the gender-specific needs of women. The policies tend to prioritize broader climate resilience goals but lack detailed strategies for incorporating women into decision-making processes or addressing how climate impacts women differently than men. For example, while counties like **Bomet** and **Busia** have comprehensive climate change policies and governance structures, they have yet to integrate specific gender-focused actions that would address women's vulnerabilities in agriculture and resource management¹⁸¹. The documents indicate that county officials and climate change committees have been trained on general climate resilience, but there is little evidence of gender-specific training or sensitization. As further illustration, only 7 of the existing 13 climate change action plans (Kericho, Nyamira, Kakamega, Siaya, Baringo, Kisii, Kisumu) include gender-sensitive actions explicitly tailored for women, particularly focusing on agriculture, energy, and capacity-building. The other counties either mention vulnerable groups broadly or do not provide explicit gender-sensitive strategies for women¹⁸².

Extension services and agricultural training programs in Kenya still often fail to reach women due to social norms, time constraints, or lack of targeted outreach. The International Food Policy Research Institute (IFPRI) highlights that “women farmers are often not reached by extension services, either because they are not targeted or because the timing and content of the services are not suitable for them”. Additionally,

¹⁷⁹ FAO, State of Agriculture : Women and Agriculture, 2011.

¹⁸⁰ Government of Kenya, CSA Implementation Strategy 2023-2027.

¹⁸¹ Based on an analysis of county development policies, 2023.

¹⁸² Based on an analysis of 13 county climate change action plans.

women farmers may face language barriers and gender discrimination from male extension workers, which further discourages their participation¹⁸³. Many counties have an overall shortage of staff, particularly in sectors critical for delivering agricultural extension and climate services. Women often face additional barriers as extension workers, due to low female representation in technical staff. This creates gaps in gender-sensitive service delivery, as male-dominated staff may lack the training and awareness to address women's unique needs¹⁸⁴.

Structural barriers in agriculture and agribusiness severely limit women's ability to achieve equity, equality, and representation. One of the most significant issues is limited access to land and financial resources. Women often lack land ownership rights, which hinders their ability to use land as collateral for loans, preventing them from expanding their agricultural operations or investing in technology. Furthermore, financial institutions tend to view women as higher-risk borrowers due to the smaller size of their enterprises and their limited access to capital, perpetuating the financial exclusion of women in agribusiness.

In addition, gender bias in leadership and decision-making roles within agricultural cooperatives and businesses further entrenches inequality. Women are underrepresented in leadership positions and key decision-making processes, often excluded by societal norms that prioritize men in these roles. This imbalance in representation prevents women from influencing policies and initiatives that affect agricultural value chains, further marginalizing their role in the sector. Cultural expectations often relegate women to informal or subsistence farming activities, limiting their participation in more profitable markets¹⁸⁵.

Lastly, access to markets, finance and agricultural extension services is another structural issue. Women frequently lack the networks and market access necessary to sell their products on a larger, more profitable scale. There is also a definite bias (both cultural and structural) against women within financial institutions, which feeds the perception that women are "less solvable" than men, for all the reasons cited above.

5.1.8 LREB Case 1: Gender Analysis of Climate Risk and Value Chains Consultation Survey

The following gender analysis draws on the Climate Risk and Value Chains Consultation Survey (completed as part of the formulation process of the project), which queried 115 stakeholders (85 males and 28 females) on several gender-specific questions across three domains: gender equality in access to climate information, gender differentiation in types of climate information needed, and gendered intra-household decision-making in crop, livestock, and poultry production. Given the sample size of the survey participants is not large enough to yield inferential statistics, descriptive statistics are reported by gender of the respondent.

Value chain representation among male and female respondents (n=115) are shown in the table below. Most female respondents were from the dairy value chain (12), followed by African leafy vegetables (6), fruit trees (5), poultry (4), and coffee (1). Male respondents were involved in all 6 value chains, with the majority in African leafy vegetables (31), followed by dairy (18), poultry (17), fruit trees (12), coffee (3), and tea (2).

¹⁸³ https://www.afaas-africa.org/wp-content/uploads/2023/05/GENDER-POLICY-BRIEF_BamanyakiP_November-2022.pdf

¹⁸⁴ County Capacity Assessment, 2022.

¹⁸⁵ <https://gender.cgiar.org/news/status-women-agriculture-and-food-systems-persistent-gaps-and-promising-solutions>

		Which food commodity do you mainly work with/have expertise in?							Total
		African Leafy Vegetables	Coffee	Dairy	Fruit trees	Poultry	Tea	Other Vegetables	
Gender	Female	6	1	12	5	4	0	0	28
	Male	31	3	18	12	17	2	3	86
Total		37	4	30	17	21	2	3	114

Equal access to climate information. For the first question, “Do you think women have equal access to climate information as compared to men during input supply, food production and harvesting?”, respondents were able to choose yes, no, or unsure. Cross tab analysis by gender revealed that most male (39.5%) and female (42.9%) respondents reported that women do not have equal access to climate information (34/86 males, and 12/28 females). Nearly one-third of male respondents (27/86) and 25% of female respondents (7/28) reported that women do have equal access to climate information. Males and females reported they were unsure whether women had equal access to climate information too, with 16.5% of males (14/86) and 14.3% (4/28) of females reporting uncertainty.

		Do you think women have equal access to climate information as compared to men during input supply, food production and harvesting?				Total
		Did not answer	No	Unsure	Yes	
Gender	Female	5	12	4	7	28
	Male	11	34	14	27	86
Total		16	46	18	34	114

Survey respondents who answered “yes” also had the opportunity to fill-in what they thought were the key different needs in accessing climate information between women and men. These responses are disaggregated by sex and listed in the table below.

Female Responses	Male Responses
Access to mobile phones, affiliation with groups.	Further trainings on how to interpret the information received
Interest and attitude.	Higher workload for women reduces time available to access climate information,
	Men dominate,
	Men have a wider social network compared to women,
Most women in my area have interest in farming than men.	Most of them [women] are not farmers,
Smart phone and register with service providers.	Most women are the producers, yet information is majorly given via smartphones which they lack.
	Most women rely on men to make decisions.
	Social media and community speakers.

	The type of crops to be planted, for example men are interested in crops that earn them money, while women will just be engaged in enterprises that thrive in their region.
	The women will know when to start preparing seeds, have stocks wood fuel, store water for future use.
	Timing and mode of dissemination.

Gender differences in types of climate information needed. For the second question, “Do you think women need different types of climate information as compared to men?”, respondents were able to choose yes, no, or unsure. Cross tab analysis by gender revealed that most male and female participants answered “no.” These results indicate that over 60% of women respondents (17/28) and almost 55% of male respondents (47/86) did not think women need different types of climate information as compared to men. Interestingly, only 4 women (14.3%) responded that women need different information, while almost 30% of men (24/86) responded that women need different information.

		Do you think women need different types of climate information as compared to men?				
		No response	No	Unsure	Yes	Total
Gender	Female	5	17	2	4	28
	Male	11	47	4	24	86
Total		16	64	6	28	114

Respondents who answered “yes” also had the opportunity to fill-in what they thought were the key different needs and capacities in assessing and using climate information between men and women.

Female Responses	Male Responses
Best to use SMS, WhatsApp group because of the excess workload on woman. And indicate the benefits of doing this both to the family and economic benefits.	Accurate and efficient weather and climate information.
Diseases are likely to occur and it's the women who stay with children and as well plan what vegetables to grow in case of any climate uncertainties (pests and diseases).	Changes in crop production.
Educated men and women have equal abilities in accessing the information. Growing of annual crops requires more immediate climate forecasting, while perennial crops are more forgiving, but soil moisture data would help.	Community speakers.
The mobility is constraint so the access to radio and phone will assist in delivering the information to women.	Education, expand more sources of information.
Women are always busy doing their productive work.	How to manage farm activities.
	Information Boards in market centers near the grain flour millers and fliers at the dispensary.
	Most women in rural areas are not as endowed as men in terms of technology access and use, with most being illiterate and often overworked with domestic

	chores. Therefore, they need climate information that's simplified, gender focused and availed in a format easily accessible to them.
	Need information affecting household livelihood.
	Organize women seminars on climate risks, deliberate target dissemination of information to women.
	The information should relate more directly to activities undertaken by women and emphasis placed on how helpful the information is to them and their activities.
	Their greatest focus usually is harvest.
	Women are at home (indoors) most of the time and radio, telephone SMS alerts are good for them. Men may need displays on billboards, etc.

Gendered intra-household decision-making in crop, dairy, and poultry production. For the third question, respondents were asked, *"In your experience, **who in the households** targeted by your organization **makes decisions** on agricultural changes using climate information?"* This question was asked for crop, livestock, and poultry production activities.

Results for the application of climate information for decision-making in crop production indicate that over 50% of both men and women participants responded, "together as a family." This is perhaps unsurprising given that the question did not further disaggregate the specific plant species cultivated (e.g., crops for home consumption versus crops for sale/commercial production). These findings could also be explained by socio-cultural norms that view intrahousehold decision-making as cooperative rather than conflictual, and/or that the question itself did not consider differences in respondents' perceptions between the process of 'consultation' between spouses in households compared to who in the household has the power to make the final decision regarding crop production.

		In your experience, who in the households targeted by your organization makes decisions on agricultural changes using climate information? [Application of climate information for decision-making in crop production]				
		No response	Men only	Together as a Family	Women only	Total
Gender	Female	5	7	16	0	28
	Male	15	18	50	3	86
Total		20	25	66	3	114

Results for the application of climate information for decision-making in dairy production were similar to those for crop production, in that the majority of respondents answered dairy production decisions are made together as a family. For both of these categories (crops and dairy), it is interesting that no women responded that decisions on agricultural changes using climate information are made by 'women only.' This could indicate that women view themselves as having less autonomy/agency in the decision-making process, or potentially have different understandings regarding the 'consultation' process between spouses, as explained in brief above. These initial results indicate that future quantitative studies must further disaggregate the types of crops, as well as follow up with more qualitative questioning related to

how producers make decisions together as a family – does this mean one family member has access to climate information and ‘informs’ the other members (e.g., tells them what to do based on their knowledge?) Or is the decision-making process more iterative, or take into consideration the decisions of other men and women farmers in their communities?

		In your experience, who in the households targeted by your organization makes decisions on agricultural changes using climate information? [Application of climate information for decision-making in dairy production]				
		No response	Men only	Together as a Family	Women only	Total
Gender	Female	6	10	12	0	28
	Male	20	20	43	3	86
Total		26	30	55	3	114

Results for the application of climate information for decision-making in poultry production show that most of both male and female respondents answered decisions were made “together as a family.” This was followed by women only, and by men only. These results echo the findings for both dairy and crop production decisions, although both women and men reported higher levels of ‘women only’ than for the other two types of production. For future lines of questioning related to gender and decision making on agricultural changes using climate information, it would be helpful to assess the level of commercialization in the household (e.g., size of production, income over the last 12 months) to see if there is a ‘ceiling for success’ related to women’s involvement in poultry production, and whether they can maintain decision-making power only if their production and sales are ‘supplementary’ to other sources of male-derived income in the home. Additionally, future studies could also clarify what types of decisions, over which nodes of the value chain are made, disaggregated by sex- and age.

		In your experience, who in the households targeted by your organization makes decisions on agricultural changes using climate information? [Application of climate information for decision-making in poultry production]				
		No response	Men only	Together as a Family	Women only	Total
Gender	Female	7	2	13	6	28
	Male	21	7	45	13	86
Total		28	9	58	19	114

5.2 Gendered participation, performance, and empowerment in the six CRLCSA priority value chains

This section’s analysis integrates data and studies from the desk review, as well as data collected as part of a qualitative study during the CRLCSA project proposal development in 2022. The selected value chains

(coffee, tea, fruit trees, indigenous leafy vegetables, poultry, and dairy) have distinct gender dynamics, with women and men having different roles and responsibilities across the value chain. Across these value chains, there are differences in the level of gender market integration across three domains: acceptability of women's participation in the value chain, their position in the value chain, and in their access to resources to support benefit from engaging in agricultural value chains.¹⁸⁶ These differences have important implications for CRLCSA's practices for gender and social inclusion.

Studies in East Africa, including Kenya have found that gender issues permeate every node of the value chain, and that existing gender and social inequalities have real consequences regarding who are the 'winners' and 'losers' in value chains, including instances where women are 'weak winners'¹⁸⁷ or are settled with win-lose dynamics via their participation in value chains.¹⁸⁸ These gendered trade-offs work to encourage or discourage women from engaging in certain aspects of value chains, including informal versus formal marketing venues, whether to pursue increasing marketization and commercialization of products, or whether to pursue entrepreneurship and small businesses.¹⁸⁹

Most pertinent to the study sites are gender issues in the commercialization of agricultural and livestock products. A recent study in East Africa found that as smallholder farmers increasingly marketed and commercialized their production (and as profits increased) women were more likely to lose out on proceeds and decision-making they had previously had access to.¹⁹⁰

Despite specificities across sectors, similar gender barriers limit the benefits women receive from agricultural and livestock production. These constraints, which occur at multiple levels, include: the invisibility and undervaluation of rural women's labor and their disproportionately heavy labor burdens, limited and precarious control over resources, discriminatory social and gender norms that hinder women's voice and influence in decision-making and governance, and exclusionary institutions such as resource-user groups and extension and data systems. Thus, to achieve transformative change in food systems, changes in each sector are required in women's agency, access to and control over resources, gender norms, and policies and governance.¹⁹¹ Such changes can improve dietary outcomes, gender equality and women's empowerment, economic and livelihood outcomes, resilience and environmental

¹⁸⁶ Rubin, D., Boonabaana, B., & Manfre, C. (2019). Building an inclusive agriculture: Strengthening gender equality in agricultural value chains. Annual Trends and Outlook Report: Gender Equality in Rural Africa: From Commitments to Outcomes, 83-96. <https://rtbfoods.cirad.fr/content/download/4186/31890/version/1/file/Building+an+inclusive+agriculture+-+Strengthening+gender+equality+in+agricultural+value+chains.pdf>

¹⁸⁷ Bain, C., Ransom, E., & Halimatusa'diyah, I. (2018). 'Weak winners' of Women's empowerment: The gendered effects of dairy livestock assets on time poverty in Uganda. *Journal of Rural Studies*, 61, 100-109.

¹⁸⁸ Ihalainen, M., Shaikh, S., Mujawamariya, G., Mayanja, S., Adetonah, S., Tavenner, K., & Elias, M. (2021). Promise and contradiction: value chain participation and women's empowerment. Advancing gender equality through agricultural and environmental research: past, present and future, 147-188.

¹⁸⁹ Tavenner, K., Crane, T. A., & Saxena, T. (2021). "Breaking Even" under Intensification? Gendered Trade-Offs for Women Milk Marketers in Kenya. *Rural Sociology*, 86(1), 110-138.

¹⁹⁰ Tavenner, K., Van Wijk, M., Fraval, S., Hammond, J., Baltenweck, I., Teufel, N., ... & Manda, L. (2019). Intensifying inequality? Gendered trends in commercializing and diversifying smallholder farming systems in East Africa. *Frontiers in Sustainable Food Systems*, 3, 10.

¹⁹¹ Njuki, J., Eissler, S., Malapit, H., Meinzen-Dick, R., Bryan, E., & Quisumbing, A. (2022). A review of evidence on gender equality, women's empowerment, and food systems. *Global Food Security*, 33, 100622. https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/9132/fss_briefs_review_evidence_gender_equality.pdf?sequence=3&isAllowed=y

outcomes. Closing gender gaps across sectors requires multipronged strategies that simultaneously engage these four change pathways to lift structural barriers to inequality.¹⁹²

5.2.1 Coffee

Coffee is responsible for an estimated 15% of employment in the agricultural sector, having declined from 21% in the mid-1990s. The national production landscape comprises of both smallholder and large-scale estates at 60% and 40%, respectively. Current estimates suggest that there are more than 700,000 smallholder farmers. These smallholder farmers have an average farm size of only 0.25 ha and per hectare yield at well below 400kgs of clean coffee. At the beginning of the chain are producers, comprising both estates and smallholders. Smallholders are usually organized into cooperatives. The co-operatives are functionally mobilization units for primary processing and marketing.¹⁹³ Gender equity within coffee cooperatives suggests that there are significant gaps between women's participation and concrete benefit from coffee production.¹⁹⁴ While women perform more than two-thirds of the work in coffee farming in Kenya, accounting for up to 70% of labor in production, starting from the farm level, cooperatives, and processing level – yet they represent less than 5% of leadership roles in coffee cooperatives in the country.¹⁹⁵

Acceptability of women's participation in the coffee value chain. While coffee is traditionally considered to be a 'man's cash crop', recent efforts have been put towards mainstreaming gender and youth in smallholder sustainable coffee in Kenya. However, women and youth participation has largely concentrated on their labor contribution to coffee production, as over 95% of coffee farms are owned by men. Thus, the groups who produce Kenya's coffee have little or no access to the income from the commodity. These dynamics have worked to create apathy among women and youth in respect to active and voluntary engagement in coffee production.¹⁹⁶

Gender Roles in Kenya's Coffee Sector¹⁹⁷

Gender Roles	Participation in the roles (% ratio)			
	Men	Women	Boys	Girls
Cultivating	5.4	74	9.3	11.3
Picking	7.1	54.4	18.5	20
Sorting at home	3.6	60	7	29.4
Sorting at the factory	9.3	45	14.3	31.4
Taking to the factor for processing	4.1	44	22.2	26.7

¹⁹² Elias, Marlène; Zaremba, Haley; Tavenner, Katie; Ragasa, Catherine; Paez Valencia, Ana Maria; Choudhary, Afrina; Haan, Nicoline C. de. 2022. Beyond crops: Towards gender equality in forestry, fisheries, aquaculture, and livestock development. Presented at the CGIAR GENDER Science Exchange, Nairobi, 12-14 October 2022. Rome: Alliance of Bioversity International and CIAT. <https://hdl.handle.net/10568/125598>

¹⁹³ Morris, A., Douglas, B., & Charles, A. (2015). Mainstreaming gender and youth in smallholder sustainable coffee supply chain in Kenya. *Journal of Economics and Sustainable Development*, 6(18), 76-86.

¹⁹⁴ Dijkdrenth, E. (2015). Chapter 7: Gender equity within Utz certified coffee cooperatives in Eastern Province, Kenya. In *Coffee certification in East Africa: impact on farms, families and cooperatives* (pp. 489-502). Wageningen Academic Publishers.

¹⁹⁵ Onyalo, P. O. (2019). Women and agriculture in rural Kenya: role in agricultural production. *International Journal of Humanities and Social Science*, 4(4), 1-10.

¹⁹⁶ *ibid*.

¹⁹⁷ Liaison Consulting. 2011. Cited in Morris, A., Douglas, B., & Charles, A. (2015). Mainstreaming gender and youth in smallholder sustainable coffee supply chain in Kenya. *Journal of Economics and Sustainable Development*, 6(18), 76-86.

Taking to the market	32.7	54.3	5	8
Collecting money from bean sales	87.4	12.6	0	0
Owning coffee farms	95.2	4.8	0	0

However, there has been progress in Western Kenya towards closing these gender gaps. For example, in Nandi County, in the 'Coffee by Women' program, women are offered training in leadership, confidence building and self-development. Women, men, and youth are trained together, as the program aims to "include as many farmers as possible on the journey towards the coffee of the future."¹⁹⁸ These types of practices can be potentially gender transformative, if the content of the trainings actively challenges social norms that reinforce gender-and age-based discriminatory practices. A transformative change in gender relations requires changes in women's attitudes and capacities in the relationship between men and women, but also progress at the institutional and structural levels.¹⁹⁹

Gendered positions in the coffee value chain. In Kisii County, age and gender norms shape gendered positions in cooperative societies. For example, the average age for coffee cooperative society members was 57 years old, and that norms around elder men's dominance hampered young women from ascending to leadership positions in the coffee cooperative.²⁰⁰ Fairtrade certification has impacted women and men farmers in Kenyan cooperatives differently due to gender roles and highly separated divisions of labor.²⁰¹ A recent study suggests that while Fairtrade positively impacted the incomes of women, there was only marginal alteration of current inequitable gender roles, nor did Fairtrade challenge women's subordination to a significant degree – thus Fairtrade interventions only partially empower women and address gender inequalities.²⁰² Women, including female-headed households, young women, and women with disabilities are disadvantaged in terms of coffee production, as these subgroups often lack the land, capital, resources, networks, and technical skills necessary to farm, and that the institutional context for coffee production is highly masculinized, with elder men predominating in terms of land rights and derivative income from the sale of coffee.

Gendered Access to resources in the coffee value chain. Institutional arrangements within coffee cooperatives in Kenya have also mediated the role of gendered market orientation. A recent case study from Kericho county found that women do not have the same market opportunities, have limited access to resources, and have less say over coffee production's planning and supply segment. Women continually face significant barriers to their equal participation, such as limited access to land, capital, and technical information.²⁰³ These barriers are caused by intersecting challenges of socio-cultural practices, land tenure system that privileges men, and discriminatory bylaws that hinder women and youth participation in cooperative membership.

Findings from CRLCSA fieldwork: "Coffee production is dominated by adult males. In Nandi County there is a growing involvement of the youth and women courtesy of an on-going program, called "Women in

198 Lauri, J., & Bäckström, H. (2019). Coffee by women: the 'duty of ethical enjoyment'. *Cultural Studies*, 33(5), 866-887.

199 Bilfield, A., Seal, D., & Rose, D. (2020). Brewing a more balanced cup: supply chain perspectives on gender transformative change within the coffee value chain. *International Journal on Food System Dynamics*, 11(1), 26-38.

200 Ngeywo, J., Basweti, E., & Shitandi, A. (2015). Influence of gender, age, marital status and farm size on coffee production: a case of Kisii County, Kenya. *Asian Journal of Agricultural Extension, Economics & Sociology*, 5(3), 117-125.

201 Rölander, F. (2016). We are Women in Coffee!: An explanatory case study of Fairtrade's gendered impact on female and male farmers of a Fairtrade certified Kenyan coffee cooperative.

202 *ibid*.

203 <https://weeffect.org/stories/empowering-women-in-the-coffee-value-chain/#:~:text=For%20a%20long%20time%2C%20coffee,level%2Ccooperatives%20and%20processing%20level.>

Coffee". Value chain activities include production, picking coffee cherries and delivering to pulping stations/factories. Primary processing is carried out by farmers' coop societies. Inputs are purchased from agro-dealers, sometimes the cooperatives buy in bulk and supply to farmers/ democratic farmer societies. Marketing contracts are made with actors registered with the Coffee Board of Kenya. Climate effects such as prolonged drought have affected productivity over time. Technologies to address this include irrigation and agro-forestry."

5.2.2 Tea

In Kenya, smallholder farmers grow tea in contract with the Kenya Tea Development Authority (KTDA). KTDA is an independent and private tea enterprise owned by smallholder tea farmers and offers management and professional services to individual factories and companies. The Tea Board of Kenya (TBK) also works together with the farmers by offering management, professional services, leadership on production and quality. However, even though women play a dominant role in tea production, there is no mention of gender issues in either the KTDA or TBK strategic plans.²⁰⁴ Despite the significant role women play in smallholder tea production, there are extreme gender inequalities in access to and control over the benefits accrued from tea. A recent study showed that women within male-headed households were discriminated against in access to and control over income derived from tea, however women-headed households had access to and control over tea-derived income.²⁰⁵ The Kenya Tea Development Authority (KTDA) listed culture, widowhood, level of education, were identified as barriers to gender equality. The study established that women from male-headed households had more roles in tea production than their husbands. The women were assigned the roles that were tedious and took many hours like tea plucking and transportation, while the men undertook seasonal roles like tea planting and pruning.

Gender roles in tea production²⁰⁶

Activity	Men	Women
Land preparation	X	X
Tea planting	X	X
Weeding		X
Pruning	X	
Picking		X
Tea transportation		X

Factors such as KTDA Policy of registering men as tea owners, community norms, lack of title deeds by women, gender-biased culture, and illiteracy made it difficult for women to materially benefit from tea production. WHH were able to access and control tea benefits because they had acquired the tea owners' rights after the demise of their husbands and/or inheritance from their fathers.²⁰⁷

Gendered Positions in the tea value chain. A recent survey in Nyamira county of 110 Tea SACCO members across five tea factories ranked barriers to women's participation as: lack of shares in the tea factory; lack

204 Njeri Kibere, E., Kimani, E. N., & Lodiaga, J. M. (2018). Gender Dynamics in the Access and Control of Benefits Accrued from Tea Farming in Kisanji Division, Gatundu District. Applied Science Reports, Forthcoming.

205 *ibid.*

206 *ibid.*

207 *ibid.*

of land ownership rights; fear to run against men in elections; low levels of education; gender roles; and lack of experience in politics.²⁰⁸ Women, including female-headed households, young women, and women with disabilities are disadvantaged in terms of tea production, as these subgroups often lack the land, capital, resources, networks, and technical skills necessary to farm, and that the institutional context for coffee production is highly masculinized, with elder men predominating in terms of land rights and derivative income from the sale of coffee.

Gendered Access to resources in the tea value chain. Land registration and contestations over land ownership between women and men have been documented for tea production in Kenya.²⁰⁹ Gender dynamics influence access and control of benefits accrued from tea farming.²¹⁰ A recent study found that intra-household gender inequalities can bring discontent to women farmers, which leads them to neglect the proper care of the tea crop. For example, manure or fertilizer which is intended for the tea cash crop is diverted by women without the knowledge of men to food crops which they feel they have control over. This affects the quality and quantity of the tea grown.²¹¹

Findings from CRLCSA fieldwork: “Tea is grown in mainly large estates and smallholder farms. Value chain actors include smallholder farmers (majorly older males) farming an average of 2 acres. Farmers source inputs from large factories on credit advanced via financial providers such as SACCOs. Climate smart technologies in practice include retention ditches to harvest water especially on sloppy land – to prevent runoff and increase infiltration. Terraces are also in place to curb erosion.”

5.2.3 Fruit Trees

Acceptability of women’s participation in the fruit tree value chain. Fruit tree, African leafy vegetables, and poultry are value chains that are traditionally viewed as appropriate for women – women customarily have had greater autonomy and decision-making power in these activities and are more able to retain income derived from marketing these products. However, this acceptability is under the assumption that the profits from these activities will remain small, and not compromise the position of the male head of household as the primary ‘breadwinner’. Thus, the social acceptability of women’s participation in these value chains is largely perceived based on the scale of production and marketization. For example, in the banana-value chain, women’s ability to participate in income sharing involved complex intra-household dynamics and relationships around decision-making.²¹² There are specific high-value fruit species that are framed as being more appropriate for men, such as exotic avocados which have a major export market in Kenya.²¹³ However, new initiatives such as the African Continental Free Trade Area (AfCFTA) could

208 Onyancha, R. M., & Wambu, C. K. (2021). Socio-Economic Factors Affecting Young Women Participation in Tea SACCOs Politics in Nyamira South Sub-County, Nyamira County, Kenya. *African Journal of Co-operative Development and Technology*, 6(2), 36-48

209 Davison, J. (2019). Who Owns What? Land Registration and Tensions in Gender Relations of Production in Kenya. In *Agriculture, Women, and Land* (pp. 157-176). Routledge.

210 Njeri Kibere, E., Kimani, E. N., & Lodiaga, J. M. (2018). Gender Dynamics in the Access and Control of Benefits Accrued from Tea Farming in Kiganjo Division, Gatundu District. *Applied Science Reports*, Forthcoming.

211 Mwangi, M. J. (2015). Role of Gender Equality in the Enhancement of Tea Quality and Quantity in Kenya: A Survey of Farmers Served by Gathuthi Tea Factory of Nyeri County. *International Journal of Academic Research in Business and Social Sciences*, 5(7), 373-380.

212 Nyabaro, V., Mburu, J., & Hutchinson, M. (2019). Factors enabling the participation of women in income sharing among banana (*Musa spp.*) producing households in South Imenti, Meru County, Kenya. *Gender, technology and development*, 23(3), 277-292

213 Muriithi, B., & Kabubo-Mariara, J. (2022). The Dynamics and Role of Gender in High-Value Avocado Farming in Kenya. *The European Journal of Development Research*, 34(5), 2272-2304.

provide gender-and youth-responsive solutions to export market barriers through targeted investments.²¹⁴

Gendered Position in the fruit tree value chain. The results of a recent study in Kenya's avocado value chain suggest that in the more commercialized and well-developed export chains, upgrading strategies vary for the different typologies of women (women-headed households versus women in male-headed households). While women in women-headed households may require limited efforts such as tailoring financial products to their needs or providing interlinked services coupled with prompt payment for their produce to allow them to produce quality fruits and access lucrative markets, women in male headed households need institutionalization of gender-sensitive policies in the governance of producer groups to enable them to upgrade as chain integrators and chain owners. In the less commercialized domestic avocado value chain, limited efforts may be required to upgrade women along the chain, but the need to change from the less marketable local variety to exotic variety is likely to alter women's position, thereby calling for the need to institutionalize gender-sensitive policies in the governance of existing organized groups and use the groups as a platform to introduce the new variety.²¹⁵

Gendered Access to resources in the fruit tree value chain. While fruit trees grown as subsistence crops have traditionally been female dominated and controlled, the commercialization of fruit trees has led to a dynamic shift in production with more men taking up active roles in the value chain (for example, bananas, avocados). This has led to the displacement of women from the high value chain with men taking up dominant roles. A study on these displacement dynamics in the banana value chain showed that the participation of women in income sharing was partially determined by the presence of off-farm income, so that women in male-headed households had a higher probability in taking part in household decision-making.²¹⁶

5.2.4 African Leafy Vegetables

Acceptability of women's participation in the African leafy vegetables value chain. Traditional food crop marketing, including African Leafy vegetables, tends to follow distinct gender roles, with women having greater social acceptability in cultivating and retaining the sales from these plants than other value chains.²¹⁷ However, recent evidence suggests that although African leafy vegetables have traditionally been considered a woman's crop, women vendors in emerging markets are not necessarily empowered to earn equal income as men.

Gendered Positions in value chain. Compared to crops that have historically been commercialized, there is less documentation on gendered positions and dynamics in the African leafy vegetables value chain. However, findings from the CRLCSA fieldwork suggest that the value chain is dominated by women over the age of 35. It is unclear how gendered production of African leafy vegetables changes with different

²¹⁴ Africa Renewal. (2023). AfCFTA: Reaping the benefits of the world's most youth and women-friendly trade agreement <https://www.un.org/africarenewal/magazine/february-2023/afcfta-reaping-benefits-world%E2%80%99s-most-youth-and-women-friendly-trade-agreement#:~:text=The%20AfCFTA%20will%20address%20the,undermining%20the%20continent's%20overall%20development>

²¹⁵ Oduol, J. B. A., Mithöfer, D., Place, F., Nang'ole, E., Olwande, J., Kirimi, L., & Mathenge, M. (2017). Women's participation in high value agricultural commodity chains in Kenya: Strategies for closing the gender gap. *Journal of rural studies*, 50, 228-239.

²¹⁶ Nyabaro, V., Mburu, J., & Hutchinson, M. (2019). Factors enabling the participation of women in income sharing among banana (*Musa spp.*) producing households in South Imenti, Meru County, Kenya. *Gender, technology and development*, 23(3), 277-292

²¹⁷ Handschuch, C., & Wollni, M. (2016). Traditional food crop marketing in sub-Saharan Africa: does gender matter?. *The Journal of Development Studies*, 52(3), 343-359.

scales of production (i.e., whether women are still primarily responsible for cultivation, harvesting, and post-harvesting activities, or whether some or all of these activities would be outsourced to hired labor).

Gendered Access to resources in the African leafy vegetables value chain. The resources required to engage in African Indigenous Vegetable (AIV) seed systems have been documented in Western Kenya.²¹⁸ Seed access was a constraint, even though most seeds used by farmers are self-produced. Income from selling AIV seed differed significantly depending on gender, with *men earning more than twice as much as women*. This study²¹⁹ demonstrates that the constraints farmers face in accessing high-quality AIV seed can vary significantly between species and over short distances. This study speaks to the importance of using localized information to develop programs for improving informal seed systems and continuing to employ gender-sensitive and transformative activities.

Findings from CRLCSA fieldwork: *“In Nyakach, vegetables such as cowpeas, spider plant, osuga (black nightshade). Cow peas are dominant followed by Osuga. Vegetables occupy at least one acre of the average 1-acre family farm. The value chain is dominated by adults above 35 years with majority being women. Production is hampered by unpredictable seasons that are dependent on rainfed production. This makes markets erratic and affects demand. Lack of cold storage facilities forces farmers to sell immediately after production (when prices are low) to avoid food loss. Climate smart technologies in place include sun drying, use of organic manure, use of certified seed and on-farm practices such as spacing.”*

5.2.5 Poultry

Acceptability of women’s participation in the poultry value chain. Women’s participation in poultry production in the tropics is significant.²²⁰ In Western Kenya, virtually every household keeps small flocks of between 5-30 chickens.²²¹ Poultry production, of both indigenous chickens and commercial varieties is crucial to the wellbeing of rural households, as production (and egg production) are important sources of inexpensive animal-based protein and cash income for resource-poor households. These species are highly adaptable and are less capital and labor-intensive than other livestock species, such as large ruminants. Customarily a feminized livestock value chain, women’s participation in poultry is highly accepted and expected of rural women. While women are increasingly taking on enterprise ownership in indigenous poultry production,²²² there are emerging challenges regarding their participation as entrepreneurs.²²³ Indeed, in line with gender dynamics in crop value chains, as poultry production becomes more commercially successful, there is a growing risk that women will lose out on previously controlled incomes and decision-making power.²²⁴

²¹⁸ Pincus, L., Croft, M., Roothaert, R., & Dubois, T. (2018). African indigenous vegetable seed systems in western Kenya. *Economic Botany*, 72(4), 380-395.

²¹⁹ *ibid*.

²²⁰ Alemayehu, T., Bruno, J. E., Getachew, F., & Dessie, T. (2018). Socio-economic, marketing and gender aspects of village chicken production in the tropics: A review of literature. *ILRI Project Report*.

²²¹ Okitoi, L. O., Ondwasy, H. O., Obali, M. P., & Murekefu, F. (2006). Gender issues in poultry production in rural households of Western Kenya. *KARI-Kakamega, Ministry of Livestock and fisheries, Kakamega*.

²²² Karanja-Lumumba, T. Gender Dynamics in Indigenous Poultry Enterprise Ownership and Newcastle Disease Control in South Eastern Kenya. *Animal Production Society of Kenya*.

²²³ Biwott, K. D., Musalia, O. J., Martha, P. C., & Awori, H. N. (2015) Role of poultry production in promoting peri-urban small-scale farm welfare and the gender effects of a shift from traditional extensive to intensive commercial system.

<https://www.internationalscholarsjournals.com/articles/role-of-poultry-production-in-promoting-periurban-smallscale-farm-welfare-and-the-gender-effects-of-a-shift-from-traditi.pdf>

²²⁴ Ngeno, V., Langat, B. K., Wendi, R., & Kipsat, M. J. (2011). Gender aspect in adoption of commercial poultry production among peri-urban farmers in Kericho Municipality, Kenya. *Journal of Development and Agricultural Economics*, 3(7), 286-301.

Gendered Positions in the poultry value chain. Women are generally responsible for the feeding and care of birds, and do not face socio-cultural or economic barriers regarding poultry ownership.²²⁵ Women's preferences thus shape the demand for poultry feed in Kenya²²⁶ and traditional gendered knowledge and attitudes shape poultry feeding practices.²²⁷

Gendered Access to resources in the poultry value chain. While women have easy access to poultry animals, they face challenges in access to production information, access to markets, extension services, and agro-vet services, including vaccines.²²⁸

Findings from CRLCSA fieldwork: *"The poultry value chain is attractive because of several factors such as its ability to involve many households and as a source of protein, hence a ready market. The improved kienyeji is the main breed that is being focused by many producers due to the shorter time it takes to mature and the ready market. Producers face the challenge of getting chicks, as well as high costs for feeds. To increase competitiveness producers such as the Victoria Kuku Cooperative are focusing on interventions such as value addition to fried chicken, boiled eggs. Training in disease control is also necessary, as well as improvement in production practices. Climate smart technologies include manure management through poultry units, water harvesting, use of incubators for hatching and making home rations."*

5.2.6 Dairy

Acceptability of women's participation in the dairy value chain. Recent reviews indicate that gender roles and dynamics greatly influence dairy production practices in Kenya. Women tend to be responsible for most management tasks around dairy animal husbandry, including fodder and water provisioning, veterinary health, knowing when a cow is in heat and requires mating or artificial insemination (AI), manure removal, and milking the cow. Yet, despite their contributions to dairy labour, women are often marginalized in the control of the resource (e.g., cow ownership), access and inclusion in veterinary services and training, decision-making associated with the animals (buying/selling), and do not receive commensurate income from the sale of milk.²²⁹ It makes sense, then, that because the burden of dairy work falls predominantly on the woman/women of the household, any mitigation intervention must consider the impacts that a new technology may have on women. Furthermore, consideration must be given to the gender roles and relationships that exist at household and community level to achieve mitigation project outcomes. This means actively engaging both women and men in the intervention

²²⁵ Ngeywo, J., Biwott, T., & Lagat, R. (2021). Gendered Participation in Chicken Feeding in North Rift Region, Kenya. *South Asian Research Journal of Biology and Applied Biosciences*, 3(1).

²²⁶ Macharia, J. N., Diiro, G. M., Busienei, J. R., Munei, K., Affognon, H. D., Ekesi, S., ... & Fiaboe, K. K. (2020). Gendered analysis of the demand for poultry feed in Kenya. *Agrekon*, 59(4), 426-439.

²²⁷ Waithanji, E., Affognon, D. H., King'ori, S., Diiro, G., Nakimbugwe, D., & Fiaboe, K. K. (2020). Insects as feed: Gendered knowledge attitudes and practices among poultry and Pond Fish farmers in Kenya. *NJAS-Wageningen Journal of Life Sciences*, 92, 100312.

²²⁸ Biwott, K. D., Musalia, O. J., Martha, P. C., & Awori, H. N. (2015) Role of poultry production in promoting peri-urban small-scale farm welfare and the gender effects of a shift from traditional extensive to intensive commercial system.

<https://www.internationaljournalofbiology.com/articles/role-of-poultry-production-in-promoting-periurban-smallscale-farm-welfare-and-the-gender-effects-of-a-shift-from-traditi.pdf>

²²⁹ Tavenner, K. and Crane, T.A. 2016. Best practice guide to socially and gender-inclusive development in the Kenyan intensive dairy sector. ILRI project report. Nairobi, Kenya: ILRI.

process.²³⁰ The dairy value chain in Kenya has been a large source of potential investment in recent years, with win-win dynamics promoted for environmental sustainability, climate adaptation and mitigation and improved livelihoods for smallholder farmers, with benefits from climate-smart agricultural practices, low-emissions dairy development to mitigate climate change, and increasing incomes from the commercialization of dairy practices. Gender dynamics have also been documented in dairy production in Kenya.²³¹

Gendered positions in the dairy value chain. Women are also customarily responsible for decisions regarding whether milk is to be consumed at home or sold locally.²³² Women have become increasingly involved in the cultivation of improved forages and feeds for cows as required for low emissions dairy development.²³³ A recent study found that young women and men often have different aspirations for engaging in dairy production and intensification, but that their opportunities for leadership are curtailed by age-related discrimination.²³⁴ Women tend to benefit more directly from their involvement in informal dairy sales, which has implications for low emissions dairy development.²³⁵

Gendered access to resources in dairy value chain. Women have been able to gain access to agricultural inputs, include improved feeds, vaccines, and other supplies through producer organizations.²³⁶ However, evidence that direct payments incentivize women's participation and empowerment in dairy development interventions is more complex.²³⁷ Furthermore, household surveys that have sought to gather information regarding women's access to resources, ownership over dairy cows, and decision-making related to cattle have highlighted the contested nature of intra-household relations, and whether there can be agreement within a household regarding 'who does what'.²³⁸

Findings from CRLCSA fieldwork: *"Many farmers are practicing dairy and there is huge potential to increase productivity. Production is challenged by poor breeds kept by farmers and low knowledge of breeding (e.g., heat detection, lack of fodder preservation). One of the successful dairy cooperatives in Nandi is Lelchego that was started by East African Dairy Development (EADD) in 2009. The cooperative uses the hub model to mobilize members, bulk milk and provide services such as AI, drugs etc to members for a fee. The cooperative promotes a range of climate smart technologies in dairy production and husbandry. These include zero grazing, fodder production, conversion of manure to biogas and silage making. Other interventions have been the change of breed to Ayrshire from the Friesian which is a heavy feeder."*

230 Tavenner, K. and Crane, T.A. 2016. Best practice guide to socially and gender-inclusive development in the Kenyan intensive dairy sector. ILRI project report. Nairobi, Kenya: ILRI.

231 Gallina, A. (2016). Gender dynamics in dairy production in Kenya: A literature review. *CCAFS Working Paper*.

232 Galiè, A., Farnworth, C. R., Njiru, N., & Alonso, S. (2021). Intra-household handling and consumption dynamics of Milk in Peri-Urban Informal Markets in Tanzania and Kenya: a gender lens. *Sustainability*, 13(6), 3449.

233 Njuguna-Mungai, E., Omondi, I., Galiè, A., Jumba, H., Derseh, M., Paul, B. K., ... & Duncan, A. (2022). Gender dynamics around introduction of improved forages in Kenya and Ethiopia. *Agronomy Journal*, 114(1), 277-295.

234 Bullock, R., & Crane, T. (2021). Young Women's and Men's Opportunity Spaces in Dairy Intensification in Kenya. *Rural Sociology*, 86(4), 777-808.

235 Tavenner, K., Saxena, T., & Crane, T. A. (2018). Gendered participation in informal milk markets in Kenya: Implications for low emissions dairy development.

236 Mwambi, M., Bijman, J., & Galie, A. (2021, July). The effect of membership in producer organizations on women's empowerment: Evidence from Kenya. In *Women's Studies International Forum* (Vol. 87, p. 102492). Pergamon.

237 Tavenner, K., & Crane, T. A. (2018). Do direct payments to women incentivize participation in low emissions dairy development interventions? Evidence from Kenya. ILRI Policy Brief, 26, 1-4.

238 Tavenner, K., Fraval, S., Omondi, I., & Crane, T. A. (2018). Gendered reporting of household dynamics in the Kenyan dairy sector: trends and implications for low emissions dairy development. *Gender, Technology and Development*, 22(1), 1-19.

5.2.7 LREB Case 2: Qualitative Gender Assessment: Gendered Participation, Performance, and Empowerment in Agricultural Value Chains in Target Sites

Qualitative data collected from leaders of cooperatives in different value chains in the targeted counties yielded important information that triangulates with the research findings from existing studies. Leaders representing two dairy cooperatives, 1 coffee cooperative, 1 African leafy vegetable cooperative, and 1 poultry cooperative were interviewed. A summary of their responses is provided below.

What are the barriers to entry and/or requirements for men's and women's active engagement at any node of the value chain?

Socio-cultural norms play an important role in determining the nature of involvement of men and women in the value chains. In general, men control the productive assets and income while women participate more in the production/management, while the youth dominate transport. In the coffee value chain, the youth were found to be largely involved in activities such as weeding, harvesting and transport to factory. For dairy, women and youth were found to participate more in taking care of the animals, milking, and transport while men largely make decisions on how the income is used.

Non-ownership of productive assets especially land by women and youth limit their participation in production and subsequently engagement at different nodes of the VC. In most cultures within LREB, productive assets such as land and cattle belong to men. Tea and coffee are capital crops practiced on land owned by men. The land itself is rarely subdivided hence the youth are unlikely to have a say on how it is used. Livestock except for chicken are also owned by men although women and youth spend significant amount of time taking care of all livestock.

Limited capital to start a business and buy inputs required at the various points of the value chain. For example, animal feed and supplements remain unaffordable to many households thereby affecting production. Poor prices paid on the farmers produce has not helped matters either.

Long-distances to the cooperatives or collection points especially for the dairy value chain is a problem that places unnecessary burden on women. This was reported in both Nandi and Trans Nzoia as a major challenge, more so in the afternoon and evening.

In terms of opportunities to serve in the management of cooperatives, it was felt that both men and women have opportunities to participate, but women are not able to fully seize the opportunities due to competing demands on their time, with the majority expected to prioritize taking care of household chores as opposed to serving in the cooperatives.

What are the disparities in men's and women's ability to maintain or improve their position in the value chain?

Most of the cooperatives visited provide several benefits to members. Included are collection and value addition; agrovet services (feed supplements, drugs, fertilizers, seeds etc.); transport; loans; access to markets; extension services including capacity building on climate smart technologies.

Due to the low income and limited access to inputs, women end up not using the recommended inputs such as fertilizers, certified seeds, feed rations, etc. This limits their ability to improve their position in the value chain as production and income generally stagnate.

What are the differences in men's and women's ability to access and control income, assets, or other facets of well-being derived from value chain participation?

While women manage the assets including land, livestock etc., men generally own and control the assets and income realized from the same. Examples given include payments from dairy, tea and coffee cooperatives which are channeled to men, yet it is the women and youth who take care of crops and livestock.

Social beliefs and practices tend to favour women as far as poultry and African leafy vegetables are concerned. The two value chains are not held in high esteem by men hence allowing women more room to make decisions on how income from such sources is used.

What steps/changes are needed so that women can control the benefits of their participation in agricultural value chains to make and carry out strategic decisions about their own lives?

Sensitization and mobilization to enable women and youth to participate more in the value chain remains critical. Awareness creation and sensitization are important in highlighting the benefits associated with being a member of a cooperative and issues that need to be addressed to overcome some of the cultural barriers to participation including leadership within the coops. In the case of Lelchego Dairy in Nandi County, sensitization and mobilization efforts saw the number of women increase from less than 10% in 2009 to more than 52% of the current membership that stands at over 8000 members. Four women currently sit on the board which consists of 11 members.

In a bid to enhance access and control of the proceeds by women in agricultural value chains, some cooperatives like Meebot are taking pro-active steps like making payments in the evenings when both men and women are at home to ensure they are aware and can participate in making decisions on how the money is used. Provision of transport especially for collecting milk from far off and difficult terrains using motor bikes is helping women, as it significantly reduces the time they would otherwise spend delivering. The male youth also benefit particularly on this part of the VC through gainful employment. The dairy cooperatives give youths loans to buy motorbikes for use in collection and delivery of milk. Capacity building is also needed to equip members with the necessary skills and best practices to enhance their production and efficiency. This may entail training on available climate smart technologies that may be adopted within their respective value chains. Flexibility in payments: Women prefer weekly payments to enable them deal with household/subsistence needs. Their participation in decision making has seen Lelchego Dairy Cooperative adopt weekly payments to members to meet this need.

Participation

Sample quotes used to inform analysis	Value Chain Interview Details
<i>"Yes, the growing and selling of vegetables is largely women's affair, while men prepare the land. Keeping chickens is also considered a women's domain, but men are largely in the marketing – buying and selling at profit."</i>	<i>African Leafy Vegetables</i> Margaret Opiyo, Lower Nyakach Women Traders Sacco, Kisumu. 16 Dec 2022.

<i>"Fewer women participate in dairy because of culture. The Nandi [culture], just like the larger Kalenjin community treats women as children. Cattle owned by men, but women spending a lot of time tending to the animals."</i>	<i>Dairy</i> Ruth Kosgey, Lelchego Dairy Co-op, Nandi. 14 Dec 2022.
<i>"There's been a lot of sensitizations on women's participation and leadership. Currently 4 of the 11 board members are women. In 2009, women were less than 10% of members, not there are 8,000 women members (52%)."</i>	<i>Dairy</i> Ruth Kosgey, Lelchego Dairy Co-op, Nandi. 14 Dec 2022.
<i>"We have a youth and gender committee that helps mobilization and sensitization; women are interested in introducing weekly payments as they have commitments in women's groups/savings; training on best dairy practices; Male youth are active in transport – given motorbikes and can get fuel from select providers using the check-off system."</i>	<i>Dairy</i> Ruth Kosgey, Lelchego Dairy Co-op, Nandi. 14 Dec 2022.
<i>"Women take care of the animals [cows] but do not get to keep the income in most cases."</i>	<i>Dairy</i> Christine Tiisa, Meebot Dairy Cooperative, Trans Nzoia. 13 Dec 2022.
<i>"Women have difficulty is getting fertilizers – men don't give them money to buy fertilizers, so their outputs [yields] remain low. So, women have no capital to start coffee business."</i>	<i>Coffee</i> Truphena Muhembi Kisikwa, Siboti Coffee Factory, Trans Nzoia. 13 Dec 2022.
<i>"At the co-op management level, women don't really have opportunities to be leaders. Because they [management] look at those who can lead, and women tend to be less visible as they are busy with other chores."</i>	<i>Coffee</i> Truphena Muhembi Kisikwa, Siboti Coffee Factory, Trans Nzoia. 13 Dec 2022.

Performance

Sample quotes used to inform analysis	Value Chain Interview Details
<i>"Women don't own land or cows, so it's largely cultural barriers" [that stop women from maintaining or improving their position in the value chain]</i>	<i>Dairy</i> Ruth Kosgey, Lelchego Dairy Co-op, Nandi. 14 Dec 2022.
<i>"Women can access feeds, supplements, and drugs for cattle using the check off system at the hub. The hub also provides extension services and loans/income advances, and artificial insemination (AI) services."</i>	<i>Dairy</i> Ruth Kosgey, Lelchego Dairy Co-op, Nandi. 14 Dec 2022.
<i>"Men take the pay on behalf of the households. Some go and drink the money without sharing/paying them back."</i>	<i>Coffee</i> Truphena Muhembi Kisikwa, Siboti Coffee Factory, Trans Nzoia. 13 Dec 2022.

Empowerment

Sample quotes used to inform analysis	Value Chain Interview Details
What types of services could the co-op offer that would contribute to women's empowerment? <i>"Capacity building on improved farming methods to improve production, value addition (i.e., drying of vegetables), and loans to buy water harvesting/storage facilities to irrigate farms during the dry season."</i>	<i>African leafy vegetables</i> -Margaret Opiyo, Lower Nyakach Women Traders Sacco, Kisumu. 16 Dec 2022.

<i>“Capacity building on new approaches and technologies in dairy farming for women to play their role without fear”</i> <i>“Access to extension services including on fodder growing and animal nutrition”</i> <i>“Exchange visits to see other co-ops with women leaders (chairwomen)”</i>	Dairy Ruth Kosgey, Lelchego Dairy Co-op, Nandi. 14 Dec 2022.
<i>“Lelchego Dairy has plans to celebrate women members during International Women’s Day in March 2023. We want to celebrate women who have come out to supply milk and serve in the cooperative; how small savings have been useful to families. Need support to also celebrate women living with disability who work very hard to supply the milk so that they feel they are a part of the cooperative.”</i>	Dairy Ruth Kosgey, Lelchego Dairy Co-op, Nandi. 14 Dec 2022.
<i>“Support to care for coffee trees, fertilizers, capital for starting business, training for women’s groups.”</i>	– Truphena Muhembi Kisikwa, Siboti Coffee Factory, Trans Nzoia. 13 Dec 2022.

5.3 Gender dynamics of cooperative membership, leadership, and capacity in the LREB by county and value chain

5.3.1 Cooperatives and producer organizations

Shifts in global agriculture have led more women into formal roles in agricultural and livestock industry as small-holder producers and cooperative members. Inclusion of women in these institutions, however, does not guarantee a change in historical power relations, or the benefits that might flow from this. A transformative change in gender relations not only requires changes in women and men’s attitudes and capacities and in the relationships between men and women, but also progress at the household, community, institutional and structural levels.²³⁹ For example, a recent study of the coffee value chain in Guatemala found that empowering women as productive cooperative members requires not only technical assistance and support, but also creation of an inclusive social and political environment that supports expanded choices for women and men.²⁴⁰

Access to information and collateral challenges are some of the other barriers women face in agricultural financing, but increasingly women are coming together in groups to try to bridge this gender gap. Women groups are common and accessible in the farming communities. Many women are organized into existing women groups, which are stable and outlive most projects and initiatives, but are largely unsupported. Directly involving women in production initiatives has the potential to increase such production.

For example, in Bomet and Nandi counties in Western Kenya, there is a strong male dominance in dairy cooperatives because cows culturally ‘belong’ to men. Even though many women are highly involved in dairy production, experts suggested that women’s representation at cooperatives is still low. Women are often less confident in their leadership abilities than men and may be hesitant to be elected as leaders of co-ops. At producer organization level, experts stated that men are often not willing to nominate women

²³⁹ For a recent compendium of promising approaches and tools in GTA, see: FAO, IFAD, and WFP. 2020. Gender transformative approaches for food security, improved nutrition, and sustainable agriculture – A compendium of fifteen good practices. Rome. <https://doi.org/10.4060/cb1331en>

²⁴⁰ Bilfield, A., Seal, D., & Rose, D. (2020). Brewing a more balanced cup: supply chain perspectives on gender transformative change within the coffee value chain. *International Journal on Food System Dynamics*, 11(1), 26-38.

for leadership positions. This leads to co-ops having training that excludes women. ²⁴¹

5.3.2 LREB Case 3: Quantitative Analysis of Cooperatives

This analysis looks at intra-gender comparisons between men, women, male youth, and female youth members among the cooperatives surveyed. 'Youth' members were defined as those between the ages of 18-34.

Analysis 1: Gendered membership rates

In assessing gendered membership rates by county (Tables 1-2), the cooperative survey found that Kisii, Bungoma, and Nyamira have the highest numbers of female members, Kisumu, Busia, and Transzoia have the lowest numbers of female members. For female youth members, Nandi, Kericho, and Bungoma were the highest and for total youth membership, Kericho, Bungoma, and Kisii have the most members. For youth membership, Kericho, Bungoma, and Kisii have the highest numbers of youth, while Homabay, Busia, and Transzoia have the lowest numbers (for both total youth and female youth).

Table 1. Gendered membership rates by county

	Frequency and % of cooperatives surveyed (n=321)	Male Members* (sum)	Female Members (sum)	Youth Members** (sum)	Female Youth Members (sum)	Total Members (sum)	Active Members (sum)
Bomet	37 (11.5%)	19,782	17,013	6,351	2,478	36,795	20,697
Bungoma	47 (14.6%)	54,822	27,532	13,982	3,808	82,354	55,970
Busia	6 (1.9%)	2,670	1,804	506	223	4,474	1,883
Homabay	17 (5.3%)	4,606	2,044	808	448	6,650	2,790
Kakamega	19 (5.9%)	3,663	2,689	1,367	450	6,352	2,471
Kericho	39 (12.1%)	104,753	20,540	14,484	4,803	125,293	67,284
Kisii	24 (7.5%)	112,432	28,946	13,341	3,554	141,378	60,255
Kisumu	8 (2.5%)	1,557	1,898	1,112	620	3,455	1,299
Migori	24 (7.5%)	13,430	7,788	3,065	1,513	21,218	7,720
Nandi	34 (10.6%)	22,603	16,762	6,311	5,016	39,365	17,818
Nyamira	18 (5.6%)	38,546	24,680	8,663	3,059	63,226	39,190
Siaya	12 (3.7%)	3,631	3,376	1,261	516	7,007	1,819
Transzoia	18 (5.6%)	2,729	1,385	447	200	4,114	2,739
Vihiga	18 (5.6%)	5,040	4,466	1,193	762	9,506	4,645
TOTAL	321 (100%)	390,264	160,923	72,891	27,450	551,187	286,580

*Male members calculated by subtracting female members from the total members column

**Unclear if "youth" is the combined total of male and female youth or just total male youth

Table 2. County rankings for highest female, youth, and female youth membership

Rankings Female Membership * County		Rankings Youth Membership * County		Rankings Female Youth Membership * County	
Rank	County No. Female Members	Rank	County No. Youth Members	Rank	County No. Female Youth Members

241 Tavenner, K. and Crane, T.A. 2016. Best practice guide to socially and gender-inclusive development in the Kenyan intensive dairy sector. ILRI project report. Nairobi, Kenya: ILRI.

1	Kisii - 28,946	1	Kericho - 14,484	1	Nandi - 5,016
2	Bungoma - 27,532	2	Bungoma - 13,982	2	Kericho - 4,803
3	Nyamira - 24,680	3	Kisii - 13,341	3	Bungoma - 3,808
4	Kericho - 20,540	4	Nyamira - 8,663	4	Kisii - 3,554
5	Bomet - 17,013	5	Bomet - 6,351	5	Nyamira - 3,059
6	Nandi - 16,762	6	Nandi - 6,311	6	Bomet - 2,478
7	Migori - 7,788	7	Migori - 3,065	7	Migori - 1,513
8	Vihiga - 4,466	8	Kakamega - 1,367	8	Vihiga - 762
9	Siaya - 3,376	9	Siaya - 1,261	9	Kisumu - 620
10	Kakamega - 2,689	10	Vihiga 1,193	10	Siaya - 516
11	Homabay - 2,044	11	Kisumu - 1,112	11	Kakamega - 450
12	Kisumu - 1,898	12	Homabay - 808	12	Homabay - 448
13	Busia - 1,804	13	Busia - 506	13	Busia - 223
14	Transzoia - 1,385	14	Transzoia - 447	14	Transzoia - 200

In assessing gendered membership rates by value chain (Tables 3-4), the cooperative survey found that coffee, dairy, and tea had that highest numbers of female and total youth membership. For female youth members, the top value chains were dairy, coffee, and poultry. For female, total youth, and female youth members, fruit trees and indigenous vegetables were in the bottom three value chains. While female and youth members' lowest membership rates were in poultry, female youth members had tea in their bottom three.

Of the six value chains analyzed, men are the overwhelming majority of members in tea (91.2%) and coffee (73.6%), and a slight majority in dairy (53.6%) and fruit trees (56.3%). Female membership is the majority in both poultry (54%) and indigenous vegetables (59.6%). Female youth membership was the lowest across all six value chains, with less than 1% of tea members being female youth.

Table 3: Gendered Membership Rates by Value Chain

Value Chains	Frequency and % of Cooperatives (n=321)	Male Members	Female Members	Youth	Female Youth	Total Members	Active Members
Coffee	141 (43.9%)	229,284 (73.6%)	82,052 (26.4%)	40,179 (12.9%)	11,891(3.8%)	311,336 (100%)	175,385
Dairy	117 (36.4%)	69,114 (53.6%)	59,815 (46.4%)	25,446 (19.7%)	12,213(9.5%)	128,929 (100%)	53,275
Fruit trees	23 (7.2%)	5,496 (56.3%)	4,274 (43.7%)	1,775 (18.2%)	901 (9.2%)	9,770 (100%)	5,851
Poultry	18 (5.6%)	2,707 (46%)	3,175 (54%)	1,336 (22.7%)	838 (14.2%)	5,882 (100%)	3,061
Tea	13 (4%)	81,105 (91.2%)	7,846 (8.8%)	2,567 (2.9%)	631 (0.7%)	88,951 (100%)	46,107
Indigenous Vegetables	9 (2.8%)	2,558 (40.5%)	3,761 (59.5%)	1,588 (25.1%)	976 (15.4%)	6,319 (100%)	2,901

Table 4: Rankings for female, youth, and female youth membership by value chain

Rankings Female Membership * Value Chain		Rankings Youth Membership * Value Chain		Rankings Female Youth Membership * Value Chain	
Rank	Value Chain No. Female Members	Rank	Value Chain No. Youth Members	Rank	Value Chain No. Female Youth Members
1	Coffee – 82, 052	1	Coffee – 40,179	1	Dairy – 12,213
2	Dairy – 59,815	2	Dairy – 25,446	2	Coffee – 11,891
3	Tea – 7,846	3	Tea – 2,567	3	Poultry – 976
4	Fruit tree – 4,274	4	Fruit tree – 1,775	4	Tea – 901
5	Indigenous veg – 3,761	5	Indigenous veg – 1,588	5	Fruit trees – 838
6	Poultry – 3,175	6	Poultry – 1,336	6	Indigenous veg – 631

Analysis 2: Gender Governance Rates

In assessing gender governance rates by county (Table 5), female board members were highest in Bomet (n=123), Bungoma (n=96), and Nandi (n=88), and lowest in Busia (n=25), Nyamira (n=37), and Kisumu (n=35). Total youth board membership was highest in Nandi (n=56), Bomet (n=46), and Kakamega (n=39), and lowest in Nyamira (n=16), Siaya (n=17), and Transzoia (n=17). Female youth board members were highest in Kakamega (n=32), Bomet (n=17), and Migori (n=17), and lowest in Transzoia (n=3), Busia (n=4), and Nyamira (n=5).

Table 5. Female, Youth, and Female Youth Board Members by County

	Frequency and % of cooperatives surveyed (n=321)	Female Board Members	Youth Board Members	Female Youth Board Members
Bomet	37 (11.5%)	123	46	17
Bungoma	47 (14.6%)	96	31	7
Busia	6 (1.9%)	25	11	4
Homabay	17 (5.3%)	67	35	12
Kakamega	19 (5.9%)	57	39	32
Kericho	39 (12.1%)	63	35	12
Kisii	24 (7.5%)	57	18	6
Kisumu	8 (2.5%)	35	22	9
Migori	24 (7.5%)	81	38	17
Nandi	34 (10.6%)	88	56	15
Nyamira	18 (5.6%)	37	16	5
Siaya	12 (3.7%)	45	17	6
Transzoia	18 (5.6%)	64	17	3
Vihiga	18 (5.6%)	73	25	10
TOTAL	321 (100%)	911	406	155

** There was no 'total board members' or 'male board members' data collected, so these comparisons were not possible.

In assessing gender governance rates by value chain (Table 6), highest female board membership was in dairy (n=374), coffee (n=283), and fruit trees (n=102). Highest youth board membership was in dairy (n=148), coffee (n=131), and poultry (n=53). Highest female board membership was in dairy (n=69), coffee (n=34), and poultry (n=28). Tea board membership was lowest across all social categories – female (n=33), youth (n=14), and female youth (n=4).

Table 6. Female, Youth, and Female Youth Board Members by Value Chain

Value Chains	Frequency and % of Cooperatives (n=321)	Female Board Members	Youth Board Members	Female Youth Board Members
1Coffee	141 (43.9%)	283	131	34
2Dairy	117 (36.4%)	374	148	69
3 Fruit trees	23 (7.2%)	102	35	12
4Indigenous Vegetables	9 (2.8%)	33	25	8
5Poultry	18 (5.6%)	86	53	28
6 Tea	13 (4%)	33	14	4
Total	321 (100%)	911	406	155

Analysis 3: Gender Staff Rates**Table 7: Female, Youth, and Female Youth Staff by County**

	Frequency and % of cooperatives surveyed (n=321)	Female Staff	Youth Staff	Female Youth Staff	Total Staff
Bomet	37 (11.5%)	30	51	26	89
Bungoma	47 (14.6%)	52	82	24	225
Busia	6 (1.9%)	9	28	7	16
Homabay	17 (5.3%)	25	22	13	42
Kakamega	19 (5.9%)	27	33	14	78
Kericho	39 (12.1%)	214	312	133	560
Kisii	24 (7.5%)	44	57	17	292
Kisumu	8 (2.5%)	2	2	1	3
Migori	24 (7.5%)	16	23	11	60
Nandi	34 (10.6%)	56	81	26	136
Nyamira	18 (5.6%)	35	45	19	156
Siaya	12 (3.7%)	15	10	9	23
Transzoia	18 (5.6%)	8	26	8	39
Vihiga	18 (5.6%)	14	15	11	23
TOTAL	321 (100%)	547	787	319	1742

Table 8: Female, Youth, and Female Youth Staff by Value Chain

Value Chains	Frequency and % of Cooperatives (n=321)	Female Staff	Youth Staff	Female Youth Staff	Total Staff
1Coffee	141 (43.9%)	179	306	94	892
2Dairy	117 (36.4%)	170	279	112	427
3 Fruit trees	23 (7.2%)	12	9	5	16
4 Indigenous Vegetables	9 (2.8%)	9	16	11	17
5Poultry	18 (5.6%)	3	2	1	7
6 Tea	13 (4%)	174	175	96	383
Total	321 (100%)	547	787	319	1742

Analysis 4: Gender Capacity Rates using presence of a gender representative as proxy

In assessing the gender capacity rates (using the presence of a gender representative as proxy) of cooperative at county level (Table 9), most counties (n=10) had a gender representative present, while most cooperatives in Vihiga, Siaya, Kakamega, and Kisii did not have a gender representative present.

Table 9: Gender representative present by County

County	Frequency and % of cooperatives surveyed (n=321)	Gender Rep Present	Not Present
Bomet	37 (11.5%)	18 (49%)	19 (51%)
Bungoma	47 (14.6%)	27 (57%)	20 (43%)
Busia	6 (1.9%)	4 (67%)	2 (33%)
Homabay	17 (5.3%)	15 (88%)	2(12%)
Kakamega	19 (5.9%)	7 (37%)	12 (63%)
Kericho	39 (12.1%)	24 (62%)	15 (38%)
Kisii	24 (7.5%)	9 (38%)	15 (62%)
Kisumu	8 (2.5%)	7 (88%)	1 (12%)
Migori	24 (7.5%)	16 (67%)	8 (33%)
Nandi	34 (10.6%)	26 (76%)	8 (23%)
Nyamira	18 (5.6%)	14 (78%)	4 (22%)
Siaya	12 (3.7%)	4 (33%)	8 (66%)
Transzoia	18 (5.6%)	13 (72%)	5 (28%)
Vihiga	18 (5.6%)	6 (33%)	12 (66%)
TOTAL	321 (100%)	190 (59%)	129 (41%)

Across value chains (Table 10), the majority (n=6) had a gender representative present.

Table 10: Gender representative present by Value Chain

Value Chain	Frequency and % of Cooperatives (n=321)	Gender Rep Present	Not Present
Coffee	141 (43.9%)	80 (57%)	61 (43%)
Dairy	117 (36.4%)	71 (61%)	46 (39%)
Fruit trees	23 (7.2%)	14 (61%)	9 (39%)
Indigenous Vegetables	9 (2.8%)	6 (67%)	3 (33%)
Poultry	18 (5.6%)	9 (50%)	9 (50%)
Tea	13 (4%)	10 (77%)	3 (33%)
Total	321 (100%)	190 (59%)	129 (41%)

Summary analysis of cooperative survey findings. Gender data from the Cooperative census reveals the existing inequalities between women's and men's participation as cooperative members. Across the 14 target counties in LREB surveyed, men constituted 70.8% of members and women only 29.2%. This gender gap is also present among youth, with 13.2% of cooperative members being young men compared to only 5% of members being young women. In evaluating gendered membership rates by value chain, only poultry and indigenous vegetables had higher levels of female membership than male membership. In formal, high-value commodity chains, men made up 73.6% of coffee members, and 91.2% of tea members. Dairy value chains had a closer gender parity with 46.4% women members. Among the less commercialized value chains of (fruit trees, poultry, and indigenous vegetables), there were greater levels of female and male youth participation. However, gender gaps still existed between youth as young women were the least represented in membership across all six value chains.

Table 11: Summary analysis of the main challenges and gender-based constraints by value chain.

Value chain	Summary of Gender-based constraints
All value chains	- Women, especially female-headed households (FHH), women living with disabilities, widowers, and young women, have limited access to land and technologies, inhibiting their capacity to adopt climate resilient value chain practices. - Women, especially FHH, women living with disabilities, widowers, and young women, have limited access to climate, agricultural, and market information, advisory and networks – including information disseminated via traditional agricultural extension agents and new digital technologies. - Women have disproportionate labour responsibilities, and are often clustered in the most time-intensive and menial tasks across the value chain from production, post-harvest, and value-addition activities - Women and youth have lower capital/incomes, locking them out of expensive farming technologies (e.g., irrigation), and limited financial resources to access climate resilient technologies - Gender norms and socio-cultural practices limit women, including married women, in income generating activities that would contest/compete with the income of men/husbands.
Commercial value chains/traditionally male-centric	
Dairy	- Dairy is a masculinized commodity, and women are discouraged from entering the most productive nodes of the value chain. - Dairy labour is predominately borne by women, even in intensified productions. - Women have limited access to fodder crops and grazing land and lack of access to veterinary services, including vaccines, supplements, and feeds. - Limited infrastructure for feed storage and transportation, and transport for milk cooling, packaging, and storage - Informal nodes of the value chain unrecognized/illegal and dangerous for women to sell milk at night. - Institutional constraints via the Kenyan Dairy Board
Tea	- Sexual harassment and gender-based violence on tea plantations (especially amongst male managers and female workers) - The women were assigned the roles that were tedious and took many hours like tea plucking and transportation, while the men undertook seasonal roles like tea planting and pruning. - Despite the significant role women play in smallholder tea production, there are extreme gender inequalities in access to and control over the benefits accrued from tea – this is especially the case for women within male-headed households. FHH were able to access and control tea benefits because they had acquired the tea owners' rights after the demise of their husbands and/or inheritance from their fathers. - Institutional constraints via the Kenya Tea Development Authority (KTDA) (e.g., Factors such as KTDA Policy of registering men as tea owners, community norms, lack of title deeds by women, gender-biased culture, and illiteracy made it difficult for women to materially benefit from tea production).
Coffee	- Land ownership as being both a prerequisite for membership, and a critical barrier for women due to structural laws reinforced by socio-cultural norms.

	Even in situations where women are widowed, structural and legal barriers may constrict their ability to assume ownership of their family land. - Women, including female-headed households, young women, and women with disabilities are disadvantaged in terms of coffee production, as these subgroups often lack the land, capital, resources, networks, and technical skills necessary to farm, and that the institutional context for coffee production is highly masculinized, with elder men predominating in terms of land rights and derivative income from the sale of coffee. - Gender equity within coffee cooperatives suggests that there are significant gaps between women's participation and concrete benefit from coffee production. While women perform more than two-thirds of the work in coffee farming in Kenya, accounting for up to 70% of labor in production, starting from the farm level, cooperatives, and processing level – yet they represent less than 5% of leadership roles in coffee cooperatives in the country. - Over 95% of coffee farms are owned by men. Thus, the groups who produce Kenya's coffee have little or no access to the income from the commodity. These dynamics have worked to create apathy among women and youth in respect to active and voluntary engagement in coffee production. - Women do not have the same market opportunities, have limited access to resources, and have less say over coffee production's planning and supply segment. - Women continually face significant barriers to their equal participation, such as limited access to land, capital, and technical information. These barriers are caused by intersecting challenges of socio-cultural practices, land tenure system that privileges men, and discriminatory bylaws that hinder women and youth participation in cooperative membership. - Institutional constraints via the Coffee Board of Kenya
Less commercialized/Food security value chains/traditionally female-centric	
African Leafy Vegetables	- Traditional food crop marketing, including African Leafy vegetables, tends to follow distinct gender roles, with women having greater social acceptability in cultivating and retaining the sales from these plants than other value chains. - However, recent evidence suggests that although African leafy vegetables have traditionally been considered a woman's crop, women vendors in emerging markets are not necessarily empowered to earn equal income as men. - It is unclear how gendered production of African leafy vegetables changes with different scales of production (i.e., whether women are still primarily responsible for cultivation, harvesting, and post-harvesting activities, or whether some or all these activities would be outsourced to hired labor). However, one study found that income from selling AIV seed differed significantly depending on gender, with men earning more than twice as much as women - Fragmented and untracked value chain, lacking a proper product classification and evaluation; informal markets due to several challenges to enter formal markets because of the poor value-addition capacities of vegetable products, the lack of research and forecasts on demand and supply trends; - Women lack information on marketing, CSA practices and technologies, access to extension officers/services, and invisibility of investments and credits.
Poultry	- Poultry is a feminized commodity, with lower input costs relative to cattle and other livestock production, the barriers to initial entry are lower. - However, the process of commercialization and increasing marketization is masculinized, making it difficult for married women and young women (living at home) to maintain income received from poultry.

	- Limited access to adequate feed and water resources, high costs of production and prices of inputs (e.g., drugs, vaccines) - Limited support by farmers' cooperatives for marketing and bargaining, and low farmers' participation in cooperatives and limited access to credit services. - While women have easy access to poultry animals, they face challenges in access to production information, access to markets, extension services, and agro-vet services, including vaccines
Fruit trees	- While fruit trees grown as subsistence crops have traditionally been female dominated and controlled, the commercialization of fruit trees has led to a dynamic shift in production with more men taking up active roles in the value chain (for example, bananas, avocados). - The participation of women in income sharing in the banana value chain was partially determined by the presence of off-farm income, so that women in male-headed households had a higher probability in taking part in household decision-making. - Low access among women, youth, and poor farmers to financial resources and credit to invest in climate-proofed technologies (e.g., greenhouses and irrigation, post-harvest facilities, cold chain technologies). - Upgrading strategies vary for the different typologies of women (female-headed households versus women in male-headed households). While women in female-headed households may require limited efforts such as tailoring financial products to their needs, women in male-headed households need institutionalization of gender-sensitive policies in the governance of producer groups to enable them to upgrade as chain integrators and chain owners.

6. Strategies to integrate gender in CRLCSA

In acknowledging the challenges and necessity of addressing existing gender and social inequalities in Kenya, all project outcomes and activities have corresponding gender outcomes and activities, presented in the GAP. To achieve the project's impact, outcomes, and co-benefits, GAP proposes activities that are gender-responsive within a broad gender-transformative approach (GTA). **Gender-responsive** activities are those which include specific actions to recognize, respond, and reduce gender and social inequalities (e.g., strategies, technologies, practices that reduce gender gaps in agriculture related to decision-making, labour burden, and access to agricultural information, finance, inputs). In the context of CRLCSA, adopting a **broader gender-transformative approach** means that those gender-responsive activities are designed around the fundamental aim of addressing the root causes of these gender gaps and social inequalities to ensure long-term project and social sustainability.

The Gender Action Plan (GAP) for the proposed project tackles gender inequalities across several priority areas in agri-climate adaptation and mitigation²⁴². These include closing gender gaps and existing inequalities in participation (at intra-household level and within producer organizations/cooperatives), workloads (prioritizing agricultural technologies and practices for adaptation and mitigation that reduce workloads and negative impacts on women), access and use of productive resources (such as agri-climate

²⁴² Adaptation requires adopting specific practices to lessen climate change impacts, while mitigation deals with addressing the root causes of climate change (i.e., Greenhouse Gas Emissions).

information, technologies, livelihood incomes, credit), and collective action (working with women's groups as platforms for enhancing access, agency, and voice in climate-smart agriculture). In closing these gender gaps, the project will use Gender Transformative Approaches (GTA),²⁴³ and more specifically, **Gender Action Learning Systems (GALS)** to address the underlying discriminatory socio-cultural and gender norms that perpetuate gender inequality and constrain women's capabilities. This requires a culturally sensitive, multi-level approach that includes women and men in all their diversity.

The proposal, which addresses gender dimensions within the project design and implementation, identifies and integrates interventions to provide gender-responsive and transformative results through the GALS approach combined with farmer field schools and other activities, particularly in the LREB where traditional social norms make it more difficult for women to participate in project and community activities. The gender-responsive approach recognizes and considers the particular needs, priorities, and realities of diverse groups of women and men, including the specific constraints they may face, and adequately addresses these in the design and implementation of the activities, resulting in equitable sharing of benefits.

The project will also ensure that gender and social inclusion issues are integrated and/or strengthened within extension programs and within any support provided to cooperatives using a gender transformative approach. A gender transformative approach (GTA) is required to address the underlying discriminatory socio-cultural and gender norms that currently perpetuate inequality and constrain women's capabilities within the six value chains targeted. A GTA approach requires a culturally sensitive, multi-level approach that includes women and men at across the project - at farm, cooperative, private sector and governmental partners, and project management levels. A GTA approach was chosen as a social safeguard 'backstop' to ensure that women can concretely benefit from the project and existing inequalities are not exacerbated so that no one is left behind from efforts to address climate change.

Specific activities will draw upon tested GTA methodologies, for example, GALS to engage in capacity building and training exercises among beneficiaries and project facilitators to discuss the issues underpinning gender inequalities at intrahousehold, core and extended value chain, and enabling environment levels, and how these can be addressed within the project. This will include designing and delivering specific modules within and in addition to agronomic training, which will include guidance on service provision to marginalized groups and the adaptation of services to persons living with disabilities.

The project explicitly seeks to redefine and transform gender norms and relationships to redress existing inequalities. For example, the project pursues to challenge and change gender roles and responsibilities and cultural/social norms and uneven access to resources to strengthen women's ability to participate and adopt climate resilient technologies. The logic for using a GTA approach is in alignment with the latest FAO Policy on Gender Equality 2020-2030²⁴⁴, which states that, *"...across regions rural women still face major gender-based constraints that limit their potential as economic agents and their capacity to reap the full benefits of their work. The root cause of these discriminations lies in social norms, attitudes, and beliefs, which shape how women and men are expected to behave, the opportunities that are offered to them and the aspirations they can pursue. Discriminatory sociocultural norms affect how policies and legal frameworks are formulated and implemented; who participates in decision-making processes and governance mechanisms; how rural institutions are managed; how service providers target their clients*

²⁴³ For a full list of publications by the FAO Joint Programme on Gender Transformative Approaches for Food Security and Nutrition see: <https://www.fao.org/joint-programme-gender-transformative-approaches/resources/publications/en>

²⁴⁴ FAO. 2020. FAO Policy on Gender Equality 2020–2030. Rome

and prioritize their needs; and, ultimately, how resources are allocated, and decisions are taken within households and communities.”

Gender transformative approaches address not only gender roles and power dynamics, but also institutional and legal exclusion as key barriers to equality, justice, and the achievement of global development outcomes. A main goal of gender transformative approaches involves the creation of an enabling social environment and more equitable inclusion in formal and informal institutions that support expanded choices for women and men. Key characteristics that distinguish gender transformative approaches from other efforts to integrate gender into agricultural development include: (1) Development of an understanding of people in their context, particularly the way social inequalities affect choices and outcomes; (2) Engagement with both women and men, as both have a role and stake in gender transformative change; (3) Commitment to addressing unequal power relations and to challenging oppressive norms, behaviors, and structures and (4) Engagement with different actors across levels in response to how the power relations and norms underlying gender and social inequality are distributed.²⁴⁵

GALS is a transformative methodology that goes beyond the symptoms of gender inequality to tackle the underlying causes of inequalities (norms, attitudes, behaviors) and generates positive change in areas of awareness, consciousness, and confidence; values, norms, and practices; and in policies, laws, and institutions.²⁴⁶ GALS was chosen as it is designed to be used at project level and has been used in projects working on value chains, agribusiness, and enterprise development, agriculture and rural finance, and in the context of nutrition, youth engagement, and climate change.²⁴⁷ For example, Oxfam and Hivos have used GALS methodology to improve the livelihoods and change gender relations among more than 200,000 women and men, mainly in Africa and Asia.

Gender transformative approaches in agriculture can be framed as seeking to foster change at multiple levels. Based on this nested framework, gender transformative approaches focus simultaneously on the level of the individual with individual capacities, attitudes, agency and actions; at the relational level with the expectations that shape relationships between people in the home, in groups and in organizations; and at the structural level with institutional rules and practices. Theoretically, shifts at each level can lead to a greater number of enhanced options for resource-poor women and men, for equitable norms and institutions, and finally for an expansion in women’s and men’s potential to contribute to and benefit from development.

Figure 1 provides a visual perspective of the potential target changes in smallholder agriculture about where efforts and resources may need to focus for the design and implementation of interventions that seek to contribute to gender transformative change processes. These are the entry points for GTA that the CRLCSA project aims to tap into:

- Exploring opportunities to engage with financial service providers, input dealers, and leaders governing formal or informal land tenure may be necessary to transform structure and dynamics rooted in institutions and the value chain.

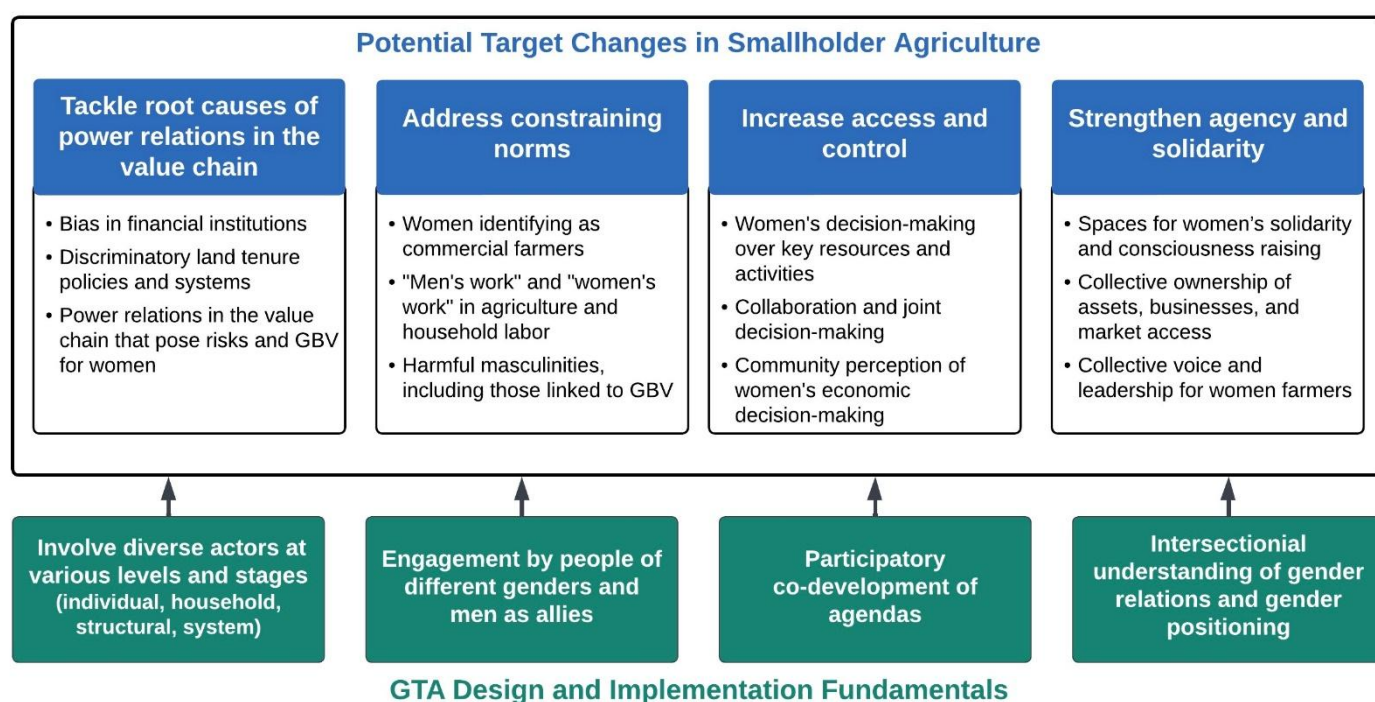
²⁴⁵ Cole, S. M., Kaminski, A. M., McDougall, C., Kefi, A. S., Marinda, P. A., Maliko, M., & Mtonga, J. (2020). Gender accommodative versus transformative approaches: a comparative assessment within a post-harvest fish loss reduction intervention. *Gender, Technology and Development*, 24(1), 48-65.

²⁴⁶ Cited from FAO, IFAD, and WFP. 2020. Gender transformative approaches for food security, improved nutrition, and sustainable agriculture – A compendium of fifteen good practices. Rome: <https://doi.org/10.4060/cb1331en>

²⁴⁷ *ibid*

- Sustaining and supporting cooperatives and group spaces where women can collectively build agency and access agricultural opportunities, services, and resources may help catalyze transformative change and especially influence gender relations across the community.
- Leverage points should be identified across the community, households, and with individuals to engage at a deep level and shift embedded norms and perceptions, such as whether women are valued as commercial farmers and economic decision-makers, the crops and livestock ascribed to certain genders, or harmful masculinities underlying possible backlash from shifts in farming and household activities.

Figure 1. Towards a gender transformative change perspective for smallholder farming commercialization. Adopted from Silvert, C. (2023). Preliminary Report for Expert Panel, IITA.



Within the project, GTA will be mainstreamed across activities to support improvements in social inclusion and women's meaningful participation in CRLCSA value chains. This activity is designed to ensure that project activities, technology transfer and support to cooperatives also contribute to the achievement of improvements in social inclusion and in the participation of women, youth and PLWD in the 6 value chains. As noted in the Gender Assessment and GAP, ensuring equitable participation and inclusion requires a more subtle approach that goes beyond fixing "quotas" and considers barriers to participation at various stages. The activity therefore aims to influence change in the way in which women, youth and PLWD participate in agriculture (beyond the primary production stage), benefit from their work (ensuring they receive appropriate remuneration and socio-economic benefits), and influence decision-making in households, farmer organizations, cooperatives and community or county policies, where gender relations and women's positions are improved, and women's entrepreneurship is boosted, contributing to SDG 5.

The project will ensure that Gender and Social Inclusion (GESI) principles are mainstreamed and integrated into all activities, trainings, materials, consultations, and processes. Gender focal points will be identified in counties, and in each EE, as well as within farmer organizations. At the start of the project, staff in the executing entities, project coordination unit, counties and financial institutions will receive mandatory training in the prevention and management of Sexual Exploitation, Abuse and harassment (SEAH), Gender-Based Violence (GBV) and in the application of the FAO Grievance and Redress Mechanism (GRM) to handle such incidents and ensure safe working conditions for women, PLWD, and vulnerable groups (sub-activity 3.1.4.1). The GRM will be survivor-centered and gender-responsive and will have specific procedures for SEAH, including confidential reporting and safe and ethical documenting. To ensure the GRM is accessible, several ways to access the GRM (such as hotline, a collection box for written and anonymous feedback, etc.) will be established and inclusive, and survivor-centered, clear information on how to access the GRM will be widely disseminated among stakeholders. The GRM will be designed to be easily accessible to all project stakeholders. This involves training of project and county officials on gender issues and increasing awareness of women, PLWD, and vulnerable groups on their rights.

The project supports the design of tailored climate information services and technologies (activity 1.1.2, 1.1.4 and 1.1.5) for women to address the unique challenges which hinder women's access to information and knowledge, and decision making power within their own household and farmers' organization over climate actions. When providing training and support to farmer organizations, the project will also ensure (under activity 3.1.1 and 3.1.2) that cooperatives and other business units have gender equality and social inclusion strategies in place to reach women, PLWD, and youth. Under these activities, the project support the formation and strengthening of women farmer group to provide a platform for collective action, advocacy and knowledge sharing. In addition, the project will deliver targeted trainings for FFS and cooperatives, following the tested Agriterro and FAO approaches to develop leadership, participation and meaningful inclusion of women, youth and persons living with disabilities (sub activity 3.1.4.2). These activities will increase women's participations in areas where they were discouraged by the gender norms, including trading and selling agriculture commodities, decision makings in adoption of new technologies and practices to address climate impacts. The approaches are the Women's Leadership-Youth leadership trainings developed by Agriterro as well as the Gender Action Learning System (GALS). These trainings will focus on all aspects of the value chain, accelerating the inclusion of vulnerable groups in production, processing and value addition, and trade. The project will support the process of building female leadership in agricultural cooperativesIf during an assessment a clear need to improve inclusivity arises, to address pervasive gender norms and increase the women's participation in leadership positions, an institutional training can be given to the cooperative.

During the Women's Leadership Workshops the role of women as members, entrepreneurs and leaders are discussed and together with male and female farmers an action plan is developed to improve the participation of women in the cooperative. Previous Agriterro Women leadership workshops have led to the following solutions:

- Reduced legal barriers to women's participation in the cooperative and changed gender biased by-laws.
- Changed the minimum amount a potential leader has to deliver to the cooperative before he/she can be elected from 1000 KGs to 500 KGs. Women often own fewer coffee trees or have smaller plots than males, therefore the threshold of 1000 kgs is a larger barrier for them to get elected than for their male counterparts.
- Reduced barriers to transfer ownership and benefits of coffee trees to women within a family
- Plan workshops at more suitable times for women/mothers

- Enable extension officers to give advice equally to men and women.
- Reduced registration prices for women in the Primary Cooperative
- Discussed cultural and social norms
- Build the capacity of current and potential women leaders of cooperatives, equipping them with leadership and management knowledge skills.
- Create a women's council to improve the position of women and source talent for leadership positions.
- Hired more women for cooperative jobs, for instance as extension officers, nursery staff and bio-composting and bio-fertilizer staff

At the end of the project, the expected result of these efforts should be that at least 10,725 women, youth and PLWD accede to roles of meaningful participation in the targeted value chains. This activity will be executed by FAO with GCF funding, in close collaboration with Agriterro and the Government of Kenya.

The project will use eight core strategies to achieve its gender and social inclusion (GESI) objectives. These are: (1) Supporting and strengthening Kenya's existing gender-sensitive legal and institutional frameworks related to gender equality, climate change, and agriculture at national and community level (including within producer organizations, county agencies, and private sector); (2) Supporting gender-responsive and socially inclusive agri-climate information and services for vulnerable smallholders in adopting climate-resilient and low-carbon production and processing practices, technologies, assets, and risk reduction mechanisms; (3) Strengthening women, PLWD, and youth representation and participation in decision-making in cooperative societies and value chains; (4) Supporting agricultural extension to disseminate and demonstrate CRLCSA knowledge, technologies and practices in ways that are gender-responsive and socially-inclusive; (5) Requiring sex-disaggregated data and relevant gender, agriculture, and climate indicators be collected, analyzed, and fed back into project activities iteratively and in a participatory manner; (6) Support GESI agri-climate finance for vulnerable smallholders and their organizations by increasing access to gender-responsive and socially inclusive financial products that support climate-resilient and low-carbon growth/Supporting gender-responsive and socially inclusive financial services, climate finance, and bundled services for enhancing women, PLWD, and youth actions towards climate adaptation and mitigation; (7) identifying and promoting gender-responsive and socially inclusive adaptation and mitigation technologies, markets, and labour practices; and (8) Promote and monitor gender and social safeguards to reduce climate risks for the most vulnerable.

The project includes strategies for addressing gender-related barriers in each output, which are highlighted under Annex 2, Feasibility Study, Chapter 5.

The process of integrating gender. The information gathered from the gender analysis/assessment should be considered in all stages of the project cycle: design, formulation, implementation, and monitoring and evaluation. In each of these stages, CRLCSA staff will be trained on gender and keep a 'gender lens' in mind throughout the project cycle and MR&E, looking at ways the project/program can address gender inequalities that emerge from the project; ensure the differential needs, interests and opportunities of women and men are addressed; ensure women and men have equal access to resources, services, and capacity development; ensure equal participation of women and men in management arrangements and as beneficiaries, partners and key stakeholders; and ensure women's equal participation in decision-making processes and leadership roles. At all stages of the project cycle, CRLCSA staff will be sensitized to the importance of addressing gender issues to achieve the gender-related project objectives via dedicated training sessions led by the Gender Specialist(s). The perspectives of women/ women's organizations will be continuously monitored (led by National Gender Specialist) and considered to ensure gender mainstreaming throughout the project implementation.

In Monitoring and Reporting, CRLCSA project staff will collect sex- and age-disaggregated baseline data at county-level that will be monitored throughout implementation (mid-line) and evaluation (end-line) surveys. Qualitative assessments will also be periodically conducted on the gender-specific benefits that can be directly associated to the project. This will be incorporated in the annual Project Implementation Report, Mid-Term Report, and Terminal Evaluation.

Quantitative indicators for the achievement of project objectives in relation to gender equality will include among others:

- Percentage of gender-responsive training materials and curricula designed and developed by a gender specialist
- Number of extension methods incorporating gender equality and social inclusion/GTA/GALS
- Percentage of county-level extension officers trained on gender-responsive and socially inclusive extension methods, disaggregated by age and sex
- Number of county governments that have policies promoting women's digital literacy in agriculture
- Number of women, youth, and PLWD that have access to extension agents trained in GALS
- Number of women, youth, and PLWD that have been consulted on types of climate services and information needed during scoping
- Number of women, youth, and PLWD that have direct access to finance for cell phones/digital services
- Number of women, youth, and PLWD trained in decentralized carbon accounting and related topics
- Percentage of databases upgraded with gender, sex- and age-disaggregated data
- Number of local administrators trained on collecting and managing sex-and gender-disaggregated data and how to integrate these data into project M&E, disaggregated by sex and age
- Number of women's and youth groups involved in generating climate solutions and county-level advocacy
- Number of women, PLWD, and youth consulted on the accessibility of knowledge platforms
- Percentage of landscape management plans with iterative gender-responsive monitoring and feedback mechanisms to address gender equality and social inclusion in implementation
- Number of women and youth-led cooperatives benefitting from project support
- Number of trainers/facilitators on GTA/GALS for FFS, disaggregated by sex and gender
- Number of women, youth, PLWD that are involved in peer-to-peer networks and exchanges
- Percentage of cooperatives and other business units that have gender equality and social inclusion strategies in place to reach women, youth, and PLWD.
- Percentage of project personnel trained on GTA, SEAH, GBV, and project GRM, disaggregated by sex and age
- Number of female leaders trained in the AgriTerra Female Leadership Programme and Masterclass, disaggregated by age
- Number of male and female youth leaders trained in the AgriTerra Youth Leadership Programme and Masterclass
- Number of business plans co-developed with women smallholders and women-led cooperatives and Micro, Small, and Medium Enterprises (MSMEs)
- Percentage of gender-responsive carbon and biodiversity schemes that have been designed and developed by a gender specialist

Qualitative Indicators for the achievement of project objectives in relation to gender equality will include among others:

- Stories of gender dynamics change from participants, household members, members of the community and private sector companies, backed by photographic records demonstrating key aspects of change.
- Monitor changes in norms and attitudes among women and men (working at different nodes of the value chain).
- Narratives of how women, men, and youth are engaging in GALS and changing behavior in different areas, for example:
 - Division of labour among household members
 - Income generation and control over income by women and men
 - Property/assets ownership by women and men
 - Reduction in domestic violence
 - Participation of women in decision making in the home, producer groups, community, and value chain platforms
 - Participation of women in accessing project and other services

Recommended Quantitative Indicators for Tracking Gender Equality and Social Inclusion in Cooperatives
(All indicators are recommended to be disaggregated by county and value chain).

- Cooperative Society Membership Data
- # Of female members and % compared to male members (#/% comparisons) disaggregated by age
- # Of male / female youth (#/% comparisons) disaggregated by age
- Governance Data
- # Female board members (#/% comparisons) disaggregated by age
- # Of male / female youth (#/% comparisons) disaggregated by age
- # Of men/women members (#/% comparisons) disaggregated by age
- # Of youth men/women members (#/% comparisons) disaggregated by age
- # Of women, youth, & PLWD chairpersons disaggregated by sex and age
- Services offered to members
- #Extension workers (women, men, girls, boys)
- #Mobile money accounts linked to coop disaggregated by sex and age of the mobile money account owner
- Capacity for women's empowerment
- #Gender representatives/champions in cooperatives disaggregated by sex and age
- #Youth representatives/champions in cooperatives disaggregated by sex and age
- # of people/stakeholders trained on GESI/women's empowerment disaggregated by sex and age

Qualitative Indicators for Tracking Gender Equality and Social Inclusion in Cooperatives

- Cooperative Society Membership Data
- Socio-cultural and gender norms and other barriers that mediate gender equitable and socially inclusive participation (which can be achieved with GTA)
- Governance Data
- Women's, PLWD, and Youth experience in leadership
- Services offered to members
- Quality of support mechanisms to build leadership and increase board membership of women, youth, and PLWD

- Types and qualities of services offered to women, PLWD, and youth
- Experiences related to barriers, bottlenecks in access to services
- Capacity for women's empowerment
- Women report feeling empowered economically, socially, and politically in agricultural value chains.
(e.g., women can make decisions on the marketization of agricultural/livestock commodities, on how income derived from value chain activities is allocated)
- Men report feelings towards women's empowerment/their relative level of empowerment
- Stakeholders at different levels (household, community, institutions) report changes in their views and beliefs on gender roles

The GAP should be further developed with the participation of women, men, girls, boys, youth, PLWD, and other vulnerable people who are most affected by the climate crisis within the agricultural sector to ensure they are gendered and meet the rights, needs, and experiences of women, men girls, boys, youth, PLWD, and other vulnerable people.

Annex 1: Details of consultation missions and meetings

Gender Assessment Field Interviews with Cooperative Leadership

County Visited	Value Chain	Date	Interviewee Details
Trans Nzoia	Dairy	13 DEC 2022	Women Representative, Meebot Dairy Cooperative
Trans Nzoia	Coffee	13 DEC 2022	Supervisory Committee Member; Siboti Coffee
Nandi	Dairy	14 DEC 2022	Treasurer and Director – Extension, Lelchego Dairy
Kisumu	African leafy vegetables	16 DEC 2022	Lower Nyakach Women Traders Sacco
Kisumu	Poultry	16 DEC 2022	Victoria Kukus
virtual	Indigenous peoples	Nov 2023	Women representative of Indigenous Peoples organizations

Various tools and venues were used to gather the concerns of women during feasibility. This included:

- Field visits and farm visits, including women farmers in all counties
- A Climate Change and Value Chain Survey with specific gender related and women-specific questions (28 women respondents out of 114)
- A cooperative census that included in-depth interviews and assessments of cooperative functioning, including gender representativity in the various roles. Among the 130 cooperative representatives, 51 women were interviewed.
- Gender-related questions in the consultations with county governments and interviews with women government staff in each county (14 counties)

A field-based gender assessment (December 2022) which was conducted as complementary to the value chain market analyses, the findings of which are included in this report.

Annex 2: Data Collection for Gender Assessment

Instrument I - Questionnaire Guide for Co-op Members

Target population: Cooperative members** prioritize the following value chains: Fruit trees, African leafy vegetables

Sample size: 2 women from each value chain targeted

Location: Kenya (counties targeted by project)

Purpose: To better understand the gender-based barriers and opportunities for women's engagement in the agricultural value chain, that influence their participation, performance, benefits, and empowerment.

Table 1: Circumstances of Interview [fill out for each interviewee]

Code number (gender, location, date – e.g., female, Bomet county, 17 Nov 2022)	
Date (dd/mm/yyyy)	
Name of Cooperative	
Rural/Urban	
Name of Interviewee	
Age	
Marital Status	
Preferred Title for Ethnic Community/Identity	
Religious Affiliation	
Type of Agricultural Value Chains involved in (e.g., dairy, fruit trees, coffee, tea)	
Telephone Number (for following up with the results of the analysis)	

A: Participation Questions

Big question: What are the barriers to entry and/or requirements for men's and women's active engagement at any node of the value chain?

A1: What type of work do women and men in the co-op typically do? Are there certain types of jobs that are never done by men or women? What about differences between older and younger women?

A2: Are some types of value chains/value chain activities considered more socially acceptable/"appropriate" for women or men? Why?

A3: If in a mixed co-op group (men and women), can women actively participate? Why? Has this changed in the past? List reason(s) for change

B: Performance Questions

Big question: What are the disparities in men's and women's ability to maintain or improve their position in the value chain?

B1: Do women access informal markets (outside the co-op) to sell or buy VC products (e.g., selling milk to neighbors, working as 'hawkers' or street vendors, selling outside of formal cooperatives)?

B2: Is there a "ceiling for success" for women in the value chain? For example, if a woman starts with a small business and it becomes more successful, will a spouse/male relative step in to assist in running the business?

C: Benefits Questions

Big question: What are the differences in men's and women's ability to access and control income, assets, or other facets of well-being derived from value chain participation?

C1: Regarding the income you make, what generally happens to the money you make? Who keeps it, decides about how to spend, or save it? Have these dynamics changed over time?

C2: Are there benefits beyond additional income that you derive from being a member of the co-op?

D: Empowerment Questions

Big question: What steps/changes are needed so that women can control the benefits of their participation in agricultural value chains to make and carry out strategic decisions about their own lives?

D1: What types of services could the co-op offer that would contribute to women's empowerment?

D2: Are you aware of any policies (national or county-level) that contribute to gender equality?

Data Collection for Gender Assessment

Instrument II - Questionnaire Guide for Co-op Leadership

Target population: Cooperative leaders

Sample size: At least 1 woman in a co-op leadership position from each value chain

Location: Kenya (counties targeted by project)

Purpose: To identify the key gender issues and inequalities within specific agricultural cooperatives, and discuss solutions towards improving women's participation, performance, benefits, and empowerment.

Table 1: Circumstances of Interview [fill out for each interviewee]

KII Code number (gender, location, date – e.g., female, Bomet county, 17 Nov 2022)	
Date (dd/mm/yyyy)	
Name of Cooperative	
Rural/Urban	
Name of Interviewee	
Age	
Marital Status	
Preferred Title for Ethnic Community/Identity	
Religious Affiliation	
Type of Agricultural Value Chains of Co-op (e.g., dairy, fruit trees, coffee, tea)	
Telephone Number (for following up with the results of the analysis)	

A: Participation Questions

Big question: What are the barriers to entry and/or requirements for men's and women's active engagement at any node of the value chain?

A1: What are the specific challenges women face in participating in the VC/co-op?

A2: Has the co-op adopted any strategies to foster the equitable participation of men, women, and youth across the value chain? If yes, what have been the challenges and gains associated with these strategies? If not, why not?

A3: At co-op management level, do women have opportunities to serve as leaders? If yes, what types of opportunities are available/are the most common?

A4: Has the co-op ever had women's leadership at 'chairman' level (i.e., head of governance)?

B: Performance Questions

Big question: What are the disparities in men's and women's ability to maintain or improve their position in the value chain?

B1: What are the types of services the co-op provides to members? Are there any services women have difficulty accessing compared to men?

B2: Do women typically own the means of production in the VC? (i.e., cows, chickens, land)

B3: Do women commonly rely on both informal and formal markets in the VC?

C: Benefits Questions

Big question: What are the differences in men's and women's ability to access and control income, assets, or other facets of well-being derived from value chain participation?

C1: Are there any gender norms that influence women's ability to benefit from being a co-op member?

C2: Does the co-op provide equitable benefit-sharing mechanisms (e.g., bank accounts, credit services) to ensure that women benefit financially and can control those benefits?

D: Empowerment Questions

Big question: What steps/changes are needed so that women can control the benefits of their participation in agricultural value chains to make and carry out strategic decisions about their own lives?

D1: What types of services could the co-op offer that would contribute to women's empowerment?

D2: Is the co-op aligned with any policies (national or county-level) that contribute to gender equality?

D3: Has the co-operative ever received gender equality/women's empowerment training? If yes, fill in contact details for organization

PART II: GENDER ACTION PLAN

1. Introduction

1 The Gender Assessment above describes the current state of women in Kenya compared to men, and to a lesser extent based on limited data, how gender intersects with age, socioeconomic status, household headship status, and disability status to influence participation and benefit from engaging in agricultural value chains in the context of a changing climate.²⁴⁸ Thus, and in alignment with current gender and development practice, it is not enough to target “women” as a homogenous group in terms of project beneficiaries²⁴⁹. From the Gender Assessment, female youth²⁵⁰, female-headed households (including widows), and females living with disabilities, are the most under-represented and vulnerable groups in CSA value chains, based on how their gender interacts with other axes of social differentiation and inequality. Thus, the project’s gender mainstreaming strategy makes a concerted effort to consider intersectional factors in its targeting metrics, to ensure that female youth, female-headed households, and females living with disabilities are equitably represented in all project activities. Given this approach, gender data is expected to be collected beyond sex-disaggregation (i.e., number and % of female farmers disaggregated by age, household headship status, and disability status vs. number and % of male farmers disaggregated by age, household headship status, and disability status).

2 The Gender Action Plan (GAP) for the proposed project tackles gender inequalities across several priority areas in agri-climate adaptation and mitigation²⁵¹. These include closing gender gaps and existing inequalities in participation (at intra-household level and within producer organizations/cooperatives), workloads (prioritizing agricultural technologies and practices for adaptation and mitigation that reduce workloads and negative impacts on women), access and use of productive resources (such as agri-climate information, technologies, livelihood incomes, credit), and collective action (working with women’s groups as platforms for enhancing access, agency, and voice in climate-smart agriculture). In closing these gender gaps, the project will use Gender Transformative Approaches (GTA),²⁵² and more specifically, **Gender Action Learning Systems (GALS)** to address the underlying discriminatory socio-cultural and gender norms that perpetuate gender inequality and constrain women’s capabilities. This requires a culturally sensitive, multi-level approach that includes women and men in all their diversity.

3 The Gender Action Plan details how the goal of gender equality will be mainstreamed in two ways: participation in activities and the content of activities. Intersecting factors will be taken into account when targeting rightsholders and when designing and implementing activities. Women, female youth, female-

²⁴⁸ While the project considers intersectional socio-economic factors, it does not take a formal intersectionality approach as outlined in: FAO. 2022. Practical guide for the incorporation of the intersectionality approach in sustainable rural development programmes and projects. Santiago. <https://doi.org/10.4060/cc2823en>

²⁴⁹ Mazurana, D., Marshak, A., and K. Spears. (2023). Sex, age (and more) still matter: Data collection, analysis, and use in humanitarian practice. CARE report. https://www.care.org/wp-content/uploads/2023/03/Sex-age-and-more-still-matter_Final-report.pdf?mc_cid=88ea8613e0&mc_eid=1f4483a062

²⁵⁰ According to the Kenyan Constitution of 2010, youth are defined as people aged between 18-34; a revision from the 2007 National Youth Policy which had previously categorized youth as those aged 15-30. However, the United Nations definition of youth is 15-24 years, and the East African Community (EAC) defines youth as those between 15 and 35 years. The proportion of the youth aged 18-34 in Kenya, constitutes 25%, and those below 15 years make up 43% of the total population.

²⁵¹ Adaptation requires adopting specific practices to lessen climate change impacts, while mitigation deals with addressing the root causes of climate change (i.e., Greenhouse Gas Emissions).

²⁵² For a full list of publications by the FAO Joint Programme on Gender Transformative Approaches for Food Security and Nutrition see: <https://www.fao.org/joint-programme-gender-transformative-approaches/resources/publications/en>

headed households, and females living with disabilities all have specific quotas that are required for each activity. Mainstreaming in terms of participation will be ensured by setting the target proportion of participants from each of these intersections to be roughly equal to the relative population sizes at the national level: 50% women, (of which 33% are FHH-16.5% of total project population), 1% for women LWD, and 25% (12.5% of total project population) are female youth (aged 18-35 years)²⁵³. The male youth participation quota is 25% (25% of total project population) and 1% for men LWD (.5% of total project population). Mainstreaming in terms of content will be ensured by integrating gender transformative approaches, more particularly Gender Action Learning Systems (GALS)²⁵⁴ into project activities and training materials. These training materials include the “gender- and youth- specific” trainings, such as the Specialized Agriterra Training Programmes in Women’s Leadership and Youth Leadership, as well as more “general” training materials on agricultural production, markets, finance, etc.

4 For gender-responsive and transformative project activities to be successfully implemented and have impact, capacity-building, awareness, and sensitization among leadership and governance, at county level and within cooperative societies and producer organizations is required. Gaining the support from local political systems is crucial in engendering project activities that have the power to transform gender inequalities and social exclusions in agriculture. Thus, capacity building will be mandated for all county and in-country partners on the importance of tackling gender and social inequalities in agriculture in the context of a changing climate, and data collection and MEL teams will receive specialized training in the collection of gender, sex-and age-disaggregated data. Given that more than half of the targeted population will be female (50%) and males with intersectional vulnerability (20% youth status and 1% disability), the successful implementation of the Gender Action Plan requires half of the total project budget. The relationships among the project objectives, actions, indicators, targets, and baselines are as shown in the table below.

5 The day-to-day implementation of the GAP for the project will be led by the National Gender and Social Inclusion Specialists, who will be recruited by the project. Given the implementation of GALS and GTAs more broadly, knowledge and experience in GTA is a requirement in the TORs for these positions.

²⁵³ Based on 2020 estimates that 24.3% of female youth are not involved in education, training, or employment, compared to 15% of male youth.

²⁵⁴ FAO, IFAD and WFP. 2020. Gender transformative approaches for food security, improved nutrition and sustainable agriculture – A compendium of fifteen good practices. Rome. <https://doi.org/10.4060/cb1331en>

2. Gender Action Plan

Gender Action Plan of the project " Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc, Kenya"				
Project Expected Results	Indicators and Targets	Timeline	Responsibility	Budget
GCF Outcome level: Reduced emissions and increased resilience IRMF Core Indicators 1, 2, 4				
Core Indicator 1: GHG emissions reduced, avoided, or removed / sequestered	GHG emissions reduced, avoided or removed/sequestered Baseline: 0 Target: tCO2eq over 30 years	In 30 years	FAO (as Executing Entity), particularly National M&E Specialist	-
Core Indicator 2: Direct and indirect beneficiaries who took part in trainings	Number of direct and indirect beneficiaries, disaggregated by sex and age. Direct: Baseline: 0 Target: 572,000 farmers (50% women, 16.5% FHH, 2% PLWD and 25% youths) Among women, 1% women LWD, and 25% female youth (aged 18-34 years). The male youth (aged 18-34) participation quota is 25% and 1% for men LWD. Indirect: Baseline: 0 Target: 2,098,140 (1,049,070 women, 41,962 PLWD and 524,535 youths)	By end of Project Year 6	FAO and Agriterro (as Executing Entity), particularly National M&E Specialist	65% of total project budget
Core Indicator 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice	Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice. Baseline: 0 Target: 2,800	By end of Project Year 6	FAO (as Executing Entity), particularly National M&E Specialist	Included in project budget

<u>Supplementary Indicator 2.5</u> <u>Beneficiaries (female/male)</u> <u>adopting innovations that</u> <u>strengthen climate change</u> <u>resilience</u>		% of the participant beneficiaries adopted at least one of the innovations that strengthen climate change resilience. Baseline: 0 Target: 60%			By end of Project Year 6						FAO and Agriterro (as Executing Entity)		Included in project budget	
GAP Expected Outcomes, Activities and Targets												Cost		
Project activity	GAP activity	Indicator	Baseline	Target	Y 1	Y 2	Y 3	Y 4	Y 5	Y 6	Responsibility	Exclusively dedicated for GAP	Included in Project Activity	
Component 1 – Enabling local government support for adaptation and mitigation														
Expected gender outcome: Component 1 addresses capacity gaps in public agro-climate service providers to support women, men, and youth farmers in proactive adaptation and mitigation actions. The expected gender outcome of component 1 is Women, PLWD, and youth farmers can access and benefit from gender-responsive and socially inclusive public agro-climate services for CRLCSA. This means building capacity on both the local administration side (e.g., county governments and agricultural institutions, extension workers, cooperative leadership), and on the side of women, men, and youth farmers/cooperative members themselves to achieve gender equality by considering intersecting social factors. These activities will ensure women, PLWD, and youth have strengthened capacities on climate-resilient agriculture and mitigation actions, facilitate leadership and entrepreneurship, and leverage gender equality advancements in Kenya while addressing specific gender issues in the LREB.														
GAP Output 1.1: Women, PLWD, and youth’s participation, leadership, and decision-making in cooperative societies and value chains is strengthened via enhanced gender-responsive and socially inclusive local administrations.														
Cross-cutting	Mainstream GESI and integrate gender-sensitization into all training materials and trainings	Percentage of gender-responsive training materials and curricula designed and developed by a gender specialist	0	100%							International/ National Gender Specialists	12,000		
1.1.1 ²⁵⁵	Build institutional capacity within local administrations and cooperative leadership to develop gender equality and social inclusion action strategies extension methods to address gender and social inequalities to ensure equity in accessing knowledge.	Number of extension methods incorporating gender equality and social inclusion	0	14 (1 per county)							National Gender Specialists, FAO, and local administration		Already included in the project budget, GAP activity not requiring significant specific budget	

²⁵⁵ 1.1.1 Develop and deploy innovative and efficient extension methods for disseminating and demonstrating CRLCSA knowledge, technologies, and practices in gender-responsive and socially inclusive ways.

	technologies, and practices within their organizations.												
1.1.1	Build capacity within county-level extension officers on gender-responsive and socially inclusive methods for disseminating and demonstrating CRLCSA knowledge, technologies, and practices.	Percentage of county-level extension officers trained on gender-responsive and socially inclusive methods (e.g., -Choose extension methods that are inclusive – training the family together as a whole, even on crops/activities traditionally labelled “men’s” or “women’s” activities.) disaggregated by sex and age	0	100% (40% women)							National Gender Specialists, FAO, and county government		
1.1.1	Ensure women, youth, and PLWD have access to extension services/methods	Number of women, youth, and PLWD that have access to extension services/ method (measured in the baseline, mid-term, and end of project survey of cooperatives)	0	50% women, 16.5% beneficiaries from FHH, 20% youth, 2% PLWD of direct beneficiaries							National Gender Specialists and county government		Already included in the project budget, GAP activity not requiring significant specific budget
1.1.2 ²⁵⁶	Consult with women, youth, and PLWD on types of climate services and information needed during scoping.	Number of women, youth, and PLWD consulted (per county) under sub-activity 1.1.2.1	0	50% women, 16.5% FHH, 20% youth, 2% PLWD of total consulted							National Gender Specialists, FAO, local administration		Already included in the project budget, GAP activity not requiring significant specific budget
1.1.2	Disseminate climate information services that are accessible to women, youth, and PLWD (i.e., radio, videos, public campaigns)	Number of women, youth, and PLWD that have access to climate information services	0	50% women, 16.5% FHH, 20% youth, 2% PLWD of direct beneficiaries							National Gender Specialists, FAO, local administration		125,000

²⁵⁶ 1.1.2 Strengthen the dissemination of climate information and services to last-mile users including women, youth, and PLWD through cooperatives and Farmer Organizations.

1.1.3 ²⁵⁷	Ensure equitable representation of women, youth, and PLWD in the development and testing of methodologies.	Number of women, youth, and PLWD represented in the development and testing of methodologies (per county)	0	50% women, 20% youth, 2% PLWD of participants in the development and testing of the methodologies							National Gender Specialists, FAO		26,000
1.1.4 ²⁵⁸	Update databases with sex-and gender-disaggregated data to ensure that marginalized and invisible groups (i.e., women living in male-headed households, widows, PLWD) are included in updating datasets.	Percentage of databases upgraded with gender, sex-and age-disaggregated data	0	100%							National Gender Specialists, FAO, local administration		Already included in the project budget, GAP activity not requiring significant specific budget
1.1.5 ²⁵⁹	Engage women, youth, and PLWD in participating in the selection, development, and testing of methodologies	Number of women, youth, and PLWD participating in the selection, development, and testing of methodologies (per county)	0	At least 50% women, 20% youth, 1% PLWD participants in each climate solution workshop							National Gender Specialists, local administration		Already included in the project budget, GAP activity not requiring significant specific budget
1.1.5	Mobilize women's and youth groups in generating climate solutions and county-level advocacy. (1.1.5)	Number of women's and youth groups involved in generating climate solutions (per county)	0	At least 2 women's groups and 2 youth groups per county (28 groups total)							National Gender Specialists, local administration		
1.1.6 ²⁶⁰	Ensure knowledge and lessons learned are accessible to women, youth, and PLWD (1.1.6)	Number of knowledge sharing events targeting all types of groups	0	15 knowledge sharing events. And at least 50% of the participants are women, 20% youth and 2% PLWD							National Gender Specialists, local administration		Already included in the project budget, GAP activity not requiring significant specific budget

²⁵⁷ 1.1.3 Develop and test methodologies for decentralized carbon accounting

²⁵⁸ 1.1.4 Upgrade and update agricultural databases, crop and productivity datasets, cooperative census

²⁵⁹ 1.1.5 Assess local climate change impacts and eligible climate solutions for the agriculture sector

²⁶⁰ 1.1.6 Share knowledge and lessons learned through existing platforms

Component 2: Sustainable resilient agricultural landscapes												
Expected gender outcome: Component 2 aims to reduce GHG emissions from the AFOLU sector and enhance resilience of ecosystems through the development, implementation, and successful monitoring of climate resilient and low-carbon management plans. <u>The expected gender outcome of component 2</u> is to mainstream gender equality and consider intersecting social factors into planning content and process of co-developing, implementing, and monitoring gender-responsive and socially inclusive agricultural landscape management plans. This will be achieved through consultations and inclusion of women, PLWD, and youth in the development of the landscape management strategy and implementation plan and building capacity among county-level, regional, and national officials on the importance of mainstreaming GESI content and creating monitoring mechanisms to support the successful implementation of GESI goals in landscape management.												
GAP Output 2.1: Agricultural landscape management strategies have robust gender equality and social inclusivity content and commitments, and are co-developed, implemented and monitored in ways that are gender-responsive and socially inclusive.												
Cross-cutting	Build capacity among county-level officials on the importance of creating monitoring mechanisms to support GESI goals.	Number of county officials trained on creating monitoring mechanisms to support GESI goals disaggregated by age and sex	0	At least 4 officials per county (56 total), including 30% women							National Gender Specialists, FAO, local administration	Already included in the project budget, GAP activity not requiring significant specific budget
2.1.1 ²⁶¹	Ensure women, youth, and PLWD are consulted in the development of the landscape management strategy and implementation plan, to ensure their specific needs are addressed. (2.1.1)	Number of women, youth, and PLWD consulted	0	50% women, 16.5% are participants from FHH, 20% youth, 2% PLWD of the participants in the consultation							National Gender Specialists, FAO, local administration	Already included in the project budget, GAP activity not requiring significant specific budget
2.1.1	GESI concerns mainstreamed in LMS by the county-level Gender and Youth leads in cooperation with county-level leadership (2.1.1).	Percentage of gender-responsive training materials and curricula designed and developed by a Gender specialist	0	100%							National Gender Specialists, FAO, local administration	Already included in the project budget, GAP activity not requiring significant specific budget

²⁶¹ 2.1.1 Develop a country climate-resilience and low-carbon agricultural landscape management strategy and implementation plan, including improved watershed management, land use planning, reforestation, and natural regeneration

2.1.2 ²⁶²	Ensure women, youth and PLWD will be consulted and involved in implementation of landscape management (2.1.2)	Number of women, youth, and PLWD engaged and involved	0	50% women, 16.5% participants are FHH, 20% youth, 2% PLWD of participants							County governments		Already included in the project budget, GAP activity not requiring significant specific budget
2.1.2	Create monitoring/feedback mechanisms so if landscape management plans are not enhancing gender equality and social inclusion in their implementation, these inequalities can be prioritized and addressed (2.1.2).	Percentage of landscape management plans with iterative gender-responsive monitoring/feedback mechanisms to address gender equality and social inclusion in implementation	0	100%							National Gender Specialists, FAO, local administration		
Component 3: Resilient livelihoods Expected gender outcome: Component 3 aims to increase smallholders' (including women, PLWD, and youth) climate resilience and production of commodities using climate-resilient, low carbon technologies. Across the 6 value chains targeted in the LREB, the key barriers and constraints facing women, PLWD, and youth in using climate-resilient, low carbon technologies, include: cultural dynamics that undermine women, PLWD, and youth's business aspirations, low levels of formal education and limited technical skills, limited access to productive assets and resources, limited access to finance, and limited access to networks and information. <u>The expected gender outcome of component 3</u> is for women, PLWD, and youth's adoption of CRLCSA production and processing practices to be enhanced by closing gender gaps and social inequities in productive resources and assets, networks and information, technologies and technical skills, and risk reduction mechanisms. The following activities will contribute towards this gender outcome: Ensuring gender equitable and socially inclusive CRLCSA production/processing assets and training among smallholder farmers, farmer organizations, and associations; Increasing access to productive resources and assets, particularly among women, youth, and PLWD using GESI strategies (e.g., Increase bundled services that provide women with insurance, credit, inputs, and technical advice, Awareness creation of agro-dealers/cooperatives on gender and inclusivity to improve access of their services and products to women, Targeting manufacturers of mechanization to tailor to women's needs and design gender friendly tools and equipment); and Prioritizing women and youth-led cooperative development. Targeted leadership programs for women and youth (e.g., AgriTerra Female Leadership and Youth Leadership Masterclass) will capacitate and empower farmers to contribute to and benefit from enhanced public agro-climate services.													
GAP Output 3.1: Women, PLWD, and youth's adoption of CRLCSA production and processing practices is enhanced by closing gender gaps and social inequities in productive resources and assets, networks and information, technologies and technical skills, and risk reduction mechanisms.													
Cross-cutting	Assign gender focal points in project coordination and implementation mechanisms	Percentage of project coordination mechanisms with an assigned gender focal point and Disaggregate the number of gender focal	0	100%							Co-led by National Gender Specialists, FAO, with input from local administration		Already included in the project budget, GAP activity not requiring

²⁶² 2.1.2 Implement and monitor climate-resilience and low-carbon landscape management plans

		points by sex and age																significant specific budget
Cross-cutting	Collect qualitative information on progress of gender activities	Stories of gender dynamics change from beneficiaries, monitor changes in norms and attitudes among women and men (working at different nodes of the value chain), narratives of how women, men, and youth are engaging in GALS and changing behavior in different areas	N/A	N/A														
3.1.1 ²⁶³	Ensure women, PLWD, and youth have access to CRLCSA assets (including inputs, materials, and equipment) and training through the FFS Approach (3.1.1)	Number of women, PLWD, and youth that have access to CRLCSA assets and training including number of female-headed households	0	50% women, 16.5% beneficiaries are from FHH, 20% youth, 2% PLWD of beneficiaries														1,000,000
3.1.1, 3.1.2	Ensure that CRLCSA FFS training materials and dissemination channels are gender responsive or gender transformative (3.1.1, 3.1.2)	Percentage of gender-responsive FFS training materials and dissemination channels designed and developed by a gender specialist	0	100%														Already included in the project budget, GAP activity not requiring significant specific budget
3.1.1	Ensure that the TOR, design, and development of questionnaire for baseline, mid and end-line surveys integrate inclusion, gender considerations	Percentage of gender-responsive tools, surveys, and questionnaires designed and developed by a gender specialist	0	100%														Already included in the project budget, GAP activity not requiring significant

²⁶³ 3.1.1 Deploy CRLCSA production/processing assets and training to smallholder farmers, farmer organizations, and associations.

3.1.1	Ensure gender considerations in the review/update of selection criteria of farming HH beneficiaries	Percentage of sets of gender-responsive selection criteria with due consideration of gender and social issues	0	100%							National Gender Specialists with support from FAO		specific budget
3.1.1	Training of trainers/facilitators on GESI for FFS	Number of staff trained on GESI to facilitate gender interventions in FFS disaggregated by sex and gender	0	100%							Led by National Gender Specialists, FAO, and local administration		
3.1.2 ²⁶⁴	Ensure each cooperative has a trained gender and youth focal point to document ongoing challenges in reaching, benefitting, and empowering women, youth, and PLWD, and that there are appropriate mechanisms (including adoption of GESI strategies ²⁶⁵) within the cooperative to prioritize addressing these challenges.	Percentage of cooperatives with a trained gender and youth focal point	0	100%							National Gender Specialists, Agriterria, FAO, and local administration		Already included in the project budget, GAP activity not requiring significant specific budget
3.1.2	Ensure peer-to-peer networks and exchanges include women, PLWD, and youth. (3.1.2)	Number of women, PLWD and youth that are involved in peer-to-peer networks and exchanges	0	50% women, 20% youth, 2% PLWD of direct beneficiaries participated in the peer-to-peer network							National Gender Specialists, Agriterria, FAO, country governments		

²⁶⁴ 3.1.2 Disseminate CRLCSA technology, knowledge, and assets to cooperative members through peer-to-peer networks and exchanges

²⁶⁵ By GESI strategies, we refer to approaches that address the root causes of gender and social inequalities, as well as the impacts of these inequalities.

3.1.2	Ensure women, PLWD, and youth have access to CRLCSA technology, knowledge, and assets (3.1.2)	Number of women, PLWD, and youth that have access to CRLCSA technology, knowledge, and assets including inputs, materials, equipment, supplies, and/or cooperative infrastructure.	0	50% women, 16.5% beneficiaries are from FHH, 20% youth, 2% PLWD of direct beneficiaries							National Gender Specialists, Agritererra, FAO, local administration		
3.1.3 ²⁶⁶	Ensure cooperatives and other business units have gender equality and social inclusion strategies in place to reach women, PLWD, and youth. (3.1.3)	Percentage of cooperatives and other business units that have gender equality and social inclusion strategies in place to reach women, PLWD, and youth	0	100%							National Gender Specialists, Agritererra, FAO, local administration		Already included in the project budget, GAP activity not requiring significant specific budget
3.1.3	Ensure women, PLWD and youth will be invited to join existing cooperatives.	Percentage of women, PLWD, and youth among farmers that have joined existing cooperatives.	0	50% women, 20% youth, 2% PLWD of the newly joined cooperative members							National Gender Specialists, Agritererra, FAO, local administration		
3.1.4 ²⁶⁷ 3.1.4.1	Training of project personnel on GTA, SEAH (Sexual Exploitation, Abuse and harassment) & GBV (Gender Based Violence) and the FAO Grievance and Redress Mechanism (GRM) to handle such incidents and ensure safe working conditions for women, PLWD, and vulnerable groups. (This involves training of project	Percentage of project personnel trained on GTA, SEAH, GBC, and project GRM disaggregated by sex and age	0	100%							FAO, supported by National Gender Specialists	29,658	

²⁶⁶ 3.1.3 Support smallholder farmer aggregation into cooperative and other business units as climate risk reduction and sharing mechanisms.

²⁶⁷ 3.1.4 Support improvements in social inclusion and women's meaningful participation in CRLC value chains

	and county officials on gender issues AND increasing awareness of women, PLWD, and vulnerable groups on their rights).																	
3.1.4 3.1.4.2	Deliver Women's and Youth Leadership trainings and GALS awareness raising to Smallholders	Number of female leaders trained disaggregated by age	0	84 (6 cooperatives per county)										FAO, supported by National Gender Specialists				
3.1.4 3.1.4.2	Deliver Women's and Youth Leadership trainings and GALS awareness raising to Smallholders	Number of women trained disaggregated by age	0	50% women, 20% youth, 2% PLWD of direct beneficiaries										FAO, supported by National Gender Specialists	160,000			
3.1.4 3.1.4.3	Monitor gender objectives through surveys and studies (WEAI)	Women's Empowerment in Agriculture Index Indicators		100%										FAO, supported by National Gender Specialists	135,099			
Project Component 4: Scaling through CRLCSA market and finance																		
Expected gender outcome: Outcome 4 aims to increase access to finance as a means of upscaling business and CRLCSA practices. <u>The expected gender outcome of Component 4</u> is two-fold: 1) for Women, PLWD, and youth to have increased access to markets and profitability of climate smart, low carbon sustainable agricultural products; and 2) Women, PLWD, and youth and their organizations are economically empowered through increased access to gender-responsive and socially inclusive financial products that support climate-resilient and low-carbon growth. The following activities towards this gender outcomes include increasing information and capacity for women, PLWD, and youth to access markets and trade opportunities; increasing access to credit and financial services for women, PLWD, and youth, and engaging with private finance lenders on how to strengthen their targeting of women and socially marginalized groups.																		
GAP Output 4.1: Women, PLWD, and youth have increased access to markets and profitability of climate smart, low carbon sustainable agricultural products.																		

4.1.1 ²⁶⁸	Ensure women, PLWD, and youth can work with buyers and aggregators to increase demand and market opportunities. (4.1.1)	Number of women/PLWD and youth-led organizations who have new partnerships/agreements with buyers	0	30% women, 15% youth, 0.5% PLWD of the beneficiaries who established new partnership/agreements							National Gender Specialists, FAO, local administration		
4.1.1	Ensure women, PLWD, and youth have equal opportunities to organize and participate in trade fairs, marketing events, awareness campaigns, and monitor markets. (4.1.1)	Number of women, PLWD, and youth attending these fairs and events	0	Out of the totally number of participants joined trade fairs, marketing events, and awareness campaigns, 50% women, 16.5% FHH, 20% youth, 2% PLWD							National Gender Specialists, FAO, local administration		450,000
4.1.2 ²⁶⁹	Ensure women, PLWD, and youth have equal access to certification and labelling schemes. (4.1.2) (also includes the simplification and increasing the information flow of CRLCSA for more women, PLWD, and youth to understand and tap into this opportunity.)	Number of women, PLWD, and youth have access to certification and labelling schemes (e.g., organic certification, fairtrade certification, TBD certification)	0	50% women, 16.5% of FHH members, 20% youth, 2% PLWD of the total number of beneficiaries who have access to certifications							National Gender Specialists, Agriterro, FAO, local administration		Already included in the project budget, GAP activity not requiring significant specific budget
GAP Output 4.2: Women, PLWD, and youth and their organizations are economically empowered through increased access to gender-responsive and socially inclusive financial products that support climate-resilient and low-carbon growth.													
4.2.1 ²⁷⁰	Make improved and/or new financial products gender responsive to the needs of women and women's groups (4.2.1)	Percentage of improved and/or new financial products that are gender responsive	0	100%							Financial institutions supported by National Gender Specialists, FAO, local administration		80,000

²⁶⁸ 4.1.1 Work with buyers and aggregators to increase demand and market opportunities for CRLCSA commodities

²⁶⁹ 4.1.2 Increase access to various certification and labelling schemes such as FairTrade or GlobalGap

²⁷⁰ 4.2.1 Develop gender-responsive and socially inclusive private finance tools, procedures, and products to promote the upscale of CRLCSA value chains.

4.2.1	Prioritize technical assistance via trainings to financial institutions on gender and inclusiveness, particularly prioritizing the use of sex and age disaggregated data and evidence in key decisions while designing and delivering financial services.	Number of technical assistance trainings	0	At least 2 trainings per each financial institution							National Gender Specialists, FAO, local administration		
4.2.2 ²⁷¹	Engage with women smallholders and women-led cooperatives on developing business plans (4.2.2)	Number of business plans co-developed with women smallholders and women-led cooperatives and Micro, Small, and Medium Enterprises (MSMEs)	0	3 business plan per county (14), minimum of 42 business plans							National Gender Specialists, Agriterra, FAO, local administration, financial institutions		900,000
4.2.2	Ensure business plans include strategies for social inclusion and gender-based access (4.2.2)	Percentage of business plans with strategies for social inclusion and gender-based access	0	100%							National Gender Specialists, Agriterra, FAO, local administration, financial institutions		
4.2.3 ²⁷²	Ensure women will be trained on carbon-based mechanisms and conservation finance	Percentage of women participation in the training	0	50%							National Gender Specialists, Agriterra, FAO, local administration		200,000

²⁷¹ 4.2.2 Support smallholders and their business units in the development of bankable business plans, with particular focus on social inclusion and gender-based access.

²⁷² 4.2.3 Facilitate smallholders access to financial incentives schemes for agroforestry